

Transitions in Undergraduate Mathematics Education

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Transitions in Undergraduate Mathematics Education

Edited by Michael Grove, Tony Croft, Joe Kyle & Duncan Lawson

UNIVERSITY OF
BIRMINGHAM



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Edited by Michael Grove, Tony Croft, Joe Kyle & Duncan Lawson

With a foreword from Professor Dame Celia Hoyles & Professor John Blake



**UNIVERSITY OF
BIRMINGHAM**

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Michael Grove, Tony Croft, Joe Kyle & Duncan Lawson

January 2015

Foreword

It gives us great pleasure to write a Foreword to this important book on *Transitions in Mathematics Undergraduate Education*, edited by Michael Grove, Tony Croft, Joe Kyle and Duncan Lawson. As educators passionate about teaching our subject, mathematics, we start by saying how delighted we have been, not only to have the opportunity to read this excellent book, but to write about our combined reflections. There is no doubt we can all learn much from this cadre of experienced authorities who provide a working guide to many of the issues that face our undergraduates.

The development of mathematics and its applications has been a dominant factor in the development of civilisation from the earliest of times through to the present, ranging from the deeply abstract and theoretical through to the more utilitarian approach in many of the applications, although also recognising that many enormously beneficial and financially rewarding developments involve the deepest mathematics.

As lecturers, tutors and teachers, we are in a privileged position to work with young minds to support their mathematical growth. The transmission of our excitement, knowledge and understanding to our students is a complex challenge that demands considerable expertise, especially given the different mathematical backgrounds of our students, their diverse mathematical needs and career goals. Students will inevitably face barriers to their learning of mathematics and this is where this volume is invaluable with its pragmatic and practical advice and excellent reference source.

The transition from school to university has long been the subject of anxiety – for students and faculty alike. In the past, the focus of scholarly endeavour was on the shift in the nature of mathematics itself towards more rigour, axioms and proofs. Now the attention is broader, including changes in pedagogy in the face of larger student groups, and the challenges of access to many courses, due to demanding entry conditions, cost and geography. The book discusses the changing and challenging nature of the demands of UK higher education more generally, motivating students, supporting students at the various transitions they face, becoming independent learners through, for example, group work, problem solving, modelling, and taking on particular subject areas such as learning statistics. Other issues discussed are the role of technology-led teaching, employability, teaching support for students with disabilities, accessible learning environments, gender issues, learning and developing new ideas from the literature and embedding a scholarly

approach into teaching. Thus the chapters of this book return to many debates and arguments but take some of them further.

The contributions range from helpful summaries of recent developments in mathematics education to more specific reflections on current issues and challenges, concerning pedagogy, methods of feedback, and how to engage a more diverse student population. The authors come from different disciplinary backgrounds themselves, so the reader is inevitably confronted with new research perspectives. Taking a new perspective can lead to rethinking past assumptions, around, for example, the use of technology, giving lectures, issues of social justice and taking account of the voice of the student in order to encourage both female and male undergraduates in mathematics. These are challenges for all subjects' teaching, but maybe particularly acute for mathematics.

In summary, this book is a wonderful read, full to overflowing with ideas, information and resources for lecturers, tutors and teachers in mathematics. Many of us will note and reflect upon the observations and feel reassured that many of our colleagues have had similar experiences. We strongly encourage you to read this volume thoroughly, perhaps later 'dipping' into it to refresh your memory or to allow you to reflect on developments in your own teaching, providing a broader perspective to your own efforts.

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Chapter 1: The changing nature of mathematics within UK higher education

Joe Kyle

Introduction

The word mathematics comes from the Greek *μάθημα* (*máthema*), which, in the ancient Greek language means, quite simply, “*that which is learnt*” and in modern Greek just “*lesson*”. Heath (1921) suggests that it was the Pythagoreans who introduced the distinction between those who had acquired the theory of knowledge in its most complete form, the mathematicians, and those who knew only the practical rules. The Liddell-Scott-Jones Greek-English Lexicon (LSJ, 2014) gives both the original meaning but also indicates that Plato and others used it in more restricted terms for arithmetic, geometry, and astronomy. Thus from the outset, knowledge and the learning of that knowledge were deeply intertwined.

More recent definitions of mathematics include the famous example:

“Thus mathematics may be defined as the subject in which we never know what we are talking about, nor whether what we are saying is true.”

Russell (1917)

And Pierce offered:

“Mathematics is the science that draws necessary conclusions.”

Pierce (1882)

All indications are that he was referring to a very wide interpretation here; much wider than the versions given by Russell.

Mura (1993) asked a number of university teachers to define mathematics. There is a bewildering variety ranging from the touchingly romantic “*Le grand amour de ma vie*” to the ineffable “*Eh?*” and the beautifully self-referential “*Mathematics is what mathematicians*

do!” which does not sound so absurd when one reflects on the extent to which mathematics has changed, and continues to change, over recent times. It is unlikely that Russell would have foreseen ethnomathematics, financial mathematics or mathematical biology as branches of the discipline he defined in 1917. The Mathematics Subject Classification, published by the American Mathematical Society, has grown from 42 pages in the 2000 edition to the 47 pages of the current version issued in 2010. This spectacular growth and glorious diversity is a matter for celebration of course, but it certainly raises issues for anyone attempting to pin down ‘mathematics’ to a definition that will endure for any length of time.

Does any of this matter? Well, in the context of this book it does. Appleby and Cox (2002) in their discussion on transition, mention that one of the sources of confusion at transition is entirely a matter of definition. We must therefore continually remind ourselves that the word ‘mathematics’ means different things to the different constituencies in and around higher education and that this must be addressed if we are to mitigate the various problems that arise in transition.

A century (or so) of change

It is argued later that transition, as an area of concern as discussed in this book, is the culmination of a number of factors that have their roots over the last hundred years or so. In 1913, the English mathematician G.H. Hardy received a strange letter from an unknown clerk in Madras, India – Srinivasa Ramanujan – and one of the most romantic and poignant episodes in the recent history of our discipline was about to unfold (Hardy, 1940).

But in Scotland, the Edinburgh Mathematical Society at the newly-instituted colloquium, was wrestling with (inter alia) the new theories of relativity and non-Euclidean geometry, and was experiencing its own reflections on the undergraduate experience:

“It was indeed a grand week. Each lecturer surpassed himself; and every one of the audience enjoyed himself or herself to the full. Professors and lecturers of world-wide reputation renewed their student days, and sat and took notes with humble zest. Professor Steggall, in moving a vote of thanks to the lecturers and to the organisers of the colloquium, hoped that this experience of taking notes for one brief week would make us feel more sympathetic with the unfortunate students who had to do the same for weeks and months.”

Knott (1913)

Incidentally, there is no indication that mathematicians, any more than others, had any insight into the impending cataclysm that would soon engulf Europe. Once the Great War was underway, there is some evidence that, whereas the Germans saw some value in deploying mathematicians for their skills, the British saw no reason to use them other than as ‘trench fodder’ (Aubin & Goldstein, 2014).

Other far-reaching changes were underway around the start of the 20th century. The few ‘ancient’ universities, having already seen Durham and London created earlier in the 19th century, were to be matched in number with the arrival of ‘redbrick’ institutions in the great provincial cities. Thereafter numbers have increased with two notable jumps in the 1960s and the 1990s. They are rising still and just the sheer increase in the numbers of

undergraduates was always going to have major consequences for how subjects are studied and taught. But history and heritage still play a part: for example, some claim that in the UK the current role of the External Examiner (not a system observed in all developed nations) owes its origins to the arrangement that, when the Durham University was granted a Royal Charter, Oxford examiners were appointed to ensure equivalent standards.

The first Professor of Mathematics at the new University of London was Augustus de Morgan, who in some ways might be described as the founding forefather of Mathematics Support as discussed in Chapter 4. His *On the Study and Difficulties of Mathematics* was written in 1831 to support “*the student of Mathematics, who has not the advantage of a tutor, ...*” As the title suggests, this was not a run of the mill textbook, of which there were many, but includes many remarks on educational questions associated with the mathematical topics. He was an excellent teacher and introduced the method of, after each lecture, giving out a number of related problems and examples. Students would hand these in and they were returned, with revisions, before the next lecture. In de Morgan’s opinion, a thorough comprehension and mental assimilation of great principles far outweighed in importance any merely analytical dexterity in the application of half-understood principles to particular cases (MacFarlane, 1916). Here we see a major change in pedagogy. Students, who previously would have ‘read’ a subject – spending most of their time and energy reading and digesting what was read, guided by a tutor – were now engaged in a teaching-learning-feedback cycle – not that anyone would have described it as such then!

Influential in leading change in the teaching of geometry was the Mathematical Association (MA). In their Report on the Teaching of Geometry (MA, 1902) they had a slightly unlikely ally in Russell (1902) who wrote scathingly of Euclid in an article that, although elegantly written, prompts one to think that it is “*extremely harsh to criticise Euclid in the way that Russell does. As someone once said, Euclid’s main fault in Russell’s eyes is that he hadn’t read the work of Russell.*” (MacTutor, 2007). It certainly is at odds with Russell’s more well-known observation that, aged eleven, meeting Euclid was as dazzling as first love.

Subsequently lamented as over-cautious, the 1902 Mathematical Association report nonetheless led the way for future reform. Reasons for the caution are discussed by Fujita and Jones (2011) who observe, rather wryly, that “*curriculum change processes are invariably complex and often subject to much politicking*” a sentiment that rumbles beneath almost every attempt at curriculum change in subsequent decades.

The Mathematical Association produced a steady stream of reports that offer, in themselves, a reflection of changing curricula. For much of the first half of the 20th century, these concentrated on geometry (often revisited), algebra and trigonometry. But in later decades we see, for example:

- *The Teaching of Analysis in Sixth Forms*; combining Analysis I and Analysis II in a single volume (1963).
- *Mathematics in Education and Industry* (1966).
- *Mathematics at the University* (1975).
- *An Approach to A-level Probability and Statistics* (1975).

- *Calculators Have Come* (1976).
- *Mathematics in a Multicultural Society* (1988).
- *Computers in the Mathematics Curriculum* (1992).

Admittedly, the Mathematical Association reports generally focus on activity in schools, but the titles do indicate the change of priorities over the years, a change that was mirrored in university mathematics curricula. However, as observed by Appleby and Cox (2002), many of the reports and commentators point to changes in school mathematics as a principal contributor to problems at the transition to university. In 1995 there was, in the words of *Tackling the Mathematics Problem* (LMS, 1995), “*unprecedented concern amongst mathematicians, scientists and engineers in higher education about the mathematical preparedness of new undergraduates.*” Now it is worth reminding ourselves that, since the days of Socrates each generation has been quite happy to bemoan the deficiencies of ‘youth today’. But the concerns of the London Mathematical Society were supported (possibly for the first time in these matters) by hard data gathered over years by Lawson and others (see for example Savage & Hawkes, 2000).

We would be remiss if we concluded this look at the 20th century without mentioning the impact of the introduction of Statistics into A-levels in 1984:

“Their content proved to be heavy for the time available and, with a much wider choice of subjects now available, the number of students taking A level Mathematics fell steadily. Alongside compulsory pure mathematics students now had a choice of statistics and mechanics questions. Many opted for statistics, finding it easier and relevant to them. One informal survey of 500 papers showed that, where there was a choice, 93% of the questions answered were statistics and only 7% mechanics.” Smith (2004) had already recognised “*that the introduction of Statistics and Data Handling may have been at the expense of time needed for practising and acquiring fluency in core mathematical manipulations.*”

Porkess (2008)

But to date, there has been no movement to implement Smith’s recommendations in this area.

Thus the 20th century ended with overcrowded curricula, a plethora of rival A-level examinations beleaguered amid accusations of grade inflation and a large, a diverse and competing family of institutions in higher education. How different from the position in the 1900s. Small wonder there are issues at the transition to university.

Current issues: demand, participation and expectations

Following *Tackling the Mathematics Problem* (LMS, 1995) the Engineering Council published *Measuring the Mathematics Problem* (Savage & Hawkes, 2000) which recommended, among other things, that there should be diagnostic tests on entry to university and that “*prompt and effective support*” be provided to aid those students found wanting by these tests.

There was also an increasing acknowledgement of a shortage of well-qualified graduates in mathematically-based disciplines. In its 2012 report *Jobs and Growth: the Importance of Engineering Skills to the UK Economy*, the Royal Academy of Engineering (RAEng, 2012)

calculated that at least an extra 10,000 science, technology, engineering and mathematics graduates are needed just to maintain the UK's current industrial position. Furthermore, in their 2013 report *In the Balance: The STEM human capital crunch* the Social Market Foundation (SMF, 2013) calculated that as a result of an increasingly ageing workforce around 100,000 vacancies will arise annually in jobs requiring degrees in STEM subjects and that there currently exists an annual shortfall of around 40,000 in the UK supply of STEM graduates. According to the 'state of the nation' report published by the Royal Society (RS, 2008), before the turn of the millennium there were approximately 56,000 students in England taking¹ A-level mathematics each year. Changes to the school curriculum and examination regime in the year 2000 proved to be disastrous for mathematics and numbers plummeted to around 44,000 in 2002. In 2005 there were 46,000. Many university departments struggled to find sufficient numbers of students to fill their courses and a number of university departments of mathematics, particularly in the post-92 universities, closed. Similar problems were experienced in other departments that relied on recruiting students with A-level mathematics – particularly engineering and physics.

Efforts to counter these problems were forthcoming. Building upon a pilot project (2000-2005) the government through the Department for Education and Skills established a national Further Mathematics Network (Stripp, 2007) which aimed to make it possible for school students across England to study Further Mathematics at A-level even when this qualification could not be offered directly within the students' own schools or colleges because numbers were too small or staff were not suitably qualified to deliver it.

In 2004, the then Secretary of State sought advice from the Higher Education Funding Council for England (HEFCE) on strategically important, but vulnerable, higher education subjects or courses. A range of disciplines were identified to be in need of support including chemistry, engineering, mathematics and physics. In response, a consortium of mathematical bodies developed the HEFCE funded proposal More Maths Grads (Grove & Lawson, 2006). This undertook, between 2006 and 2010, a range of pilot discipline-based interventions that were specifically designed to increase demand for, and to widen participation within, the mathematical sciences in higher education.

The Institute of Mathematics and its Applications established a website to promote further mathematical study by highlighting the career opportunities for those with mathematical qualifications. It shows mathematics in ways that school students may never have thought of. For example, it shows how quadratic equations and probability are linked to football and how logarithms are used in our analysis of earthquakes. Alongside this, it shows how mathematics can be fun, as students themselves will be able to perform – and understand – magic tricks (see Chapter 2 where Easson & McOwan develop this theme).

In 2009, the three-year National HE STEM Programme was established which, while addressing a broader agenda, integrated the four discipline-based interventions and worked to ensure their learning was transferred more widely and with a specific higher education focus. Through the Programme, universities were encouraged to adopt and embed a range of new

1 The Royal Society describes "total entries in mathematics A-level from all secondary schools and FE sector colleges in England." (RS, 2008)

approaches to working with school and college age students and many examples of effective practice are available in the mathematical sciences (see for example Grove & Jones, 2013).

As important as these positive initiatives are, it is equally important to ensure that every effort be made to ensure that talented students who choose to study mathematics are not turned-off the subject.

The under-representation of women in the mathematical sciences is receiving increasing attention. Despite advances in recent decades with a rising proportion of women earning bachelors, masters and doctoral degrees, there are still differences. In Chapter 17 Rodd looks at some aspects of the issue of gender and studying mathematics at university level. Others have written extensively about the ways in which mathematics is differentially experienced by women undergraduates and believe that institutional change and increased awareness of potential bias are ways of increasing the flow of talented women through the sciences. The report for the London Mathematical Society *Advancing Women in Mathematics: Good Practice in UK University Departments* (LMS, 2013) shows graphically how the proportion of women participating in mathematics drops off. For example, whilst 40% of candidates for A-level Mathematics in 2011 were female and 42% of those graduating in mathematics in 2010/11 were female, the proportions continuing with further study reduce substantially (32% Masters, 19% doctorate). To promote equity and adequate representation of women in the STEM subjects the Athena Swan scheme recognises university and departmental commitment to its agenda through Bronze, Silver and Gold awards. Many university mathematics departments are working towards these awards and it is becoming increasingly recognised that an Athena Swan award is vital for the image and continuing health of a department.

Those aspects of teaching, learning and support which actively address any gender issues through the provision of more opportunities and the enhancement of the student experience are likely to impact positively upon students of both genders. Young women who show an interest in mathematics need to be encouraged and not deterred, albeit inadvertently, from embarking upon further study or careers in mathematics.

Finally, we return to the situation in the schools. There remains in the UK a dire shortage of mathematics teachers. The report, *Making Mathematics Count* (Smith, 2004), noted:

“The serious magnitude of the current problem can be appreciated from the fact that to solve the problem of shortfall we would need to attract into teacher training over 40% of the annual UK output of mathematics graduates for each of the next several years.”
(Smith, 2004)

Able mathematics undergraduates need to be encouraged to consider teaching as a possible career. There are particular efforts in this respect. For example, the Institute of Mathematics and its Applications currently administers a scheme on behalf of UK government that awards scholarships to able candidates who will undertake recognised Initial Teacher Training (ITT) courses in England. All of us have a part to play to ensure potential teachers among our students finish their mathematics degrees with an enthusiasm for the subject that has been boosted and not depleted during their undergraduate studies.

Signposts to the Chapters

In Chapter 2, Easson and McOwan consider ways in which we might motivate and inspire future generations. After surveying a number of UK initiatives that looked in various ways to stimulate interest in mathematics, they then move on to describe current strategies that might assist in engaging learners from all demographics. In particular they offer examples of the use of conjuring and the role that modern celebrities might play in motivating and inspiring.

Pritchard, in Chapter 5, asks us to re-evaluate the oft-cited criticisms of the traditional lecture. This is a fascinating area for debate, as despite having attracted many enemies, the traditional lecture is proving remarkably resilient and particularly hard to kill off. Pritchard offers some convincing arguments that might go some way to explain this. As recently as 2013, we read of colleagues wrestling with the issues of lecturing to large classes in the UK (Moriarty, 2013) and elsewhere (Martschink, 2013). So, for the foreseeable future, particularly within the mathematical and cognate disciplines, new undergraduates will have to come to terms with the lecture as part of the transition to higher education and so Pritchard provides some pointers for managing this transition in a positive and constructive manner.

It is argued earlier in this chapter that mathematics support has been deployed in an *ad hoc* fashion since at least the time of de Morgan, but Mathematics Support (in title case) as an organised evidence-based intervention has its roots in the work of Croft, Lawson and others. As a part of the teaching and learning landscape it is greatly admired and the methods are widely adopted throughout the UK and beyond. In Chapter 4, Lawson makes a clear case for the important role support can play in retention and a number of other important areas that are key to securing successful transition.

The need to instil in students the skills and principles of independent learning has taken an increasingly central role in UK higher education, but like so many similar concepts, it can be a frustrating task to pin down exactly what is meant by the term. In Chapter 6, Walker provides a detailed discussion of the concepts surrounding independent learning. This is followed by a deep analysis of the implications. There are several models of practice and a detailed description of one particular scheme that has been in operation at the University of Manchester. There is no doubt that transition to independent learning is seen to be one of the main challenges both upon entry to higher education and between levels within the undergraduate curriculum and Walker's chapter is a valuable addition to the debate.

Of course an important vehicle for bringing a richer experience of other modes of learning is the use of group work. In Chapter 7 Challis explores two fundamental questions: why should we use group work and how best can we use group work with MSOR students. He then embarks upon a more nuanced discussion of the pedagogical issues surrounding group work and an exploration of various means of implementing it with our students.

Few deny the importance of feedback, but it is easier to say this than to describe, especially within the mathematical disciplines, exactly what constitutes effective feedback, how best it should be delivered and how we may encourage students to make constructive use of feedback. In Chapter 12, Robinson tackles these questions and, drawing upon a recent

National HE STEM Programme project, presents a number of interesting conclusions on a number of different forms of feedback.

The teaching of statistics crops up in many university disciplines, many of them loosely related, if at all, to those in mathematics. In Chapter 10, Hewson focuses on the issues confronted by those students following a degree in the mathematical sciences – mathematics majors to use Hewson's term. As he points out, much of the literature concerning the pedagogy of statistics concentrates on learners using statistics as a tool in their main discipline. For this reason, and because of the number of contemporary references, this is a welcome and valuable addition to the literature.

There is a well-documented increase in the number of neurodiverse students entering higher education and, but not enough, good practical resources for those working with such students within the mathematical sciences. In Chapter 15, Trott discusses four neurodiversities: dyslexia, dyspraxia, attention deficit hyperactivity disorder (AD(H)D) and Asperger's syndrome. After reminding us of the legal framework that governs the area and providing appropriate definitions she goes on to discuss relevant strategies and provides an abundance of sound practical advice, interspersed with a number of vignettes that bring life and reality to the discussion. Trott's chapter concludes with a checklist that will be a useful resource for both institutions and individuals.

The transition that most often gains attention is that from pre-university study (often school) to first year undergraduate programmes. In addressing this issue, there is a danger of displacing many of the problems to later stages of the curriculum. Croft and Grove, in Chapter 13, provide a well-researched analysis of the problems, especially those of demotivation, that bedevil transition from first to second year. Schemes such as peer assisted learning initiatives are documented and analysed. Bringing together findings from a number of very recent studies, the authors offer strong evidence which colleagues may reflect upon when considering what they might, in practice, do to alleviate these particular transitions.

Even with all the caveats that should be mentioned, a simple search using a propriety search engine leads us to believe that the word 'employability' would hardly be recognised 100 years ago, certainly in discussing graduates. Now it is seen as an essential ingredient. And we can no longer assume (although a few still do) that study of mathematical topics per se addresses the employability question. In Chapter 14 Hibberd and Waldock, both of whom have worked intensively in this area, provide detailed analyses of the issues and tested strategies for the development of these skills in the mathematical curriculum.

To paraphrase Mogens Niss, teaching practice uninformed by mathematical education risks becoming narrow-minded and inefficient; mathematical education that ignores practice risks becoming barren. The two chapters by Treffert-Thomas and Jaworski and Williams offer the insights of contemporary experts in mathematical education. Williams presents findings and reflections that are informed by the large-scale Transmaths project that investigated students' trajectories in and through mathematics programmes from compulsory school, through college to higher education. Treffert-Thomas and Jaworski provide the reader with an up-to-date and informed tour of the existing literature in mathematical education, chosen with an eye to enhance practice in higher education. At

the heart of their chapter is a substantial report on a research study into the teaching of linear algebra.

The distinct, but related, question of embedding a scholarly dimension to the work of the university teacher is addressed by O'Donoghue and Fitzmaurice. Their principal contention is that a scholarly approach to university teaching will enhance the overall experience. The various facets of the professional work of the university lecturer are explored. In some sense, the authors describe another type of transition, or journey as they put it, and a case study is presented for readers who are invited to reflect how this might inform their own practice.

As noted already, institutions have a legal requirement to ensure that students with disability do not suffer discrimination and enjoy equal access to programmes. But what exactly are the resources available and the interventions that are successful? In Chapter 16 Cliffe, drawing upon a wealth of experience, discusses specific measures to help those who cannot read standard print effectively, those with hearing impairment and those who cannot effectively use pen and paper. Moreover, the discussion takes great care to consider the special factors that come into play when mathematical disciplines are involved.

In the 1890s, Philippa Garrett Fawcett caused a stir by matching the performance of the top male student (Senior Wrangler) in the Tripos at Cambridge (Siklos, 1990). The male student was made a Fellow of St John's College and awarded the university prize for mathematics; for neither of these was Fawcett eligible. One hundred years later, this would be inconceivable. But there are still issues in gender equality to be addressed and Rodd, in Chapter 17, explores a number of the more nuanced factors. Written for university lecturers, it is punctuated with a number of student quotations and invites colleagues to reflect more deeply on any gender assumptions they may be making. Few now would assert, as they did in the 1890s, that women were mentally incapable of scaling the heights of mathematics, but Rodd reminds us that other less visible assumptions might be raising unnecessary (even if unintended) barriers to transition.

Almost every university in the UK proclaims that it produces graduates with well-honed problem-solving skills. But when pushed, few can give a robust definition, let alone explain how students can be helped to develop these skills. In Chapter 8, Hawkes tackles the issue head-on. After discussing the matter of definition, he considers what we might describe as a 'good problem'. Sprinkled with plenty of examples and case studies from current practice in UK universities, the discussion covers the relevance and value of problem-solving skills. Savage and Grove in Chapter 9, consider another form of problem-solving that has its roots in the essential skills that are required when modelling, especially in the sense that used to be applied to conventional classical mechanics.

The extent to which technology and particularly the World Wide Web has influenced teaching and learning could barely be imagined 100 years ago. And one imagines that further as yet unimagined changes will unfold over the next century. Sangwin in Chapter 11 presents a scholarly analysis of the general principles that should inform our use of technology, particularly in assessment. Taking a long historical perspective and an inclusive use of the term, we are invited to consider how the existence of technological tools may well lead us to re-examine assessment design and practice, whether or not this might involve technology.

Transition in transition

Leaving aside changes in content and teaching styles, there is one other big difference between now and 100 years ago: transition to higher education is much more complex and multi-dimensional now than it was in the past. Obviously, there is a natural apprehension that attends any change in life. Our contention is that there are much larger transitional issues, and that these are complex and multifaceted. There is, of course, the important transition into higher education, but there are also transitions at the other end of our programmes – transition into employment, research and the transition to active participation in society at large. Whatever our roles, we each face a transition of ‘professionalisation’: the late 20th century brought with it instruments of management and audit that are unlikely to disappear. Alongside these, changes to funding mechanisms, including the introduction of substantial student fees, add to the complexity of transition. And, of course, the unstoppable evolution of technology transforms all and everything.

A century ago, transition to undergraduate mathematics followed a straightforward path. Oxford and Cambridge demanded Euclid as an entry requirement; almost every schoolmaster was a graduate of Oxbridge, and therefore taught Euclid. And thus a stable system was established for most of the 19th century until the first expansion of the university sector in the early part of the 20th century. As programmes developed and expanded, transition to, within and from undergraduate mathematics has confronted us with greater challenges and it is the purpose of this work to help meet those challenges.

Transition can be thought of as akin to transshipping at sea: in calm waters, although some care is needed, the task should be quite feasible and may not even require much thought! In the presence of greater turbulence, one must be much more sure-footed and take much more care if there are to be no casualties. Reports referred to above indicate that we live in relatively turbulent times and all need to work together to address the issues that surround transition. The chapters that follow collect together expertise from across the UK higher education sector and this chapter has already provided a brief indication of the menu on offer but, as with all tasters, this is certainly not as satisfying as the full meal. So turn a page or so and – *bon appetit*.

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Chapter 2: Using mathematics to motivate and inspire future generations of learners

Vivien Easson & Peter M. McOwan

Abstract

This chapter contains a review of selected initiatives from the last ten years in the UK to encourage pupils to study mathematical subjects at university. These range from major national funded programmes to individual efforts online and through social media and phone apps. A recurring theme in initiatives and approaches discussed is that of achieving a balance between relevance, imaginative play and critical thinking. We explore lessons learned and legacy resources to allow others to pick up and use their outputs, ideas and findings. We also summarise key outcomes from the growing body of empirical research examining some misconceptions around the subject, its cultural profile and the reasons why pupils choose to study mathematics in higher education. The chapter also contains consideration of what might be termed 'mathemagical outreach' as an example of one form of motivating mathematical learning. Personal experiences of practitioners are included to allow readers an insight into effective deployment of good practice. The chapter concludes with some final thoughts and challenges to enhance activities in this important area.

Introduction: Still difficult, boring and irrelevant?

Making Mathematics Count (Smith, 2004), the report of the government initiated inquiry into post-14 mathematics education, considered factors influencing students' decisions on whether to continue their study of mathematics and gave the following summary:

"Factors influencing student choice are complex and not well understood, although certain themes emerge anecdotally from focus groups:

- *the influence of the teacher is clearly important; in particular, poor teaching is likely to turn students off mathematics;*

- *the perceived difficulty of mathematics relative to other subjects is also important both to schools (concerned with league tables) and to individual students (concerned with university entrance);*
- *separate from perceived difficulty, the content of the course may be perceived to be boring or irrelevant, or insufficiently stimulating or challenging;*
- *lack of awareness of the link between career options and subject choices may also play a role, both for teachers and students."*

Smith (2004:13)

Despite this negative list, between 2003 and 2012, A-level Mathematics numbers increased by 69%: from 50,602 to 85,714. A-level Further Mathematics numbers increased by 149%: from 5,315 to 13,223 (MEI, 2012). Higher Education Statistics Agency (HESA) statistics (HESA, 2013) show that, over the last five years, the number of full-time undergraduates within the mathematical sciences has increased by 28% from 22,770 in 2007/08 to 29,065 in 2011/12. The number of part-time undergraduates increased by 31% from 6,225 to 8,180 over the same period. These increases exceed the average: mathematical sciences students formed 1.9% of all undergraduate enrolments in 2011/12 compared to 1.6% in 2007/08. The proportion of all mathematical sciences qualifications gained by women has risen slightly from 38.8% to 41.1% over the same period.

In the following sections of this chapter we will explore what has happened in the past decade that may have contributed to overturning perceptions that mathematics is difficult, boring and irrelevant.

Choosing mathematics

One of the outputs of the More Maths Grads project (2006-2010), discussed in more detail later, was a report into the practice and provision of mathematics degree courses in higher education (Robinson, Challis & Thomlinson, 2010). This report contains results from a survey of mathematics students on arrival at university. These results show that career considerations and interest/ability in the subject are the most important positive factors when choosing to study for a mathematics degree. Mathematics is regarded as a subject that keeps career options open, as well as one which can lead to careers in the traditional areas of finance or teaching. Few students expect to use advanced mathematical ideas in their work, but they do expect to use key mathematical and analytical skills.

The report also discusses various barriers to starting a degree in mathematics, in particular that of local availability. It references Steele (2007) who showed that there are parts of the UK, namely Eastern England, Wales and the central and western parts of Southern England, with no mathematics degree locally available for students obtaining good (that is up to a grade B in A-level Mathematics) but not excellent qualifications. When this finding is aligned with the observation that students in the More Maths Grads survey whose parents work in skilled trades/service/sales or are not in paid employment are five times more likely to live at home than students whose parents work in managerial, professional or technical occupations, it shows that there are considerable accessibility barriers to higher education mathematics for some student groups. Robinson et al. (2010) point out that the Open University provides an excellent alternative to local study, as well as providing

flexibility of study for those who wish to combine it with a full-time job. The Open University is currently leading on the development of a large UK venture into massive online open courses (MOOCs), FutureLearn. We discuss MOOCs in more detail later. Such developments may provide further opportunities to extend access to mathematics learning in under-represented groups, particularly adult learners.

Gender and mathematics

Another perceived barrier to mathematical study is gender. In 2011/12 42.2% of undergraduate mathematics students were female, with 58.0% of undergraduates overall female. This is around the middle of the range for science, technology, engineering and mathematics (STEM) subjects: engineering & technology (14.3%), computer science (17.4%), physical sciences (42.6%), medicine/dentistry (57.6%), biological sciences (60.8%). Although the gender balance amongst mathematics undergraduates is not far from equality, the situation is rather different amongst university staff and professors:

“...girls make up 40% of A-level maths students (though fewer take further maths), and the proportion studying for a first degree in maths is even a little higher, at 42%. But the subsequent trajectory runs broadly downhill, with numbers falling away to just 29% of female researchers, 19% at doctorate level, and the 6% handful who bag a professorship.”

Ward (2013)

The International Review of Mathematical Sciences (EPSRC, 2010:48) noted that *“action about gender diversity is not a sufficiently high priority in the UK”*.

Many popular science authors have written about mathematics and gender. One typical approach is biographical, recording the struggles and successes of female mathematicians either historical (Osen, 1974) or modern (Murray, 2000). Others explain imbalances by proposing theories based, for example, on *“the priestly culture in physics”* (Wertheim, 1997) or *“the extreme male brain”* (Baron-Cohen, 2003). A consequence of past and current imbalances is that books on mathematics tend to reference a large number of male mathematicians: 125 men are listed in the index to *On the Shoulders of Giants* (Lines, 1994), but no women. *The Music of the Primes* (du Sautoy, 2003) manages to reference 11 women out of 261, but achieving this requires a detour specifically to mention Germain and Noether.

While some teenagers are influenced by research mathematicians or popular science writing, it is likely that many more are influenced indirectly and unconsciously by popular views of mathematics and by gender roles. Sociological gender theory suggests that gender need not be viewed as static – that a person is male or is female – but instead that different actions are considered masculine or feminine and the individual performing these actions is performing masculinity/femininity. For example, it may be perceived that a girl discussing cosmetics is performing femininity but a girl discussing mathematics is performing masculinity.

Mendick (2006) explores the interplay between performing gender and doing mathematics via a focus on discourses: *“structures of language and practice through which objects come into being”*. She explains that this:

“...produces a figure/ground switch, whereby the problem of gender and maths is not located within individual girls who make the “wrong” choices or who understand their performance “wrongly”, but in the discursive context of gender, schooling and maths. This context makes the patterns of identification that produce these girls’ choices and performances possible and indeed sensible.”

Mendick (2006:18)

Mendick (2006:156-163) suggests that mathematics needs “a makeover” which emphasises, for example, projects, play, imagination and uncertainty rather than “a certain, objective, external collection of absolute truths”. On a practical note, girls may feel uncomfortable displaying mathematical ability if they perceive it as unfeminine to do so, and outreach practitioners need to find ways of managing situations to reduce this discomfort, for example by encouraging collaborative working and a supportive environment, by emphasizing processes of working towards a solution rather than just achieving a correct answer and/or by drawing attention to a variety of people doing mathematics for many different purposes.

A more extensive discussion of issues relating to gender and mathematics can be found in Chapter 17 of this book.

More Maths Grads

More Maths Grads was a project funded by the Higher Education Funding Council for England (HEFCE) to develop, trial and evaluate means of increasing the number of students studying mathematics in higher education and of encouraging participation from groups of learners who have not traditionally been well represented in higher education (More Maths Grads, 2010). In the words of its Project Manager, Makhan Singh (2013):

“The outcomes from the project were above and beyond what anyone could have expected – a series of excellent legacy resources (multimedia and hard copy leaflets/booklets) were created for all maths teachers in secondary schools and FE colleges; and also for university Mathematics and Widening Participation/Outreach departments; all nicely packaged into a cardboard briefcase aptly titled ‘Maths in a Box’.”

Singh (2013)

The pilot areas included London boroughs Hackney, Newham, Tower Hamlets and Barking and Dagenham (Easson 2010, 2011) as well as Yorkshire and the West Midlands, all of which contain substantial areas of social deprivation. Piloted initiatives were taken up in Wales using funding from the Higher Education Funding Council for Wales (HEFCW) and later further promoted by the National HE STEM Programme (2009-12). Resources remained in demand after the end of the project and can be found via 'Maths in a Box' on the Maths Careers website¹.

A key lesson from More Maths Grads, and supported by National HE STEM Programme case studies (HE STEM, 2012), was that student ambassadors have a strong positive impact on encouraging school/college students to continue to study mathematics and/or

¹ <http://www.mathscareers.org.uk>

consider applying to university. Schemes such as the Undergraduate Ambassadors Scheme (UAS)² support universities providing a credit-bearing module in which students receive training and spend ten half-days in local schools/colleges. This placement meets a common requirement for entry into PGCE courses (namely time spent in schools) and thereby also improves students' employability. Of the 142 departments listed on the UAS site, 29 are mathematics departments. A more expensive option which many universities adopt is to pay an hourly rate to students to undertake such ambassadorial roles.

A major part of the More Maths Grads programme was the delivery of an extensive programme of outreach activities in schools. When delivering such events, it is essential to develop good relationships with teachers. The More Maths Grads team recommended events aligned with the curriculum and within the scheduled timetable. It is vital to use teachers' local understanding of their students when trying to engage them by appealing to 'popular' culture. For example, at the time of More Maths Grads, the Indian film actors Aishwarya Rai and Shahrukh Khan were easily recognised by students in Stepney Green (Tower Hamlets) but unknown four miles away in Stoke Newington (Hackney), where the best 'hook' was David Tennant (then starring in the TV show Doctor Who). Football analogies made in a classroom seemed to turn off as many students as they engaged.

What proved universally effective was emphasising that studying mathematics keeps career options open. Resources explaining how mathematical techniques are used in different jobs were popular. Similar free resources have since been developed for mathematics undergraduates, including the booklet *Where the maths you learn is used* (Easson & Yates, 2010) and a series of videos *Advice from employers to mathematical sciences students* (Mathscareers, undated).

Other significant UK initiatives

More Maths Grads provided a relatively short, time-limited injection of funding and staffing to complement a number of significant longer-term initiatives. The initiative which has done most to increase take-up of A-level Further Mathematics is the Further Mathematics Support Programme³ now funded by the Department for Education (Searle, 2012).

The National Centre for the Excellence in the Teaching of Mathematics (NCETM)⁴ was set up in 2006 to ensure all teachers of mathematics have access to high-quality continued personal development. An independent report commissioned in 2011 concluded that NCETM funding had provided a "*valuable external catalyst*" for teachers and schools to take action and develop mathematics education projects and form professional networks (Gouseti, Potter & Selwyn, 2011).

The initiative Computing at School⁵ promoted by Microsoft and Google, has been successful in developing, supporting and upskilling a national network of computing

2 <http://www.uas.ac.uk>

3 <http://www.fmnetwork.org.uk/>

4 <http://www.ncetm.org.uk>

5 <http://www.computingschool.org.uk/>

and mathematics teachers. It has also successfully lobbied government for changes to the national curriculum to include a more rigorous study of the mathematical fundamentals of computing and the teaching of programming in schools.

The NRICH Project⁶ based in Cambridge aims to develop interventions to enhance the mathematical experiences of all learners, over a range of age groups and abilities. One particular innovation is to support and train teachers wishing to embed rich mathematical tasks into everyday classroom practice. NRICH is part of the Millennium Mathematics Project (MMP)⁷, a highly successful mathematics education and outreach initiative for ages three to nineteen and the general public.

One particularly high profile and significant thread of work related to the raising of interest in mathematical sciences, is around the understanding and assessment of risk in life, championed by Professor Sir David Spiegelhalter⁸ and promoted via the MMP Risk Roadshow. Though mathematically challenging, an understanding of risk and how it can usefully be calculated is arguably one of the most important skills to allow the public to critically evaluate claims of safety or danger, and other stories in the media.

Using magic to engage mathematics learners

Conjuring tricks have proved a popular method to engage students with mathematical concepts for many years. The popularity of UK television shows focusing on street magic and high profile celebrity magicians has given this particular pedagogic approach a new mystique with pupils.

The uses of magical effects as a stimulus to deep learning of mathematical concepts range from computer science fundamentals⁹, number theory (Kruse, 2003), discrete mathematics (Simonson & Holm, 2003) and even fractals and Fermat's Last Theorem (Diaconis & Graham, 2011). For teaching statistics and probability, magic can provide an ideal and engaging point of entry for initiating classroom study (Mansfield, 1989). There are a number of seemingly magical effects, for example based round the exploration of apparent tests of extrasensory perception (ESP) or seeming cheating in gambling, which rely on statistical principles to operate. Lesser and Pearl (2008) identified magic as one of the 20 modalities of fun identified for motivating students learning statistics.

Outreach practitioners using magic in the classroom should consider cultural, age and social issues, particularly if choosing to perform tricks where claims of supernatural powers or strategies for cheating are initially employed to present the trick. Lesser and Glickman (2009) also emphasise the need to create a situation where the *"intensity of surprise is 'just right,' so that students will feel engaged to want to dig deeper to discuss and understand, rather than feel unduly tricked or deceived."*

6 <http://nrich.maths.org/>

7 <http://mmp.maths.org/>

8 <http://understandinguncertainty.org/>

9 <http://www.cs4fn.org/magic>

Card tricks can be used to introduce mathematical modelling, proof and verification to classroom audiences (Curzon & McOwan, 2008). Research into the mathematical properties of the perfect (or Faro) shuffle led to a patent for high speed dynamic computer memory (Morris, 1998). One critical element of a successful transition to university mathematics is the realisation that proof is necessary. In the context of a magical performance this can be motivated by needing to be certain that a trick will always succeed. This also carries over easily into discussion of bugs in computer software, especially safety critical software.

High-profile celebrity mathematicians/magicians can provide background human interest. Young (2011) describes how Stanford professor Persi Diaconis left home at 14 to tour with famous sleight-of-hand magician Dai Vernon. Having become increasingly interested in scams used by crooked gamblers, he completed a PhD in mathematical statistics at Harvard. He has since published extensively on randomness and probability. He also co-wrote *Magical Mathematics* (2011) with San Diego professor Ron Graham (a former professional juggler and trampoline acrobat): this book contains a history of mathematical magic/magicians.

Using magic to teach mathematics is not limited to self-working card tricks. Arthur Benjamin has achieved an international reputation in another field of mathematical magic, lightning mental calculations. Like Diaconis, Benjamin has strong academic credentials and currently works as a Professor of Mathematics at Harvey Mudd College in California. His contribution below gives a useful reflection into how teaching and magic performance skills can support each other:

“...as a professor of mathematics, my intended reaction from the audience is very similar to my goal as a magician. I want my students to say “That’s really cool! Why does that work?”
Benjamin (2013)

Using magic in the classroom to enhance the teaching of mathematics aligns with Mendick’s conclusion, cited earlier in the chapter, about the importance of imagination in mathematics. One mathematics teacher (Davidson, 2013) described the value of magic as follows:

“What is the point of magic? It is a chance for us to make believe. It is a chance for us to feel childish again. It offers people living in a world filled with “routine” and the “everyday” to believe just for a moment that perhaps the impossible is possible. Mathematics provides similar moments of beauty. The real difference with mathematical magic though is that once the secret is out now, the moment can still hold magic in it.”
Davidson (2013)

This section has presented the research evidence and personal insights of practitioners into the utility of appropriately used magic effects in the classroom to enhance deeper learning and engagement with a range of mathematical sciences. However no discussion of this topic could be complete without a mention of perhaps the most influential mathemagician of all time, the late Martin Gardner. Gardener’s books and Scientific American articles, Gardner (2005) and Gardner (1956), are frequently cited by mathematicians as a key formative inspiration.

Using new technology for learning mathematical sciences

Technological advances are providing new opportunities for learning and new ways in which learners can study. In this section, we discuss two particular forms of technology, apps and MOOCs, both of which have the potential to attract and enable more people to study mathematics to a high level.

Applications (apps) are cheap/free software for use on smartphones and tablet devices. This modern cultural phenomenon opens up new possibilities and challenges for the promotion of skills development in the mathematical sciences. People can learn both in and outside the classroom. Apps encourage self-paced interactive learning and provide a new way to attract prospective mathematical recruits.

The app market itself has interesting properties driving different types of software developers. The ease of entry and access to a global distribution platform means that rather than just being the exclusive domain of professional commercial developers, amateur developers can now enter the market. Apple with its proprietary iOS system, and Google with its Android system dominate the current market, but with different business models. Submitting an app to the iTunes store requires the software to conform to design guidelines and undergo quality assurance by Apple. In contrast the Android system is deliberately designed to be open source and submitting to Google Play is available without a quality checking stage. This has led to many enthusiasts learning programming and submitting apps, several of which have a mathematics tutoring element.

Pelton and Francis-Pelton (2011) discuss design rules for allowing the development of skills in mathematical procedures through play and exploration. There is a range of apps specifically targeting the development of functional maths skills in the very young child. These are often in the form of games that involve mathematical puzzle solving to progress through the levels. An example is the popular Mystery Math Town¹⁰ where players help a friendly and curious ghost rescue fireflies that are hidden in various rooms, which can only be entered by solving a range of mathematics puzzles.

Mathematics-based apps aimed at older users tend to focus on mathematical toolsets. For example, the WolframAlpha app¹¹ consists of a sophisticated mathematics and problem-solving engine supporting mathematics up to advanced calculus, statistics and physics. A middle ground is struck by Tunetrace¹², influenced by the Game of Life (Conway, 1970). In this app, simple but undisclosed rules control the sound and movement of graphical sprites as they traverse a graph formed by a processed photograph of the user's drawing. This approach combines creative play, music, mathematics and programming.

The prevalence of the web in today's information society is changing the access to and focus of education, with facts, figures and high-quality, bite-sized teaching resources readily available online. Mathematical skills for the future must focus on how to critically interpret, manipulate and apply this data.

¹⁰ <http://www.mysterymathtown.com/>

¹¹ <http://www.wolframalpha.com/about.html>

¹² <http://www.qappsonline.com/apps/tunetrace/>

In 2006 the non-profit educational website Khan Academy¹³ was launched. This ground-breaking initiative, created by educator Salman Khan, has a stated mission to provide “*a free world-class education for anyone anywhere*”. Mathematical science topics are well represented, and provide learners with self-study micro tutorials in topics of their choice.

The recent upsurge of MOOCs which support platforms on which traditional universities now provide free, large-scale, interactive participation in lectures, tutorials and even in some cases assessment via the web is set to further challenge traditional models of mathematical education. They have led to the idea of the flipped classroom. This is a blended learning approach where students watch lectures online and work face-to-face on problem sets with other students in class, allowing teachers to spend more time interacting with students instead of lecturing (Martin, 2012).

Driven by the power of social networks, MOOCs have high registration rates, often in the millions. They also have low completion rates. Nevertheless they provide easy exposure to elements of the higher educational experience, which may raise aspirations in many and allow teachers to selectively use these resources in the school classroom. While the business models for MOOCs are yet to be tested over time, those taking part in the first waves are motivated by their wish to raise their institution's profile and enhance potential recruitment to campus courses, to experiment and innovate in using new learning technology and, in some cases, through a civic duty to promote knowledge to all.

Concluding thoughts

In this chapter we have undertaken a brief overview of the many and varied techniques and projects that have been employed to excite and educate learners to delve more into the possibilities of further study of the mathematical sciences. In these concluding remarks we reflect on the important effect that media and popular culture can have on interest in, and motivation to undertake, studying at a higher level.

Strong traditional media figures like Professors du Sautoy, Hawking and Cox, presenting popular landmark television series on mathematics and related science, have been joined by a new generation of social media mathematics promoters. Examples include Matt Parker (@standupmaths), Sara Santos¹⁴, James Grime (MMP) and Katie Chicot (Open University), who promote their love of mathematics with a contemporary sense of humour. All have usefully raised the profile of the subject. The ‘Brian Cox effect’ (Paton, 2013) has been described as a significant factor causing the 50% rise, over eight years, in students studying physics at university, a rise to more than 40,000 in 2011. In years to come many who chose to study mathematics may cite social media mathematics promoters rather than television personalities as role models.

The popularity of the award winning comedy TV series The Big Bang Theory, following the adventures of a group of young graduate physicists, has also been important in supporting

13 <http://www.khanacademy.org/about>

14 <http://mathsbusking.com/>

the growing youth culture of geek/nerd chic, where technical and mathematical skills are prized and flaunted (Konzack, 2006).

Educators need to take advantage of these changes in modern youth culture to inspire and make the subject relevant. But they also need to be able to effectively teach the ‘hard stuff’ too. Parker (2013) says that many students are caught off-guard, suffering from “*proof stress disorder*” when the focus shifts from learning techniques to proving them, but:

“warning students of this paradigm shift not only allows them to brace themselves, but acts as an active promotional tool. The gift of a maths degree is not the ability to do large sums, but to think in a critical and mathematical manner.”

Parker (2013)

Achieving a balance between relevance, imaginative play and critical thinking is challenging and needs to be recognised as such, particularly where issues of mathematics and gender are involved. This chapter has described research into best practices and detailed effective resources to help and assist in the task.

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Chapter 3: Mathematics education and the transition into higher education – Transmaths demands better learning-teaching dialogue

Julian Williams

Abstract

The research basis of this argument comes from six years of mathematical education research in colleges and universities in further and higher education through a series of Transmaths¹ projects funded by the Economic and Social Research Council. The findings involved a synthesis from large-scale surveys, longitudinal student interviewing, and case study work focused on teaching and learning mathematics in post-16 colleges and across science, technology, engineering and mathematics (STEM) and other mathematically-demanding programmes in higher education. The research developed a critique of ‘transmissionist’ teaching – which finds its ultimate in the ‘university lecture’, but which is also found in colleges. However, we found good quality communication which we associate with the kind of ‘connectionist’ teaching found in previous research in schools and colleges; and we were able to define and measure it in opposition to transmissionist, lecture-style and ‘delivery’ models of teaching. The connectionist, dialogical forms of learning-and-teaching were found in many situations – especially where students worked together solving problems in small groups, where teaching was designed in response to assessment of the students’ needs, such as in university mathematics support centres and sometimes in classrooms, workshops and tutorials. On the other hand, lectures were often seen as iconic of a university education, and lecturers and new students see the lecture as a prestigious event (compared to other teaching-and-learning activities). This raised questions about the identity and status of the university lecturer. Recommendations based on the findings are offered for universities to address the mathematical transition for learners.

¹ See <http://www.transmaths.org>

Introduction

The aim of this chapter is to summarise for students, lecturers and course managers the findings of the Transmaths projects about mathematics education in relation to transition from school or college into university.

Traditionally, mathematics education has conceived of the problem of transition as mainly a cognitive one: the kind of mathematics learned at school or college (up to so-called A-level) is understood to be rather instrumental, while the new demands of ‘advanced mathematical thinking’ at university are more conceptual, rigorous, and formal. While accepting this is one issue for some students in transition, particularly for those entering more demanding mathematics courses, we view other aspects of transition as also problematic: the social and ‘cultural’ aspects of transition involve changes in the whole system in which students are expected to learn. This social and cultural perspective leads us to look at the learning practices, the institutional learning-teaching system, the motives of mathematical activity, and the academic communities and identities of those involved. For instance, from this point of view, the fact that engineering students study mathematics in separate academic lectures while they learn to practise engineering in laboratories, placements and workshops is a significant division in the teaching-learning system; significant because of what it tells us about these students’ mathematical identities and motives, and hence their likely engagement or disengagement in mathematical practices.

Furthermore, our² Transmaths research examined a broader range of learning outcomes than the usual cognitive performances, including dispositions to study more mathematics in future, and students’ mathematical self-efficacy, which we found can play an especially important role in post-compulsory students’ choices, and their engagement in learning mathematics. This involved making special measurement instruments for the surveys, and of course, focusing attention on these aspects in interviews and case studies.

The Transmaths projects investigated students’ learning as they progressed from school to college and into mathematically-demanding programmes in higher education. We aimed to understand transitional practices and their effects on students’ mathematical identities, their choices and the associated learning outcomes. We used a mixed methods approach, analysing large scale survey data alongside findings from longitudinal sets of student interviews, and case studies of teaching-and-learning in field work in colleges and universities. This combination allowed us to establish some patterns across large samples of students, but also to explore qualitatively the reasons for such patterns in individual students’ accounts, and how practices might develop that could make a difference to the quality of teaching and learning.

² ‘We’ and ‘our’ refer to the Transmaths research teams whose work is utilised as the basis of this chapter.

Findings

Key effects of teaching on learner disengagement

A key result from the analysis of data from college and university first year students concerned the effects of ‘transmissionist’, teacher-centred teaching, which we found to be associated with negative or declining attitudes. Generally across the post-compulsory samples, students’ attitude to studying more mathematics was in decline, and of course such students increasingly drop out of mathematics. In the 6th form/Further Education College (6fFEC) study we found transmissionist teaching (as self-reported by the teachers) correlated negatively with their students dispositions, that is their attitude declined more sharply than those in less transmissionist classrooms.

Figure 3.1 shows the results of one analysis, as reported in Pampaka et al. (2012a, 2012b). The right hand side of the Figure shows the distribution of 6fFEC teachers’ scores on our self-reported transmissionist teaching instrument, and you can go online and find your own score on this scale if you are a classroom mathematics teacher³. The graph on the left of Figure 3.1 shows the negative relation between the disposition of the students towards further mathematical study in higher education at the end of the year and the average of their teachers’ scores on the same transmissionism of pedagogy instrument after many other factors have been taken into account (such as prior dispositions at an earlier point in time, previous examination grades, etc.).

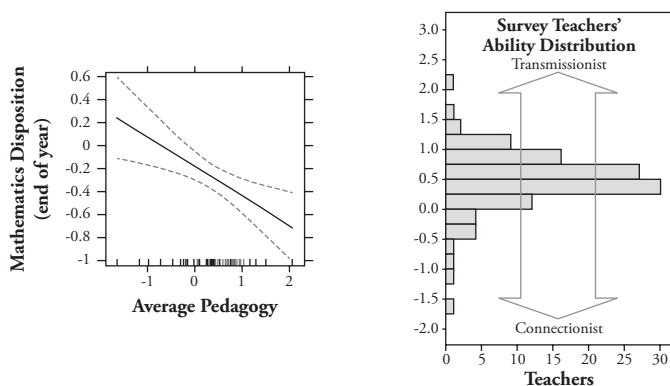


Figure 3.1: The transmissionism of teachers’ pedagogy (right) and the regression of dispositions to study mathematics on the transmissionism of their teachers (left).

Additionally, the students in first year university courses reported more negatively about their transition the more they felt that their university teaching was transmissionist (Pampaka, Williams & Hutcheson, 2012b). So: what is ‘transmissionism’ and ‘transmissionist teaching’?

Essentially transmissionist teaching to A-level in school involves the common practice involving teacher-directed ‘telling’ of mathematics: learners are expected to mimic the teacher’s examples, practising exercises, etc. Teachers dominate talk, and the whole is

³ <http://www.transmaths.org/teachers/pedagogy>

usually directed at the upcoming test. In university, transmissionism is most clearly shown in the monologue of the ‘lecture’ in which there is very little interaction, and students copy notes and examples.

An alternative more learner-centred teaching might be ‘connectionist’ teaching in which the mathematics being taught is ‘connected’ to mathematics (or other knowledge) the students already know, and there is a dialogue between teacher and learners; the teacher tries to connect with the students’ previous experience and understanding, perhaps through formative assessment, and the learners are involved in the process of deciding how they learn. A connectionist teacher whose classroom we studied made sure there was plenty of time in her classroom for the students to talk and work together in groups on their conceptual understanding; likewise in some university workshops and modelling classes students work together on problems and call on the teacher when they need help: so teaching tends to be responsive to students’ needs. Even in formal lectures some teachers make space for students’ questions.

In Figure 3.2 one can see that this is important for students’ sense of positive feelings about their transitional experience. Here first year university students (approximately 1700 across five universities and thirteen mathematically-demanding programmes) were asked about their feelings regarding their transition from school to university (classified as ‘positive’, ‘mixed’ or ‘negative’); they were then also asked to compare the opportunity to ‘ask questions’ and ‘discuss ideas’ at university compared with school. In Figure 3.2a one sees, for instance, that 36.3% of the students said they had negative feelings about transition and that they felt they had less opportunity to ask questions at university compared to school, while 26.8% responded that they had positive feelings about transition and felt that they had more opportunity to ask questions at university than at school. On the other hand, very few students who had positive feelings felt they had less opportunity for questioning at university than at school, while very few who had negative feelings and said they had more opportunities to ask questions at university. Similar patterns hold when the students are asked about their opportunity to discuss ideas (see Figure 3.2b) and we conclude that negative feelings are generally associated with a perception of poorer dialogue when students enter university programmes, and vice versa.

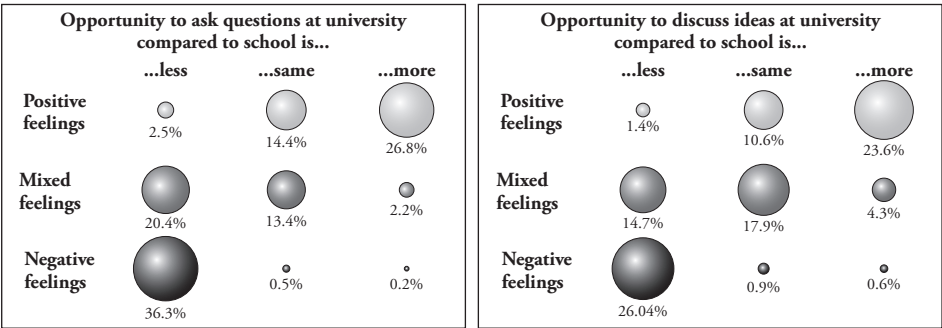


Figure 3.2a, 3.2b: Opportunity to ask questions / discuss ideas at university compared to school (see Pampaka, Williams & Hutcheson, 2012b).

These statistical findings were well supported by interviews and by what we saw in case studies: interactions with lecturers were often fewer and less engaging than students had expected, and this was associated with a negative view of the university experience and transition in particular. University lecturers were inclined to see this as 'dependency'.

Furthermore, we found that teaching that was responsive to learner-centred work was widely appreciated by students and was associated with progress in learning. We saw this for example in workshops where the A-level students worked on their own coursework tasks, or when university tutors used tutorial time to discuss the learners' work, problems or misconceptions. Of course, this is typically what happens in mathematics support services and centres where the agenda is student-led.

In the next section we will consider how some of these more connected practices were managed/encouraged in practice in the cases we studied.

University systems and practices analysed and discussed

Lectures or classes

The lecture, as already suggested, was observed habitually to become a monologue; in one lecture that we videoed a lecturer spoke-and-wrote continuously without interruption for the whole lecture as the students tried to copy what was being written on the overhead projector. In another case the lecturer similarly talked while directing students through an electronic projection of the lecture notes they had been given. Of course it is possible that students can learn something even from such monologues: they hear the words pronounced, they may hear some meta-commentary on reasons and explanations, in the former case as the mathematics is actually being written down, and so on. At minimum, students told us they went to lectures to get these notes and so find out what was 'on the course syllabus', and what types of problems were likely to 'come up' on the test. Lecturers sometimes concurred, saying the reason they did not make their notes available to students was that they thought they would not attend their lectures if the notes were otherwise available.

Some lectures could be inspirational. In one course that adopted a Problem Based Learning (PBL) approach, lectures were rare but well attended. We observed senior figures in the medical profession talk about the significance of what the students were learning for real cases: students participated in trying to interpret x-rays, and were encouraged to consider the bravery sometimes needed to intervene in emergency medical situations. On the other hand these students were taught their mathematics from a computer programme.

We noted this gap between how students were taught mathematics and how they learnt 'their own discipline' in most STEM subjects. There were some efforts made to help engineering students understand how the mathematics they were being taught would be used in engineering, for example, with at least one question in an engineering context on each weekly problem worksheet. Mathematics majors were of course taught by lecturers who could be inspirationally enthusiastic about their mathematics, though mathematics lecturers do not always see the mathematics they teach to first years as inspiring of enthusiasm. To be enthusiastic about such elementary mathematics requires a lecturer/teacher who is interested in the process of learning, and indeed the process aspects

of mathematics, such as modelling and problem solving heuristics. In one first year mathematical modelling course (described in Chapter 9), the lecturers were evidently deeply engaged in the students' work on open-ended problems and in some of the phenomena that their practical work produced.

Some lectures were held with reasonably small groups and were more like college classrooms, sometimes even taught by experienced college teachers, and these lectures could be more interactive and even dialogical. The experienced teacher who has many years of interacting with learners in informal situations anticipates where their difficulties are likely to arise and can sometimes use this to teach more dialogically. Sometimes lecturers give students notes which are partially blank that they need to fill in as the lecture proceeds – so the lecturer can spend a little time checking that students have adequately 'taken on board' what is necessary to follow the lecture. New technologies using clickers to collect student responses to diagnostic questions can be used to do the same thing but can involve all students in a large lecture.

However, very few of the 'large' mathematics lectures we saw were anything other than transmissionist, and almost none were obviously dialogical and connectionist. Most of the dialogue about mathematics was seen in smaller groups, workshops, tutorials, or between students themselves in informal group work.

Workshops and tutorials

Workshops and tutorials, on the other hand, opened up opportunities for dialogic teaching-and-learning, allowing learners to really engage in dialogue with lecturer input. Sometimes, such tutorials reduced to another mini-lecture, where the lecturer spends the time more or less monologically working through the problems for the students at a board, perhaps those chosen by the students as the ones that caused difficulty for them.

However, in one workshop we observed, the students were expected to attend to work on their problem sheet in groups with their peers, and the tutors were present (including graduate students) to monitor and help only when called upon. The tutors had plenty of opportunities to observe students at work, and would also collect students' work and hand back with comments: this allowed the tutors and lecturers opportunities to learn about their students' difficulties and mathematical limitations, which we suggest is a vital part of the dialogue necessary to effective teaching and learning.

Interestingly, in many cases the lecturers delegated much of this vital work of tutoring, marking and feeding back to postgraduate assistants and tutors, while themselves keeping their role as presenters and lecturers. This division of labour can be unfortunate, as it tends to break down the two-way communication of listening-and-explaining that dialogic teaching and learning requires.

However, in cases where teachers carefully observed, listened to, or read students mathematics, a deeper engagement with learners' needs became possible. In one case study (Jooganah & Williams, 2010) a mathematics lecturer expressed surprise at their students' work on a problem sheet where they were required to find the inverse function of $f: \mathbb{R} \rightarrow \mathbb{R}$, defined by $f(x) = 2x + 3$. The first year students had written $y = 2x + 3$, rearranged to make x the subject of a formula like $x = \frac{y-3}{2}$, and finally swapped the x and the y , to obtain something

equivalent to $y = \frac{x-3}{2}$. In a tutorial he asked the students why they were doing this and a lively discussion ensued. The students were adamant that ‘this was how we were taught’ and an argument broke out as to whether a graph (x, y) might represent a function (defined by x maps to y) *as well as* its inverse (defined by y maps to x). After our analysis of this event we replayed some of the video and interviewed the lecturer about it. He concluded:

“Well, it was very, very clear wasn’t it what I learnt from that particular class. One of the 3 things I learnt was that something has changed about the way students are taught to write functions at A-level and their inverses, and I damn well better understand what that is, and why it is.”

This is exactly the point here: effective teaching and learning requires such moments where the teacher is as engaged with the learners in the sharing of mathematical conceptions and ideas. This occurred because the lecturer both taught the course and had clear expectations of the learners but also crucially had marked their work and was interested in and wanted to diagnose a problem. Such events can arise in many ways, and an expert teacher who is well prepared and understands their students’ mathematics can make this happen even in quite large lectures and in other apparently unlikely places. But a thoughtful design of the learning-teaching system will plan to enhance such opportunities for dialogues, while some common practices we observed might make such events less likely.

Modelling and problem solving courses

There is a strong case for teaching mathematics in special modelling and problem solving course units that focus upon the application and real problem solving with mathematics, whether this is taught by mathematicians who are sensitive to the main discipline or by the main discipline lecturers themselves (i.e. engineers, science lecturers, etc.).

Our findings from our college research was that such modelling as part of A-level coursework allowed a critical shift in the relations between students and teacher: we argued this allowed for a more dialogical relationship in which students’ mathematics could become focal to the dialogue (Hernandez-Martinez et al., 2011). Similar observations have been made in classes in the university context as in the workshops mentioned above: when the students’ own work becomes focal then there emerges a real opportunity for the lecturer/tutor to engage with the students in mathematical dialogue. This is quite apart from the obvious higher-level aims that such problem solving and modelling courses sometimes address, but arguably is essential to the effective teaching of these higher order objectives.

The support centres and diagnostics

Most universities we studied offered some kind of support services (see Chapter 4), and some had invested very seriously in mathematics support services and centres. These were places where student-tutor interactions were typically led by a student’s perceived needs: this was a great place for dialogic teaching-learning events to begin, in principle. We saw many examples where this appeared to lead to good learning, and many students were pleased to have such a facility.

In one case for example a trainee student teacher called in to a support centre we were visiting to ask for advice about fractions: she was concerned that she felt when she began

teaching practice that her perceived deficiency in fractions might cause her problems. This anticipation was actually quite a remarkable event, and signalled perhaps that her course had helped her diagnose and anticipate a difficulty even if it had not planned a solution with her. In a number of courses diagnostics were planned and there were planned remediation events for students with resulting identified needs; often these were online resources, but sometimes tutorial events were planned around students' diagnosed needs.

In many cases, students said they had found difficulty with a course, had decided they had not understood it well enough, and came for support with their problems before an upcoming examination; in other cases they had done badly in an examination or test and come belatedly to a centre for help. The support centre therefore offers an important service. In some cases however their work went further and became potentially developmental: the centres sometimes used these experiences to feed back to course lecturers what they were finding, thus enabling development teams to build on this knowledge. Arguably such work might be much more effective: a powerful mathematics support service, that is one with expertise and institutional support, might be a really effective change agent in such universities.

There are some well-resourced support centres which have the expertise to take on such a wide remit, and some have mathematics education research expertise and programmes of work that are helping to improve mathematics education within higher education. Although this investment might seem extensive, in reality their research budgets remain modest as a percentage of the actual expenditure on teaching by their higher education institutions, and more universities could be encouraged to fund these centres.

Informal student working groups

Early in our research we began to appreciate from student interviews and from our case study observations how important peer groups (see Chapter 6) were to students' learning. For instance, students who were perplexed by their lectures would meet in smaller groups to try to work out what was going on, and weekly problem sheets or upcoming examinations would particularly focus peer group support. We often saw groups of students at work in coffee bars after or before a class, social areas outside and between lectures, etc. One group of joint honours mathematics-with-economics met every Friday lunchtime in the pub to 'sort out' the mathematics for the week – we joined this group and felt this was very powerful for those involved, even though we were not totally convinced of the environment's efficacy!

We also saw formal groups structured by PBL approaches: these systems formalise the good things that students otherwise often produce informally. There is a vast literature on both peer learning and PBL which encouraged us to focus our attention on this and which encourages us to focus recommendations for development in this area (see for example Topping, 1998).

Interview evidence suggested these informal contacts and communities were crucial to many students' learning. In many cases these peer groups were subgroups of the courses students were following, and the workshops where students were formally required to work together on problem solving perhaps had encouraged these; but sometimes these informal groups involved students from different year groups and different courses altogether.

Thus one science student told us they got tutorial help with their mathematics from a mathematics student from a more advanced year. In general these groups form for social reasons and are not generally consciously planned by the students: thus a group may form more because the students share a place of residence or because of their shared ethnicity than for consciously thought-out reasons.

Many students reported that these sessions with peer groups were crucial to their progress. We argue that such groups should be supported and encouraged by the system and by individual lecturers, both by helping to form such groups in their classes, by programmed peer support structures, and by managing the timetable appropriately. There are some notable structures that may discourage such groups forming: for example in some places students work very intense and full days with few gaps between formal lectures; students say they like this as it means they can have whole days 'off' the course when they can take on paid work (despite being full-time students). However, this intense timetabling leaves few serendipitous times and congenial spaces for informal groups to work together. In such cases perhaps groups need to be more formally organised and supported, perhaps with the aid of communication technology.

Thinking through the idea of creating 'time and space' for students to work together: we note that others have identified mathematics support centres as one such resource (see Solomon, Croft & Lawson, 2010); we should further add formal classrooms and courses where students are required to work together in cooperative groups in solving problems as part of the coursework (see Chapter 7). We argue that it is important to repeatedly create such opportunities, not only in induction but in each semester of a course, so that students are naturally given multiple opportunities for making use of these informal learning systems throughout their courses.

The teachers/lecturers 'teacher identity'

Looking back and reflecting on what the previous remarks suggest about encouraging teaching and learning, the reader might be perplexed: isn't all this pretty obvious, once one has the general idea that teaching and learning is a joint, dialogical activity involving students with academics?

Indeed, our research involved interviews with many in higher education that recognised this point; many of these interviewees were teachers who had experience of education outside the mainstream of university practice – they had taught in schools or colleges, or had experience of tutoring in adult education such as at the Open University. Many were employed as support tutors on teaching-focused contracts, sometimes in support centres or tutorial roles, and were sometimes regarded as relatively low status employees on temporary contracts. These people often identified themselves as 'teachers' or tutors in contrast to their university academic colleagues, who were labelled lecturers or professors.

Yet, as our argument above has suggested, their functions are important and their understandings may be of crucial value to the teaching-and-learning system and its development. If they are given low status, this will no doubt limit the capacity of the university system to learn from their expertise. On the other hand there are some signs

that teaching expertise is gaining some value in academe, and in some quite traditional institutions promotion is significantly affected by teaching quality criteria.

One problem for higher education to address is what many refer to as a conflict or contradiction between teaching and research priorities (or between these and other important duties in administration). Can these tensions be resolved; can and should academics become 'teachers'?

However one way through this may be to see teaching-and-learning as more related to research and other duties. Every act of research involves communication of the work to others, an essentially pedagogic activity. Similarly, teaching can involve a sharing of mathematics, albeit apparently between non-equals. In the teaching of problem solving for instance, teachers and learners can share experiences of problem solving strategies and emotional accompaniments of intense efforts, and even considerations of the aesthetics, for example of why one solution might be judged more elegant, pleasing or functional compared to another.

Even the problems themselves can quickly become challenges for both teacher and student in problem solving courses. In the example of the mathematical modelling and problem solving course at the University of Leeds discussed in Chapter 9, several of the problems used with the students would stretch many university teachers, and some perhaps open up new research problems and tasks. There is for instance a deceptively simple bicycle problem, discussed at a recent seminar by lecturers in STEM for whom the principles were far from obvious! An external force is applied to the pedal as shown, tending to rotate the pedal clockwise and so turn the wheel clockwise, that is in the opposite direction to the force applied. Which way does the bicycle move, and why?

Modelled as a particle under the action of just two forces, with friction assumed to be no greater than the applied force, one concludes the bicycle can only accelerate in the direction of the applied force, thus implying the gears rotate in the anticlockwise direction: counterintuitive? Indeed: it is possible to extend the point of application of the external applied force in such a way as to impel the bicycle in the other direction, when friction is greater than the applied force. Just think: the bicycle is not a particle, and when cyclists pedal the only external force is in fact frictional, while the cyclist is part of the system that is generating a torque on the wheels, with friction tending to oppose the rotational effect of the pedalling on the wheels. Thus one is forced to reconsider the model and a new modelling cycle (no pun intended) begins.

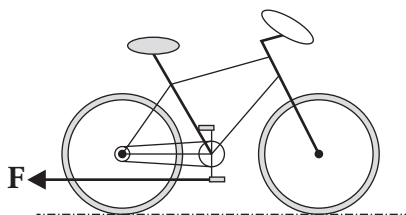


Figure 3.3: The bicycle problem with a 'backward force' being applied to the 'bottom' pedal.

Conclusion

Teaching practices in universities as well as colleges and schools are habitually transmissionist, as is symbolised in the iconic status of the monologic lecture in the university. Additionally, teaching, and ‘learning to teach’ is not always accorded the importance one might expect in an academic institution, and lecturers’ identities are often led by other activities in administration or research.

Yet university lecturers often – even usually – expect that learners have developed or will develop conceptually advanced mathematics, and learn how to become independent learners. These are two highly-demanding, high-level learning outcomes, implicating learners’ attitudes and metacognition, and require careful attention to pedagogy. We have found classrooms, workshops and tutorials where dialogic teaching-and-learning is tailored to the needs of students, and we found that students’ often informal peer group collaboration could be a very significant factor in students’ mathematical development. But too often the mathematics teaching transmits facts and procedures, and assumes that higher-level skills have been learned.

We therefore concluded that opportunities for students to experience dialogic teaching-and-learning should be made prominent in the development of university practices and systems, as elsewhere. This might be done through course design, influencing all the teaching offered in formal programmes, without neglecting the importance of informal settings especially involving peer group work. In the next section some recommendations for achieving this will be made.

Discussion and recommendations

For transitional activities

Students in transition would benefit from more preparation or even information concerning the change in learning systems to be expected when they arrive at university: what lectures are for, how to manage lectures, how to work around the system to get the support they may need, and the importance of making social contacts and informal support networks.

We would add to this the possible benefit of providing students in school/college with some more experiences of university-style working, for example lectures in college might be given and discussed; students might be given some experience of reading mathematics texts on their own, working on some topics in informal groups, etc.

In the special case of Further Mathematics, some students we interviewed had experienced distance learning through the Further Mathematics Network, which can involve some of these useful experiences. A strong case might be made for all students to have some experience of a course such as this, so that becoming independent, or better, ‘autonomous’, does not come as quite such a shock in transition.

For university lecturers teaching mathematics

The messages from the findings discussed above can be summarised in the mantra: ‘listen to and try to understand your students and their problems’, and more effective dialogic

teaching will follow. This involves trying to create situations in which the learners' work can be observed and reflected upon; and additionally the creation of situations where dialogue about mathematics can occur. It may seem facile to say so, but it also requires that the lecturer is actually interested in the students' mathematics, and so in hearing what students think about mathematics – an endlessly fascinating experience for those who adopt an identity as a mathematics educator and teacher.

Some small steps could include the following: asking open ended questions that might stimulate a range of answers/conceptions to be discussed; collecting student work and using it to take issues back to class; creating some workshop-time when students work together in groups, during which time the lecturer might circulate and listen without comment; encouraging such groups to meet and do tasks together between sessions; selecting a group of students to act as 'researchers' who collect answers from the whole class, or even interviewing class members outside the public arena; identifying 'isolates' in the class and trying to engage them in group tasks, etc.

Above all, perhaps teachers could try to find some like-minded colleagues who are interested in teaching, visiting one another's classes and giving each other constructive feedback. Perhaps the group might plan a 'lesson study' research together (see Hart, Alston & Murata, 2011).

For management

The preceding argument seems naturally to suggest lines of action for leadership and management. There are questions about:

- a) How to value teaching in your institution and how to make this visible alongside the many other priorities and requirements of lecturers.
- b) How to encourage and facilitate teachers' work on self-improvement of teaching quality; in particular the value that might result from a research investment, whether through an enhanced mathematics support service or other pedagogic/educational research centres. There are some current initiatives involving students themselves in research and design related to teaching that could help.
- c) How to facilitate time and space for students' informal learning together in various groups (See Chapter 7).

Some of the lessons identified above could be written into students' handbooks and lecturers' preparation and in-service training courses or peer evaluation agendas. There were also some caveats: for example to beware timetabling errors like filling up the day with formal events that exclude time for informal work, and otherwise responding to what students say they want without carefully considering the educational implications.

For students

Students new to university courses may be surprised at the different learning contexts in which they are placed, being expected to attend classes or lectures that sometimes seem like their school classrooms, but which are often very different. Students are recommended to get some experience of the kind of learning expected on the course they aim to take, whether through studying through the Further Mathematics Network, through visiting

first year students at university, or through some summer school type events. At the least students should engage in online fora with existing students to find out how they found the transition in the previous years.

Such students need to think about how they are learning and what, if anything, they need to make progress: it is not too early to start thinking about this while at school. Many students get benefit from informal 'problem solving' homework groups and this would be a good preparatory experience for most A-level students.

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Chapter 4: Mathematics Support at the transition to university

Duncan Lawson

Abstract

There is considerable focus in higher education on the retention and success of students. Research evidence shows that a positive start by students influences the results of their entire university education and for this reason there is a particular emphasis placed on the transition into higher education. Key generic factors that enhance this transition are effective induction engendering a sense of belonging and curricula designed to promote retention. Although many issues apply to all students, there are some discipline-specific issues that mathematics undergraduates face. In many ways, the nature of the subject is different at university from what they have experienced at school/college with a new focus on rigour and proof. Additionally, the majority of mathematics undergraduates have achieved high grades at A-level Mathematics and are accustomed to being able to cope with new material. Consequently, they have had little opportunity (or need) to develop ‘mathematical resilience’. It is therefore important to provide mathematics students with support at the transition. This chapter outlines ways in which many universities offer mathematics support provision, principally through mathematics support centres. Although such support was often originally conceived as a way of offering support to struggling non-specialist students, it plays an important role in supporting specialist students who (as a cohort) are often major users of the support facility. The chapter also describes the development of a community of mathematics support providers and its development from a community of practitioners to a community of scholars.

Introduction

Thomas (2012:7) reasoned that *“Regarding ethics and social responsibility it seems reasonable to argue that if an institution admits students to HE it has an obligation to take reasonable steps to enable them to be successful”*. In addition to ethical appeals, she also presents economic

arguments relating to the loss of income to institutions when students withdraw before completion of their target award.

Whether ethics or economics (or a mixture of both) has been the driving force there has been considerable work undertaken in recent years to promote student retention and success. Interest in increasing student retention and success is not a parochially British phenomenon, but is truly international. Indeed, the National Audit Office has identified the United Kingdom as one of the more successful of OECD (Organisation for Economic Co-operation and Development) countries in terms of student ‘survival’ rates (National Audit Office, 2007).

The first year has been identified as a key time for supporting students as they make the transition into higher education. A network of interested academics, European First Year Experience¹, was established in 2006 to consider the first year experience at universities across Europe. This group aims to explore ways to address students’ needs and share and develop strategies to optimise student success. They are linked with the US National Resource Centre for First Year Experience and Students in Transition². Although this Centre considers all transitions in higher education, in view of its importance, there is a particular focus on the first year experience and the Centre has organised annual conferences focused on the First Year Experience since 1982. A key aim of the Centre, declared on its website, is *“to set a standard of excellence for supporting student transitions and facilitating educational success for a diversity of students in the 21st century”*.

In the UK, the Higher Education Academy, working in collaboration with the Paul Hamlyn Foundation has undertaken a major study, the *What works?* programme (Thomas, 2012). This has identified good practice relating to student retention and, in particular, the importance of creating a sense of belonging in students (this issue is considered from the perspective of mathematics undergraduates in Chapter 6). The programme established that there is a large volume of activity being carried out throughout the sector to increase student retention.

Rather than describe these generic activities, this Chapter will focus on issues that are specific to students making the transition into higher education to study mathematics. Mathematics students encounter the same challenges as students of all disciplines as they begin their studies in higher education and the generic measures will no doubt be helpful to them. But there are also some subject specific issues that mathematics students must face. The National Audit Office (2007:24) reported that *“When science, technology, engineering and mathematics students are considered together, they are less likely to continue to a second year of study than students following other subjects”*. The report goes on to show that, when determining continuation rates by HESA JACS (Higher Education Statistics Agency Joint Academic Coding System) codes, Mathematical and Computing Sciences have the lowest continuation rate apart from Combined Subjects.

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1 <http://www.efye.eu>

2 <http://www.sc.edu/fye>

Engstrom and Tinto (2008) asserted that “*access without support is not opportunity*”. In this chapter we will focus on one particular support mechanism, namely mathematics support, which many universities have introduced in an attempt to help ease the transition of students into higher education and thereby promote greater student retention and success. Before focusing on mathematics support, we will give a short description of some of the mathematics-specific issues that new undergraduates must come to terms with, many of which are addressed further in other chapters of this book.

Transition issues specific to mathematics

There is no doubt there are unique features about every discipline which affect the ease with which students make the transition from study of the discipline in school or college to study of it in higher education. However, many mathematicians argue that there are several factors about mathematics that make this transition more difficult than in virtually every other subject.

As subjects at A-level (the primary route into mathematics degrees), Mathematics and Further Mathematics are amongst the subjects with the ‘best’ results. Table 4.1 shows data from summer 2013 A-level results, giving the cumulative breakdown by grade for a range of subjects with at least 10,000 entrants. Table 4.2 shows the grade frequency in Mathematics and Further Mathematics.

As can be seen from Table 4.1, Further Mathematics (28.0%) and Mathematics (16.8%) have the highest proportions of A* grade amongst all subjects. For only one other subject (Art & Design subjects) does this proportion exceed 10%. This pattern is repeated for the proportions achieving grade A or better where once again Further Mathematics (56.5%) and Mathematics (43.0%) are the highest.

Subject	Number sat	A*	A	B	C	D	E	U
Mathematics	88060	16.8	43.0	65.2	81.3	91.9	97.6	100
Further Mathematics	13821	28.0	56.5	78.5	89.9	95.6	98.5	100
Art & Design subjects*	44069	12.2	28.6	57.8	83.2	95.6	99.1	100
Biology	63939	8.0	28.3	52.8	73.7	89.0	97.5	100
Chemistry	51818	8.3	33.6	60.7	79.5	91.4	97.8	100
Economics	26139	7.0	32.2	61.6	83.1	94.3	98.9	100
English	89442	6.2	20.7	48.6	79.4	95.8	99.5	100
French	11272	6.5	38.5	68.8	86.7	95.8	99.2	100
History	52149	6.5	26.0	56.3	82.7	95.6	99.2	100
Physics	35569	9.2	31.0	54.6	73.9	88.0	96.9	100
Psychology	56088	4.0	17.0	43.1	69.9	87.9	97.4	100

*These titles cover a range of related subjects.

Table 4.1: 2013 A level results, cumulative percentages of subject results by grade (source: JCQ, 2013).

Subject	Number sat	A*	A	B	C	D	E	U
Mathematics	88060	16.8	26.2	22.2	16.1	10.6	5.7	2.4
Further Mathematics	13821	28.0	28.5	22.0	11.4	5.7	2.9	1.5

Table 4.2: 2013 A level results for Mathematics and Further Mathematics by grade (source: JCQ, 2013).

As can be seen from Table 4.2, the modal grade for both Mathematics and Further Mathematics is grade A (although for Further Mathematics grade A* is almost as frequent). A variety of explanations have been advanced as to why the results in A-level Mathematics are so ‘good’. These include, the modular nature of A-levels and the opportunity to resit modules to improve grades. However, these factors apply to all disciplines. A more ‘mathematics-centred’ explanation is that, since A-level Mathematics is perceived to be one of the more difficult subjects, schools and colleges are quite selective about the students they admit on to A-level Mathematics courses (Matthews and Pepper, 2007). It is the norm that students who have not achieved grade B, or better in GCSE Mathematics are not permitted to enrol to study the subject at A-level; indeed in many schools, the barrier is set at grade A at GCSE. In most other subjects the qualifying entry level is typically grade C. It is postulated that this higher entry hurdle disbars the weaker students and contributes to the skewed distribution of final A-level grades. With A-level Further Mathematics, this argument about the ‘self-selecting’ nature of entrants has probably even greater strength. The vast majority of students who are willing to study two A-levels in mathematical subjects are likely to have been high achievers in mathematics thus far. In addition, for many students Further Mathematics is their ‘fourth subject’ at AS-level. Some schools/colleges give students the choice at the end of Year 12 about maintaining four subjects or dropping down to three subjects for A-level. Those who have struggled with Further Mathematics to AS-level are likely to drop it, whilst those who have achieved well are likely to continue studying it through to full A-Level.

Another suggestion is that a significant proportion of the marks on A-level Mathematics examinations are awarded for procedural precision. This, allied with the relative predictability of the examinations, allows candidates to be ‘drilled’ until they are skilled at accurately applying the procedures, without them necessarily having understood the processes they are using or the concepts that underlie them.

Whatever the root cause, the distribution of A-level Mathematics results means that the majority of students recruited to undergraduate mathematics degrees from an A-level background have achieved at least grade A. This has some potentially wide-reaching consequences for both students and staff.

Most undergraduate mathematics students believe that they are good at mathematics and their A-level results appear to confirm this. Furthermore, they often find that the amount of work required to achieve their good results in mathematics is much less than that required in the other subjects they are studying at A-level. This is not necessarily because mathematics is easier or because they are particularly talented at mathematics, but because of the nature of different subjects at A-level. As noted above, mathematics at A-level is principally procedural. Compared with most other disciplines, in both sciences and humanities, there is a relatively small amount of content to be remembered and applied. In all subjects, past papers are used

to help candidates prepare for their examinations, but in most subjects it is necessary for the candidates to revise, often substantial, amounts of content before they can begin to attempt the questions. In mathematics this is rarely the case.

This belief in their mathematical ability can often be a reason behind students' decisions to choose mathematics as a subject of study in higher education – their positivity towards mathematics is sometimes based on their apparent ability in the subject. As one student, from an unpublished study (Solomon, undated), has put it:

“It [mathematics] was the only thing that I thought I liked at the time – I confused it with being able to do it.”

(Solomon, undated)

The outcomes of this can be significant. Firstly, many students have not been required to develop persistence in solving a mathematical problem. This is not to say that they are not prepared to work hard; but for many, their experience of working hard has been simply one of practising more examples (which they invariably do correctly) rather than of being faced with problems that they cannot solve relatively quickly and which they have to ‘wrestle with’, trying a variety of approaches until they find one that works. Secondly, when some no longer find mathematics as straightforward as they found it at A-level, they start to realise that their positive disposition towards mathematics was based primarily on the affirmative emotions generated by success rather than because of a deep interest in the subject. One of the key findings of a report focusing on more able school students, Ofsted (2013:9), is that

“Students did not do the hard work and develop the resilience needed to perform at a higher level because more challenging tasks were not regularly demanded of them.”

Ofsted (2013:9)

Johnston-Wilder and Lee (2010) have outlined ‘mathematical resilience’ as a positive affective stance to mathematics which leads to perseverance in the face of difficulties. They claim that:

“Currently it seems that mathematical resilience develops by accident if at all.”

Johnston-Wilder & Lee (2010:1)

The phenomenon of lack of resilience prompted by a misunderstanding of the nature of mathematics was observed by Liston (2008) in her PhD study in Ireland. According to one student she interviewed:

“Once I was getting through it like usually the longest I’d spend on a maths problem might be only five or ten minutes. If you hadn’t got it out by then like there’d be something wrong like.”

Liston (2008:215)

This lack of mathematical resilience allied to less intrinsic liking of the subject than they realised can have strong de-motivational effects on students if (when) they encounter difficulty with the subject in higher education.

The high A-level grades in Mathematics and Further Mathematics of most entrants to mathematics degrees can also cause academic staff to have unrealistic expectations of their

students. A significant proportion of mathematics lecturers are from overseas, particularly from Eastern Europe and East and South-East Asia, countries which mainly achieve significantly better results in international comparisons than the UK (Guardian, 2013). Where international staff are unfamiliar with A-level Mathematics and its grade distribution, they may assume that those achieving high grades may have similar competencies to those achieving the highest grades in their home countries – something which is rarely the case.

In all disciplines, there is a step up from school/college into higher education. However, many mathematicians argue that the step in mathematics is much larger than any other discipline. It appears to some undergraduates that the nature of the subject has changed completely – the aforementioned focus on procedure is replaced with an emphasis on rigour and proof. Commentators such as Savage, Kitchen, Sutherland and Porkess (2000) ascribe the change from GCE O-level to GCSE as bringing about a catastrophic change in students’ conceptual understanding of proof:

“The GCE [O-level] examination was replaced by GCSE which, for Mathematics, brought a decline in students’ concept of proof and in their technical fluency and understanding of algebra. At a stroke A-level Mathematics was undermined.”
Savage et al. (2000:2)

In particular, the removal of large amounts of Euclidean geometry (including the proof of several theorems) from pre-16 mathematics appears to have contributed to a significant weakening of the understanding of the importance and nature of mathematical proof (Healy & Hoyles, 1998).

The procedural nature of mathematics at A-level has already been highlighted. A consequence of this is that many students arrive at university believing that pure mathematics is about such things as selecting and using the appropriate rule to compute a derivative or integral rather than about rigorous proof. Their first encounter with Analysis featuring such things as ϵ,δ approaches to continuity and differentiability can often be a very challenging experience (Alcock & Simpson, 2002) for which they are unprepared.

The start of the 21st century has witnessed a significant growth in the number of students studying the mathematical sciences in higher education. DIUS (2008) showed that over the period 2002-2007, when overall acceptances to higher education courses rose by 12%, acceptances to mathematics degrees rose by 54%. Table 4.3 shows enrolment to mathematical sciences courses in higher education for the period 2007/08 to 2011/12.

	07/08	08/09	09/10	10/11	11/12
Full-time	22,770	24,315	26,225	27,530	29,065
Part-time	6,225	6,885	7,465	7,715	8,180
Total	28,995	31,200	33,690	35,245	37,245

Table 4.3: Student enrolment on undergraduate courses in the mathematical sciences (source: HESA, 2013).

The data in Table 4.3 show a 28% increase in total enrolment on undergraduate mathematical sciences courses. Many factors, including schemes such as the Further

Mathematics Support Programme³ and initiatives like More Maths Grads⁴ have contributed to this growth in numbers studying mathematics in higher education. Many in the higher education mathematics community regard mathematics support as a vital provision which is complementary to initiatives such as these, designed to make the study of mathematics in higher education more attractive and accessible.

These mathematics-specific issues at the transition from pre-university to university require a subject-specific response. One such response is Mathematics Support which is explored in the following section.

Mathematics Support

The nature of Mathematics Support

Mathematics support began as a result of the belief of mathematics lecturers that incoming undergraduates on non-specialist courses (so-called ‘user programmes’ or ‘service teaching’) were not as well prepared as had previously been the case. Three academics expressed their view thus:

“Many students of science subjects arrive at university with little facility and less interest in mathematics.”

This could easily be taken from a contemporary piece of writing, but is, in fact, taken from a paper published in 1973 (Baker, Crampin & Nuttall, 1973). The paper outlines a way of supporting students making the transition to studying mathematics in higher education.

It is impossible to identify the first occurrence of mathematics support since one-off initiatives such as that of Baker et al. (1973) have been taking place for many years on an ad hoc basis. However in the early 1990s a more co-ordinated approach to mathematics support began to develop. The Mathematics Support Association was formed in 1994; this organisation attempted to promote sharing of practice between institutions. Although the Association itself disappeared in 1999, the move towards systematic provision of mathematics support across institutions and throughout the sector has continued.

Initially, the primary motivation for establishing mathematics support was as a means of addressing high failure rates on mathematics modules amongst students of other disciplines, often engineers. For example, in 1991 the then Coventry Polytechnic secured funding from the BP Engineering Education Fund to establish the BP Mathematics Centre. This Centre sought to assist engineering students who were in danger of not progressing on their courses because of the difficulties that they were encountering with mathematics.

³ The Further Mathematics Support Programme is a scheme funded by the Department for Education which aims to provide all students with the opportunity to study A-level Further Mathematics (see <http://www.fmnetwork.org.uk/>).

⁴ More Maths Grads was a 3-year Higher Education Funding Council for England (HEFCE) project to increase the number of students studying mathematical sciences courses in HE (see <http://www.nationalstemcentre.org.uk/elibrary/collection/58/more-maths-grads>).

In applying for funding from BP, Coventry Polytechnic set out two principal objectives for the putative centre. Firstly, it would provide early identification of problems and, secondly, it would provide on-going support to enable students to address any problems identified. The early identification of problems would take place through diagnostic testing of potentially at-risk students on entry to higher education. The on-going support would be provided through a drop-in centre that students could access at times and frequencies of their own choosing.

This illustrates two fundamental characteristics of mathematics support: it is additional to the mainstream teaching in mathematics that students receive and accessing the support is voluntary. So, for example, students studying engineering degrees at most universities will study modules called Engineering Mathematics. These modules will be delivered by a variety of teaching methods which may include lectures, tutorials, problems classes, online materials, etc. All students will be expected to participate in these learning sessions. However, for a variety of reasons, for some students this will not be enough; they will struggle to make progress and be in danger of failing their Engineering Mathematics module and most likely other modules that rely heavily upon mathematics. Mathematics support was originally created for such students: to make available additional help that may assist such students in coming to terms with learning mathematics. But participation in mathematics support is, generally, not compulsory. Students need to choose to access the support available.

Although originally targeted at students of other disciplines, most providers of universal mathematics support report that mathematics students are major users of their provision. Solomon, Croft and Lawson (2010:421) report that *“an unforeseen consequence of the support centres was the mathematics students’ colonisation of the physical space”*. The same authors in a later study report that, particularly for female students, mathematics support centres represent ‘neutral ground’ which allows a power shift in the tutor:student relationship enabling one-to-one discussions that would not take place in a formal class (Solomon, Lawson & Croft, 2011).

The Mathematics Learning Support Centre at Loughborough University is one of the largest drop-in mathematics support centres in Britain. Table 4.4 shows the usage of this centre broken down by the students’ home department for the academic year 2011/12.

Department	% (Students)	% (Visits)
Mathematics	26.2%	48.0%
Mechanical & Manufacturing Engineering	13.2%	8.3%
Physics	8.0%	12.2%
Electronic & Electrical Engineering	6.8%	5.5%
Civil & Building Engineering	6.5%	3.1%
Chemical Engineering	4.9%	3.6%
Economics	4.9%	3.4%
Sport, Exercise & Health Sciences	4.9%	1.7%
13 other departments	24.6%	14.2%

Table 4.4: Usage of Loughborough University Mathematics Learning Support Centre, 2011/12 (source: Mathematics Education Centre, 2012).

The data in Table 4.4 show that mathematics students comprise the largest proportion of students using the Mathematics Learning Support Centre, with over one quarter of the student users being from the mathematics department. In addition, nearly half of the visits to the centre are made by mathematics students showing that mathematics students make more return visits to the Centre than students of other disciplines.

Before coming to university, many mathematics undergraduates, particularly those who studied A-level Further Mathematics, will have been taught in relatively small groups and have benefitted from a significant amount of one-to-one attention. In the vast majority of universities, this is not available in the normal scheduled teaching that students receive; large lectures are the norm and in many cases tutorials and workshops are much larger than the groups the students have experienced previously. Mathematics support centres provide students with an opportunity for one-to-one interaction with tutors in a manner that is reminiscent of their pre-university experience. Solomon et al. (2010, 2011) present several quotes from mathematics students about the reasons they value mathematics support so highly:

“When they are in maths support you know they [tutors] are there to help you and you’re not bothering them.”

Solomon et al. (2010:426)

“... they [tutors] will sit down and work an example, and if you don’t get it, they will try to be as helpful as they can...”

Solomon et al. (2011:580)

“I don’t like to ask public questions myself, I would rather go to maths support afterwards and be able to ask it myself...I think it’s because I don’t want to look stupid in front of the rest of the group.”

Solomon et al. (2011:580)

Some of the characteristics that Johnston-Wilder and Lee (2010) identify in students that have ‘mathematical resilience’, in addition to persevering in the face of difficulties, include working collaboratively with peers, having the language skills to express their lack of understanding and knowing that the more they work at mathematics the more successful they will be. Mathematics support centres provide opportunities for students to develop these facets. They provide an informal environment where students are encouraged to work together to develop their mathematical understanding. Students from Solomon et al.’s study (2010) reinforce this:

“If you don’t understand it, they [fellow students] can explain it to you more on your level rather than the way the lecturer would ... that’s why it [the support centre] is so helpful, because you can sit next to people at the round tables and you can explain things in a way that makes sense to you.”

Solomon et al. (2010:427)

"It's sometimes if there's two of you and you're both struggling but you've both got half of the answer, then you work together you can put your half answers together and get the right answer."

Solomon et al. (2010:427)

It is also worth pointing out that, although the original motivation for the provision of mathematics support was to assist non-specialist students who were at risk of failing, more recent research (Pell & Croft, 2008) has identified that mathematics support has moved from 'remedial support to enhancement'. Their data show that mathematics support is used more by better students seeking excellence than by less able students trying to avoid failure. Again, the students interviewed by Solomon et al. (2010) confirm this:

"I used it [the mathematics support centre] a lot because a group of us who tend to get fairly good marks used it a lot."

Solomon et al. (2010:428)

Delivery of Mathematics Support

The most common form of mathematics support is the drop-in centre. As reported earlier, Coventry University has one of the longest established mathematics support centres and this has been used as a model for the development of the majority of mathematics support provision throughout the country. The basic principle behind drop-in centres is that during their opening hours they are staffed by tutors who are willing to offer support to any student who drops in. Whilst such centres often develop extensive publicity to encourage students to take advantage of the support on offer, the ultimate responsibility for using the centre rests with the student.

In principle, all that is needed to operate a mathematics support centre is a tutor and a location. However, in practice mathematics support centres are usually more sophisticated. A range of learning resources such as handouts, textbooks, videos and online materials are usually made available. Students can use these resources whilst waiting to consult the tutor and, on occasion, the tutor may point students to specific resources.

Support may be provided in a variety of locations. The larger, well-resourced centres have a dedicated location. Often this may be situated in the Library or in a Skills Centre. In other institutions, where the budget for mathematics support is not as large, the provision may be delivered from a teaching room which is timetabled for mathematics support for a few hours each week.

The single location drop-in centre is the most common, but not the only mode of delivery of mathematics support. The University of Portsmouth has a long-established Mathematics Café. The name of this mathematics support provision is apt as this began as a peripatetic support provision that toured the University's student eating venues (Pevy, 2010). Following a widely publicised timetable, a tutor equipped with a trolley loaded with a laptop and a collection of handouts goes to a café and 'sets out his/her stall' on a table and waits for students to come seeking assistance.

The mathematics café approach has some similarities with the drop-in centre model in that staff are available to provide support at times and in places that are publicised in advance.

Students can choose to ‘opt in’ to the support at any time when it is available. A rather different model is that pioneered at St Andrews University (Marr, 2010). In this model, students must book appointments in advance with the mathematics support tutor. This mode of providing mathematics support has a number of advantages. It does not need a large room and the students are provided with support at the appointed time without them having to wait for the tutor to become available. In drop-in centres, at busy times students may experience a considerable wait before a tutor is available to see them. However, there are disadvantages too. The absence of a centre means that there is no opportunity for peer group work (unsupported by tutors) which is often a feature of drop-in centres (Solomon et al., 2010) and which frequently produces deep learning. Furthermore, there can be times when students may have to wait for days before there is a vacant appointment at a time when they are available unlike a drop-in centre where, although they may have to wait for a while, they will receive assistance at the time of their visit.

Further case studies of mathematics support are given in *MathsTEAM* (undated) and Lawson (2012).

Resources for Mathematics Support

Although it is generally accepted that the resource that students most value in mathematics support is the availability of one-to-one interaction with tutors (Croft (2000) and Lawson, Croft & Halpin (2003)) most mathematics support centres additionally provide a range of other resources. These resources are available to promote independent learning by students so that the most expensive resource, staff time, can be used most effectively.

The increasing interest in mathematics support during the early years of the 21st century led to national funding, principally from the Fund for the Development of Teaching and Learning (FDTL) for development of resources that would be shared across the higher education sector. Notable resources produced include *Helping Engineers Learn Mathematics* (HELM)⁵, *Mathematics for Economics: Enhancing Teaching and Learning* (METAL)⁶ and **mathtutor**⁷. Whilst the first two of these projects had a specific ‘user-subject’ focus, many of the resources that they produced are useful to students of other disciplines including mathematics itself. The resources produced in the third project are generically mathematical and focus particularly on material at the interface between pre-university and university mathematics.

Unlike these three time-limited projects, the website **mathcentre**⁸ is an on-going initiative which provides a wide range of resources that are useful for mathematics support. These resources include facts and formulae leaflets, short one-or-two side handouts, longer ‘teach yourself booklets’, interactive exercises and instructional videos which can be viewed online or downloaded for use on mobile devices. In addition, the site is increasingly being used as a repository for resources produced by other projects. All resources on the site are freely available and can be accessed directly by students. Alternatively, staff managing support

5 <http://helm.lboro.ac.uk>

6 <http://www.metalproject.co.uk>

7 <http://www.mathtutor.ac.uk>

8 <http://www.mathcentre.ac.uk>

centres in their own institution can download and print bundles of resources to have them available in hard copy in their support centres. The site also contains resources for staff such as key reports and links to research about the effectiveness of mathematics support.

In addition to resources for students, several resources have been produced by **sigma** (see the following section) that are intended to assist staff in the establishment and delivery of mathematics support in their own institution (for example Green (2012) and Mac an Bhaird & Lawson (2012)). These staff resources are freely available from the **sigma** Network website⁹.

The Mathematics Support Community

As outlined earlier, it is impossible to date the origin of mathematics support as individual ad hoc unreported initiatives have been taking place for many years. The Mathematics Support Association ran from 1994 to 1999 and attempted to bring some national co-ordination to mathematics support activities. Although the Association disappeared in 1999, mathematics support continued to gather momentum. In 2001, the Learning and Teaching Support Network (LTSN) Maths, Stats & OR Network sponsored a study into mathematics support and produced a related good practice guide (Lawson, Croft and Halpin, 2003).

In 2005, a joint proposal by Loughborough and Coventry Universities in the Higher Education Funding Council for England (HEFCE) Centres for Excellence in Teaching and Learning (CETL) programme was successful and this led to the establishment of **sigma**, Centre for Excellence in university-wide mathematics and statistics support. Although much of the work of **sigma** was internal to the two host universities, unlike many CETLs, **sigma** had a considerable programme of activity externally and working with the Higher Education Academy's Maths, Stats & OR Network initiated the annual CETL-MSOR conference in 2006. This conference series has played a significant role in facilitating the bringing together of the mathematics support community for the sharing of good practice and development of the field.

In many institutions, mathematics support practitioners can be quite isolated. They may be the only mathematician in a central study skills/learning support service. Or they may be the only support provider in a research-focused mathematics department. There is therefore a particular need for the establishment of a community of practice in mathematics support.

As part of the CETL programme, **sigma** initiated a pilot 'regional hub' covering South-West England and South Wales. The aim of a regional hub is to act as a focus in a locality, bringing together mathematics support practitioners for mutual support. Following the success of this pilot regional hub a second hub was set up covering North-East England.

In 2009, the National HE STEM Programme¹⁰ commissioned **sigma** to undertake a strand of work to further promote mathematics support in universities. This involved managing funding calls for universities wishing to establish their own support centres and then mentoring the successful proposers; creating a comprehensive network of regional hubs

9 <http://www.sigma-network.ac.uk>

10 <http://www.hestem.ac.uk>

covering the whole of England and Wales; maintaining the CETL-MSOR conference and promoting the transfer of good practice between institutions.

The importance of the work of **sigma** was acknowledged by the Minister for Universities and Science in 2013:

“Another important initiative is “sigma”, a Hefce [sic] funded project. It is establishing approachable maths support services at institutions across the country.”

Willetts (2013:51)

In 2013, HEFCE funded **sigma** to undertake a three-year programme of activity to firmly establish the fledgling mathematics support community that had been initiated as part of the National HE STEM Programme¹¹. This included funding for the establishment of additional mathematics support centres, the maintenance of the network of regional hubs, the continuation of the annual CETL-MSOR conference and the development of additional mathematics support resources.

In the Epilogue to *Responding to the Mathematics Problem*, Kyle (2010) begins with the statement that:

“I probably regarded mathematics support as a form of cottage industry practised by a few well meaning, possibly eccentric individuals...”

Kyle (2010:103)

then concludes with the reflection:

“Mathematics support came of age in the first decade of the 21st century. What might have been described as a cottage industry now plays a respected and widely adopted role in Higher Education.”

Kyle (2010:104)

Mathematics support has developed from a practitioner-focused activity into one underpinned by a considerable amount of scholarship. A number of recent PhD and MRes theses have explored mathematics support (for example Symonds (2009), Patel (2011), Grehan (2012) and Bhakta (2012)). A recent literature review conducted by Matthews, Croft, Lawson and Waller (2013) cited 56 publications related to the evaluation of mathematics support centres. Samuels and Patel (2010) argue that mathematics support has become a discipline in its own right. We might therefore reasonably conclude that the mathematics support community is no longer just a community of practitioners but is also a community of scholars.

Concluding comments

Mathematics support should not be viewed as a panacea that will cure all ills and enable every mathematics undergraduate to succeed. The amount of usage of mathematics support centres by some mathematics undergraduates is an indication that they find value

11 See: <http://www.sigma-network.ac.uk>

in attending the centres. The nature of this value may differ from student to student. For some, the primary benefit is the opportunity to receive support with their specific area of difficulty. For others, the main reason may be because the 'neutrality' of the support centre liberates them to ask questions and enter into discussion with tutors that they feel unable to do in either classes or lecturers' offices. For yet others, the key attraction may simply be the dedicated physical space: a place where they can engage in peer learning with other motivated students and where the availability of a tutor is very much a secondary bonus.

However, there are many students who do not visit mathematics support centres. For some, the support provided by such centres is either not needed or does not align with their preferred learning style and their non-engagement is quite sensible. But for others, it seems highly likely that they would derive benefit from attending and yet they choose not to do so. When questioned such students often give shallow responses, such as not being aware of the centre's location, for their non-engagement. However, research indicates that there are sometimes deeper reasons underlying their behaviour (Symonds, Lawson & Robinson (2008) and Grehan, Mac an Bhaird & O'Shea (2010)).

The National Audit Office (2007) identified the most commonly cited reasons given by students for their withdrawal as:

- Personal reasons
- Lack of integration
- Dissatisfaction with course/institution
- Lack of preparedness
- Wrong choice of course
- Financial reasons
- To take up a more attractive opportunity

These are principally generic reasons and most institutions have measures in place to seek to address at least some of these. Mathematics support can make a contribution to help mathematics undergraduates in three of these areas in particular: lack of preparedness, lack of integration and dissatisfaction with course. It is clear that mathematics support can offer students assistance in dealing with lack of preparedness. Indeed, a significant proportion of the resources available on the **mathcentre** website are focused specifically on material that ideally students would have mastered before commencing their course. In terms of lack of integration, mathematics support centres provide a venue where mathematics students can meet, interact and engage in collaborative learning with others from their course. Finally, mathematics support can also help address dissatisfaction with the course if that dissatisfaction comes as a result of the difference in the nature of the subject in schools/colleges and in higher education as they offer students the opportunity to spend time, in a tutor-supported environment, wrestling with the new conceptual approach towards the discipline that they are required to take.

Mathematics support is increasingly widespread in higher education. Surveys undertaken in 2000, 2004 and 2012 indicate an upward trend in the percentage of responding institutions offering mathematics support from 48% to 65% to 85% (Perkin, Croft & Lawson, 2013).

David Youdan, Executive Director of the Institute of Mathematics and Its Applications, is quoted in Fletcher (2013) underlining the importance of mathematics support:

“Mathematics and statistics support has now attained a critical mass and overcome the significant hurdle where universities worry whether offering such support is an indication of modest aspirations. The accepted position is now that it is a student’s right to receive support with the mathematical content of their degree.”

Fletcher (2013:49)

Mathematics support practitioners are generally motivated by a desire to help all students to succeed and consequently the community of practitioners tends to be open and collaborative rather than closed and competitive. Extensive sharing of good practice and a considerable amount of informal mentoring take place alongside formal schemes within funded projects. Colleagues wishing to initiate or enhance mathematics support within their own institutions will generally not find it difficult to identify experienced colleagues from other institutions willing to offer them support and guidance.

For many years, the biggest issue facing mathematics support providers was securing on-going funding. Mathematics support is staff intensive and therefore can be quite expensive. Over recent years, the increasing publicity that mathematics support has received and the value that it is perceived to have in enhancing the student experience has led many institutions to commit on-going funding to this provision and indeed, to record this in their Office For Fair Access (OFFA) Access Agreements. National initiatives such as the **sigma** Network have maintained the high profile of mathematics support. The removal of uncertainties over funding enables mathematics support providers to focus on what they do best – providing support to students.

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Chapter 5: Lectures and transition: from bottles to bonfires?

David Pritchard

Abstract

Despite much criticism, lectures remain a common and highly valued element of university mathematics teaching, with the potential to contribute significantly to learning. This potential, however, may not be realised because students' and lecturers' expectations, especially at the start of a degree course, are often very different. This chapter gives an overview of the difficulties and suggests ways in which lectures can be used to ease the transition from school to university mathematics.

Introduction

The title of this chapter comes from a classic essay about education:

“For the mind does not require filling like a bottle, but rather, like wood, it only requires kindling to create in it an impulse to think independently and an ardent desire for the truth. Imagine, then, that a man should need to get fire from a neighbour, and, upon finding a big bright fire there, should stay there continually warming himself; just so it is if a man comes to another to share the benefit of a discourse, and does not think it necessary to kindle from it some illumination for himself and some thinking of his own, but, delighting in the discourse, sits enchanted...”

Plutarch (1927)

The recipient of Plutarch's advice was facing the ancient equivalent of the school–university transition. He had come of age, left his tutor's guidance, and become an independent student attending public lectures in philosophy. The essay warns him that he will encounter both bad lecturers and good, that the crowd-pleasers and charlatans among them outnumber the true philosophers, and that in order to learn one must cultivate the ability to benefit from listening to lectures. Almost two millennia later, although much

of education has changed unrecognisably, two points remain the same. Lectures remain, especially in mathematics, a key element of higher education; and many students find the transition from tutelage to independent learning very hard.

This chapter will first summarise the debate about lectures: what, if anything, are they good for? It will then look at the challenges that lectures present when students first encounter them, and suggest how they can help equip students for university mathematics.

The debate around lectures

In contrast to their enduring use in mathematics, lectures are regarded in many disciplines as outdated and ineffective, and have come under sustained criticism since the work of Bligh (1972). Although a variety of criticisms have been levelled at them (see for example Gibbs, 1981), the fundamental objection is that lectures are essentially transmissive: they are simply a mediaeval technology for equipping students with slightly inaccurate versions of the lecturer's own notes. This is troubling both because better reproduction technologies now exist and because mathematics is learned actively, by tackling problems, proving or disproving claims, and testing the logic of arguments. It is argued that we cannot therefore expect our students to learn from lectures.

In fact, lectures in university mathematics form one component of a wider process, such as the traditional cycle of lectures, homework and tutorials. If this cycle is provided, and if the students engage with it, then the question to ask is not whether lectures are a good way of transmitting the material, nor whether students can be expected to 'learn from lectures', but whether lectures help students to start the learning process. In this view, an effective lecture is one that equips or inspires students to engage in further genuine mathematical activity, and it cannot be considered in isolation.

When seen this way (Pritchard, 2010a,b), lectures have several strengths and can perform three main roles. They can be an effective way to communicate ideas, to model the ways in which mathematicians think and work, and to motivate students and shape their attitudes to mathematics. Although lectures are likely to be most effective when their multiple strengths are consciously recognised, traditional practice has evolved to exploit many of them implicitly (LMS, 2010). The resulting benefits may help explain why mathematics students continue to attend lectures even when alternatives are provided, and why those students who rely on lectures in preference to other resources tend to be the most successful (Cretchley, 2005; Inglis, Palipana, Trenholm & Ward, 2011; Trenholm, Alcock & Robinson, 2012).

The merits of lectures, however, are only potential merits. If students have no opportunity to apply the ideas they meet in lectures to fresh problems, and to receive feedback on their attempts, then the best lectures imaginable will be of little benefit. If a student turns up diligently to lectures only, in Plutarch's phrase, to 'sit enchanted', that student is not likely to learn much – and is liable to feel less enchanted come the examination.

Ultimately, the success of the lecture–homework–tutorial cycle depends both on students' participation and on some shared understanding between students and lecturers of what is meant to happen. A lecturer may present three examples to illustrate a principle and find that a student interprets them as three distinct problem-solving templates, concealed

within a bran tub of ‘theory’. A lecturer can spend a lecture motivating the proof of a major theorem, explaining its importance and sharing a sense of its beauty, only to be derailed by the question ‘*Is all this going to be in the exam?*’ What any form of teaching can give students is inseparable from what students are prepared to take from it.

A useful way to express this is in terms of the “*didactic contracts*” that develop between students and teachers (Mason, 2002:166–168). These contracts are generally never made explicit, and indeed, they may contradict the explicit expectations that students and teachers profess to have of each other. Rather, they are negotiated implicitly as each party responds to the behaviour of the other. We will see examples of these contracts, and how they may be negotiated, in the following sections.

Challenges around transition

As lecturers we tend to be painfully familiar with our students’ mathematical weaknesses when they arrive at university. These weaknesses fall into three categories (LMS, 1995): lack of technical facility (see also Gibson, Goldman & Grinfeld, 2005); inability to cope with extended problems; and failure to appreciate the logical nature of mathematics. Whether or not the picture is improving, concern about these weaknesses has continued (Royal Society, 2006), and there is still frequently a gap between what we would like to be able to expect from our incoming students and what they can immediately do.

This gap is important, but it is not the whole story. If we see our students’ difficulties solely in terms of the mathematics that they have failed to learn before they reach university, this may suggest *what* we ought to teach them, but it will not suggest *how* we ought to teach them. To approach this problem, it is necessary to ask more elusive questions: what expectations about mathematics and about learning do our students carry into the school–university transition, and what effect does the transition (see also Chapter 3) have on them?

Threats to the existing contract

Many students who choose to study maths at university do so principally because they were ‘good at maths’ at school (Wiliam, 2005:3; Robinson, Challis & Thomlinson, 2010:\$2.1). In practice this means that such students were good at carrying out complex but tightly specified procedures, in an environment where such procedural tasks were set regularly and frequently, and that their enjoyment of mathematics comes largely from performing these tasks efficiently (Wiliam, 2005; Quinlan, 2009). Effectively, these students and their schools have negotiated a contract in which the students undertake to follow accurately the procedures supplied by their teachers and textbooks; in return, the teachers undertake to provide regular affirmation of the students’ performance, and also to provide them with enough procedures to pass the assessments. A secondary element in many schools is that collaborative learning and communication are heavily emphasised, while classes are ‘kept interesting’ by frequent changes of focus. Although this emphasis develops useful skills, the corollary is that individual responsibility for learning, and sustained attention to problems, receive less attention.

When students arrive at university, the familiar contract is threatened in several ways. First, most students discover that they are no longer among the top performers, and may feel for the first time that they are struggling with mathematics (Goulding, Hatch & Rodd,

2003:376–377; Wiliam, 2005:3–4). Students who are motivated by performance can be very vulnerable to failure, especially if they also believe their mathematical ability is innate and unalterable (Dweck & Molden, 2005). They are prone to stress, may be keen to blame their anxiety on external factors (Rodd, 2009), and may even fall into self-handicapping behaviour in which they follow ineffective study strategies that provide ready excuses for failure (Rhodewalt & Vohs, 2005).

A second threat is that the nature of the subject has changed. Procedures are less important; their place has been taken by argument and explanation, which students may perceive as discursive or directionless and thus less ‘logical’ than following procedures (Daskalogianni & Simpson, 2001). A related threat is the higher level of abstraction, which students may see as irrelevant to real life (Robinson et al., 2010:§3.1) or – to quote one of my own disgruntled students – “making maths seem academic”.

Underlying these threats is a more fundamental one, which is that university education assumes a new balance of responsibility between students and teachers. The authority of a teacher or a textbook is no longer absolute, and students are expected to make their own judgments of validity and relevance (Mason, 2002:177). The classroom is no longer where all learning happens but where it starts: new ideas may be supplied very rapidly and without expecting everything to be immediately understood (Mason, 2002:9). Homework is typically not compulsory and so requires more self-discipline, in addition to the ability to concentrate on single problems for extended periods. Even tutorials, though superficially similar to school classes, require more willingness to take the initiative (Alcock, 2012:§10.2) and to deal with genuinely open questions to which the questioner may not know the answer (Mason, 2002:76).

The demands that the lecture–homework–tutorial cycle makes on students are, at their best, what makes the cycle effective, but they are also dangerous. One danger is that students may break under these demands; another, paradoxically, is that they may not realise the demands are there. Students who are used to being strongly directed often find it hard to appreciate that homework and independent study are essential to learning even when they are not compulsory (Macrae, Brown & Bartholomew, 2003:58–59; Wiliam, 2005:7). The contract that they deduce from the lack of compulsion is one in which university education is largely transmissive and transmission mainly occurs in lectures; accordingly they turn up, in Plutarch’s phrase, like empty bottles waiting to be filled. When this approach inevitably fails, such students feel this as a breach of contract: they become confused and either angry or apathetic (Macrae et al., 2003; Rodd, 2009), and it is very hard for them to have a fulfilling mathematical education.

Developing new contracts

Many students, of course, do not give up following a difficult transition to university mathematics. During the transition period, they develop new habits of study and settle into new implicit contracts with their teachers, and once these are established, their confidence and enjoyment generally recover (Robinson et al., 2010:§3.1). The first few weeks or months of university are therefore a crucial time – the more so because some of the contracts that form are no more appropriate to university than those that were inherited

from school. To understand them we need to examine both what students say about lectures and how they behave.

When asked why they attend or value mathematics lectures, students give fairly consistent responses (Cretchley, 2005; Hubbard, 2007; Slomson, 2010). Lectures are described as a good way to meet new material and acquire an overview of an area; to identify what lecturers think is important; to gain motivation and feel a sense of community; and, even since the advent of online resources, to acquire good notes. These are generally responses with which lecturers would concur – although it is possible that ‘what lecturers think is important’ is merely code for ‘what will be in the exam’. Such findings suggest that students’ expectations are not hugely different from lecturers’.

Students’ behaviour, however, is not always consistent with their responses in surveys. In particular, although they appreciate lectures they prefer to behave very passively during them, avoiding interaction with each other and with the lecturer. Yoon, Kensington-Miller, Sneddon and Bartholomew (2011) suggested that this was due both to fear of public embarrassment and to a belief that the purpose of lectures was for the lecturer to “*to cover the allotted content*”. This suggests a view of lectures as distinctly transmissive or even ritual: a contract where students’ main duty is to turn up because “*you are supposed to*” (a view expressed by 40% of respondents in Hubbard, 2007).

Many of the students interviewed by Yoon et al. (2011) claimed they expected to have to learn the material outside class. Self-reported study patterns, though, suggest that this expectation is optimistic. Figures from England (Bekhradnia, 2012) suggest that mathematics students typically work a little under 30 hours a week, of which half is spent in class: this ratio of self-study to class time suggests that many students rely on lectures for a large portion of their learning, whether or not they actually expect to learn much from them. Like many educationalists, these students have fallen into the trap of trying to make sense of lectures in isolation from other activities.

The implicit contracts between students and teachers are not, of course, created solely by the students, and their confused expectations of lectures may reflect our own. As noted earlier, it is common to represent lectures as essentially transmissive, and this attitude must communicate itself to the students. It is also common to use ‘lectures’ as shorthand for all university teaching, just as we use ‘lecturer’ as shorthand for ‘university teacher’, and it is common when asking for student feedback and when trying to improve our teaching to focus on what can be done to improve or replace lectures. This focus is probably misplaced: as noted above, students’ performance is not improved if they rely on alternatives to lectures; and even innovations such as classroom response systems which appear to make lectures more effective have little measureable effect on achievement (King & Robinson, 2009). Although lectures lie more within our direct control than other aspects of the teaching cycle, the real problems may lie elsewhere.

All this is not to say that lectures are unimportant in the school–university transition. The fact that some students develop unrealistic expectations of lectures, and that we unconsciously collude with them in this, suggests that it is worth examining how our use of lectures shapes our contract with our students. At the same time, lectures can be used directly to address some of the threats arising from transition.

Addressing transition-related challenges through lectures

The preceding section argued that new university mathematics students face difficulties that arise both directly through weaknesses in their mathematical education and indirectly through the educational ‘contracts’ that they inherit from school (cf. Daskalogianni and Simpson, 2001). Some of these difficulties may be addressed most effectively outwith lectures, but must be allowed for when we lecture. Others can be tackled directly using lectures alongside other elements of our teaching. This section will first look at what lecturers can do to help students deal with threats and weaknesses, and then at ways in which institutions can support or hinder lecturers in doing so.

The role of lecturers

Accommodating lack of technical facility

Of students’ mathematical weaknesses, lack of technical facility is the one that lectures can do least about. What is needed, as many students realise (Gibson et al., 2005), is practice. Homework and tutorials, supplemented with diagnostic tests and extra support where necessary (see Chapter 4), may be the best way to address this weakness. Another promising approach, which lecturers can sometimes employ, is to revisit school mathematics from different angles, reinforcing basic material without presenting it as remedial (Robinson et al., 2010:\$7).

This weakness does carry implications for effective lecturing. Because some students see basic algebraic processes as major operations in themselves, they are easily disconcerted when these processes are passed over swiftly as part of a larger argument. This is particularly the case if students are used to looking for local rather than global understanding of a mathematical argument (Lithner, 2003). Surrendering to the demand to fill in all the details is time-consuming and reinforces the perception that these details are difficult and important. An alternative is to leave gaps for students to fill in later: it is essential in this case to advertise that this is what is intended, and to anticipate that some students will not do so.

Addressing inability to solve extended problems

Although the inability to tackle extended problems also results largely from lack of practice, it has an extra dimension because one of the purposes of lectures (Pritchard, 2010b) is to provide students with a model of how mathematicians think. Faced with an extended problem, we naturally try to break it into smaller problems and tackle them systematically: by demonstrating and narrating this process, we may help students to break through the mental block that descends on them when they are first confronted with unfamiliar problems. Such modelling will not be effective if students do not realise it is happening, so it is important to be explicit (cf. Mason, 2002:56–57) and to advertise the occasions when we put the notes to one side and work from reason rather than from memory. Students also readily confuse what we write as part of a solution with the comments we make about the solution process, so it can be helpful to distinguish between these – for example, using a black pen for the solution, and a blue pen for comments and explanations – and encouraging students to do likewise in their own notes.

Using lectures to demonstrate problem-solving is popular with students (Robinson et al., 2010:\$5.1). The trap that it presents is the “*tyranny of examples*” (Barton, 2011), a pernicious contract resembling that inherited from school, in which the lecturer undertakes to provide templates for every question that could appear in the exam, and the students undertake to memorise and reproduce them. It is necessary to resist the pressure – expressed persistently and plaintively through student feedback – to succumb to this tyranny, which arises because students misunderstand what mathematics is about.

Clarifying the nature of mathematics

The lecturer’s modelling role is even more important in addressing students’ perception of mathematics as a set of ritual procedures rather than a process of creative logical argument. In many cases, they misconceive what real mathematics looks like because they have never or rarely seen it. We can help to address this weakness, and to make the change of perspective less threatening, by doing mathematics, live, in front of them: “*thinking and struggling*” through problems (Atkinson, Derry, Renkl & Wortham, 2000); even, at best, providing a performance that engages their imaginations like those of a theatre audience (Rodd, 2003; Körner, 2013). Of course it is unrealistic to expect every lecture to be an inspiring piece of theatre, but most courses provide a few opportunities to share something of the intellectual fire inside the subject.

The dangers of presenting lectures as performances are that we focus too much on putting on a show, while students come to treat the lecture as a form of entertainment. Like those caricatured by Plutarch, they will enjoy the warmth without kindling fires for themselves. A related danger is that students may treat a charismatic lecturer as the source not just of social but of intellectual authority, and so never come to rely on their own reasoning. Part of the solution may be for lecturers to advertise our own fallibility, which brings us to the next point.

Mitigating anxiety and vulnerability

The anxiety and vulnerability experienced at the school–university transition probably cannot be eliminated, but they may be reduced by demonstrating that all mathematicians are fallible and by giving mathematics a human face. Wood, Joyce, Petocz and Rodd (2007) argue that the threat that abstraction presents to new students can be reduced when “*a lecturer personifies the content of his/her lecture*”, making an emotional connection to abstract concepts and connecting multiple representations of ideas. Lecturers can also reduce the threat presented by students’ encounter with failure and difficulties by “*being human*” (Mason, 2002:57–58) and sharing our own experiences. Rather than presenting mathematicians as uniquely gifted geniuses, we can demonstrate that we are human beings who meet difficulties and struggle through them, both making and correcting errors on the way.

Tackling problems live on the visualiser or chalkboard, rather than writing up a solution from notes or memory, naturally involves us in bringing multiple resources to bear on problems and in dealing with difficulties and errors. Once this pattern is established, even common errors that students make can plausibly be presented as errors that one has made oneself in the past (whether this is true or not). If not overused, this ploy can generate a real air of relief in the classroom. The disadvantage of live performance is that it involves risk: Körner (2013) likens lecturing both to a conjuring trick and to a high-wire act. Some students will interpret any mistakes we make as evidence of poor preparation or

incompetence, and it needs some self-confidence to shrug off their criticism. It is certainly safer not to expose our own weaknesses in class, but it may be necessary to do so if we are serious about inviting students into a real experience of mathematics.

Establishing the purpose of lectures

The final challenge identified above was that students readily misunderstand the purpose of lectures and how they relate to other activities. This manifests itself in passive behaviour during lectures, and in a tendency to forget that homework and tutorials are necessary to develop an understanding that merely begins with lectures.

Like Plutarch, we can start to address this misunderstanding by explaining to our students what lectures are for and how to get the most out of them. Alcock (2012:\$9) provides a good example; it might be worth going one step further and providing an overview of the entire lecture–homework–tutorial cycle, a pattern that seems so obvious to us that it may be invisible to students. Just explaining to students how to use lectures will certainly not be enough, however: we need to find ways to ease them into a new way of learning. The difficulties here are to introduce familiar elements without compromising the strengths of lectures, and to provide a smooth transition from strongly directed to more independent learning.

Every guide for new lecturers (for example Mason, 2002:\$2; Cox, 2011:\$3) contains suggestions of how to combine lectures with complementary activities. An increasingly popular approach is to set short questions and poll the class on them using a classroom response system (King & Robinson, 2009); a two-stage process (Nicol & Boyle, 2003), in which students first work and vote on questions individually and then revisit them in small groups, can be particularly effective. Such exercises help to vary the pace of a lecture and provide immediate feedback on students' difficulties, though they can degenerate into another 'tyranny of examples'.

An especially thorny question is whether students should be required to take notes in lectures or have full notes supplied online. The more that is supplied to students, the more accurate the information that they have and the less of their attention has to be devoted to copying from the board. Equally, the more that is supplied, the more likely students are to try, unsuccessfully, to manage without lectures (Cretchley, 2005; Inglis et al., 2011; Trenholm et al., 2012), and the less attention they will pay when they are there. Mason (2002:64) gives a nice summary of the possible compromises: for many courses, some form of partial notes, which may state the theorems or the questions but leave the proofs or solutions to be constructed in class, seem to strike a reasonable balance.

Other practical issues with notes are less obvious. One is that many students arrive at university unable to read mathematical texts (Robinson et al., 2010:\$5.1; Shepherd, Selden & Selden, 2012), so supplying written notes merely replaces a note-taking problem with a reading problem. Another is that contemporary students, used to keyboards, often write much more slowly than those of earlier generations, while computer slideshows tempt us to present material far too fast. The traditional way to control pacing is to write by hand everything that students are expected to copy: on a visualiser, it is helpful also to use a pen that slows one's writing down. (No pace, though, will be slow enough to satisfy some students, and it may be necessary to resist their protests.)

Using the lecturer's authority

Finally and very generally, a crucial feature of lecturing is that it lends the lecturer an artificial social authority, as the figure on whom attention is focused. This authority means that a lecturer's behaviour can do a lot to shape the contract with new students. The points raised above concerning the nature of mathematics are one reflection of this: if we teach mainly set procedures we reinforce the procedural contract inherited from school; if we teach genuine mathematics we may start to renegotiate this contract.

Another aspect is classroom management. As schoolteachers know, this is especially important when dealing with a new class (see for example Cox, 2011:§3.7, §3.9): by arriving promptly, dealing courteously with students and requiring the same behaviour of them (cf. Alcock, 2012:§9.6), lecturers can establish the attitude and the levels of attention that successful learning requires. By using lectures to set or distribute homework and to respond to matters raised in tutorials, one can also make it clear that different elements of the course are expected to reinforce each other, and that all of them are necessary.

The danger of a firm classroom management style is that it can reinforce students' impression that they are expected to be entirely passive: seen in class but never heard. To mitigate this, a lecturer can use the "*mandate to interact*" that their status gives them (Yoon et al., 2011), for example explicitly instructing students to discuss a question with their neighbour. It is also wise to lay down some house rules that encourage interaction – such as the rule that 'there is no such thing as a stupid question' – reiterating and acting on these until students come to believe them. At the end of the day, different lecturers have different personalities and will handle classes in different styles, but any successful lecturer implicitly uses her authority to shape the expectations of her students.

The role of institutions

The principal responsibility for lectures lies with lecturers, but our institutions can support or hamper us. Especially in first year, it is important that homework and tutorials are properly resourced, so that students are not forced to rely exclusively on lectures. Similarly, it is necessary to provide properly equipped and suitably designed rooms for lectures. Of course this is desirable for all lectures, but the large and uncertain student numbers in first-year classes tend to breed timetabling problems, while these students are simultaneously much more vulnerable to disruption because their study habits are still being formed. Many lecturers have had the experience of trying to muddle through the first few weeks of term in undersized or ill-equipped rooms; most of us will testify that students who have been introduced to university in this way can take a long time to settle down.

Institutional policies can undermine lectures in unintended ways. It is particularly important that student demands for online material are not answered by a blanket policy of providing everything online, regardless of the damage that this can do (Inglis et al., 2011). Another policy matter concerns student feedback and the attitudes to learning that it encourages. Surveys that treat students as consumers imply, and may reinforce, a sense that their educational contract places all the responsibility on the teachers; surveys that invite students to consider their own engagement with a course may instil a different expectation.

(Multiple-choice questionnaires seem to encourage particularly unreflective responses, as well as providing absolutely no useful suggestions for improvement.)

Perhaps the most important point to recognise is that lecturing to new undergraduates, whether on mathematics degree or service courses, is difficult. Mathematicians sometimes assume that the difficulty of teaching a course depends only on its content, so the most inexperienced staff should teach the most elementary courses. As the discussion above indicates, this is a mistake. To lecture effectively to new students, one needs the confidence to tackle problems live, the performance skills to motivate students without simply playing to the gallery, and the experience to sense and anticipate their technical difficulties. Any department that expects newly appointed lecturers to possess these qualities has itself to blame should students emerge from first year unprepared for their degrees.

Summary

There are both good and bad reasons for lecturing and for attending lectures, and corresponding good and bad uses that staff and students can make of them. Two fundamental errors are to see lectures principally as a way of transmitting the contents of notes or textbooks, and to expect them to be effective in isolation from other learning activities. As lecturers, we sometimes make these errors; evidence suggests that our students are still more likely to make them, especially when their expectations are formed by their experience of strongly directed learning in school. Lectures are thus vulnerable to students' misconceptions of what mathematical learning involves, and if used unreflectively they may exacerbate these misconceptions.

Nevertheless, lectures have a positive role to play, in which they are not readily replaced by other forms of teaching. In particular, they provide an invaluable opportunity to draw students into the world of university mathematics by modelling the processes of mathematical thinking, demonstrating that mathematics is about creative and logical thought rather than applying memorised algorithms to highly stereotyped problems. When they are properly used and adequately supported, lectures can be a powerful tool for rewriting our students' expectations and helping them make the transition to university mathematics.

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Chapter 6:

Enabling students to become independent learners

Louise Walker

Abstract

One of the greatest challenges facing students entering undergraduate mathematics programmes is adapting to the change from intensively supported learning to independent learning. Students have to deal with a huge increase in class sizes and, at the same time, a reduction in support from teachers and parents. To survive at university they have to develop strategies for self-motivation and an ability to make maximum use of teaching and learning support. This chapter will focus upon the use of peer assisted study, facilitated learning, and online support to help students become active learners and to equip them with the study skills they need to become independent learners. The benefits of peer support and facilitated learning have been widely researched (see for example Schön, 1987; Topping, 1996; Boud, Cohen & Sampson, 2001). The use of techniques, focusing upon peer assisted study, postgraduate facilitators of group project work and the use of online discussion groups and computer aided assessment to support large lecture courses will be discussed in the particular context of the School of Mathematics at the University of Manchester.

Introduction and context

We need to begin by defining what we mean by the phrase ‘independent learners’. One could argue that there is no such thing as a true independent learner because the process of learning relies on interaction with external sources, whether they are peers, teachers or learning materials. The term *independent learner* is widely used within higher education and while we feel we may recognise one when we see one, it may mean different things depending upon the subject and level of study. Students often report that one main difference between school and higher education is the fact that they are expected to be much more independent at university. Candy (1991) suggests four characteristics of independent study:

1. *“Independent study is a process, a method and a philosophy of education: in which a student acquires knowledge by his or her own efforts and develops the ability for inquiry and critical evaluation;*
2. *it includes freedom of choice in determining those objectives, within the limits of a given project or program and with the aid of a faculty advisor;*
3. *it requires freedom of process to carry out the objectives;*
4. *it places increased educational responsibility on the student for the achieving of objectives and for the value of goals.”*

Candy (1991:13)

From this we could deduce that an independent learner is someone who is able to take control of, and manage, their own learning, aided by academics, within the constraints of their programme of study. In this context, *independent* does not mean *isolated*. An independent learner will effectively use the support of peers and experts as well as a range of resources and they will be in control of how and when they learn.

On entering a higher education mathematics programme, students may find it difficult to adjust to the more rigorous approach afforded to the subject. The London Mathematical Society’s report *Tackling the Mathematics Problem* (1995:9) states that *“Most students entering higher education no longer understand that mathematics is a precise discipline in which exact, reliable calculation, logical exposition and proof play essential roles.”*

Students will be expected to spend substantial time outside formal classes trying to understand difficult and unfamiliar course material and will have to deal with less obvious examples and open-ended problems. Walkden and Scott (1980) identified three attitudes which contribute substantially to the mathematical difficulties encountered by undergraduate students:

1. *“an expectation to assimilate new ideas without mental effort;*
2. *a reluctance to devote time to study and to practice;*
3. *the lack of persistence necessary to tackle exercises of a non-trivial nature.”*

Walkden & Scott (1980:45)

In a study of first year mathematics students, Anthony (2010:9) found that *“successful students placed more importance than failing students on... understanding rather than rote learning, and the ability to work independently.”*

How do we help our students to develop the skills and mindset to become independent learners? We shall consider, and then explore in more detail, three factors that contribute to this process:

1. Motivation
2. Autonomy
3. Collaboration

Motivation

There are many factors affecting motivation to study. Psychologists break motivation into two main types – intrinsic and extrinsic motivation. Intrinsic motivation is created by interest and pleasure in undertaking a task and is not influenced by external factors. Extrinsic motivation is driven by external factors such as reward or fear of punishment. It has been suggested (Ryan & Deci, 2000) that learners who are motivated intrinsically will engage more deeply with their learning and have a desire to fully understand new ideas. On the other hand extrinsically motivated learners are goal driven and will study strategically with minimum effort to achieve their desired grades. In higher education, students are continually told that success equates with gaining high marks in examinations, and they are also very aware of the benefits of obtaining a first-class or upper second-class honours degree in terms of their future employment. This encourages students to take a strategic, surface approach to learning (Ramsden & Entwistle, 1981). Students often do not see the value of the material they are studying beyond using it to pass examinations. This means that they have little incentive to retain knowledge and skills for later use. In a study of third year mathematics undergraduates Anderson, Austin, Barnard and Jagger (1998) found that there had been poor retention of first year material, even by students who had performed well later in their programme of study.

Of course there is nothing wrong with students wishing to gain a good degree class and strategic learning can be a successful way to do this due to the predictability of assessment, especially examination papers. We are all aware of students who have *learned* a course by memorising solutions to past examination papers. These same students may accuse lecturers of being unfair when the examination questions are different from those in previous years. Anthony (2000) found that failing students blamed their lack of success on the fact that assessments contained unfamiliar questions.

Students motivated by an intrinsic love of their subject will enjoy studying and are often more successful in assessments than their peers because they will learn material above and beyond that required to tackle standard examples and will be able to attempt unfamiliar questions in examinations. Their enjoyment of learning means that they will not mind spending time studying and practising examples. They will work hard throughout the year, rather than simply cramming for examinations and the knowledge they acquire will be retained for future courses rather than being instantly forgotten after the examination. When asked what helped them to become an independent learner, one student at the University of Manchester replied *“the most important thing is interest. Interest can drive me to learn something deeper by myself.”*

Lecturers are likely to be interested in mathematics for its own sake and may wish that all students were intrinsically motivated with a similar deep interest and passion for their subject. In reality, we have to accept that many students will be extrinsically motivated and see a degree solely as a mechanism for obtaining a good job. We should try to develop teaching, learning and assessment methods that encourage more intrinsic motivation. This will lead to students who are driven to succeed because of the rewards of a qualification but also who engage more deeply with their subject and enjoy the process of learning.

Student-led activities such as peer learning, project work and group work encourage students to engage more deeply with the course material. Enthusiastic, passionate lecturing will inspire students to see the beauty and importance of the ideas they are learning. If students are able to see how their course material relates to other subject areas and real world problems they may be more interested in gaining a deeper understanding.

Self-efficacy is another important factor in motivating students. Self-efficacy is a belief in ones ability to be successful. It has been widely studied by psychologists, most famously by Albert Bandura (1994). Zimmerman (2000:82) states that *“self-efficacy has emerged as a highly effective predictor of students’ motivation and learning”*. Therefore we should try to increase students’ confidence in their ability to succeed, and as Parsons, Croft and Harrison (2009) note:

“The findings suggest that having attended to the mathematics syllabi, lecturers could seek to boost student confidence in their ability in mathematics as a further means to improve student performance at university.”

Parsons, Croft & Harrison (2009:53)

This may be particularly important for programmes where students have a range of entry qualifications; in particular where some students have taken Further Mathematics A-level while others have only studied a single A-level in mathematics. Parsons, Croft and Harrison (2011) discovered that the self-confidence of engineering students was affected by their mathematical background. Hoyles, Newman and Noss (2000) found that at a university where over half the mathematics intake have studied some Further Mathematics, those first year undergraduates who had not taken Further Mathematics often believed they were disadvantaged.

Motivation, whether it be intrinsic or extrinsic, is essential for students to become independent learners. Students who have no desire to study for interest or reward will quickly lose interest in their programme.

Autonomy

Autonomous learners have control over *what*, *when* and *how* they learn. Within the specifications of a mathematics degree programme there will be constraints on *what* a student has to learn. However most degree programmes give students some choice, at least in their latter stages. This enables students to personalise their learning and develop a broad range of knowledge and skills. Students who are able to design their programme by making informed choices are more likely to be motivated and independent. It is important that, where choice is available, students are properly advised about the knowledge and skills they will acquire from different modules. Detailed information about the syllabus, teaching methods and assessment need to be freely available and students should be able to discuss this with academic staff to ensure they make appropriate choices.

Independent learners need to be able to manage *when* they study. Although most mathematics degree programmes are structured around a fixed timetable, students should be able to access materials and gain feedback on their work outside of formal teaching hours. Mathematics support centres (see Chapter 4) give students a chance to seek help when they are ready to make the most of it, rather than at a fixed time during a tutorial

or through a lecturer's 'office hour'. Online resources and computer-aided assessment (Chapter 11) can provide an opportunity to learn at a time that suits the student and if these resources also provide feedback, students will be able to continually review their progress. A blend of face-to-face teaching and online resources enables students to control their learning and consolidate their knowledge.

It has been known for a long time that every student has a unique way of learning and there has been a significant body of research by educational psychologists into learning styles; see for example Entwistle (1988). Students should be able to experience a range of teaching, learning and assessment practices to enable them to learn effectively and independently. Traditional lectures tend to favour those learners who have visual and auditory learning styles. By introducing more problem-based learning, group work and projects into our curricula we allow students with different learning styles to 'play to their strengths'.

Collaboration

As previously mentioned, independent learning is not the same as learning in isolation. Independent learners are able to make effective use of collaboration with other students and academic and support staff. The advantages of peer learning and support have been well documented by Hill, Gay and Topping (1998) and Topping (1996). Glynn (1985) argues that students learn *how* to learn from their peers and that the social interaction between students gives them more control over their learning. The use of later year students as peer tutors for those in their earlier years benefits both sets of learners. Duah, Croft and Inglis (2013) found that peer assisted learning helped students achieve higher marks in their final examinations and prevented 'cooling off' where students lose motivation and interest in their chosen subject. The earlier year students are able to ask questions they may feel afraid to ask their lecturers. They pick up practical study skills from their more experienced peers and learn how the material they are studying is used later in their programme of study. Working with students who have recently been through the same experience also helps increase their confidence and self-efficacy. The peer tutors also gain a great deal from their tutees. Their knowledge of, and confidence in, their subject increases as they try to explain the mathematical details to others. They feel part of a community of learners and are more likely to reflect upon their own learning style. In their review of good practice in mathematics peer support, Sinka and Kane (2011) gathered quotes from both peer tutors and tutees:

"Student mentors know what we are going through. (1st year mathematics student)."
Sinka & Kane (2011:12)

"You understand something when you can explain it to someone else." (PASS leader)."
Sinka & Kane (2011:12)

Collaboration also occurs between undergraduate and postgraduate students and between undergraduates and academics. Facilitated learning, where an expert guides rather than teaches, enables undergraduate students to take more control of their own learning and engage more deeply with the material (Brockbank & McGill, 1998).

Tutorials, problem-based learning, individual and group project work and mathematics support centres all provide opportunities for facilitated learning. In a facilitated learning environment, students initiate the learning process and have a greater sense of ownership of the knowledge and skills they acquire. The facilitator's role is to guide students through the learning process and to reassure students that they are on the right track rather than telling students exactly what to do. The use of postgraduate students as facilitators can be particularly fruitful; they act as role models of successful independent learners that undergraduates can more easily relate to than academic staff.

Discussion and evidence

In this section we will look at various schemes and initiatives that have been developed at the University of Manchester and other higher education institutions that have helped students to become more effective independent learners.

Peer assisted study and support

Peer assisted study has been used to great effect in several mathematics departments. The schemes may have different names such as Peer Assisted Learning (PAL) (Waldock, 2010) and Peer Assisted Study Sessions (PASS) (Walker, 2010) but they each provide opportunities for later year students to facilitate the learning of earlier year students through regular meetings. Feedback from students suggests that both sets of students gain in confidence and motivation through this interaction. Other, more informal, peer support networks have been developed by staff and students, aided by social media (Goodband & Samuels, 2010). There are a wealth of opportunities to develop online student support networks using virtual learning environments and social media sites. A comprehensive guide to peer support in mathematics undergraduate programmes within the UK has been produced by Sinka and Kane (2011).

At Manchester a PASS scheme for mathematics undergraduates based on the Supplemental Instruction (SI) model developed at the University of Missouri in the 1970s and brought to the UK by Jenni Wallace (1992) in the 1990s is run. Second and third year undergraduates volunteer to become PASS leaders. They receive extensive training in group facilitation and communication techniques. PASS leaders work in pairs or threes with a small group of around ten first year students. They meet weekly to discuss course material and share advice on studying at university. The scheme is not meant to replace any formal teaching but instead offers a chance for students to discuss mathematics and develop study skills in an informal, non-threatening environment. The PASS sessions are planned by the students to suit the needs of the group and at a time convenient for everyone involved. They may choose to hold their session in a student common room, library or classroom. This autonomy means that students remain actively involved in PASS throughout the semester. Peer assisted learning is an excellent way for students to develop as independent learners. By discussing problems, students develop self-efficacy and are reassured when other students are also challenged by some of the course material. As Zimmermann (2000) notes:

"The effect of prior math experiences on math problem solving was mediated primarily by self-efficacy beliefs, but self-concept played a small but significant role."

Zimmermann (2000:85)

In a study of peer assisted learning at Dublin City University, Parkinson (2009) found that:

“The tutored students progressively increased their performance at in-house tests in calculus compared to the untutored students, their examination marks in chemistry and calculus substantially improved (>13%) and failure rates were cut dramatically. Student progression was substantially improved.”

Parkinson (2009:381)

Arguably it is the PASS leaders who gain more from the scheme than the first years attending PASS. PASS leaders gain confidence in their own mathematical skills and knowledge as they experiment with different ways of explaining difficult concepts and ‘link together’ different aspects of mathematics. The University of Leeds has recently introduced a scheme similar to PASS in their School of Mathematics. One of their PASS leaders commented:

“I have become more confident, not only in what I know, but also in being able to express what I know to other people. It’s a good experience for anyone who wants to improve their self-confidence.”

As they experience the way other students learn, leaders are able to reflect upon their own strengths and weaknesses as learners. The PASS leaders are motivated in their own studies by the feeling of achievement they gain from helping other students to learn. Students who attend PASS as first years are more likely to become PASS leaders as second years and to form their own informal peer support groups. By encouraging more collaborative learning we are helping students to develop the skills they need to be successful on their degree programme and in later life.

Facilitated learning

In this section we explore some of the learning environments that have been developed at the University of Manchester and elsewhere to allow students to learn independently under the guidance of staff and postgraduate students. Facilitated learning comes in many forms but a common theme is that students take more responsibility for, and control of, their learning. The facilitators may be peers, postgraduate students or academic staff. Postgraduate student demonstrators are used widely in mathematics undergraduate teaching and often make excellent facilitators because they can identify with undergraduates’ learning processes and make them feel at ease.

Tutorials, group work and project work provide an environment for facilitated learning, allowing students to identify what they need to know, to discuss problems in more depth and to have the opportunity to learn from their mistakes. In this way they enable students to take control of their learning.

The Mathematical Workshop is a course for first year mathematics undergraduates that focuses upon developing problem solving skills and introduces students to the process of mathematical modelling. It is taken by about 300 students who work in groups of five with a postgraduate facilitator. The groups are given problems to discuss in weekly two-hour sessions following a lecture introducing the background material. Some of the problems involve the analysis, solution and validation of mathematical models of real world phenomena, others focus upon open-ended problem solving. The use of real world

examples motivates students to engage with the problems. The postgraduate students are present to ensure that the groups work effectively; they make sure that all group members are engaged in the discussion and provide hints if one of the groups get stuck. The group work gives weaker, less confident students a chance to benefit from the discussion of problems they would find difficult to tackle alone. The stronger students are able to practise communicating mathematics effectively by explaining their ideas to the rest of the group. In this way, all members of the group gain in confidence and mathematical fluency. Students' feedback reflects the importance of the social interactions in the group work:

"All the projects were fun and stimulating to do and it was a useful way to meet people on the course."

"Basically this course made me lots of friends and also tells us how to work with a group."

Students become more intrinsically motivated by the pure enjoyment of doing mathematics together.

Another environment where facilitated learning can be used effectively is in tutorial classes. Again, postgraduate facilitators are particularly effective because the undergraduate students may feel more at ease with them than with a lecturer and therefore are more likely to actively engage in discussion. In a good example of this, rather than directing the students' learning, one of our recent postgraduate demonstrators allowed the students to take control of the tutorial. Using the chalkboard the students took it in turn to explain their solutions to the problem sheet to the other members of the group. The postgraduate had little input beyond asking a few questions to deepen the discussion. The end of course questionnaires highlighted this approach as a particularly helpful way to engage with the course material. In a study of the effectiveness of group work in first year mathematics at the University of Manchester students reported that:

1. Group work helped them to consolidate and further explore material covered in lectures.
2. Peer discussions facilitated their understanding of the lecture material.
3. Group work helped them to retain their concentration levels.
4. They felt more able to ask for clarification or help from their peers than from the lecturer.

Mathematics support centres, such as the Mathematics Learning Support Centre at Loughborough University (Croft, 2010) are being increasingly employed by universities to provide a space for students to manage their learning. This environment allows more autonomy than traditional lectures and support classes. As well as seeking help from experts, students are able to access resources and work together in a more informal setting than a timetabled class. Solomon, Croft and Lawson (2010) found that students feel more comfortable coming to the neutral ground of a support centre rather than approaching staff in their offices. There are also online support centres, such as **mathcentre**¹, which provide even greater freedom for students to choose when and where to study.

¹ <http://www.mathcentre.ac.uk>

Project work is another way in which students develop independent learning skills, facilitated by a project supervisor, and are a feature of many mathematics undergraduate programmes. Projects can be used to develop higher-level study skills such as research and analysis and increase intrinsic motivation through a deeper engagement with a mathematical topic.

Online support materials

The introduction of computer aided assessment (CAA) and virtual learning environments (VLE) in mathematics teaching has provided an opportunity for students to take more control of their learning and study independently. CAA is particularly useful for independent learning when it used for formative assessment. The ability to access vast question banks and gain instant feedback on their solutions means that students can use these resources to practice techniques and learn from their mistakes. As well as being an easy way to access online lecture notes and exercises, VLEs have enabled students to interact with lecturers and other learners through discussion boards, blogs and wikis.

There are many examples of good practice in CAA within the mathematics higher education community (see also Chapter 11), such as STACK (Sangwin, 2013) and Maths e.g. (Greenhow, 2008), both of which are open source and so freely available; there are also commercial packages available with similar functionality such as Maple T.A.². STACK is able to deal with algebraic and symbolic answers and so it can be used to test a wider range of skills and knowledge than multiple choice quizzes. Maths e.g. has huge question banks formed by taking standard question types and randomising parameters and has been used to support mathematics courses across a range of disciplines. What both systems have in common is the ability to provide instantaneous, detailed feedback to students taking the tests. This feedback is crucial if students are using the system independently for formative assessment and not as part of a timetabled class or summative assessment. Students using Maths e.g. have been found to spend substantial amounts of time using the feedback as an independent learning resource. When investigating the effectiveness of CAA in a mathematics course for engineering students, Pollock (2002:249) found that *“most students preferred this method of assessment to traditional examinations, because it increased confidence and reduced stress levels”*. Wadsworth, Husman, Duggan and Pennington (2007) demonstrated that online mathematics tests could improve levels of motivation, concentration and self-efficacy.

As well as improving confidence and fluency in basic techniques, CAA systems help students to identify what Sangwin (2013) calls *“question spaces”* made up of questions of the same basic type. Students learn how to apply the same methods to solve different questions in the same question space. This may help students to develop higher-level skills such as abstraction and generalisation and increase intrinsic motivation to understand underlying mathematical properties. At Manchester we have made wide use of STACK to develop formative assessment to support mathematics courses for engineering and science students. We have also used STACK to provide follow-up materials for the diagnostic test given to all first year mathematics undergraduates which is also supported by online resources such

² <http://www.maplesoft.com/products/mapleta/>

as the HELM materials (2005). These materials give students the autonomy to consolidate their knowledge of A-level mathematics at their own pace.

At Manchester use has been made of the VLE, Blackboard 9³, to develop formative and summative assessments for several undergraduate mathematics courses. This has been used successfully in our first year probability course. This course is taken by nearly 400 students and in the past it has been difficult to provide students with quality feedback through traditional tutorials. The course lecturer developed Blackboard quizzes that were used in the tutorial classes. Students were given practice questions to work on in their own time and then the latter half of the tutorial class was used to take an online test. The lecturer used the in-built discussion boards on the VLE to enable students to help each other and interact with postgraduate demonstrators and the lecturer outside of the timetabled classes. All the lectures for the course were recorded as podcasts, with the lecture slides accompanied by an audio recording of the lecturer. These innovations have proved popular with the students, who appreciate the flexibility in how and when they learn as well as the chance to get continual feedback on their progress. Randomised question banks enable students to have access to hundreds of examples and also mean that lecturers can set summative assessments where each student receives a different set of questions or students repeat an assessment until they gain a particular score. Both approaches encourage more engagement with the course material (Moriarty, 2013).

Conclusion

In this chapter we have explored some of the many factors influencing independent learning and we have discussed some of the ways in which these have been addressed in mathematics undergraduate teaching and support. We conclude by offering some practical advice for staff who would like to encourage their students to develop independent learning skills:

- Develop opportunities for peer learning and support – this may be achieved formally by setting up a PASS or peer mentoring scheme or by encouraging students to form their own peer support groups. It helps if there is dedicated group study space for students to access outside classes.
- Introduce facilitated learning into courses, either through tutorials or group project work.
- Extend access to resources and support either online or through mathematics support centres.
- Do not overload the curriculum but instead give students more time to engage fully with the course material.
- Increase intrinsic motivation by explaining the importance and relevance of course material, linking different areas of mathematics and showing how mathematics is used to model the real world.

³ <http://www.blackboard.com>

- Increase extrinsic motivation by helping students to articulate the skills they develop from learning mathematics and by describing the opportunities available to mathematics graduates.
- Help students to take control of their learning and influence curriculum design and delivery by taking part in discussions with academic staff.

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Chapter 7:

Group work within undergraduate mathematics

Neil Challis

Abstract

This chapter contains a discussion of some of the issues which arise when one employs or considers employing a group work approach within an undergraduate mathematics programme. Both formally assessed group work and informal or low stakes work are considered. The chapter consists of two distinct, but related, sections: Initially the question “Why use group work?” is addressed, with discussion focusing upon enhancing graduate employability, forming a community of learners and some of the educational reasons why group work is considered valuable. Later the matter of how to deploy group work is considered, and some recent work in this area is explored to see what may be learned.

Introduction and context: the ‘why’

What has working in groups got to do with learning mathematics? Doesn’t the word ‘collaboration’ carry an overtone of cheating? Why would one take on all the potentially complicated issues that can arise when dealing with other people when you don’t have to?

Over past years, various answers have been offered to questions such as these, with both positive and negative views of the usefulness of group work vehemently expressed. It is an educational topic which engenders strong feelings in the mathematics community, amongst both academic staff and students.

To provide a structure for a discussion of the issues surrounding group work, it is helpful to categorise these issues, and one has a degree of choice here. For instance one could choose to distinguish between formal and informal group work. Formal group work might be defined as group activity stipulated as a required part of a course, often involving for example mathematical modelling projects. Such work raises issues of how to assess fairly, requiring decisions on assessing individual contributions to the group, possible opportunities for peer assessment, and a clear-headed approach to the matter of plagiarism. Indeed the most

controversial issues arise from the assessment of formal group work. Informal group work on the other hand may be defined as work wherein assessment is not crucial or high stakes, and students often form groups themselves or with the help of tutors.

Alternatively, group work may be categorised by its potential purpose or usefulness. Thus one may for instance focus this discussion of group work by asking three questions. Can it enhance or improve mathematical learning? Can it play a role in developing the employability skills of a graduate? Finally, and by no means least, can encouraging or requiring some working in groups help students to develop the notion that they are part of a community of learners of mathematics, or indeed of a community of mathematicians, with the consequent positive feelings that may emerge?

These categorisations are clearly inter-related and both provide useful parts of a framework for exploring and addressing the issues.

A chapter on the topic of group work sits easily within a book which has an underlying theme of transitions. When they arrive at university, many students will not have experienced working in groups to develop mathematical ideas and concepts or to solve mathematical problems; indeed they are also unlikely to have encountered the notion of mathematical modelling, an activity often supported at university via group work activities (see for example Hibberd, 2002).

At the other end of their career as an undergraduate mathematician, most graduates will make the transition to the world of work, often applying for jobs for which they are in competition with graduates from other disciplines. Employers repeatedly state (for example CBI, 2012) that they value graduates who know how to successfully function as part of a team; it is useful for students to have some experience with which they can enhance their curriculum vitae, together with the ability to articulate their skills in this respect, for example when responding to interview questions.

Between these transitions lies the fostering of a spirit of community, of which more later.

Much has already been written about the place of group work in undergraduate mathematical education. Over the last ten to fifteen years, developmental work in this area has been stimulated by the Maths, Stats & OR Network, firstly as part of the Learning and Teaching Support Network (LTSN) and latterly as part of the Higher Education Academy. Although now sadly defunct, elements of the work of the Maths, Stats & OR Network have been continued through the re-organised Higher Education Academy¹. In more recent times, a further burst of activity has arisen through the Higher Education Funding Council for England (HEFCE) funded More Maths Grads initiative and what may be considered in some areas its successor, the National HE STEM Programme².

Two works in particular form a useful place to start. They are separated by nearly ten years, both gathering data via interviews and/or surveys of both students and staff and then analysing and reflecting upon those data.

1 See <http://www.heacademy.ac.uk/disciplines/maths-stats-or>

2 <http://www.hestem.ac.uk>

The earlier work of MacBean, Graham and Sangwin (2001) was reported in an excellent and still relevant document entitled *Guidelines for Introducing Group Work in Undergraduate Mathematics*. This document raises many of the issues discussed within this chapter, although perhaps inevitably categorises a number of the points differently, around so-called ‘Key Questions’, and over the intervening time period, some emphases within the national higher education landscape and agenda have shifted.

In fact their first Key Question was a composite one: “*Why choose to use group work? What are the benefits?*” They begin their answer by saying, “... *many people view mathematics as an individual mental pursuit*” (MacBean et al., 2001:2) It is indeed easy to find stories which reinforce this in some respects stereotypical view. For instance Andrew Wiles, who proved Fermat’s last theorem, was labelled “*The Attic Recluse*” by Simon Singh in his book *Fermat’s Last Theorem* (Singh, 2002). However there is far more to mathematics within universities than the individual mathematician working alone.

Forming the community

The discussion in MacBean et al. (2001) develops a wider focus. They gleaned answers to their key questions by interviewing staff and students, and brought answers to their first, composite key question “*Why choose to use group work? What are the benefits?*” together under the headings of social benefits, support, consolidation of prior work, introduction of new concepts or topics, and “*skills*” development.

It is interesting that MacBean et al. (2001:2) mention social benefits first: “*One of the benefits is social, that is building a community of peers...*”. This chimes significantly with the third category of usefulness of group work identified earlier, that of fostering feelings of belonging to a community of learners of mathematics. The later work, undertaken by Robinson, Challis and Thomlinson (2010), identifies the same theme. This work, entitled *Maths at University: reflections on experience, practice and provision*, is the final report of their work as part of the More Maths Grads project.

Indeed Robinson et al. (2010) identified “*a great benefit for both students and staff in our fostering a mathematical community in a university*”. Benefits listed include the observation that “*Students with a strong network of peers have a ready source of both personal and academic support...making them happier overall.*” Further, they go on to say:

“...there are less tangible but arguably more important advantages of students having a strong sense of belonging. Students who feel part of a wider community, and who feel valued by their peers and by staff, are generally happier students, and as such are more likely to enjoy their studies, work hard, succeed in their studies, and leave with a positive memory of their time as university mathematicians. With a more cynical eye on outside measures like the National Student Survey, they are more likely to rate us highly and forgive us our trespasses.”

Robinson et al. (2010)

They recommend that “*effort to ease the ‘transition’ to university study should focus particularly on helping a student to develop the social and support networks amongst each other and develop a good working relationship with staff.*” How can group work assist with this particular task?

The employability skills agenda

A notable change of emphasis in the period since 2001 concerns the key skills, or graduate employability agenda. MacBean et al. (2001:3) identify one of the answers to their Key Question 1 (“*Why choose to use group work? What are the benefits?*”) as “*skills development*”, but they then choose to say “*No attempt will be made here to enter the debate about transferable/ key skills, as it is beyond the scope of these guidelines*”. They observed that “*... This topic invoked the most opposing viewpoints. Indeed some of the staff and students interviewed questioned the assumption that the purpose of a mathematics degree was to develop such ‘skills’ at all.*”

However it is interesting that within a year, in a book (Kahn & Kyle, 2002) which is in many ways a precursor of this current volume, a chapter would appear entitled “*Developing transferable skills: preparation for employment*” (Challis, Gretton, Houston & O’Neill, 2002). At the same time, the first version of the Quality Assurance Agency (QAA) for Higher Education Mathematics, Statistics and Operational Research (MSOR) Subject Benchmark Statement (QAA, 2007) was being drafted. This document is widely supported within the mathematics community for the way it captures the essence of courses in the mathematical sciences but at the same time honours their diversity. It states in its section on “*General Skills*” that students in the discipline area “*...will also have general communication skills, typically including the ability to work in teams, to contribute to discussions...*” (QAA, 2007:16).

Notwithstanding this Subject Benchmark Statement, the question of whose role it is to help students to develop their skills in this respect remains controversial. The whole issue of graduate employability and the workplace is discussed more widely elsewhere in this book (See Chapter 14), and while that discussion does not need duplicating here, it may be worth considering a couple of points which are relevant to the case of group work.

The graduate skills agenda has gathered momentum in recent years, particularly as the UK has moved from having an elite to a mass higher education system; tuition fees are now substantial with responsibility for their payment now placed upon the individual student. It should therefore come as no surprise that many students are genuinely interested in enhancing their employability prospects, and in how their time at university can directly contribute; findings from the More Maths Grads project confirm this. For instance, Robinson et al. (2010) reported that of 223 first year students they surveyed early in their first year, 93% rated the statement “*I need a degree to get a good job*” as quite or very important, and 95% rated “*Improving my employability*” as similarly important.

Of course one may argue cynically that some universities contribute to their graduates’ employability simply through their reputation, and perhaps argue, with some justification, that simply having a mathematical sciences degree offers an advantage. However, mathematics graduates will be in competition with graduates from other disciplines for jobs, and employers make it very clear (see for example CBI, 2012) what they are looking for in terms of graduate skills and attributes. A particular skill frequently quoted relates to being able to successfully demonstrate that one can contribute positively and appropriately in a team-working situation.

These issues go to the heart of what the real purpose of a degree course programme is in these days of mass higher education. There are many more graduates than there are places

for postgraduate study in university research programmes, important though this is for the development of the mathematical sciences as a discipline. Most of our graduates enter the world of employment (Prospects, 2013). There are many indicators that employment prospects and skills development are regarded as an important dimension of study at higher education level. The Confederation of British Industry (CBI) repeatedly emphasise the importance of graduates who have amongst their skill set a developed ability to work in teams, universities claim to make their graduates employable, and the QAA Subject Benchmark Statement mentions a graduate skill set “*typically including the ability to work in teams*”(QAA, 2007:16).

In these circumstances, whose role is it to develop such skills? While it is important that students learn and indeed enjoy their mathematics, does it not also behove us to respect our students’ long-term aspirations to develop their employability and to earn a good living, and the expectation they might, quite reasonably, have that their course will contribute towards satisfying these aspirations? The examples of implementation which follow later, ranging from 2002 to the present day, demonstrate the conclusion that many academics are now coming to realise.

What about the mathematics education agenda?

The third category of reason for deploying a group work approach is educational: what contribution can working together, in various different ways, make to the effectiveness of the educational process for both learners and academic members of staff? A well-designed group work situation will contribute to all three purposes. If there is a group activity which is built around a mathematical topic or a mathematical modelling case study, it is not possible to draw a line between skills developed to help tackle the mathematical problem and negotiating skills developed which may prove to be useful in later employment, or indeed when called upon to answer a difficult interview question such as ‘*give us an example of when you have managed conflict in a team?*’ Similarly one cannot separate out when an activity is community-forming.

Discussion: the ‘how’

Having considered the pros and cons, and made a decision to implement an approach which utilises group work, one must face the question of how it can be successfully implemented as part of an undergraduate mathematics programme. One does not, of course, want to reinvent the wheel or indeed to repeat the errors of others, but there is a potential issue with regards to the extent to which practice is transferable: universities are as prone to the ‘not-invented-here’ syndrome as other organisations. However, an important part of the National HE STEM Programme was ‘Practice Transfer Partnerships’; collaborations based upon the principle that successful practice developed in one place could be learned from and adapted for implementation elsewhere.

There is a selective and annotated list of questions which must be faced by any practitioner planning on using group work approaches. These questions are informed by those identified by MacBean et al. (2001), work which retains a significant degree of relevance despite how UK higher education has progressed since then (see Chapter 1 for a summary).

The questions are followed by selected cases of recent experience, to enrich the discussion and exemplify responses.

First let us state and annotate the selected questions:

- **What are the main inherent difficulties which commonly arise, and what ways have been found to deal with them?**

It may seem strange to begin with such a negative question, but given the strong feelings that exist about this topic, perhaps not. Themes which seem to emerge regularly as difficulties often directly or indirectly involve formal assessment. Particular issues include how best to form groups, and how to deal with variable individual levels of engagement, participation and contribution.

- **How should one form the groups?**

This can be a major source of friction in group work, particularly if the task is formally assessed and 'high stakes' in terms of its contribution, or perceived contribution, to overall student marks. Should the tutor allow students to form groups themselves, or should groups be imposed by the tutor? If the latter, should selection be random, which one might perhaps expect to be more 'realistic' in an employability sense, or 'engineered' in some way? Two cases discussed more fully below have interesting points to make here. Bradshaw (2008) implements a novel hybrid method of group formation, and Chadwick et al. (2012) identifies the essential artificiality of feasible group formations in university group working exercises.

- **What support can tutors offer in the functioning of each group?**

A tutor could, for instance, offer to act as an arbiter in group disputes, or indeed if one sets up a peer assisted learning scheme, whereby students in later years are trained to mentor groups in earlier years (See Chapter 6), then the student mentors can act as arbiters in the first instance. However, as Challis et al. (2002) hint, one powerful way for academic members of staff to assist is by supporting students to become aware of group dynamics by use of something such as the Belbin methodology (Belbin, 2013). This can help students understand the various roles individuals assume in a group, and thus one might hope that they will not only be able to act in a more mature and informed way, but will also be better able to articulate their skills in respect of working with others.

- **How should group work be assessed?**

Such a question presumes that the task tackled by the group will be assessed, although the task may be low stakes as described in Waldock (2010) or higher stakes as described by Hibberd (2002). In some schemes students might work together but submit work separately, or make a partially joint submission as seen in Jaworski, Matthews, Robinson and Croft (2012) and Jaworski and Matthews (2011). One matter which has to be faced concerns plagiarism, not only how it manifests itself in group work situations, but indeed what constitutes plagiarism in such situations. Not unrelated to this is the matter of assessing how each student contributed to the overall task, which creates not only issues of fairness, but of ensuring that each student has the opportunity of demonstrating that they can meet every learning outcome when tasks are distributed.

Possible responses are various, but may include allowing part of the marks to be allocated for analysing how well the group functioned, and integrating into the mark scheme some element of peer assessment related to individual levels of engagement and contribution. Interesting points in this regard are raised by Cliffe and Bradshaw (2012) and Martin and Cliffe (2012), and are discussed further below.

It is now appropriate to explore just some of the teaching interventions, many of which have been cited during consideration of the preceding questions, that utilise group work in a range of ways as part of undergraduate mathematics programmes.

Hibberd (2002) identifies the teaching of mathematical modelling as providing an ideal opportunity to practice and develop group working skills. In the context of what we now see as an emergent graduate skills and employability agenda, he identifies that *“The sophistication of problems that can be addressed and delivered...can be increased through making the modelling activities a group activity. This also has the significant advantage of providing a platform for introducing team-working skills.”*

Challis et al. (2002) agree that *“there are particular opportunities for skills work in the practical mathematical modelling units more than in the underpinning techniques or foundation units.”* In this report of successful experience at two universities, they comment in their conclusions that *“It is not sufficient to put students into groups and ask them to undertake tasks. They must be shown that a group can operate much more effectively than an individual. They must also know how to assign roles to group members and how to plan and monitor the work during the course of a group project”.*

Further developments arise in the intervening years, and two case studies reported as part of the More Maths Grads project exemplify how thinking has broadened. Waldock (2010) describes a peer assisted learning scheme at Sheffield Hallam University, whereby new first year Mathematics students are allocated to small groups of about seven or eight, and are supported and mentored by a final year student who has received appropriate training in facilitating student groups. The group is focused around a formal but low stakes mathematical activity through which it is intended that the new students will develop their group working experience and skills. However, a major purpose of the activity is social, helping break the ice at the time of transition, and friendship groups and informal work groups have been observed to persist throughout the course and beyond. Walker (2010) describes a Peer Assisted Study Sessions (PASS) scheme (see also Chapter 6) at the University of Manchester. Once more, new first year students are organised into groups which are facilitated by students from later years, and this provides an opportunity for all students to discuss mathematics and course related issues in a safe and supportive environment.

Bradshaw, Parrott, Lakin, Mann and Sharp (2012) describe a development called the Maths Arcade (see also Chapter 13 for a more detailed discussion of this initiative), which aims to encourage students otherwise reluctant to access support mechanisms, to attend regular maths games, talk and puzzle or problem sessions. Within the current discussion this may be seen as part of the community-forming aspect of support work. Successful practice developed at the University of Greenwich in this area has been successfully spread to several other institutions through the National HE STEM Programme.

In discussing this theme of forming a community by offering social support in a mathematical context, practitioners must be aware of the ‘inclusive practice agenda’ which has developed and, quite rightly, continues to gather considerable momentum (see also Chapters 15, 16 and 17). Of particular interest here is the impact of the use of group work on those students with autism spectrum disorders including Asperger’s syndrome, students not unknown in the world of mathematics. Cliffe and Bradshaw (2012:45) identify one typical challenge as *“difficulties with social relationships”*, with barriers including *“anxiety regarding how to approach the situation, establishing a group with which to work, understanding the ‘rules of engagement, the unpredictable nature of other group members, ...and being misinterpreted by others”*.

Clearly anyone considering deploying group work, whether formal or informal, needs not only to be aware of this dimension, but to be aware of the discussions on how to minimise the barriers. For instance, Cliffe and Bradshaw (2012) propose a starting point for a framework to achieve this. They suggest providing explicit educational rationales, and scaffolding opportunities to build group work skills over time. They suggest using a carefully designed group allocation method, and including group working skills as learning objectives. They suggest that explicit rules of engagement are produced, and that transparent and fair marking include an element of peer reporting. Finally they encourage explicit reflection on how well the group work has gone.

To pick up on one of these points, one decision faced by the designer of a learning opportunity involving group work is how to allocate groups. Does one do so randomly, arguing that that is the way it is in the workplace? Does one allow self-selection, with the dangers of exclusion of individuals? Does one indulge in social or mathematical engineering? The pitfalls are many, and group formation can be an emotive topic. An interesting approach is proposed by Bradshaw (2008), whereby students have the opportunity to self-select a group of three, and then the lecturer combines these together to form groups of six, using a combination of mathematical, educational and social engineering. Some success is reported.

The Engineering Students Understanding Mathematics (ESUM) project (Jaworski et al., 2012; Jaworski & Matthews, 2011) provides an interesting story. It aims for engineering students to *“develop a more conceptual understanding of mathematics through their participation in an innovation in teaching”*, which includes *“small group activity”* and *“an assessed group project”*. It provides an example of deployment of group work not through the vehicle of mathematical modelling with its natural skills base, but in a usually traditional, conservative and techniques focused area. The researchers report *“greater engagement and higher summative results than with previous cohorts”* (Jaworski et al., 2012).

The National HE STEM Programme guide edited by Waldock (2011) contains seventeen case studies reporting various ways in which development of graduate skills is embedded in mathematics programmes. Nine of these explicitly involve the development of team working skills. In a further National HE STEM Programme publication (Waldock & Rowlett, 2012), case studies of current practice relating to employer engagement within undergraduate mathematics programmes have been collected. Chadwick, Sandiford and Percy (2011) describe student teams tackling industrial style problems, but note negative

feedback from students about group dynamics, particularly concerning perceived unequal contributions. There is some discussion here of ways of dealing with that, but the issue seems common. Further exploration of team working/group working takes place in a case study about how realistic work-related learning should be (Chadwick et al. 2012). A discussion of the role and implementation of team working drew an interesting distinction between graduates who can work in an industrial team, as opposed to those who have merely taken part in a university group project. The point is made that workplace teams *“are often drawn from different departments with members not necessarily knowing each other beforehand. Often there is a hierarchy, so some team members can be more senior than others”* (Chadwick et al., 2012:49). This sets a challenging problem if realism is the goal, and there are inevitably practical difficulties in setting up multi-disciplinary or multi level teams in an undergraduate context.

Concluding comments

The recent burst of national activity relating to group work within undergraduate mathematics has yielded some interesting stories. The guidelines issued 13 years ago (MacBean et al., 2001) still provide a useful framework, but higher education has moved on.

We close with two thoughts, both arising from work undertaken through the National HE STEM Programme. Inglis, Croft and Matthews (2012) took the brave step of asking recent graduates *“to reflect upon the knowledge/skills they believed that they developed during their mathematical study”*. Resisting the temptation to discuss all the fascinating data in this report, we will restrict ourselves to identifying comments on teamwork: *“...fewer students believed that their studies had developed generic non-cognitive skills such as...team working”,* which was considered *“useful in the workplace”*. Inglis et al. (2012:5) noted that when asked, *“incoming undergraduates expected to develop these non-cognitive generic skills ... suggesting that there is a mismatch between students’ expectations and outcomes.”*

If such reports, together with the arguments rehearsed in this chapter, prompt the reader to consider where and how they might make constructive use of group work, they may be considering reading the guidelines produced by MacBean et al. (2001), but at the same time wondering how to make sense of the more recent issues arising. Various informal or low stakes initiatives have been reported here, which one hopes might provide inspiration. The pitfalls appear when one approaches group work which is to be formally assessed. One might recommend that, in dealing with the issues raised in this chapter, one could do worse than starting with the framework for group work suggested by Cliffe and Bradshaw (2012), which we repeat here in paraphrased form for consideration:

- Be prepared to justify and defend the use of group work, and have evidence ready to support the rationale.
- Support students to build group work skills over time, initially through low stakes activities.
- Allocate groups carefully, with appropriate student input.
- Treat group working skills as something to be learnt, and therefore taught.

- Set rules, or support each group to set rules, as to how a group is to operate, and support the enforcement of those rules; be prepared to act as or support a mentor or advisor in the event of difficulties.
- Mark transparently and fairly, and include opportunities for peer assessment of relative contributions to the task.
- Encourage explicit reflection on how the group work has gone, and what has been learned from it for the future.

Although this framework is proposed by Cliffe and Bradshaw (2012) in an inclusive practice context, particularly concerning students with autism spectrum disorders including Asperger's syndrome, their list forms a useful framework which should benefit all students involved in group work.

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Chapter 8: Problem solving in a mathematics degree

Trevor Hawkes

“Mathematics has always had two surface aspects, often called skills and problem solving. Skills are seen as a well-defined set of procedures for transforming numbers, symbols or shapes; the role of the ideal performer is that of an automaton, and the role of the student is that of learning to be a reliable one. Problem solving, in contrast, involves tackling tasks that are significantly different from those one has learned ‘by heart’; a major part of the challenge lies in deciding how to tackle the problem, and which bits of one’s toolkit of mathematical skills will help. A problem solver needs a rich, connected understanding of mathematics and the ability to see patterns of similarity and association, as well as the skills to carry out the planned attack and to check that the results make sense in the context of the problem. No wonder these are called ‘higher level skills.’”

Burkhardt & Bell (2007:395)

Abstract

Many students arrive at university to study mathematics with little or no experience of problem solving. They lack the requisite “*rich, connected understanding of mathematics*” noted by Burkhardt and Bell (2007:395). The chances are they have been ‘taught to the examination’ and led by the hand in small steps through examination questions – no risks are taken. A university degree should attempt to make amends for this barren experience by introducing ‘higher level skills’ early on and reinforcing them at every opportunity. Doing this will offer students a fresh, stimulating, and sometimes challenging, experience of their chosen subject and transform their ideas of what mathematics is really about.

In this chapter we attempt to answer the following questions:

1. *What is problem solving?* In particular, what constitutes an authentic problem? We describe the nature of a genuine problem through a dozen desirable features, give some examples, and provide support for our views on problems and the process of solving them in the literature.

2. *Why is problem solving important?* We analyse the transformative impact of problem solving on a student's intellectual development, attitudes to mathematics and enhanced life skills.
3. *How can students be encouraged to engage in problem solving?* We devote a section to the problem solving opportunities open to students before they arrive at university and then discuss the theory and practice of incorporating significant problem solving within a mathematics degree.

What is problem solving?

Problem solving in mathematics is a complex interaction between the human mind and mathematical knowledge. It has many manifestations and, unlike a mathematical concept, it can't be pinned down with a precise definition. Problem solving makes heavy demands on the intellect and the emotions, calling for insight, invention, imagination, self-monitoring, perseverance, and a supple, adventurous mindset prepared to change viewpoints and explore a landscape of connections. Further, the problems *themselves* have to embody certain characteristics that authenticate the activity. As Burkhardt and Bell (2007) go on to say: “*Every theorem was a problem once.*”

Many lecturers will claim that problem solving is a central part of their teaching, that their weekly assignments are full of problems, and that most students work through them to prepare for their exams. But closer scrutiny reveals that their ‘problems’ are often ‘five-finger exercises’, rehearsals of familiar mathematical riffs, or variations of examples and routines worked through in their lecture notes. In the sense of the opening quotation of this chapter, these involve *skills* and are not strictly *problems* in the sense we will understand them.

A good problem

For the purposes of this chapter, we offer a loose description, rather than a formal definition, of what constitutes a *genuine mathematical problem* within the meaning of the act. Here goes:

A problem is a mathematical question for a learner to strive to answer. Its worth is measured by the presence of special qualities, in both the problem and the solver, affecting the process of finding a solution. For the problem itself these qualities include a selection of features drawn from the following list:

1. *Novelty*: This is the one essential ingredient in the characterisation of a problem. The question must be genuinely new to the learner and not just a variation on themes and arguments that are already familiar. If the learner already knows how to proceed to a solution, the novelty in the problem is missing, and, *for that learner*, it is not a genuine problem.
2. *Challenge*: The solution is not obvious or routine. Indeed Weber (2005) takes this as a defining feature: “*A mathematical problem is a task in which it is not clear to the individual which mathematical actions should be applied, either because the situation does not immediately bring to mind the appropriate mathematical action(s) required to complete the task or because there are several plausible mathematical actions that the individual*

believes could be useful.” The learner might be expected to respond with comments like “I don’t even know where to begin” or “I’m stuck!”

3. *Consequence:* The result leads somewhere interesting, suggests some new questions, or sheds new light on its surroundings. Isolated puzzles with clever solutions but no wider significance won’t usually count as problems.
4. *Connection:* The problem’s statement links disparate topics, or its solution brings to bear ideas or techniques from different areas, making connections that are not obvious or initially apparent, thereby witnessing the unity of mathematics.
5. *Extension:* The questions and/or methods of solution augment or extend the learner’s established knowledge.
6. *Elegance and Economy:* Problems often admit many different solutions. A solution with an original approach or an incisive idea that quickly gets to the ‘nub’ of the problem is preferable to one that follows an obvious but laborious routine or uses the type of case-by-case analysis known as ‘proof by exhaustion’.
7. *A Eureka Moment:* A flash of inspiration, a sudden insight, a surprising twist – these moments of heightened perception give real intellectual pleasure and can yield a solution that cuts to the chase and gets to the heart of the problem.
8. *General Principle:* There are well-known and often powerful methods of argument that turn up time and again in mathematical discourse. A problem that succumbs to one of these adds a valuable weapon to a learner’s armoury.
9. *Illumination:* The volume of mathematics covered in a lecture course (or module) is limited by its length and by convention. The lecturer rarely has time to include all the material needed to convey fully the significance and ramifications of the theory. Problems that provide additional consequences, side results, examples and counter-examples can enrich the material and improve the learner’s understanding and appreciation.
10. *Experimentation and Conjecture:* A problem might require the solver to experiment (by hand, with a calculator, or even with a computer program) and make conjectures based on the empirical evidence they obtain. The experience of doing this gives the learner an insight into mathematical creativity.
11. *Application:* We have already mentioned the desirable property that a mathematical problem should *lead somewhere* and perhaps apply in other situations. Mathematical modelling is another type of application. The established process to model a real (or imagined) scenario related to everyday life involves a number of steps and iterations. At some stage a mathematical formulation emerges and calls for a solution that will be used to predict or confirm behaviour. We will not attend to modelling in this chapter, it is well addressed in Chapter 9, but we concede that there are excellent problems arising from it.
12. *Affective Response:* A problem that evokes a pleasurable reaction, like curiosity, excitement, aesthetic satisfaction, or sheer joy, has a lot going for it. When I asked one of my sons studying mathematics what characterises a good problem, he replied: “One that makes you smile when you come upon the solution.”

Some examples of good problems

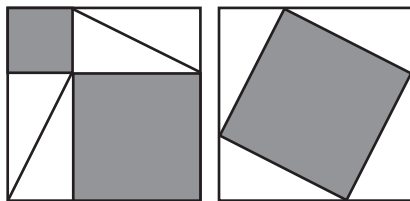
We will describe some elementary problems to illustrate our notion of what constitutes a worthwhile problem in terms of the twelve criteria just listed.

But first we must emphasise the role of the learner in the nature of a problem. We regard the first criterion, *novelty*, as indispensable, and novelty is a function of the learner. If he or she has already seen the problem (or something similar) or knows the way to solve it from previous experience, then, as a problem, it is disqualified. What might be a good problem to one learner can be just a routine exercise of skill to another. Thus ‘novelty to the learner’ is a *sine qua non* of a valid problem.

Example 1

Let us begin with the ancients and imagine Pythagoras has recently hit on the theorem that now bears his name, and further, that he wants to test the mettle of an aspiring acolyte. He might have proposed the following problem:

- i) The eight white right-angled triangles shown in these two large squares are the same (congruent). Compare the combined area of the two grey squares inside the large square on the left with the area of the single grey square inside the large square on the right.



- ii) If the lengths of the sides of the white triangles are $a \leq b < c$, deduce that $a^2 + b^2 = c^2$.

As the theorem is likely to be new to the acolyte, it would satisfy the *novelty* criterion. Because the proof is presented, it should not be a serious *challenge*, but it certainly has *consequence* (think of Fermat), *connection* (between algebra and geometry), perhaps embodies a *general principle* (using pictures), certainly shows *elegance*, prompts a *Eureka moment*, admits *application* (for instance, to surveying), and evokes an *affective* response (in the author at least). Hence, we would conjecture, it is a ‘good problem’.

Example 2

Pythagorean triples regularly turn up in courses on elementary number theory and usually involve a verification of the identity:

$$(r^2 - s^2)^2 + (2rs)^2 = (r^2 + s^2)^2$$

The following related questions could be included as genuine problems on a module assignment:

For any non-negative integer n , consider the Diophantine equation $D(n)$:

$$x^2 + y^2 = z^2 + n$$

Show that for $n = 1, 2, 3$ and 4 , the equation $D(n)$ has integer solutions with $M < y < x$ for arbitrarily-large M .

Show that $D(n)$ has arbitrarily-large solutions for infinitely-many values of n .

Problems like these are unlikely to appear in textbooks and would almost certainly be novel to an undergraduate mathematician. Although they can be solved with elementary algebraic

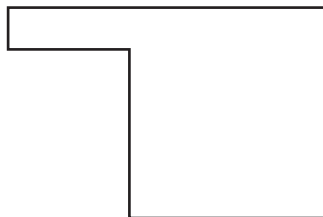
identities like the one above, the identities are not obvious and would therefore present a *challenge*, but one within the reach of a beginning student. The problem would introduce and extend the idea of Diophantine equations and show that, unlike the equation:

$$x^n + y^n = z^n \text{ for } n \geq 3,$$

they can have elementary solutions, which might provide some *illumination* for number theory as a whole. The solutions could well be *elegant* and give the satisfaction of a sudden *Eureka* insight. They would almost certainly involve *experimentation* and might lead to further *conjectures*. And finding solutions independently should induce in most students a warm glow of success (namely an *affective* response!).

Example 3

A farmer has an L-shaped field, which is a rectangle with another rectangle removed (see diagram). He has two daughters and wishes to bequeath each of them an equal share of his land. Given that his only measuring tool is a straight edge in the form of a long rope that can be stretched taut (and a couple of pegs), how can he divide his field into two connected halves of equal area?



The solution depends on the fact that any line through the point of intersection of the diagonals of a rectangle divides its area in half. This raises a number of interesting questions:

- Which 2-dimensional shapes (or, specifically polygons) have a point with this property?
- More generally, given any shape, how could one characterize the envelope of the area-bisecting lines?
- What happens in three and higher dimensions?

Example 4

Five points are drawn strictly inside an equilateral triangle of side 2. Show that at least two points are less than 1 apart.

This is the kind of question where, if you haven't seen its like before, you might have no idea how to begin. But:

- Dissect the triangle into four equilateral triangles of side 1.
- Observe that 2 points inside one of them are less than 1 apart.
- Apply the pigeon-hole principle.

The result is clear – elegant, with an ‘Aha!’ moment, and showcasing a general principle of wider application – definitely a worthy problem.

Example 5

In rowing races the boats normally contain 1, 2, 4, or 8 rowers, and sometimes a coxswain. Derive a relationship between the time T taken to cover a fixed distance and the number N of oars in the boat, using the following assumptions:

1. The boat experiences quadratic friction, proportional to the submerged cross-sectional area A of the boat.
2. Each rower has the same weight and the same power output throughout the race.
3. The relative weight of the boat is small and can be ignored.
4. The displaced volume is proportional to N (Archimedes Law).

This problem succumbs to dimensional analysis, which may be a new method, or failing that, a novel *application* of the method. Therefore this example will tick the *novelty* box, as well as presenting a *challenge*, involving a general *principle*, manifesting *economy and elegance*, involving *experimentation*, and having *consequences* for other problems of similar ilk.

For the record, dimensional analysis yields the formula $T \sim N^{-\frac{1}{9}}$, which turns out to be close to the relative times for single sculls, pairs, fours and eights found in tables of Olympic records, especially if the formula is modified to allow for a cox, weighing half the weight of the rowers.

Validating the process of solving a problem

We now change focus from the nature of a *genuine problem* to the nature of the process of trying to solve it. What qualities does the solver need in order to engage in the activity of problem solving and to have a chance of success?

In his reflections on progress since his seminal book (Schoenfeld, 1985), Schoenfeld (2013) asks how to judge whether an instance of problem solving has been a success or failure and recalls the four conditions, which, he asserts, are necessary and sufficient. They are:

- a. *“The individual’s knowledge.*
- b. *The individual’s use of problem solving strategies, known as heuristic strategies.*
- c. *The individual’s monitoring and self-regulation (an aspect of metacognition).*
- d. *The individual’s belief systems (about him- or herself, about mathematics, about problem solving) and their origins in the students’ mathematical experiences.”*

Schoenfeld (2013:11)

In other words, if a solver knows the requisite mathematics, has some suitable strategies to hand, can stand back and reflect, and has self-belief and is confident of success, then problem solving can take place – presumably irrespective of whether the outcome is a valid solution. We accept that these criteria are necessary conditions, but would challenge Schoenfeld’s claim that they are also sufficient. We believe that a vital missing ingredient is the temperament of the solver: such qualities as intellectual vigour, imagination, perseverance, ambition, the will not to be defeated, these are essential in some measure to confront the challenge a problem may present. An extreme example of these qualities is found in Andrew Wiles, who became fascinated with the Fermat Conjecture as a child and finally published the proof over thirty year later, devoting all of his research time over a period of six years to tackling this problem in almost total secrecy. We contend that without sufficient drive and persistence, effective problem solving cannot take place.

This view is reinforced by Hilton (1988) in his article “*Reminiscences of Bletchley Park, 1942 – 45*”. In the concluding section, Hilton analyses the reasons for the “*astounding success*” of their enterprise, and argues the team of “*quintessentially pure*” mathematicians were well equipped to solve the applied problems they encountered at Bletchley Park:

“To get people effectively to apply mathematics, the essential ingredients are (1) a strong education in mathematics; (2) the ability to think mathematically, to understand how to formulate a problem in precise mathematical terms (what Speiser called “mathematische Denkweise,” or “mathematical way of thinking”), and (3) a strong motivation to solve a given problem or problems.”

Hilton (1988:299)

The key words here are “*strong motivation*”; without that, the effort would be short-lived and the desire to give up all too easily yielded to. We will have more to say about “*the mathematical way of thinking*” later.

Why is problem solving important?

There are many reasons why it is important that students on a mathematics degree should engage in problem solving; here they are in brief:

- It teaches deeper understanding of mathematics.
- It develops the “*mathematical way of thinking*” (Hilton, 1988:299).
- It helps students work effectively with logic and proof.
- It engages students in a creative mathematical process, and gives them a taste for ‘doing mathematics’ and ‘being a mathematician’.
- It promotes ‘ownership’ of the material.
- It is a desirable graduate skill that can be used throughout one’s life in a range of applications.

Let us now explore these claims more closely.

Deep understanding

*“Some that have deeper digg’d mathematics’ mine than I,
Say, where her centric happiness doth lie.”¹*

John Donne (1572-1631)

Bloom’s well-known ‘taxonomy of learning objectives’ (Bloom, Engelhart, Furst, Hill & Krathwohl, 1956) does not apply well to mathematics without some significant re-interpretation. There have been a number of attempts to update and extend it, for example, and rather trivially, by replacing Bloom’s nouns (knowledge, comprehension, application, analysis, synthesis, evaluation) in the so-called *cognitive* domain – hierarchies for the *affective* and *psychomotor* domains are also proposed – with the corresponding verbs in

¹ We have taken liberties with several of the (mis)quotes in this section.

order to emphasise the view that cognition is a process rather than a state. A more serious objection was the linear nature of the cognitive hierarchy, with its implication that learning at the higher levels is dependent on having already attained the required knowledge and skills at lower levels; educators disagreed about the ordering and many felt that the top three levels should be seen as acting in parallel.

We find the two-dimensional representation proposed by Anderson and Krathwohl (2001) better suited to the complexity of mathematical problem solving. It uses levels of knowledge along one axis and of cognitive processes along the other. Tables 8.1 and 8.2 give brief indications of the two hierarchies in left-to-right progression and adapted to mathematics:

Remember	Understand	Apply	Analyse	Evaluate	Create
Identify and retrieve relevant skills and information from memory.	Make mathematical sense of concepts, examples, processes and statements.	Use procedures to frame arguments, explain them, or give instances.	Follow logical steps, analyse a proof, check a manipulation, devise a strategy.	Judge validity of a calculation, the correctness of a proof, the fitness of an approach, the value of a theory.	Forge new arguments, make new connections, discover original results, develop new theories.

Table 8.1: On the cognitive processes (horizontal) axis (Anderson & Krathwohl, 2001)

Factual Knowledge	Conceptual Knowledge	Procedural Knowledge	Metacognitive Knowledge
Have facts, such as definitions, examples or statements of results, committed to memory.	Know the meaning of terms, concepts, examples, structures and processes.	Know algorithms, methods of proof, algebraic, analytic and geometric procedures.	Be aware of one’s own cognition, learning, strategies and motivation.

Table 8.2: The knowledge (vertical) axis (Anderson & Krathwohl, 2001)

Of course, this representation only makes sense if one assumes that a given point on an axis includes all the lower levels in its compass. With this proviso, a case could be made for defining ‘deep understanding’ in mathematics to be represented by a suitably-high top-right region of the rectangle formed by the two axes, say to the right of “Analyse” and above “Procedural Knowledge”.

Thinking mathematically

“How do I think of thee? Let me count the ways.”
Elizabeth Barrett Browning (1806-1861)

Many mathematicians would claim to understand what it is to think mathematically, but might also agree that it is hard to tie down precisely what is meant by it and particularly difficult to explain to non-mathematicians. It would seem that one needs to have practised mathematics extensively in depth and breadth in order truly to know how to “*think mathematically*” Hilton (1988:299). Mason, Burton and Stacey (1982) mention three factors that influence how effective one’s mathematical thinking is:

- Competence in the use of the processes of mathematical enquiry.
- Confidence in handling emotional and psychological states and turning them to advantage.
- Understanding of the content of mathematics and, if necessary, the area to which it is being applied.

Although a thorough knowledge of the content and methods of a particular area is essential for the exercise of mathematical thinking, the book concentrates on the first two of these points: processes of enquiry and emotional states. Amongst the essential processes put forward are: specialising, generalising, conjecturing, and convincing. These are all important when properly ‘doing’ mathematics, in particular during problem solving, and should be located in the higher levels of the taxonomy discussed above. Significantly, Mason, Burton and Stacey (1982) devote a long chapter (entitled “*Something to Think About*”) to over sixty elementary but interesting problems which are amenable to these processes. Doing (and thinking) is of the essence.

The importance affective qualities, such as determination, motivation, anxiety, confidence and concentration, in learning to think mathematically are also given prominence by Mason, Burton and Stacey (1982). Negative emotions can seriously hamper one’s ability to carry out the processes required for mathematical thought. Having self-awareness, the ability to reflect on one’s state of mind, and a temperament to exercise control over it, is a great advantage in liberating the thought processes and allowing the ideas to ‘come to mind’.

Working with logic and proof

“‘Contrariwise,’ continued Tweedledee, ‘if it was so, it might be; and if it were so, it would be; but as it isn’t, it ain’t. That’s logic.’”

Lewis Carroll (1832-1898)

The need for *logic* in mathematics is not in contention, and it is hard to imagine any kind of problem solving that did not demand respect for it. Whereas students embarking on a mathematics degree would hardly challenge this, they might well be unconvinced of the need for rigorous *proof*. This is not surprising, for like problem solving, proof has hardly been part of their previous mathematical experience. In contemporary school curricula in the UK, mathematics is presented as a body of received knowledge and unchallengeable facts (formulas, theorems, processes, algorithms), along with assessment routines to be exhaustively practised for exams. This sterile approach leaves its mark on students’ beliefs and attitudes to the subject; the idea that mathematics could be an area of adventurous enquiry to be explored and re-created is alien to them, and not surprisingly, they baulk when asked to change their habits of thought and practice.

This is where the experience of genuine problem solving can come to the rescue. Weber (2005) argues that proof construction may be viewed as a problem solving task, with the implication that a student who has acquired proficiency in problem solving will be able to transfer the skills to constructing proofs. Not only does problem solving encourage students to engage with mathematical thinking, but it also requires them to explain their solutions and justify their arguments. This requirement is more powerful if they also have to present their solutions to a peer group, who in turn are encouraged to question the reasoning and develop criteria for what constitutes a valid argument, in other words, a proof.

Being a mathematician

“A mathematician is a device for turning coffee into theorems.”

Paul Erdős (1913-1996)

“A mathematician, like a painter or poet, is a maker of patterns. If his patterns are more permanent than theirs, it is because they are made with ideas.”

G.H. Hardy (1877-1947)

What mathematicians do with their time is not widely understood by the laity. In earlier periods of history, they have been viewed with suspicion and even described as ‘dabblers in the dark arts’. Admittedly things have improved. The University of Oxford has a Chair in the Public Understanding of Science that is now occupied by a mathematician. The University of Cambridge has the Millenium Mathematics Project, one of whose educational aims is *“enriching everyone’s experience of mathematics”*². There are many excellent books, radio and television programmes, and online resources which aim to explain the nature of mathematics and the behaviour of its practitioners to ‘the man and woman in the street’.

But it is not really that hard to dispel the aura of mystery and magic surrounding mathematicians, for mostly what they do is to pose and solve problems; and sometimes, by doing enough posing and solving, they may develop a theory. Therefore, when students are given the right kinds of problems to solve, they become mathematicians too. The mathematical beliefs and habits of students are transformed as soon as they become problem-posers and problem-solvers, and their attitudes to what is served up in the lectures they attend are changed forever. The study of the development of mathematics as the history of ideas can have a similar, if not so dramatic, effect.

Gaining ownership

“In mathematics you don’t understand things. You just get used to them.”

John von Neumann (1903-1957)

In the context of learning mathematics, ‘ownership’ means possessing a full understanding and mastery of a mathematical entity (a concept, a solution, a result, a proof) that comes from deep immersion in its circle of ideas, a full awareness of internal relationships and logical structure, as well as an appreciation of its ramifications and wider implications in

² <http://mmp.maths.org>

areas around it. Or, following von Neumann's tongue-in-cheek quotation, you live with it long enough for it to become a familiar part of your intellectual surroundings. A good measure of whether you have acquired ownership might be this: if all other evidence and knowledge of a given piece of mathematics were to be removed from the record and destroyed, then you would be able to restore it fully to the public domain. The sense of ownership creates a sense of confidence and 'being in charge' of the material and having got to 'the heart of the matter'; such a state of mind is a fertile environment for creative processes to kick in.

Valued graduate skills

"I have nothing to declare but my graduate skills."

A latter-day Oscar Wilde at Customs Control in New York perhaps!

Governments these days profess to value trained mathematicians and have policies to provide for more of them, but mostly they concentrate on short-term utility and perceived worth to the economy. One of the metrics used in the Research Excellence Framework 2014 is reach and impact, and one of its three stated purposes is *"to produce assessment outcomes ... that provide accountability for public investment in research and produce evidence of the benefits of this investment"*³. Of course, the public has every right to know how well their taxes are being spent. The drawback of stressing the commercial capital of university teaching and research, however, is that it leaves out other important benefits that are harder for politicians to explain and justify.

There is no doubt that the average citizen is better equipped for life if he or she is comfortable with arithmetic (with and without a calculator), can make sensible financial decisions, has a feel for approximation and orders of magnitude, knows not to be fooled by spurious statistics and can intelligently judge levels of risk. Indeed, these skills should be inculcated by the end of secondary education, soon to be compulsory to the age of 17. It is also true for certain professions, such as accountancy, economics and actuarial science, that a vocational training at university is a beneficial and bankable preparation. But outside these specialist areas, why should an employer hire a mathematics graduate? Here are just three reasons of many, three qualities often found in the mind-set of someone with a good mathematics degree (see also Chapter 14): Abstract thought and structural insight; intellectual rigour, vitality and flexibility; and, mathematical modelling and problem solving ability.

Abstraction: It is impossible to study degree-level mathematics without dealing in depth in ideas that live entirely in the realms of conceptual thought; and most areas of modern mathematics, such as analysis, algebra, topology and geometry are developed as coherent hierarchical structures that travel far from everyday experience. High-level decisions in large organisations involve complicated structures with many variables, relationships and degrees of risk or uncertainty. Being able to model and conceptualise the decision problem and reach an optimal solution has much in common with the experience of studying abstract structures in mathematics.

3 <http://www.ref.ac.uk/>

Rigour and Vigour: In each year of a mathematics degree programme, an undergraduate will choose modules covering 8 – 10 different topics, each of which can present a new area of uncompromising intellectual endeavour, calling for logical precision, attention to symbolic detail and a capacity to grasp large-scale conceptual architecture. Having the ability and mental energy quickly to absorb new knowledge and master unfamiliar ideas is of significant value in a professional or commercial environment constantly in flux.

Creativity: The ability to identify, formulate and solve problems amenable to mathematical analysis is a creative faculty that transfers well to a business or industrial enterprise in dealing with day-to-day front-line questions of production and services up to the level of organisational structure, policy formulation and strategic development. Experience of, and confidence in, applying techniques of modelling and problem solving to real-life situations is an asset recognised by employers of graduates (see for example CBI, 2011).

Pre-university experience of problem solving

Before students can realistically be required to engage in mathematical problem solving when they reach university, it is essential for those setting the tasks to know what previous experience they have had and what their attitudes to it are likely to be. To paraphrase Ausubel's dictum (1968:vi) for teachers: *"First find out what learners know and teach them accordingly."*

Mathematical problem solving presents a serious challenge to students who have known only a traditional 'cook-book approach' to the subject, engendered by an assessment and accountability regime that rewards 'teaching to the exam', to an exam with predictable questions to be answered in small guided steps. To make the leap into the unknown of solving challenging problems, many will need careful preparation and plenty of support in order to build up confidence and start to derive satisfaction from success. We now consider a short account of the kind of problem solving some of the more fortunate or more enterprising students might have encountered by the time they take their A-Levels.

Problem solving in the National Curriculum

Key Stage 2 of the UK National Curriculum, which covers the last 4 years of primary school, contains 23 mentions of the phrase 'problem solving'. But most of the illustrative examples of this activity involve working out simple sums extracted from verbal descriptions of the following kind:

"Ahmed bought 5 apples and Sara bought 2 pears. If the apples cost 12 pence and pears 10 pence, how much did they spend between them?"

But continuous repetition of these routine and formulaic problems seriously risks boring most 10-year olds and making mathematics a very tedious endeavour. First-hand experience of enrichment classes confirms our view that many children are capable of, and excited by, far more challenging tasks than these.

At secondary school, things get a little better, especially after the 'catch-up' Year 7, although in contrast to Key Stage 2, the curricula in Key Stages 3 and 4 of National Curriculum for Mathematics include only two references to 'problem solving', and then only in passing and without explaining what it means.

Current support for problem solving in schools and colleges

In spite of the paucity of problem solving in the UK National Curriculum, however, there are a number of organisations offering excellent services to enterprising schools and engaged students.

The United Kingdom Mathematics Trust is a charity that organises a wide range of problem solving competitions and team challenges to students in the 11-18 age range, typically reaching 600,000 pupils from 4,000 schools⁴ each year. They also help to train competitors for Mathematics Olympiads.

The Millennium Mathematics Project, a collaboration between the Faculties of Mathematics and Education at the University of Cambridge, is a family of complementary programmes including the successful NRICH website, the online mathematics magazine *+Plus*, and an extensive outreach programme with schools and the public.

Mathematics in Education and Industry (MEI) is an independent charity for mathematics education with emphasis on the UK Key Stage 5 in the 16 – 19 age range. It is a partner in the National Centre for Excellence in the Teaching of Mathematics (NCETM) Consortium with responsibility for the secondary mathematics strand, and it also manages the government-funded Further Mathematics Support Programme (FMSP).

University mathematics departments that have more applicants with the highest A-Level grades than places on offer have to find other ways to select who they admit to their degree programmes. Some use interviews and references, others use additional examinations, such as the following:

- The Advanced Extension Award (AEA) in Mathematics is an additional A-Level paper that *“aims to challenge the top 10% of candidates nationally and help differentiate between the most able candidates”*.
- The Mathematics Admissions Test (MAT) is a two and a half hour problems-based test for applicants to the University of Oxford’s mathematics-based undergraduate degrees.
- The Sixth Term Examination Paper(s) (STEP) have been used since 1987 by the University of Cambridge to filter applicants to mathematics degrees.

These examinations are all based on the core A-Level syllabus but put much greater emphasis on original and challenging problems. Although many schools are not equipped to prepare students for AEA, MAT or STEP, help is increasingly available. This year, the Further Mathematics Support Programme is offering professional development for teachers of problem solving in Year 12, as well as problem solving workshops for sixth-form students. NRICH has extensive online resources geared to STEP preparation, and the University of Cambridge organises an annual residential summer school, funded by the Sutton Trust, for students from state schools doing STEP.

4 To put these numbers in context, in January 2013 there were 3,281 state-maintained secondary schools in England, attended by 3.2 million pupils. In addition, there were 790 independent (private) secondary schools attended by 320,000 pupils.

We have seen that there is already a rich vein of problem solving experiences to be had by all ages and conditions of school and sixth-form students, but whether they can take advantage of these depends largely on the commitment of their schools and encouragement from their teachers. There are also good prospects for further progress in bringing new problem solving opportunities to the secondary sector. It is well known that the UK Secretary of State for Education sought ways to raise students' aspirations and standards of achievement, in particular in mathematics. For example, In September 2013 he wrote:

"We must replace the modular GCSE treadmill with exams that encourage the skills universities and employers want, such as essay writing and mathematical problem solving."
Gove (2013)

All of the above, when taken together, indicate a reasonable likelihood that a student embarking now on a mathematics degree will previously have had some experience of problem solving of the kind we are discussing in this chapter. Even if only a modest fraction have benefitted from the experience, they can form a seeding population in the freshman cohort, for we will argue strongly below in favour of problem solving being encouraged to take place in a group situation, with students working together, presenting results to each other and defending their solutions in front of their peers.

Therefore heads of mathematics departments, teaching committees, and individual lecturers should not be timid about introducing a significant amount of problem solving right from the start of their degree courses. Not only does the current educational climate make it reasonable and desirable to do so, but there are good prospects for an even greater engagement in problem solving in secondary education in the future. Now is a good time to capitalise on the momentum being created.

Current university practice in problem solving

The Problem Solving Handbook

In this section we rely heavily on work undertaken through a project entitled Problem solving in Undergraduate Mathematics (henceforth 'the Project'), supported by the National HE STEM Programme in 2012. Its findings and advice can be found in Badger, Hawkes and Sangwin (2012).

At the start of the Project in autumn 2011, a short online questionnaire was sent via email to 59 heads of mathematics departments offering an undergraduate maths degree in England and Wales. Of the 59 institutions we approached, 34 completed the questionnaire. The responses generally indicated an absence of formal problem solving in the mathematics degrees in England and Wales. Twelve of the respondents acknowledged outright that they had no problem solving in either compulsory or optional modules, even though, in half of these cases, problem solving was mentioned in the published aims of their programmes.

Of the remaining 22 institutions, five offered an optional problem solving module to students, and 17 stated that problem solving was included in one or more compulsory modules. However, further investigation suggested their interpretation of problem solving was different from ours: sometimes it was a general claim that problem solving was an

inherent part of any mathematics degree programme even if not tied to any particular module; and sometimes problem solving was taken to mean mathematical modelling or optimisation, even numerical analysis or statistics.

After taking into consideration answers to other questions, just six modules, either compulsory or optional, could be deemed to include a substantial amount of problem solving in our sense. But we should stress that, where we were at odds with respondents' interpretations of the level of problem solving in their programmes, it was mainly because we were interested in specific modules where problem solving contributed a substantial amount to the students' work. We freely concede that it is unlikely that any student could make it through an entire mathematics degree course without encountering some problem solving of the type we have in mind.

The case studies

As a result of the questionnaire, we identified six mathematics departments that offer a mathematics module with problem solving at its heart and deliver it in varied and interesting ways; we were grateful that they agreed to be interviewed and to allow their institution to be identified in the Project Report. Here we give just a brief overview of the case studies and refer the reader to the Project Report, where they can be found in their entirety (Badger, Hawkes & Sangwin, 2012).

University of Birmingham: Problem solving is taught to first-year students in their first term using the Moore Method (see Chapter 18). There are no lectures, and the only material students are given are the problems they are required to solve and definitions they might need to solve them. No examples or model answers are available to students. There are two groups covering different topics, geometry and set theory, and students do not get to choose which group they are in. They meet regularly to present and discuss their solutions. As the module progresses, the students gain confidence and develop critical judgment about the nature of proof and how best to present and justify their arguments. They are assessed on their best solutions and presentations. The module is optional for BSc students but now compulsory for students on the four-year Mathematics MSci programme.

Durham University: Again delivered in the first term of the first year, Durham's compulsory module, entitled Problem solving, is based on *Thinking Mathematically* (Mason, Burton & Stacey, 1982). It is taught using both lectures and problems classes, in which students work through problems in groups, recording the progress they are making by using rubrics.

University of Leicester: Leicester's compulsory second-year module, Investigations in Mathematics, sees students working in groups of around a dozen on different mathematical topics at an appropriate level. The module is taught in parallel by several members of staff and each of them takes a slightly different approach, although for all students solving problems is the core activity.

University of Manchester: The Mathematics Workshop (see also Chapter 6) is a first-year, first-term module again using *Thinking Mathematically* (Mason, Burton & Stacey, 1982), though to a lesser degree than at Durham. Students begin the year working in computer labs, before concentrating on modelling and problem solving after the mid-term break. The module is compulsory and comprises both lectures and classes.

Queen Mary, University of London: The only third-year module in our case studies, Queen Mary's Mathematical Problem Solving is an optional module taken by around a dozen finalists each year. Each student has a different set of problems from a range of topics in pure mathematics. Although there are no lectures and only a single class each week, students may seek help from members of the teaching staff at other times. The help sought and given is monitored and used to moderate the summative assessment.

Warwick: Warwick's first-year Analysis 1 module has been taught using problem solving for the past 15 years. This unique example among the case studies is a core module taken by all 300+ first-year mathematics students and taught entirely through problem-based learning. Students have one survey lecture together with four hours of group work in problems classes each week. Each student receives a weekly workbook of problems, inspired originally by Burn (2000), and builds up a portfolio of work over the term, some of which is presented to the students and tutor in their group. The formal assessment includes evaluation of the portfolios and a traditional timed exam.

Ways to introduce more problem solving into a mathematics degree

If, like we, you fervently believe that problem solving should be at the heart of every student's experience of a mathematics degree, and if, as a lecturer, you feel that your existing curriculum does not already achieve this, then you will want to consider how to change it. We mention several options, in increasing order of difficulty:

- Introduce significant problem solving into the modules you currently teach and attach summative module credit to it.
- Build support for change by persuading like-minded colleagues to do the same.
- Get permission to develop a problem solving module that you (and perhaps one or two colleagues) will deliver, maybe following one of the successful approaches described in the case studies above.
- Mount a campaign to make problem solving a central part of the degree; invite seminar speakers who have problem solving experience, lobby the teaching committee, consult the external examiner, and encourage your head of department.

Proselytise on a wider front: share any progress you make, attend conferences and give talks. There are probably more potential believers out there, disposed to be converted, than you imagine.

Integrating problem solving into an existing course

The Project (Badger, Hawkes & Sangwin, 2012) suggests three ways by which problem solving may be integrated into an existing course:

- Include one or two significant problems on each weekly assignment and promise that at least one of them will be in the final exam.
- Devote a weekly lecture and support class time explicitly to problem solving geared to the module material.

- Base a whole module on problems-based workbooks covering the whole syllabus with regular automated summative assessment.

The Project also provides a list of books that contain a large number of mathematical problems, both general and topic-specific, but mainly suited to first-year modules. A recent example of a rich source of problems is Zeitz (2007) whose first 4 chapters are about general strategies, specific tactics and principles, and bringing ideas from different branches of mathematics. The remaining 5 chapters engage separately with problems in algebra, combinatorics, number theory, elementary geometry, and calculus.

Other ways of creating a favourable climate for problem solving in a mathematics department range from posting a weekly problem for each student cohort, with or without a prize for the best solution, to running a weekly problems drive or seeking funding for summer school on undergraduate research. Student mathematics magazines provide another medium to stimulate interest in problem solving, and if your department runs an outreach programme that includes enrichment in schools (for instance, providing STEP support or supporting UK Mathematics Trust activities) students are usually more than willing to volunteer as helpers or tutors; explaining and teaching is a good way to reinforce learning.

Summary

Having discussed at length the true nature of mathematical problem solving, both in terms of what constitutes a genuine problem and what is required of the solver, we have made the case for putting problem solving at the centre of a mathematics degree. The benefits that students thereby accrue include:

- A gain in confidence, understanding and enjoyment, with spin-offs for their degree performance.
- The experience of learning to ‘think mathematically’ and getting a flavour of what it is like to be a mathematician.
- The acquisition of analytical and creative intellectual skills that will serve them well in later life.

We encourage the reader to download our Project (Badger, Hawkes & Sangwin, 2012) and to seek enlightenment within. We conclude with a quotation which is contained at the start of the Project but which is equally applicable to this chapter:

“Our purpose in this Guide is to argue the case for putting problem solving at the heart of a mathematics degree; for giving students a flavour, according to their capabilities, of what it means to be a mathematician; a taste for rising to a mathematical challenge and overcoming it. Our purpose is also to make it easier for colleagues who share our vision to find ways of realising it in their own teaching.”

Badger, Hawkes & Sangwin (2012:7)

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Chapter 9: Mathematical modelling and problem solving in real-world physical situations

Mike Savage & Michael Grove

Abstract

There are many well documented issues concerning the mathematical preparedness of students making the transition to higher education within not only the mathematical sciences, but also chemistry, physics and engineering. In this chapter we consider a further emerging aspect of this problem where, even with an increased numbers of students applying to study these disciplines, many of whom have the highest grades, students often struggle to apply the mathematics they have learned to model physical situations and solve unfamiliar problems.

While this may be attributed to the mathematical preparation students receive prior to university entry, the issue in question is how universities help their students develop these skills to aid both progression within their programmes of study, and their transition to employment or research. Here we discuss the evidence of incoming undergraduates lack of experience of mathematical modelling and problem solving before focusing upon a particular example of how the University of Leeds has sought to address this issue via a new first year undergraduate mechanics module.

Introduction and definitions

As has been observed (Schoenfeld, 1992) ‘problems’ and ‘problem solving’ have had multiple and often contradictory meanings through the years. In this chapter we focus on real-world physical problems and in this section we set out our definitions¹.

¹ Chapter 8 discusses problem solving from another perspective.

Mathematical modelling and problem solving (MMPS) are two distinct, but vitally important skills: ‘mathematical modelling’ and ‘mathematical problem solving’. These are essential skills for problem solvers who address real world problems such as:

- How high above the earth should a geostationary satellite be positioned?
- Determine the dynamics of Felix Baumgartner’s world record skydive from 39km above the Earth.

These examples, which originally formed genuine research problems, can only be tackled successfully by attempting to understand a physical situation – in essence developing a model (mathematical modelling) and then trying to understand that model through a rigorous mathematical analysis (problem solving). The skills of mathematical modelling and mathematical problem solving are not only essential for success within the undergraduate mathematical sciences, and indeed science, technology, engineering and mathematics (STEM) programmes, but they form skills that often must be mastered for students to function successfully within the workplace:

“The development and application of mathematical models is very common and occurs across a range of industries.”

ACME (2011:19)

Mathematical modelling and problem solving can perhaps be best understood through a modelling and problem-solving cycle, illustrated in Figure 9.1, which represents a systematic way of undertaking a mathematical investigation. The cycle comprises three fundamental stages: 1) Set up a model; 2) solve the mathematical problem; and 3) interpret and validate the solution – following which the model may be refined and the cycle repeated (Savage & Williams, 1990).

The starting point for the cycle is a real-world problem, for example in nature, business, or industry, which has not yet been transformed into a mathematical problem. This transfer from a real-world problem into a mathematical problem takes place through the modelling process. This simplifies the problem by making assumptions, while ensuring the essential features are retained. Indeed, determining exactly what the essential features are, for any given situation, is itself a fundamental skill that can only be developed through experience. This results in a mathematical description that can then be solved using mathematical or computational techniques.

Once a solution of the mathematical problem has been derived and interpreted, it is then validated against real data or experiment as appropriate. Should this solution prove to be inaccurate then the model has to be refined by changing one or more of the assumptions and a more accurate solution sought by going around the modelling cycle once again.

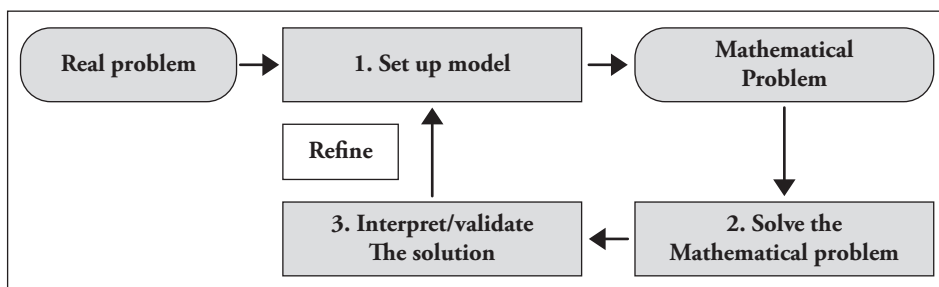


Figure 9.1: Modelling and problem solving in mathematics, physics and engineering: The modelling and problem-solving cycle.

The modelling and problem-solving cycle brings together three processes shown in Figure 9.1 (set up a model; solve the mathematical problem; and, interpret and validate the solution) and represents a coherent collection of skills expected of any graduate from the mathematical sciences. It is how these skills are applied as a whole and in a systematic manner that is of interest to us here and indeed that forms the very basis of modelling and problem solving. For example, the history of mathematical problem solving and investigations goes back to Polya's (1945) original work on heuristics (discussed in Chapter 8), and can exist independently of a mathematical modelling process. To explain what may appear to be a subtle difference, consider the example from Grove and Kyle (2014):

“...one of the most famous mathematical problems of all times is Fermat's Last Theorem which states that no three positive integers a , b , and c can satisfy the equation $a^n + b^n = c^n$ for any integer value of n greater than two. Fermat's Last Theorem is already expressed in mathematical terms, and so the modelling process can be omitted entirely. However, consider the problem that fascinated a 17-year-old Galileo: determine the period of oscillation of a bronze lamp suspended from the ceiling of the cathedral at Pisa. Before attempting a mathematical solution, an appropriate model for the oscillation of the lamp needs to be established, including assumptions, which can then be represented in mathematical terms.”

Grove & Kyle (2014:129)

While a mathematician may engage with aspects of the modelling and problem-solving cycle, they will not necessarily follow each of the individual processes in a structured and coherent manner, as their starting point may be different. When they do, this forms what we define to be modelling and problem solving. To an extent, this definition may help us recognise a key difference between pure and applied mathematics: applying mathematics to a real-world problem requires the development of a model that seeks to describe the 'real-world'.

There are many ways in which one might model a particular real-world problem and each would result in a different mathematical problem. The mathematical problem (assuming it satisfies certain conditions) will usually have a unique solution or a well-defined set of solutions. In practice, numerical approximations to theoretical solutions may be needed. Then these solutions or approximations need to be interpreted within the original physical context. Often the outcome may be insufficient or inaccurate, most likely due to the assumptions that were made at the outset, and the modelling and problem-solving

cycle needs to be repeated to build a mathematical description of the situation that is increasingly more realistic and representative of real-world behaviour.

Modelling and problem solving is therefore a collection of skills that need to be developed through experience. Getting started, once a real-world problem has been formulated, requires an ability to build a model of the real world that is relevant to the problem being investigated. This is the core modelling skill since without it the real problem is not converted into a mathematical problem and the investigation grinds to a halt. In the next section we present evidence which indicates that students' ability to solve physical problems in secondary education has declined substantially over the past 20 years.

Background: The cause for concern

A detailed understanding of exactly what constitutes or contributes to what has come to be known as 'The Mathematics Problem' is relatively recent. In 1995, the London Mathematical Society led the production of a report, *Tackling the Mathematics Problem* (LMS, 1995), that made one of the first attempts to do this. It noted three areas for concern:

"The serious problems perceived by those in higher education are:

- i) a serious lack of essential technical facility—the ability to undertake numerical and algebraic calculation with fluency and accuracy;*
- ii) a marked decline in analytical powers when faced with simple problems requiring more than one step;*
- iii) a changed perception of what mathematics is – in particular of the essential place within it of precision and proof."*

LMS (1995:3)

A subsequent report, *Measuring the Mathematics Problem* (Savage & Hawkes, 2000) identified the existence of a Mathematics Problem within the disciplines of mathematics, engineering and physics, and highlighted two contributing factors in particular: "*insufficient candidates with satisfactory A-level Mathematics grades for the number of degree places available*" and "*the freedom of A-level students to choose Statistics as an alternative to Mechanics*".

Since 2000 we have seen a substantial increase in the number of students studying mathematics at A-level, and in the percentage of those awarded the highest grades (Grove, 2012). From the growth in both A-levels and university courses (see, for example Chapter 4) within mathematics and disciplines that contain a significant mathematical component it is thus evident that there are now more learners who have higher pre-entry qualifications in mathematics than previously. Despite this, the mathematics problem persists. The Institute of Physics report *Mind the Gap: Mathematics and the Transition from A-levels to Physics and Engineering Degrees* (IoP, 2011) indicates that many physics and engineering academics still feel that new undergraduates are underprepared as they commence their university studies due to a lack of fluency in mathematics.

Measuring the Mathematics Problem recommended that "*Prompt and effective support should be available to students whose mathematical background is found wanting*" (Savage & Hawkes, 2000), and while it is evident that the mathematics problem persists, progress

has been made. An ample supply of free, good quality resources is available to help any students serious about remedying their shortcomings, and to help academic and support staff who aspire to assist students struggling at the school-university interface. In addition, a significant proportion of universities has invested in the provision of support mechanisms (see Chapter 4).

It can be argued that there are now several facets to the mathematics problem within higher education, one of which is: *“A lack of fluency in applying mathematics and specifically an inability to model and solve (unfamiliar) problems”* (Grove, 2012). This aligns with our experience of teaching first-year mathematics and physics students at two research intensive UK universities. An important issue appears to us to be an inability to apply mathematics as well as an inability to undertake the ‘routine processing’ of calculations. In particular, students struggle to identify the techniques and approaches required to solve problems. This should not, however, be taken to indicate that issues do not exist with students’ core knowledge in these disciplines, but merely that the mathematics issues we are facing within higher education are evolving and may require different approaches to their solution.

The Institute of Physics report (IoP, 2011) found that many of the academics surveyed believed that mathematics and physics provision at A-level leads to students learning by rote rather than developing conceptual understanding. The report *Newton’s Mechanics – Who Needs It?* (Savage & Stripp, 2009) highlighted similar concerns.

While this Chapter is perhaps not the place to dwell upon why such a problem exists, the fact that there is compelling evidence that the essential skills of modelling and problem solving are lacking amongst new undergraduates in mathematics, engineering and physics, implies it must be in some way linked to the pre-university curriculum and has been noted by the Institute of Physics:

“...changes in the nature of mathematics courses at school level have led to students being less proficient and confident in the mathematical skills required by physics degree courses”
IoP (2001:5)

Mathematics is an A-level subject where there have been numerous significant changes over the last fifty years, with more taking effect, or currently being planned. The nature of these changes are discussed extensively elsewhere (Smith, 2004; Gordon, 2005; Savage & Stripp, 2009:10; Lee & Dudzic, 2014), but a key feature has been a reduction in the opportunities that exist within A-level for students to apply their mathematical knowledge to solve problems. Perhaps the most drastic changes took place after ‘Curriculum 2000’ (Abramsky, Porkess, McColl & Sangwin, 2001) which ultimately led to a sharp decline in the number of students studying A-level Mathematics (Royal Society, 2008:63). As a consequence, revisions were introduced to both A- and AS-Level Mathematics qualifications for first teaching in 2004 (Porkess, 2003) in an attempt to address the issues that had been introduced and represented *“a serious attempt to raise the numbers taking mathematics in sixth form to previous levels”* (Porkess, 2003:13).

A key feature of the A-level changes introduced for 2004 was changing the balance between pure and applied mathematics: *“At present A Level Mathematics consists of 6 units; usually 3 of these units are pure and 3 applied. In the new scheme it will still consist of 6*

units, but 4 of them will be pure and 2 applied” (Porkess, 2003:13). A consequence is that students are now typically studying less mechanics prior to university entry, particularly if they do not study A-level Further Mathematics, and this has observed implications when they enter higher education:

“Currently 30-40% of schools and colleges enter their students for at most one unit of mechanics. As a result, many students arrive at university to study physics, engineering and mathematics having studied at most one unit of mechanics. The reduction in the amount of Newtonian mechanics studied by many students as part of A-level Mathematics has resulted in a decline in students’ modelling and problem solving skills”
Savage & Stripp (2009:17)

It is not only changes to A-level Mathematics that have contributed to this problem, there have been changes to other A-level subjects including Physics. A-level Physics and Mathematics are subjects that many students study in combination. For example, for examinations taken in 2011, almost a third of those studying A-level Mathematics also studied A-level Physics, and just over 80% of those studying A-level Physics also studied A-level Mathematics (Gill, 2012). By studying A-level Physics in conjunction with Mathematics, students had the opportunity to reinforce and apply their mathematical skills by tackling problems in physics. However, in recent times the requirement that A-level subjects be independent of each other has led to a reduction in the mathematical content of Physics A-level.

In the late 1990s, the Department of Physics at the University of Liverpool set about exploring the apparent paradox *“that incoming undergraduates have better qualifications but seem to know less than previous years”* (Milstead, 2002). They analysed the changing mathematical content of A-level Physics noting:

“A-level papers of one of the largest exam boards were selected. These dated from 1983 to the present day. The most striking fact was the purging of maths. The average number of mathematical steps needed to answer a question was constant at about four until the early 1990s. Two mathematical steps are typically needed now.”

(Milstead, 2002)

They explored the change in difficulty of the, at the time, revised Physics A-level, by setting their undergraduate students in-class tests:

“Older (pre 1992) and more recent (post 1997) questions were selected. Both sets of questions covered the same topics and required the same basic knowledge. Scores were on average 1.6 times higher for the recent questions.

We also repeated the exercise with second year students. As expected, they performed much better overall, although even they managed the older questions less well. It is hard not to conclude that the questions are not as difficult as they used to be.”

(Milstead, 2002)

In 2011, attempts were made to tackle the inability of undergraduates to model and solve problems by a collaborative project through the National HE STEM Programme

involving thirteen chemistry, engineering, mathematics and physics departments from eight UK universities. While each department adopted a slightly different approach, the underpinning philosophy throughout was to address the inability of incoming undergraduates to apply mathematical techniques to unfamiliar exercises, problems or situations. The interventions implemented during the first year of university study focused upon enabling students to model situations, solve problems, and have opportunities to engage with the application of mathematics in a variety of contexts.

Mathematical modelling and problem solving at the University of Leeds

A new skills module, 'Mathematical Modelling and Problem Solving' (MMPS), was introduced into the first year of the undergraduate physics course in 2011/12. The aim was to enable students to understand the key role that mathematical modelling plays in the investigation and solution of physical problems – by developing two specific modelling skills through practice and reflection. The ability to 'set up a model of the real world' is the core modelling skill; its function is to transform a physical problem into a mathematical problem. Mastering this skill is an essential prerequisite for developing a second, more advanced skill – the ability to perform a full mathematical investigation by means of a multi-stage modelling and problem-solving cycle. In its present form, following a pilot in 2011/12, this new MMPS module has been offered as an elective to first-year physicists in 2012/13 and again in 2013/14 with uptake amongst the student cohort rising from 14 to 40 students.

The underpinning philosophy is to use a four-step strategy to develop students' core modelling ability. From the outset students are presented with, and challenged by, a wide range of problems in Newtonian mechanics designed so they can practice and develop their skill in setting up a relevant model of the real world prior to solving a mathematical problem. This process of converting a real physical problem into a mathematical problem is called 'modelling' or 'modelling the real world' and it has a simple structure consisting of 4 distinct steps that can be remembered by the acronym ADVaL:

1. **Assumptions:** List the assumptions relevant to the problem.
2. **Diagram:** A diagram is constructed taking account of the assumptions.
3. **Variables:** Variables are introduced onto the diagram.
4. **Law/principle:** The application of a law or governing principle will give rise to equation(s).

Module structure

A set of preparatory notes were sent out to students advising them to prepare for the module by becoming aware of 1) its aims and objectives; 2) the module structure; and, 3) key concepts/questions such as 'what is a model?', 'what is modelling?', and 'what has modelling got to do with problem solving?'. This provided a suitable focus for questions and discussion at the start of the first meeting as students were keen to understand the module structure.

The module itself was split into two parts with *Preparation* (weeks 1 – 5) followed by two *Investigations* (weeks 6 – 11). An initial five-week period of preparation was necessary to allow students to address a variety of physical problems, individually and in small groups,

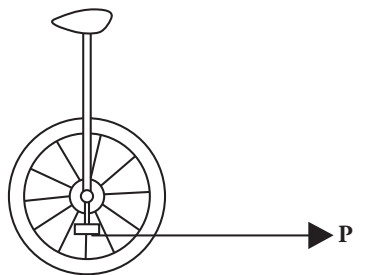
with sufficient time to practise and develop a core modelling skill – the ability ‘to set up a model of the real world’. Experience shows that such preparation is essential in order to not only master this core skill but also to understand its significance in *transforming the physical problem into a mathematical problem*. Following the period of preparation, students were then in a position to undertake two mathematical investigations – by following the various stages of the modelling diagram (Figure 9.1) and in each case writing an individual report for assessment.

Preparation

The primary aim of the first meeting with the students was to get them actively engaged with a number of interesting and challenging problems drawn mainly from Newtonian mechanics. A set of problems were introduced on the basis that they would provoke thought, generate discussion and identify misconceptions. ‘Pulling a unicycle’ is a classical problem with which students can get actively involved right from the start.

Example: The unicycle problem

A string is attached to the lower pedal of a unicycle. With a colleague or student pressing a hand on the saddle so that the unicycle remains upright, a string that is then attached to the lower pedal and a force P applied. In which direction does the unicycle begin to move?



To students beginning an undergraduate mathematical sciences or physics degree this will be a genuine problem (see Chapter 8). It is worth allowing students to think about and discuss what will happen. This can be followed with a demonstration, which in itself generates great surprise from most of those who have not seen it before, and then provides an opportunity to discuss what has happened by exploring the key conceptual issues.

Setting up a model: A simple strategy

The primary aim of the second teaching session was for the students to discover for themselves a strategy for setting up a model of the real world by tackling a simple modelling problem such as ‘lunar exploration’.

Example: The lunar exploration problem

Imagine you are a scientist involved with a lunar exploration programme that requires the building of a track or pipe that circumscribes the moon. One member of the team identifies the need to raise the track 3 metres above the lunar surface – how much extra length of track would be needed for such an approach?

The focus here is not upon determining a correct solution, but how students make an attempt at obtaining a solution. This leads nicely into introducing the ADVaL strategy which can then be applied to increasingly more sophisticated problems: For example, reviewing experimental laws often used in mechanics such as 1) Hooke’s law and the related concept of a light and inextensible string and 2) the law of static friction and the related concepts of a smooth pulley and the angle of friction.

Modelling and problem solving in mechanics

The aim of this five-week period of preparation was to give students plenty of practice in tackling unseen problems in mechanics. For each problem they used the ADVaL strategy to set up a model of the real world and then find a solution to the resulting mathematical problem. On this basis a *modus operandi* emerged for how the sessions were to operate and be delivered. Each week, the workshops began with students being given an unseen problem and invited to work with a partner so as to interact/discuss the setting up of a model and finding a solution to the mathematical problem. During this time the tutors, consisting of a mix of academic staff and postgraduate students, circulated, ready and willing to respond to student questions and queries. The final 15 minutes of each teaching session were left for tutors to go through the modelling and analysis with the whole group – inviting questions and drawing attention to key points/subtleties and any potential pitfalls.

The second workshop followed a similar pattern and concluded with students being handed an assignment consisting of two unseen problems to be worked on individually and submitted the following week.

Taking stock: student feedback (after five weeks)

When faced with a previously unseen problem, students soon come to realise that ‘setting up a model’ of the real world is the link which transforms the physical problem into a mathematical problem and then, and only then, is one ready to do mathematics and find a solution. Students come to see ‘tackling a physical problem’ as a 2-stage process, and as one student noted:

“You can’t hope to solve these problems unless you can model.”

After five weeks of initial activities, student feedback was collected from the 2013/14 cohort. Almost all the students reported they had developed the confidence and ability to address unseen problems and there were signs that they were starting to enjoy the problem-solving experience. Since 2013/14 was the first time that a large group had elected to study this module, student comments were invited on all aspects of the delivery they had experienced: from practical demonstrations to workshops, worksheets, and the ADVaL strategy:

Demonstrations: *“These are highly valued – They work well”*

“Please ensure that this part of the teaching is not overlooked as it really helps to consolidate knowledge and initial ideas”

Workshops: *“I do enjoy the workshop style of teaching”*

“A relaxed atmosphere with groups of students working together is good”

Worksheets (ADVaL): *“These are interesting and very helpful – The questions are stimulating and varied”*

“Going into such depth with each problem was enjoyable”

“The ADVaL strategy is a new concept that helps to cement the ideas for modelling – It has changed the way I think about problems”

Module (so far): *"I feel that I have learned a lot in the first five weeks"*
 "I gained much confidence and insight from doing the worksheets"

Mathematical investigations

By the start of week 5, the students had addressed a range of previously unseen physical problems by building, in each case, a suitable model, deriving a mathematical solution and, where possible, interpreting the results. They were therefore ready to be introduced to the idea of a more open-ended investigation tackled by means of the multi-stage modelling and problem-solving cycle (Figure 9.1). This was undertaken through small groups of three or four students.

The key feature was a much greater emphasis on interpreting and validating the solution. This was achieved by considering how their developed model aligned with observation, experiment or published data/evidence. After progressing once around the modelling and problem-solving cycle, students were encouraged to reflect on the strengths and weaknesses of their basic model and identify significant disparities between theoretical prediction and experimental or observational data. This then provided the motivation for them to set up a more refined model and continue as before.

Whereas the problems tackled over the previous five weeks of the module were, in some senses well defined, the investigations were much more vague and open to interpretation. The investigation 'brief' broadly defined an area for students to explore and so there were potentially many routes by which students could do this, or ways in which they might refine and extend the task. Students were offered the opportunity of identifying their own area for investigation which could then be developed in conjunction with the tutors.

Example investigation: Imagine you are members of the Red Bull team of scientists and engineers responsible for overseeing Felix Baumgartner's world record skydive from 39km above the Earth. You have been assigned to investigate the dynamics of his motion from the stratosphere. In particular you need to predict his speed both as a function of time and distance fallen.

On the face of it, this may appear to be a trivial problem involving freefall under gravity. However, while this may be the starting point for a simple model, it is a great example of how a problem can be refined and developed through the modelling and problem-solving cycle. Students quickly realised that the problem is more complicated. For example, while the assumption that air resistance is negligible will hold outside of the Earth's atmosphere, this will no longer be true as Baumgartner enters the atmosphere. This introduces the concept that his acceleration can no longer be regarded as a constant and introduces the idea of terminal velocity. Further, the density of the atmosphere varies with height from the Earth's surface, and so this means that air resistance once inside the atmosphere will vary as a function of position (height). The organisers of the record attempt have made a range of data available for individuals to utilise (Red Bull, 2012) including NASA data giving the variation of density of the Earth's atmosphere with altitude, all of which was used by students to refine their mathematical models.

For their investigations students were instructed to adopt a specified structure when writing their project reports. This helped to introduce them to the idea of scientific

writing. Their reports should begin with an abstract specifying clearly the problem(s) investigated and outlining ‘*what they set out to do*’, ‘*what they discovered*’ and ‘*their conclusions*’. Their reports needed to provide a systematic account of the models investigated – from the basic to the most refined – in each case following the stages of the multi-stage modelling diagram (Figure 9.1).

End of module student feedback and results: 2013/14

A total of 38 students completed the module; all passed, with 27 awarded firsts and 6 achieving 2(i) classifications. While this is clearly positive, of most significance were the student comments which indicated they had appreciated the methods of learning that they had experienced, particularly through the investigations:

“I enjoyed the investigations; they made me think about problems in a more realistic way.”

“Sometimes it’s unclear what the tasks are – what to do next! However, i [sic] appreciate its more about developing a new way of thinking.”

“I have learned new skills for applying maths to real world problems!”

For the 2011/12 cohort, when this module was first introduced in its pilot form through the National HE STEM Programme, a parallel educational research study was undertaken to explore the impact of the approaches to helping students develop their modelling and problem-solving skills. This analysis (Pampaka & Williams, 2012) concluded:

“The course managed to re-introduce Mathematical Modelling & Problem Solving (MM&PS) into the undergraduate student experience in an arguably ‘authentic’ course in mechanics in first year Physics... Students’ dispositions (and probably their awareness) of Mathematical Modelling significantly increased between measures ‘before’ and ‘after’ the course... The best students certainly revealed in their work a convincing grasp of modelling, the modelling cycle and model-refinement...”

Pampaka & Williams (2012)

Concluding comments

This chapter has highlighted a particular aspect of The Mathematics Problem facing students as they make the transition to studying the mathematical sciences, and indeed other science, technology, engineering and mathematics subjects, within higher education. Students can arrive at university either unprepared, or lacking confidence, in being able to model physical situations and tackling less well defined problems. This can have implications for progression through their undergraduate programmes (see Chapter 13), and for their transition into the workplace where modelling and problem solving are skills expected by employers of STEM graduates.

Through a large-scale National HE STEM Programme, a number of universities became involved in tackling this issue and from this work we have presented one example of how higher education institutions are tackling this ‘modelling and problem solving’ issue. The example is provided as an attempt to provoke thought and debate as to how other universities might recognise and address this problem facing their undergraduate mathematical sciences students.

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Chapter 10: Learning statistics at university

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Abstract

This chapter considers the opportunities and challenges of teaching statistics to mathematics majors at university level. The pedagogy of teaching statistics to non-mathematics majors represents a large and active research area, reflecting the recognised challenges associated with this set of skills as well as the additional challenge of mathematics phobia amongst many such learners. Whilst some of this work clearly has a broader relevance, generally there is rather less developed work concerning the teaching of statistics to mathematics majors, where many different issues arise. For example, we will highlight the challenges of mathematics students who have been ‘taught to the test’ and expect a similar approach to learning at university. Clearly at some stage of mathematics education this is an inappropriate expectation (consider for example mathematical modelling (Chapter 9)) but we suggest that from the outset statistics poses the need for judgement in a way that is not required when mastering many mathematical techniques. Learning statistics at its best fundamentally challenges a mindset that expects objectively verifiable answers. Whilst the place of statistics within mathematics presents particular difficulties it also affords opportunities for the development of mathematics students’ reasoning abilities. Similarly, the broader (non-mathematics major) statistical pedagogic effort has identified a number of concepts that are vital in order to apply statistical thinking. This chapter will set these within the frame of a ‘threshold concept’. Many concepts within mathematics can be considered using the framework of threshold concepts (imaginary numbers being one of the text book examples) but calculus is another example. It is possible to appear mathematically successful by completing the tests set in a calculus course. Now consider applying calculus to problems in probability; specifically when related to the likelihood. Finding the point that maximises a likelihood is an important estimation problem. Appreciating that a sharper peak in the (log)-likelihood implies a more precise estimate moves a learner into conceptually different territory. But the critical dependence of these properties on theoretical assumptions is not necessarily clear. A modern approach to teaching statistics can improve the learning across a broad range of

subjects by shifting attention away from procedures and on to concepts. This is clearly valuable in statistics and areas such as mathematical modelling but we would suggest it is generally valuable to encourage students to think around a problem. Insights from teaching statistics to non-mathematics majors therefore provide additional means of developing a deeper understanding of mathematical techniques amongst all students.

Introduction

Currently there is plentiful hype describing ‘Big Data’ (Mayer-Schonberger & Cukier, 2013); a rather generic term used to describe massive data sets with perhaps most of the interest being in their commercial potential. Famous examples of this phenomena are the kind of data social network providers and Internet search engines acquire about our on-line behaviour. The most obvious value in these data would be to better understand our susceptibility to marketing activity. The report by Manyika et al. (2011) represents a typical example of the commercial hype around this phenomenon and seeks to encourage more businesses to better understand the data they hold on their customers (or secondary data they could acquire on their customers). What is currently striking is that the graduates servicing this activity are unlikely to be statistics graduates.

It is difficult to obtain definitive figures on the supply of graduates, but in the United States a decade ago figures were estimated by Scheaffer and Stasny (2004) who suggest that in 2000 just under 11,000 Mathematics Majors, under 5,000 Mathematics education majors, just over 500 Statistics majors with slightly over 400 Actuarial Mathematics and Statistics majors and less than 200 Mathematics and Statistics majors graduated. Although the situation in the United States is changing, based on course offerings in the United Kingdom the relative proportions of mathematics and statistics students would appear comparable. It therefore seems the case that the main route to becoming a working statistician will continue to be via a mathematics degree, with or without a statistics minor. One reason therefore for considering the statistical education of mathematics majors is indeed that they might find themselves either working in a role better described as ‘statistician’ or at least using skills that one associates more with the field of statistics. However, mathematics graduates are valued not merely for their technical skills but rather for generic qualities such as problem solving and the ability to abstract and generalise. It is also the case that a modern statistics education provides a wide range of generic skills that are valuable in many areas beyond those that could be described as ‘being a statistician’. We will go on to suggest that developing a conceptual understanding of mathematics is a skill which can be acquired within a modern statistics course and hence reinforce a better mathematical intuition.

Whilst there is a thriving body of pedagogical literature specifically directed towards statistical education it remains the case that much of it at university level is directed towards how best to teach statistics to non-mathematics majors. One exception is Ko and Lee (2011) who review evidence that more mathematically talented students can attain mastery of statistical concepts sooner; highlighting that there are differences in the way one might approach their education when compared with non-mathematics majors. In general though, there is rather less discussion that is immediately relevant to how we should approach the statistical education of mathematics majors. Certainly, we find ample evidence, such as Connors, McCown and Roskos-Ewoldsen (1998), that mathematics

anxiety can act as a barrier to learning statistics. One would hope that if mathematics majors struggle in their statistical development it is for other reasons. Statistics may often be taught to mathematics majors at university level as a branch of mathematics within a school or department of mathematics. However, the skills required to engage with the subject are somewhat different to those required in mathematics and this is made very clear by the broader statistical education literature. Bessant and MacPherson (2002) show how statistics pedagogy has emerged as a separate field. More recently, del Mas (2008) reviews the development of this broader statistical pedagogy. They repeat the assertion that non-mathematics majors can be challenged by mathematics phobia. However, whilst they suggest other challenges peculiar to statistics education they make one intriguing statement, specifically that if students equate statistics with mathematics they “*expect the focus to be on numbers, computations, formulae and only one right answer*”. On this topic, Moore and Cobb (2000) provide an insightful quote:

“Mathematical theorems are true; statistical methods are sometimes effective when used with skill.”

Moore & Cobb (2000)

Skill is of course an interesting word to choose. Merriam-Webster¹ give a definition for skill as “*The subtle or imaginative ability in inventing, devising, or executing something*”. Mathematics certainly requires a large degree of skill in execution; statistics requires rather more of the ‘imaginative ability’ although perhaps we would use words like ‘select and interpret’ rather than ‘invent or devise’. There are clearly potential challenges to mathematics majors as they make the transition to university if they regard statistics as being more about the proof of procedures and hence solely about execution skill rather than requiring the ‘subtle and imaginative’ skills to use these procedures. To give one concrete example throughout this chapter, we suggest that really appreciating the implications of the *assumptions* sometimes hidden in mathematical statistics is vital to successfully engaging with statistics at university level. To be more specific, we consider linear regression. Assume we have n observations $\{y_i, x_{i1}, \dots, x_{ip}\}_{i=1}^n$. We can do nothing to fit *statistical models* to any such data without making several assumptions including the assumption that it possible to approximate the relationship between the variables at least locally by a linear function such as:

$$Y_i = \mathbf{x}_i^T \boldsymbol{\beta} + \varepsilon_i,$$

where the terms ε_i are unobservable error terms and Y_i is the random variable for which we have a realisation y_i and $\boldsymbol{\beta}$ is a p vector of regression parameters. For brevity, writing the $n \times p$ design matrix $\mathbf{X}$, and the $n \times 1$ vector $\mathbf{y}$ we note that it is relatively easy to get students to show that the least squares estimator for $\boldsymbol{\beta}$ is given by:

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}.$$

It is somewhat harder for students to appreciate the role of the error assumptions. Simple examples would be to note that if we make some assumptions (errors are zero mean, independent, and constant variance) we believe this estimator to be the best linear unbiased

¹ <http://www.merriam-webster.com>

estimator. If we make a further assumption that the unseen (and unseeable) errors are Normally distributed we have maximum likelihood estimators. If we do not wish to make any assumptions we still get some kind of line of best fit which has some nice least squares geometric properties that might be enough for a simple data description if that is all the application requires. The mathematics of the situation are usually rather more accessible to students than the statistical modelling assumptions. Yet it is the plausibility of the assumptions that makes or breaks any attempt at modelling and it is statistical modelling that we need if we wish to make statements that have validity beyond that particular dataset.

The challenges for learning statistics at university could be deeply entrenched by the time learners arrive at university. If a student has studied the Binomial distribution as a model they might be very well versed in computing the probability $P(X = x | n, p) = \binom{n}{x} p^x (1 - p)^{n-x}$ but not appreciate that the Binomial distribution is no more than a convenient mathematical fiction that in practice may not fit a given data set well. Being well versed in getting the right answer to such problems might not equip a student well on arrival at university where they could get taught by a statistician who has an understanding of context from research collaborations or consultancy. They may get taught by a statistician with an understanding of the uncertainty present in the specification of a statistical problem as well as the various approaches that may be taken to address it. We then have a transition from 'getting the right answer' to a very different environment. If one were fitting generalised linear models to responses assumed suitable for modelling by such a Binomial distribution, most data show over-dispersion. The 'correct' answer is not enough and one needs to adjust the model accordingly – and more worryingly, there is now no obvious 'correct' answer to the problem of modelling over-dispersed data.

Statistics is no longer about finding the right answer to a carefully posed problem but is now a (subtle or imaginative) *skill which must be developed*. Learning this kind of skill is a different process to the mathematical skills developed early in an undergraduate programme. It requires the development of subtle judgement and imagination and consequently the feedback process is very different. It is entirely possible that one gets all mathematical aspects of a problem correct, yet perhaps has chosen the wrong tools, or missed aspects of their interpretation. Regression models can be fitted to data, but if the assumptions required are not appropriate the interpretation varies. For simple linear regression, we still have a least-squares geometrically optimum fit to the data, which may suffice for some problems. But for other problems, a statistical model may be needed and would not be appropriate without some further work.

What has become clear from work with non-mathematics majors is that statistics throws up a number of tricky concepts. The subject is murkier without a clear understanding of these concepts. If one must teach frequentist statistical methods, we note it is possible to drill people to pass an exam in which they reject null hypotheses because the p-value is below 0.05. It is harder to get them to appreciate that this ritual is one which lets us make statements about a (theoretical) population based on evidence in a sample of data. Hence considerable effort in teaching non-mathematics majors could fruitfully involve informal inference (Makar & Rubin, 2009). We say more about this later, but for now note that this work is designed to ensure that students appreciate that the goal of formal statistical inference is to make a statement about a population based on data in a sample. It is not

necessarily the case that statistics courses on mathematics programmes develop a conceptual understanding of the subject in such ways that are becoming commonplace among non-mathematics majors. Indeed, some of the work involves language drills (“My sample of data tell me this so I may infer that in the population...”) designed to make the procedure clear but obviously taking some space in the curriculum. It is easier with mathematically capable students to rush to mathematical derivations of sampling distributions than to deal with these non-mathematical reasoning concepts. We therefore consider a framework for considering some of the concepts that must be addressed in order for students to interpret statistical problems.

Background theory: threshold concepts

In order to make sense of data subject to uncertainty, it is necessary to juggle several concepts which at best are challenging and at worst are counterintuitive and additionally misunderstood by many teachers and lecturers. In order to reduce the problem we could consider these within the framework of threshold concepts. Meyer and Land (2005) state that mastery of such threshold concepts provides a *“new and previously inaccessible way of thinking about something”*. One mathematical example of such a concept is imaginary numbers, however the statistical example given by these authors is that of a sampling distribution: the distribution of a test statistic under hypothetical repeated sampling. As noted in the previous section, a core concept of conventional statistics courses is the idea that we use data taken from a sample to make inference about the population from which it came. In a theoretically ideal world, random sampling is key to making valid inference on that population. In theory, we believe we could repeat the data collection exercise many times each time obtaining different realisations of the random variables under examination. Each time we could compute a particular statistic to summarise the data and sampling theory tells us something about the distribution of that statistic. Firstly, we are required to appreciate that the observed data are only conceptually related to the underlying assumed random variables. But then in practice, the relationship between population and sample is not always clear. Much administrative data (such as the census) is the population – concepts such as meta-population are too abstract for most introductory statistics books – and so it is not immediately clear how the concept of a sampling distribution applies when we have the whole population. Likewise, real surveys suffer from non-response; or it may be that the relationship between the sample and the intended population is not as clear as it should be (if using telephone sampling during the day one is less likely to speak to those out at work). Thus, there is a lot of judgement required to actualise one simple concept or relating a sample to a population even though the latter is a theoretical construct. In applied statistical practice, these differences are critical and yet we suggest again that it is tempting in a crowded curriculum to neglect much of this and cover the more mathematically interesting topics.

One advantage of covering this particular threshold concept with mathematics majors is that we can directly use mathematics. Whilst many concepts are either vague or difficult to understand, mathematics students should arrive with the advantage of quickly being able to understand what we mean by finding the probability $P(a < X < b) = \int_a^b f(x) dx$ for some continuous random variable X and two real constants $a < b$ given some density function

$f(x)$. So, given an appreciation of the concept of a sampling distribution as the behaviour of a test statistic given an assumed null hypothesis, understanding classical inference becomes a little clearer a little quicker for a mathematics major, provided sufficient attention is given to conceptual understanding. Nevertheless, there is a risk that a student seeking mathematical techniques could navigate much of this material without ever fully appreciating the importance of a sampling distribution in terms of inference and, vitally, fail to realise that the very idea of a sampling distribution is ultimately a mathematical fiction applicable in practice only by the use of carefully specified assumptions.

Perkins (1999) suggests mastery of threshold concepts involves transitions across “*troublesome knowledge*”; material that is difficult or counter-intuitive. Yet having transformed one’s way of thinking, like riding a bike, there is meant to be no way back. Yet whilst mathematics education might not be so comfortable with constructivism, the idea of a concept that is counter-intuitive and requires careful exploration is familiar territory; interest in probability is regularly piqued by apparent paradoxes which can be resolved by careful exploration. Where we have difficulties in undergraduate mathematics education is that attention can be focused on the mathematics of the problem (finding the area under the curve of a probability distribution in the early stages, applying the delta method in later stages) rather than on the challenging concept represented by the idea of a sampling distribution. The distribution of a test statistic under the null is a piece of fiction at several levels; de Finetti (1970) reminds us that probability does not exist and many of the assumptions we make in order to define null distribution in a particular context can be somewhat unrealistic. In this context, it is worth spending time revisiting the work that has been done on educating non-mathematics majors in statistics as focusing on these threshold concepts identifies issues that merit attention even amongst mathematics majors on a statistics course. Mathematics does not afford the only way to approach the topic. Randomisation tests afford a different way of approaching the concept (Tintle, VanderStoep, Holmes, Quisenberry & Swanson, 2011). This requires less mathematical skill; rather all that is required is sufficient appreciation of the computation to master the concepts. Indeed, randomisation tests may not only be suitable as an introduction for mathematics-phobic students, but a way for mathematics students to understand the concepts represented by the mathematics they are working on. Indeed, we note later that there are longstanding suggestions that greater use of Monte Carlo methods within mathematics education could be conceptually very helpful. For now, having suggested that it is possible to identify ‘threshold concepts’ that require some attention within statistics education we return to non-mathematics major pedagogy to examine catalogues of possible concepts that need to be mastered within statistical training.

Introductory statistics: the GAISE Report

Cobb and Moore (1997) carefully consider differences between mathematics and statistics as they are relevant for educators, in particular clearly emphasising the importance of context. Indeed Langrall, Nisbet, Mooney and Jansem (2011) highlight the importance to students of understanding context in order to understand the statistical material they are trying to learn – unfamiliar contexts having the potential to make learning difficult. Statistics is therefore not just working with numbers, but keeping their context carefully

in view. The importance of context is clearly articulated in the American Statistical Association GAISE report (Aliaga et al., 2010). While this framework has been proposed for the undergraduate statistics education of non-mathematics majors, it articulates one view of core concepts vital to making sense of statistics and this articulation is worth considering when considering courses aimed at mathematics majors.

One simple way of illustrating the importance of context is to highlight the role of study design. A well known recent example of this can be taken from clinical trials for hormone replacement therapy. It is possible to examine the incidence of side effects using two different study designs. One can carry out a study using a randomised controlled trial, where subjects are unaware which treatment they will receive (hormone replacement therapy or an inactive control). One can then compare the difference in the occurrence of some phenomena between the treated and untreated groups. Alternatively, one can examine women with the phenomena of interest retrospectively and try to find a comparable group similar in all respects other than not having the phenomena of interest. Given these two groups one can determine whether or not they were taking hormone replacement therapy. In principle, one ends up with four numbers either way, a cross-tabulation of with/without treatment and with/without phenomena of interest. However, the interpretation of these numbers is utterly different due to the differences in study design – a point well developed by Barton (2000). Given only the numbers, we cannot distinguish the context. Given the study design, we are entitled to draw very different conclusions. Therefore, one learning outcome from the GAISE specification states that learners must appreciate the importance of distinguishing between randomised allocation of controls where we can more directly make claims of causality, in specific studies and observational studies.

The GAISE document currently contains 21 further learning outcomes for an introductory statistics course; careful examination of these reveal the necessity of developing careful reasoning throughout statistical education. Hence there is good reason to claim that appreciating context is a key issue required in order to master statistics. The challenge in education is that it is difficult to anticipate the range of contexts that may be required. Rather, one has some generally applicable skills and growing experience in using judgement in a few contexts. This is likely to be challenging to a mathematics major student expecting to be able to verify that they have found the correct answer. Whilst we are starting to map out some of the concepts that might give students specific difficulties we really meet another challenge – the development of judgement. If statistics can be described as the science of uncertainty, we have to acknowledge that few undergraduate mathematics students are enthusiastic about working in such an uncertain learning environment.

Current problems in statistics pedagogy

The recent reforms in statistical education are important both for the non-mathematics majors to whom they are primarily directed as well as, to some extent, mathematics majors. Nevertheless, there is no clear consensus on the way that the evidence base should inform statistical teaching (Beyth-Marom, Fidler & Cumming, 2008). Moreover, one could argue that most introductory statistics textbooks are based on 20th century problems. A review of several recent excellent introductory statistics textbooks does indeed reveal clear critical thinking around many of the threshold concepts described above. Nevertheless, when it

comes to inference it is mainly formal and usually concerns the comparison of two groups, more than two groups as well as regression and contingency table analysis. However, with the exception of Zieffler, Harring and Long (2011) who use randomisation and bootstrap methods, most textbooks still cover methods developed in the 1920s and 1930s. The problems considered are arguably those facing researchers in the 1920s and 1930s.

Much modern science relies on large and complex mathematical models; the goal is no longer to conduct experiments and collect data by fairly direct methods, rather experiments are conducted to explore predictions from these models. The Higgs Boson is one example where the models and the predictions to be tested were developed long before they could be tested (Dittmar & Dreiner, 1997). The problems alluded to in the introduction regarding 'Big Data' and testing the predictions made by advanced mathematical models are not yet being addressed. Glymour, Madigan, Pregibon and Smyth (1997) highlight problems with 'Big Data' that are conceptual as well as computational, including the fact that we have data for the entire population which alters the conceptual framework for statistical inference.

Cohen (1994) notes that with large enough sample size, all null hypothesis tests are rejected which is just one illustration of the statistical-theory problems that emerge in current practice. Hence we suggest that it no longer makes sense to fix a statistics curriculum in terms of commonly applied techniques; whether we are teaching non-mathematics students to apply techniques using a computer or mathematics majors the mathematical derivation of such procedures. Modern science and data science require that we develop students who can reason their way around problems that are rather different to those envisaged by many current text books. It could be argued that a focus on threshold concepts equips students to reason in situations beyond those that are commonly taught. If one has a clear conceptual understanding of population, one is better equipped to conceptualise a meta-population and hence have a framework for statistical analysis of whole-population datasets. If one has an understanding of the behaviour of standard errors as $n \rightarrow \infty$ one can better appreciate the likely problems with inference on massive datasets.

Whilst there may be some pressing issues in statistics education, there are some very striking success stories. Two examples given here include informal inference referred to earlier, and the use of computers, not for data analysis, but rather to implement computationally intensive methods. Firstly, Makar, Bakker and Ben-Zvi (2011) provide a compelling argument for the place of informal inference within statistics education. They note that in order to make informal propositions about a population based on data in a sample, students need to access a number of interrelated concepts as well as integrating this with information on the problem context. Informal inference offers the potential to master several threshold concepts in statistics without the additional overhead of formal inference. They further demonstrate that this lends itself to an enquiry-based environment. Whilst this approach now seems natural in statistical education, it is noteworthy that the Moore method in mathematics education has been proposed and used for some time (Pfannkuch, 1977). Indeed, Schwartz and Martin (2004) highlight how student-led inquiry-style learning can lead to greater insights into formulae. Schoenfeld (1988) has shown that even on well taught mathematics courses, there can be too much attention paid to applying formulae with little understanding. Cuoco, Goldenberg, and Mark (2010) talk about

habits of mind in mathematics development and within statistics education there is indeed potential for this.

Secondly, Simon, Atkinson and Shevokas (1976) proposed greater attention be given to Monte Carlo methods; an approach in use at the University of Illinois since 1965. They did not propose this as an alternative to traditional methods, but as a means of understanding the concepts. Interestingly, Pfannkuch, Forbes, Harraway, Budgett and Wild (2013) discuss exactly how bootstrapping has been used within the New Zealand school curriculum. It is striking to watch this focus on concepts – and that it is not new. Both proposals are clear in suggesting that we deal with concepts that require less mathematical underpinnings, yet mathematically trained students who can synthesise the concepts with the mathematical principles obtain a deeper understanding of the subject.

UK data handling and probability curriculum

At the time of writing, the DFE (2013) are currently proposing changes to the UK mathematics and statistics curriculum. Notable features are a separation of probability and statistics (Edwards, 2013), removal of probability from primary school and the reduction in importance within the mathematics curriculum of the data handling cycle. Key ideas that might feature at GCSE level are given below:

- *“describe and compare univariate empirical distributions through: appropriate graphical representation involving discrete, continuous and grouped data and appropriate measures of central tendency and spread.*
- *describe simple mathematical relationships between two variables in observational and experimental contexts.*
- *identify appropriate questions, data collection, presentation, analysis and interpretation to conduct exploratory data analysis, including in science and geography.”*

DFE (2013)

There is some concern regarding the apparent demotion of the statistical problem solving cycle. The “Plan-Do-Check-Review” cycle has a long history prior to its use in statistical education but is clearly recognised as a cornerstone of statistical learning (Wild & Pfannkuch, 1999). The wider statistical pedagogy referred to in the introduction is quite clear that this should be a core tool when learning statistics. The problem solving cycle is an important experiential aspect of learning statistics – the first learning outcome in Aliaga et al. (2010) is to know that we need to use data to answer problems. We can only expect learners to attain this learning outcome if indeed knowledge grows as a result of using statistical methods. If statistical methods are only ever applied to problems with known answers we are unlikely to produce students who appreciate the value of statistical science in advancing knowledge – it reduces to some procedure for confirming knowledge we have acquired by some other route.

Removal of this aspect of statistical education from the school mathematics curriculum may of itself not be a problem. Smith (2004) suggested it was better taught within the context of the scientific method in other subjects and whilst it may have been in the curriculum it may have been little assessed in the past. The risk though, reinforced by the separation of

probability and statistics, is that the curriculum presents probability as a rather pure branch of mathematics, and statistics is presented as some procedures for summarising data. So the first bullet point, isolated from probability has the potential to merely teach some statistical procedures with notional contextualisation and limited links to probability. This is not the place to debate the GCSE curriculum. However, the implications for university educators are clear. It is difficult to appreciate why we are interested in visual description of univariate data distributions unless we wish somehow to match these to a probability model (however indirectly that may be done). Frequency displays are also notoriously difficult to understand; Bakker, Biehler and Konold (2005) highlight several conceptual difficulties and so it will be of great interest to see whether the curriculum emphasises construction or understanding. Whilst it is easy to assess aspects of the completion of these procedures it seems to miss the point to separate this activity from the related sense that in order to understand the world what we ultimately wish to do is to model uncertainty. Kahneman (2011) clearly demonstrates what humans do in the absence of probabilistic reasoning – we invent heuristics that seem to work simply but are often wrong.

Whilst it is important to be realistic about what can be achieved in a school curriculum, it is also important that the assessment will be what drives learning. If assessment is focused on easily examinable procedures then even the limited potential listed by these points will fail to be realised. If assessment does not drive linkages between subjects, then students will not make the linkages and will arrive at university with limited expectations of statistics. This concern is amplified when examining the second bullet point. It does refer to one contextual issue regarding study design (which we referred to above when describing hormone replacement therapy) but it misses rather broader points about context. There is more than applying a tick-box to appreciating issues around study design. Psychologists and epidemiologists readily work with the concept of confounding or lurking variables; this requires that consideration be given to a third variable. If we have an association between two variables, the association may not be causal; it may be due to a third variable. A rather trivial example is weekly gas sales and death rates of the elderly. These variables are apparently correlated but the lurking/confounding variable is ambient temperature. Indeed, this illustrates a great challenge with statistics education – do we work with trite examples that reduce to simple two variable associations or do we work with multidimensional problems that require a broader range of techniques.

In brief, the concern with this curriculum is that mathematics majors will arrive at university with some tools to understand probability, but little appreciation of the linkages between probability and statistics and could be distressed by a statistics education which concerns itself with context, judgement and interpretation rather than application of procedures. However, the pedagogic literature for education of non-mathematics majors does provide ample mature material which informs the ways we can engage students with these rather more open-ended activities. We would also suggest that in the workplace, even in a mathematical modelling project, many of these more open-ended skills will be vital for professional success and therefore careful attention to statistics education can do much for our students.

Conclusion

There is still plenty of uncharted territory in statistics pedagogy caused by issues such as massive data sets and the role of complex mathematical models. However, there is already much to learn from the literature regarding the teaching of statistics that can inform our practice. We can highlight both missed opportunities from the school curriculum as well as missed opportunities from developments in computing. Cobb (2007) refers to the “*tyranny of the computable*” – our fixation with certain procedures largely because they were once hand-calculable. Even with the advent of computers we merely compute such procedures electronically; the only real change in over half a century of statistics education has been to use computers to do these computations with supposedly real data. The point being missed is that modern computing renders such mathematical approximations as the t-distribution for the sampling distribution of the sample mean under the null, unnecessary. We could use randomisation tests which bring students closer to the concept. If we subsequently introduce the mathematics having engaged students with the concept we are then working with students who have developed an insight into the formula and who are not just technically competent in manipulating a formula. When working with the t-distribution in this way they can appreciate that it is a mathematical approximation only valid when certain assumptions are valid. Assumption checking ceases to be an afterthought, but becomes part of the learning activity. This is a skill that is valuable to mathematics students beyond its value in their statistical education.

The gap between statistics and mathematics is fascinating. It is possible to find literature that suggests some mathematicians do not value statistics within a mathematics degree, and that statistical educators regard the subject as something entirely different to mathematics. However, the potential of enquiry-based learning to afford an insight into formulae seems to argue that even within the most narrow and exclusive goals that could be articulated for a mathematics degree it would remain valuable to take a course in statistics. Taking a simple example, where we have (fictional) random variables Y_i , for $i = 1, \dots, n$ taken as Poisson independent and identically distributed so that:

$$Pr[Y_i = y_i] = \frac{\lambda^{y_i} e^{-\lambda}}{y_i!}.$$

At some stage in a statistics course we will find the maximum likelihood estimator by forming the log-likelihood for λ given $\mathbf{y}$ namely:

$$l(\lambda) = -n\lambda + \sum_{i=1}^n y_i \log \lambda - \log \prod_{i=1}^n y_i!,$$

then taking the first derivative and solving to see that $\hat{\lambda} = \frac{\sum_{i=1}^n y_i}{n}$. Mathematically we can check that this is indeed a maximum as the second derivative of the log-likelihood is $-\frac{\sum_{i=1}^n y_i}{\lambda^2}$ which is clearly negative (unless all $y_i = 0$). This quantity will reappear in formal likelihood theory where we evaluate the second derivative at the maximum of the likelihood function (yielding $\sum_{i=1}^n y_i = n\hat{\lambda}$). Before developing this theory we could compare the formula with that for the variance of $\hat{\lambda}$. If indeed we have independent and identically distributed Poisson random variables (where the variance of any Y_i is the same as any other) we get:

$$Var(\hat{\lambda}) = Var\left(\frac{Y_1 + \dots + Y_n}{n}\right) = \frac{1}{n^2} (Var(Y_1) + \dots + Var(Y_n)) = \frac{nVar(Y_1)}{n^2}.$$

The key statistical point here is that we ‘hard wired’ a rather large assumption in this formula, namely that we have independent and identically distributed random variables. But in this hard wiring, we have made the assumption rather explicit – the final step on the right hand side clearly requires that all variables have the same variance. If we tidy this expression a little, and note that for Poisson random variables $Var(Y_i) = \lambda$ we get:

$$Var(\hat{\lambda}) = \frac{nVar(Y_i)}{n^2} = \frac{\lambda}{n},$$

which clearly shows us that the variance of the estimator decreases as the sample size increases. This seems like an intuitive property for an estimator. What we can then do is look at the clear relationship between $Var(\hat{\lambda}) = \frac{\lambda}{n}$ and the value of the second derivative at the maximum $-\frac{n}{\lambda}$. We do need some reassurance that this is not just a lucky coincidence (which comes from later asymptotic theory on the distribution of maximum likelihood estimators) but for now it is intriguing to note that the second derivative is measuring the curvature of the log-likelihood at the maximum. There is some relationship (albeit we need to reverse signs and take a reciprocal) between this value and the variance of the estimator $\hat{\lambda}$. It would appear that there is an intuition to be explored here. There is a link to some nice statistical theory – that the likelihood is more than just a method for finding point estimates but rather the *function* contains information on some properties of those estimators. But the most critical point of all of this is that we clearly required some rather large assumptions. By examining the variance of the estimator first, we reinforce the fact that we are making key assumptions about our random variables. If we subsequently apply these formulae to inference problems involving data, we have hopefully made it very clear these results only apply if we can entertain the assumption of independent and identically distributed. In finding the variance of $\hat{\lambda}$ there are steps in the mathematics that cannot be taken without making an assumption.

Taken to one logical conclusion, statistics courses can serve an epistemological purpose, where both new knowledge is generated from a case study at the same time as training and teaching students in the methods and concepts they used to develop that knowledge. Nolan (2003) highlights how case studies can both teach skills and practice as well as illustrating mathematical and statistical concepts but does also indicate that such courses are not without their own challenges. If appreciating context is a vital part of statistics then Langrall, Nisbet, Mooney and Jansem (2011) suggest that a degree of context expertise can be necessary to understanding a statistical problem. Kaplan (2009) reinforces this point and goes on to define “*Belief Bias*”; the context has become so important that when a student obtains different results from those they were expecting they can assume the problem is in their statistical implementation rather than their understanding of the context.

Nevertheless we argue that whilst much of the development in statistical pedagogy has focused on non-mathematics majors, it currently makes a number of proposals to consider the way we educate mathematics majors. Mathematics graduates are highly valued not just for the mathematical techniques but for a range of problem-solving skills. Current thinking in statistical pedagogy highlights a number of reasoning skills that can be developed – all of which are valuable skills for mathematics majors to develop and take them beyond technique excellence (important though that is). It seems timely to be careful in ensuring that mathematics majors are exposed to the skills highlighted by modern

statistical educational thinking. Current proposals for the UK school curriculum clearly refer to the concept that by considering data from a randomised controlled study we can assert causation in a way that we cannot readily do if the same set of data arose from a case controlled study. It is entirely natural for a scientist to master these ideas; it is less obvious that mathematics major statistics courses cover this kind of material in many courses on mathematical statistics. And given the proposed separation of statistics and probability in the school curriculum there is a concern that undergraduates will appear with less exposure to statistical thinking.

We conclude with two problems that are particularly relevant for mathematics education in the UK. Firstly, Rotberg (2010) reviews arguments around the damage done to the UK education system by “*teaching to the test*”. There are clear dangers to mathematics education where indeed ‘right’ answers are possible. In this context, modern statistical education provides real opportunities during the transition to higher education. With an emphasis on open ended evaluation of a problem we require new undergraduates to start to use their judgement and to focus on development of skills which require use of that judgement. Thus statistics courses within a mathematics degree have the potential to move students away from an expectation that every problem has a right answer (and a right method of getting that right answer). Maturity in this area is valuable across the whole of their learning. Another problem faced across mathematics education at all levels is relevance beyond the classroom. One fruitful area in this regard is noted by Gould (2010) who shows that we meet statistical concepts outside formal education; indeed the presence of statistics education within liberal arts colleges in the United States (Schield, 2012) suggests that many statistical concepts can be learnt without probability calculus. This presents an opportunity and a challenge for statistics education – if we wish to set relevant problems in a real world context we need look no further than a newspaper. Thus it is possible within a mathematics degree to both demonstrate real-world relevance and the need for a more mature, judgement-based approach. How far we can do this in a system that rates student satisfaction, where students are likely to prefer courses where they can get ‘right’ answers, remains a challenging question for the whole sector.

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Chapter 11: Technology-led teaching: The role of mathematical software

Chris Sangwin

Abstract

In this chapter we examine the nature of mathematical technology, arguing for a broad definition that includes software such as computer algebra, and also the underlying algorithms and techniques such as place value. We define the *fundamental tension of educational technology* as the extent to which understanding a concept is necessary for the effective and successful use of an algorithm, or automatic procedure. Conversely the effective use, initially without understanding, influences the formation of concepts. Arguments about the effect of technology on teaching and learning mathematics are at least four centuries old, and continue to centre on this fundamental tension. We examine these arguments and discuss the difficulty in devising assessments for students in the context of contemporary mathematical software. The chapter then looks more specifically at the uses to which mathematical software is currently being put. This includes experimentation, calculation, assessment, communication and live data capture. We also consider the extent to which mathematical technology gives correct and complete answers. We briefly consider challenges for the future, in particular the current difficulty in encoding a complete mathematical argument that includes a mix of calculation, logic and informal analogies.

What is technology?

The word *technology* has a number of related meanings, associated with *practical art and craft*. It is principally used to represent a branch of knowledge in mechanical and applied sciences and the practical application of this knowledge. Technology can also mean the artefacts developed. Most commonly these artefacts are tools and machines, and in modern usage it often refers to electronic devices and software. Sangwin (2013) defined mathematics as “*a systematic way of structuring thought and arguments, which is tied closely to a coherent body of associated knowledge*”. Mathematics is traditionally presented as definition-theorem-proof; essentially deductive reasoning. Science is a methodology

for investigating real-world phenomena using controlled experiments, and scientific evidence is inductive. Technology, on the other hand, provides methodologies and tools for achieving specific practical goals. Mathematical technology is most commonly associated with computation and algorithms, and the practical implementation in software such as computer algebra systems (CAS).

Hence, place-value is also a form of technology which enables efficient numerical computation. Logarithms, specifically tables of logarithms, enhanced this technology significantly. Indeed, it was the invention of logarithms which promoted the acceptance of place-value in Europe in the early 1600s. Algebra, as an extension of arithmetic, significantly enhances the range of computations possible and calculus is a form of technology which enables us to avoid *ad hoc* infinitesimal arguments (see Edwards, 1979), replacing them instead with coherent systematic methodologies. Viewed in this way, modern computer algebra systems are not a sudden imposition of technology on mathematics but rather part of a continual development, albeit at a rapidly increasing pace.

2014 is the 400th anniversary of the publication of the invention of logarithms by John Napier (1550-1617). This was the most important advance in practical calculation since the introduction of the decimal number system. For example, it provided Kepler with the *technology* he needed to undertake the calculations which provided evidence for the elliptical orbits of the planets. Napier also made numerous other practical contributions to calculation, for example Hawkins (1988) and Napier (1990), but the logarithm was his most significant.

The word technology is more commonly used in a more restrictive sense of *artefact* such as software or physical object, rather than *concept* such as logarithm. Indeed, physical models remain an important component particularly in elementary mathematics education, for example Bartolini Bussi, Taimina, and Isoda (2010), as learning technology. Furthermore the logarithm also gave rise to a practical calculating device in the form of a slide rule. A slide rule juxtaposes two logarithmic scales so that the addition of physical lengths corresponds to the multiplication of the numbers represented on the scales. They came in a number of forms, for example straight, circular and spiral, with a variety of scales, for example logarithms of trigonometrical functions, to facilitate specific calculations and were in regular use from the 1630s until the 1980s.

The effect of technology on teaching

All this would merely be of historical interest were it not for the argument which arose over the priority for the invention of the slide rule between William Oughtred (1573-1660) (see Cajori, 1916) and Richard Delamain (1610-1645) (see Taylor, 1954:§122). It was concluded by Cajori (1920:205) that Oughtred was the inventor of the first slide rule but that Delamain published first. Oughtred was a well respected teacher, and had already published a very influential book on algebra, *Clavis Mathematicae* ('Key to Mathematics') (Stedall, 2000); Isaac Newton's annotated copy of this book survives. Why did Oughtred not publish his invention first?

Surprisingly perhaps, we have evidence for this in the form of the dedication contained in Oughtred's book (Oughtred, 1634), written by William Forster. Of Oughtred's attitude, Forster says the following:

“He answered, That the true way of Art is not by Instruments, but by Demonstration: and that it is a preposterous course of vulgar Teachers, to begin with Instruments and not with the Sciences, and so instead of Artists to make their Schollers only doers of tricks, and as it were Juglers: to the despite of Art, losse of presious time, and betraying of willing and industrious wits unto ignorance and idlenesse. That the use of Instruments is indeed excellent, if a man be an Artist; but contemptible, being set and opposed to Art. And lastly, that he meant to commend to me the skill of Instruments, but first he would have me well instructed in the Sciences.”

Oughtred (1634)

Oughtred emphasises that a grasp of theory will lead naturally to the mastery of practice. Since Oughtred had invented many mathematical instruments, for calculation and astronomy, this is entirely consistent with his experience. Oughtred goes further, clearly stating that a mastery of the principles involved inevitably leads to an ability to *re-invent* them for one’s self:

“But Delamain was already corrupted with doing upon Instruments, and quite lost from ever being made an Artist: I suffered not William Forster for some time to so much as to speak of any Instrument, [...] And this my restraint from such pleasing avocations, and holding him to the strictness of precept, brought forth this fruit, that in short time, even by his owne skill, he could not onely use any Instrument he should see, but also was able to delineate the like, and devise others.”

This is the start of the *theory vs. practice* debate, which is still with us. Richard Delamain took a very different view:

“But, for none to know the use of a mathematical instrument, except he who knows the cause of its operation is somewhat too strict, which would keep man from affecting the Art, which of themselves are ready enough everywhere, to conceive more harshly of the difficultie, then it deserves, because they see nothing but obscure propositions.”

Delamain, not always easy to understand¹, believed that theory and practice should co-exist, and that introducing mathematical aids sooner would motivate the theory (or Art).

“the practike being better understood by the doctrinall part, and this better explained by the Instrumentall, making precepts obvious unto sense, and the Theory going along with the Instrument, better informing and inlighting the understanding etc.”

Furthermore, Delamain acknowledges a distinction between different groups of students:

“All are not of like disposition, neither all (as was sayd before) propose the same end, some resolve to wade, others to put a finger in onley, or wet a hand: now thus to tye

1 Oughtred complains bitterly of this: *“I borrowed and perused that worthless Pamphlet, and in reading it (I beshrew him for making me cast away so much of that little time remaining to my declining years) I met with such a patchery and confusion of disjoynted stuffe, that I was stricken with a new wonder that any man should be so simple, as to shame himself to the world with such a hotch potch.”* (Oughtred, 1634)

them to an obscure and Theoricall forme of teaching, is to crop their hope, even in the very bud."

He also introduces the affective domain into the debate:

"And me thinkes in this queasy age, all helpees may bee used to procure a stomacke, all bates and invitations to the declining studie of so noble a Science, rather then by rigid Method and generall Lawes to scarre men away."

This debate has continued ever since:

"Some friends of mine assert that no boy or man ought to be allowed to use logarithms until he knows how to calculate logarithms. They say this, knowing that the calculation is a branch of what is called higher mathematics, and that the average schoolboy, after six years at mathematics, finds it hopeless to even begin the study of the exponential theorem. It is a hard saying! [...] We need not believe a craftsman when he tells us that we cannot utilize his results without practising his trade."

Perry (1900)

In her review of research on calculator use in primary school students, Duffin (2000) found that *"It was quickly discovered that using a calculator appeared to have a good, rather than inhibiting effect on the children's ability to calculate mentally"*, supporting Delemain's view. The same debate arose, and continues, with more advanced students:

"This is not to say that this matter is trivial, or that all teachers will find that computer algebra solves their most pressing problems. Indeed, it is reasonable to predict that computer algebra will cause high school and university teachers at least as much trouble as calculators caused their colleagues teaching younger pupils. Since computer algebra systems can tackle a wider range of more sophisticated problems, they will probably cause more havoc. But this cannot be avoided. For better or for worse, if the technology is there, the pupils will use it. [...] Hence the availability of computer algebra systems is, like the calculator, both an opportunity and a threat to the teacher. Opportunity because he can teach much more, threat because so much that he teaches is, or seems to be, redundant."

Davenport (1986)

Until the end of the nineteenth century a *calculator* was a profession. Now that we have electronic devices to perform calculations this profession has become obsolete, and mathematical teaching has also evolved. Nobody learns how to use logarithm tables or a slide rule for pragmatic purposes, and the mathematics underlying the construction of tables, for example finite differences, has quietly been dropped from curricula. Repetitive calculation, or indeed any mindless repetitive task, is not fit work for a creative human being. Only a small group of connoisseurs will be genuinely interested in the internal details of the algorithms but many more will need to make use of the results they produce. However,

"...the whole point of a mathematics education is to understand precisely why the ideas work."

Krantz (1999:23)

The central and enduring question remains: why *are* we teaching something now, to this group? Two common answers are (i) because we want to obtain and use the answer, or (ii) because we want to understand the underlying process. The focus of disagreements in the debate is the extent to which (ii) is needed for (i). We define the *fundamental tension of educational technology* as the extent to which understanding a concept is necessary for the effective and successful use of an algorithm, or automatic procedure. This tension, when it becomes political, can be divisive. For example, the difference between reform and traditional mathematics instruction, which at its heart has much in common with this tension, caused the so-called *math-wars* (Schoenfeld, 2004).

The uses to which mathematical technology is put

The original use of technology was as a direct aid to calculation. This remains important, but in addition technology also enables experimentation, assessment, data capture and supports communication. In this section we examine contemporary uses to which mathematical technology is put and discuss their effect on teaching.

Experimentation

The ability to automate calculation, particularly repetitive calculation, enables effective *experimentation* to take place. A particularly powerful form occurs in graphical visualisation, specifically graph plotting, and in dynamical mathematical software which enables manipulation and *dragging* to take place. It is difficult to capture the fleeting moment at which something is really understood.

One example from the author's own experience concerns the mechanical linkages shown in Figure 11.1. These consist of three movable bars, fixed to a base which constitutes a fourth bar. We are interested in the path of the pen, fixed to the middle bar for all physically realistic positions of the linkage.

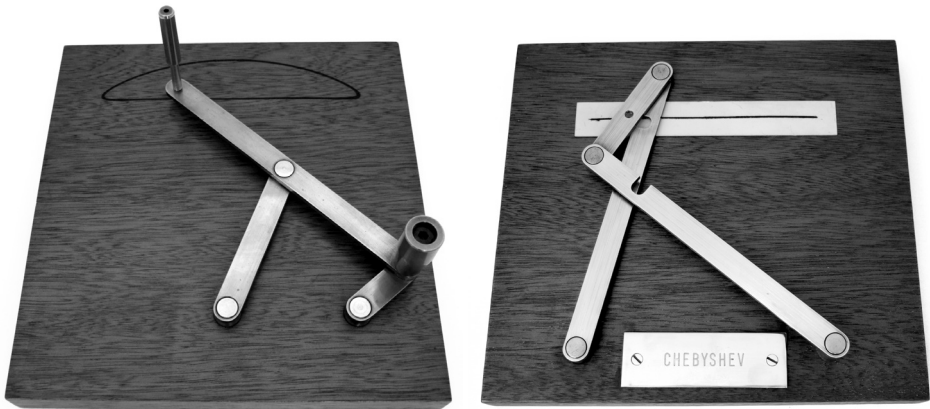


Figure 11.1: The Chebyshev linkage.

Such linkages form important parts of mechanisms, and continue to be important particularly in robotics. The goal is to prove that the two mechanisms shown in Figure 11.1

generate the same curve. By superimposing the two linkages on a single diagram using the dynamic geometry package GeoGebra² the dynamic nature of the diagram enabled the linkages to be moved together, and the *relative movement* was key to understanding the similarities involved. This led to an elementary geometrical proof. The static nature of this publication cannot recreate that here, and you will have to try this for yourself. Recreating situations in which students are likely to have the same experience is one of the goals of the teacher, and technology has demonstrably shown its value here.

Does technology give me the right answer?

As Davenport (1986) says “*this matter is not trivial*”. Hidden within classical algorithms are traps:

“For example, as discussed by Jeffrey (2004), most publications containing the Cardano solution of a cubic equation do not mention that his formula is not always correct for non-real coefficients. Consequently this formula has been misused by many people, including some computer algebra implementers, such as me. The consequences can be disastrous.”

Stoutemyer (2011)

The domain of validity of a technique may not be clear, or fully understood. This has always been the case, for example Euler discussed the previous attempts to define the logarithm of a negative number in some detail (Euler, 1978), and discusses some of the apparent paradoxes which result. Collections of similar false reasoning are readily available, for example, Northrop (1945) and Maxwell (1959). Furthermore, mathematics itself often advances by the use of analogy, induction and the application of a technique to a case where, strictly speaking, it does not apply. Plenty of examples are given by Polya (1954), together with a discussion of why it is important for *students* to appreciate the importance of this to mathematical problem solving and research. Calculation is a form of *compression*, and inevitably this must include obfuscation of some of the detail. We refer back to the fundamental tension: how much do you need to understand something in order to make effective use of it? This has, naturally, been the focus of educational research in this area. For example, how computer algebra can be used to promote an understanding of algebraic concepts, with a focus on *parameters*, is discussed by Drijvers (2003). He concluded that “*it is important to anticipate computer algebra use, to be explicit about the changing didactic contract, to orchestrate individual and collective instrumentation, to have students compare CAS techniques with paper-and-pencil techniques and to have students reflect on the way CAS works.*” (Drijvers, 2003:330).

Answers you don't understand

One of the reasons for studying calculus techniques extensively is to become sufficiently familiar with tutorial examples that when a real situation is encountered it is *recognised*.

However, the issue of recognition also really is non-trivial. Consider a *delay* differential equation of the form:

$$\dot{x}(t) + ax(t) + bx(t - \epsilon) = 0, \quad a, b \in \mathbb{R} \quad (1)$$

² <http://www.geogebra.org>

for some $\epsilon > 0$. Without the delay (that is $\epsilon = 0$) this would be a standard problem, and we would simply set $x = e^{\lambda t}$, and solve for λ . Doing this anyway, we obtain:

$$\lambda + a + be^{-\lambda\epsilon} = 0.$$

Since this equation contains a mix of polynomial and exponential terms it could not be solved (for λ) by any of the methods commonly taught. However, Maple does return a solution to this equation:

$$\lambda = -a + \frac{1}{\epsilon} W(-\epsilon be^{\epsilon a}).$$

Here W is the Lambert W function which is defined as the multivalued inverse of the function:

$$w \rightarrow we^w$$

(just as the complex logarithm is a multivalued inverse of $w \rightarrow e^w$). In general there will be an infinite number of values λ_n which satisfy this, and the solutions of (1) can then be written as the sum of these:

$$x(t) = \sum_{n=1}^{\infty} a_n e^{\lambda_n t}.$$

Here the a_n are chosen to fit the initial conditions on the interval $t \in [-\epsilon, 0]$ (see Corless, Gonnett, Hare, Jeffrey & Knuth, 1996). There are numerous other elementary applications of the Lambert W function. This function has not always been recognised by mathematicians since its first use around 1758 and Corless et al. (1996) concludes as follows: *“Names are important. The Lambert W function has been widely used in many fields, but because of differing notation and the absence of a standard name, awareness of the function was not as high as it should have been.”*

What software, such as CAS, does is reverse the normal pedagogic order. Traditionally we learn and practice many techniques in the anticipation that they will be useful. However, now the software ‘gives the answer’, and the user has to find out what it means, and the extent to which the result is a valid solution of the underlying problem. This requires other kinds of sophisticated behaviour, and a certain kind of confidence and tenacity.

Calculation

Calculation remains the central use of technology. It has opened up new areas of enquiry, particularly in statistics where the computational load previously made the practical application of theory impossible. Technology has also changed the nature of the acceptability of solutions, for example in some areas of applied mathematics numerical solutions of differential equations are now routinely published where previously only analytical solutions were acceptable. Computer generated proofs are less common, but this is an active area of research (see Beeson, 2003). Many professional mathematicians make routine use of technology for calculation. Lavicza (2010) undertook a survey of 4,500 university mathematicians in the UK, the United States and Hungary. Of the 1103 respondents, more than two-thirds reported they at least occasionally used CAS in their own research:

“The analysis of the data revealed that the majority of mathematicians believed that CAS is becoming an important element of mathematical literacy and will become [an] integral part [of] mathematics teaching and learning in the future.”

Lavicza (2010)

However he also found that “*the changing characteristics of mathematical knowledge are a polarising issue for mathematicians*”.

Assessment

Given the pace of change, and the increasing availability of computing resources, particularly mobile devices, tablets and online access to potentially very powerful computational machines, teachers are having to reassess their curricula and assessments. However Monaghan (2000) found that “*it is easier to write questions which bypass the technology than it is to write questions which embrace it*.” There is something of a paradox here: if you can automate the process you are teaching then why teach the internal details? One resolution is tied closely to our fundamental tension. To what extent do you need to understand to operate?

Algebraic calculators, with symbolic manipulation capabilities, are routinely in use in school mathematics and are, or have been, permitted in examinations in a number of countries including Austria, France and the USA. Currently CAS calculators are *not* permitted by default in UK school examinations:

“*Calculators must not be designed or adapted to offer any of these facilities:*

- *language translators;*
- *symbolic algebra manipulation;*
- *symbolic differentiation or integration;*
- *communication with other machines or the internet*

[... or] have retrievable information stored in them – this includes [...] mathematical formulas;”

JCQ (2012)

However, “*Candidates may be allowed to use resources other than calculators for some examinations, as set out in the instructions on the question paper.*” (JCQ, 2012) enabling specific examinations in mathematics to use CAS as appropriate.

It is very difficult to separate curricula and assessment issues. An assessment is *valid* when it tests what it is supposed to. It is *reliable* when it gives similar results under varying circumstances and *practical* when it does not overstrain resources to the detriment of the assessment. The *conditions* under which it takes place have to be taken into account, including the extent to which technology is available.

The extent to which a CAS calculator could be used to answer questions on undergraduate examination papers was examined by Pointon and Sangwin (2003). As part of this work, they devised the scheme for classifying mathematical tasks shown in Table 11.1 In the papers examined, nearly half of the questions required *only* routine calculations, the majority of which could be performed by the CAS by typing in the question, and reading off the answer. The availability of a CAS to students sitting these examinations would have rendered the examinations invalid. Independently Smith, Wood, Coupland and Stephenson (1996) devised a scheme which is remarkably similar. Note that for some tasks, particularly those at the bottom of the table, an ability to calculate is not sufficient.

1. Factual recall
2. Carry out a routine calculation or algorithm
3. Classify some mathematical object
4. Interpret situation or answer
5. Proof, show, justify – (general argument)
6. Extend a concept
7. Construct example/instance
8. Criticise a fallacy

Table 11.1: Pointon's question classification scheme (Pointon & Sangwin, 2003).

It is currently very unusual for students to be permitted to use CAS technology in high-stakes examinations. Unless the assessments change, technology will render many of the questions pointless, that is the computations being assessed can be done largely automatically, with perfect accuracy in almost no time. This renders the assessments invalid because they are not being undertaken under the envisaged conditions.

One exception is the optional Further Pure with Technology (FPT) module in the A-Level Further Mathematics specification offered by Mathematics in Education and Industry (MEI). It is assumed that students have access to a computer with software for the teaching, learning and assessment that features a graph-plotter, spreadsheet, computer algebra system and programming language. This is assessed by a timed written paper with the first examination held in June 2013. The subject matter includes investigations of curves and the use of sliders and CAS with parameters. Functions of complex variables are included, as is using a CAS to solve equations and differentiate/integrate. A spreadsheet may be used for iterations. In addition, students are expected to write simple computer programmes to solve problems in number theory. A specimen paper is available online, and more details are contained in Button and Lee (2013).

Live data capture and large data sets

For applied mathematics, at least, contemporary technology offers a large, and growing range of tools for live data capture. Equipment is relatively inexpensive and is transforming the science classroom. Indeed, most smart phones have a GPS, accelerometer, and some have a thermometer, barometric altimeter and other sensors from which data can be captured, stored and analysed. When combined with modelling, these are set to transform applied mathematics (and physics, chemistry and geography) education. A thorough discussion is given by Heck (2012). In statistics the ability to obtain and manipulate large data sets has transformed statistical practice over the last fifty years, and hence statistical pedagogy. Visualisation tools, for example Gapminder³, enable non-specialists to interact with data in a qualitative way and provide the potential for more specific hypotheses to be formulated and tested.

³ <http://www.gapminder.org>

Online assessment

Another common use for networked computer technology is for online assessment, where the computer sets questions, assesses work, accepts and marks answers and reports the outcomes to the teacher. While automatic assessment is commonly associated with multiple choice questions (MCQ) the current state of the art has progressed considerably further. Algebraic expressions are routinely accepted as answers, and the algebraic equivalence with a correct answer is established automatically. Many such online assessment systems use computer algebra to support the assessment process in the following ways:

- Random versions of questions are generated in a structured mathematical way.
- The system accepts answers from students which contain mathematical content, rather than MCQs.
- It establishes the mathematical properties of those answers.
- It generates outcomes which fulfil the purposes of formative, summative and evaluative assessment.
- All attempts are stored at one question, or by one student, for analysis by the teacher.

Systems differ in the extent to which assessment is balanced with tutoring, and to which free-form answers can be accepted. In more elementary parts of mathematics, for example reasoning by equivalence to manipulate simple expressions, greater effort has been made than in more advanced topics with a focus on dealing with proof.

The key point is that such systems establish the mathematical properties of an answer in an objective manner. While the prototype is to establish that (i) the answer is algebraically *equivalent* to the correct answer and (ii) the student's answer is in the appropriate *form*, (for example factored), it is the relevant properties of expressions which matters and answers are often non-unique. Where the student's answer does not satisfy all the properties, the teacher is able to encode feedback. Potentially this is specific to the answer and directly related to possible improvement on the task. This is precisely the kind of feedback which research such as Kluger and DeNisi (1996) has suggested is most effective. A recent survey of this area was published as Sangwin (2013). An example question is shown in Figure 11.2.

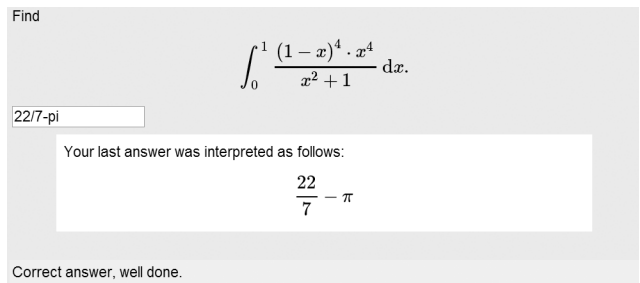


Figure 11.2: A typical online assessment.

Programming

Digital computers also provide an entirely new activity, programming, which shares much in structure and logic with mathematics. For example, the analogy between subroutine and

function, with the notions of bound variables, is strong. Indeed, computer science is a branch of mathematics. As early as the 1980s there was a criticism of online testing, for example:

“In most contemporary educational situations where children come into contact with computers the computer is used to put children through their paces, to provide exercises of an appropriate level of difficulty, to provide feedback, and to dispense information. The computer is programming the child.”

Papert (1980)

This dissatisfaction led directly to the Logo programming language, which was specifically designed to give the child the opportunity to programme the computer. The editor’s comment following Hollingsworth (1960), an early paper on automatic grading of students’ work, comments that *“while there is a great deal of research being accomplished on ‘teaching machines’, many computer educators have not realized that when teaching the use of the computer they have access to the finest ‘teaching machine’ of all – the digital computer.”* The first International Commission on Mathematical Instruction (ICMI) study into issues in mathematics education was dedicated to *“the influence of computers and informatics on mathematics and its teaching at university and senior high school level”* (see Howson & Kahane, 1986). The role of programming as an activity within mainstream mathematics, rather than as a separate subject, is not clear and deserves further attention.

Future mathematical software

It is clear, at least to the author, that in affluent countries wireless access to online resources will become ubiquitous in the near future, and that everyone (adults, students and even children) will have essentially unlimited access through a variety of devices from portable hand-held screens (such as mobile phones) through to working environments with multiple screens serving groups of users. Immersion in 3D shared virtual worlds also seems likely. This will enable communication between individuals and access to resources.

What is not clear is the means by which mathematics will be communicated. LaTeX is the defacto standard for mathematical typesetting, but it does not encode any meaning (Knuth, 1979). It is slightly surprising that currently there is no widely adopted mechanism through which we can write mathematics research papers, encode the meaning of lecture notes, experiment with colleagues’ equations, or evaluate the correctness of students’ algebraic reasoning, etc. This hampers basic communication. All the purposes discussed here, for example calculation, experimentation, and assessment rely on the ability to encode a complete piece of mathematics. However, the informal, often sloppy and incorrect, statements made by humans sit ill at ease with formal encodings such as OpenMath or MathML (see Chapter 16). Devising a format, creating intuitive interfaces, and persuading people to adopt it remains a significant challenge for the future.

Conclusion

The fundamental tension of educational technology remains, perhaps as the single biggest challenge to be negotiated in a practical teaching situation. The problem is not the technology, the problem is the attitude of people, including teachers, students and

institutional stake holders. Some teachers insist students need to maintain fluency in tasks which have *recently* been made obsolete by technology, mainly in the (well-founded) fear that students will no longer be able to do them. However, the question of why people still need to do them is often not addressed. What is not clear is the extent to which they need to continue to do these tasks, even for the specialists such as university mathematics students who focus more on understanding than obtaining the outcomes. Under the current circumstances of rapid technological change it is reasonable to be conservative, and such conservatism does not automatically equate with thoughtlessness or lack of engagement. Students vary in their motivations and some take the path of least resistance. Just as light takes the path of least resistance, this is perfectly natural. Students need, and deserve, engaging tasks which make the most of the available technology to reach clearly specified educational goals. In the face of rapid technological change institutions are slow, or struggling, to respond and one symptom is the retention of the exclusivity of exams without modern technology.

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Chapter 12: Providing effective feedback

Mike Robinson

Abstract

Feedback in some form is an essential part of education, but it is clear from the National Student Survey (NSS) and staff experience that this can be problematic. This chapter considers what makes effective feedback. In particular, effective feedback needs to be timely, good quality, and crucially it should provoke some action from students to promote further learning. A snapshot of current typical practice across university mathematics departments is presented, and some of the ways in which staff try to reconcile the competing demands for high quality feedback and rapid turnaround are discussed. It is notable that much mathematical feedback focuses upon two aspects: providing exemplars, typically in the form of model solutions, and identifying student errors, but these by no means address all the possible feedback which students need, and in themselves they will not prompt all students to take action. This chapter also reports some more innovative approaches intended to promote student engagement with the feedback, both in terms of increasing the students' awareness of what they ought to do, and by developing specific activities which prompt a student reaction to feedback.

Introduction

Few would doubt that students need an effective means of evaluating the work they do. Without this, they can neither have the self-awareness to recognise their strengths and weaknesses, nor take the steps necessary to improve in the future.

Common sense, experience – and numerous studies – tell us that a fundamental part of this is the feedback available to students. For example, Black and Wiliam (1998) review over 250 studies across a range of subjects, levels, situations and student abilities and show that formative feedback provides a positive benefit for students' learning and performance in almost all circumstances.

Whilst the importance of feedback (in its widest sense) cannot be doubted, there are questions about how effective this currently is. In recent years, the National Student Survey (NSS) has highlighted this and concentrated more attention on feedback to students. The two statements relating to effectiveness of feedback (*“feedback on my work has helped me clarify things I did not understand”* and *“I have received detailed comments on my work”*) score poorly compared to others, in most institutions and across disciplines (HEFCE, 2012).

Similarly, in the More Maths Grads Higher Education Curriculum work (Robinson, Thomlinson & Challis, 2010a), students who were asked about how their course could be improved cited aspects relating to feedback and coursework above all other areas.

The More Maths Grads project also found that staff have concerns around feedback; these concerns include whether students actively engage with the feedback which is available in a constructive way, the amount of time taken to mark student work in the face of poor staff:student ratios (especially where marking is done primarily by lecturing staff), and the quality of feedback given. Staff mentioned the latter in respect of their own feedback being less than ideal because of time constraints, as well as questioning the quality where coursework marking is undertaken primarily by postgraduate students.

This chapter considers what is meant by feedback; what constitutes good feedback; and the views of students and staff and how this affects what staff do, or should do; and how one might improve the feedback available to students. In part it draws on the work of the author and colleagues as part of the National HE STEM Programme (University of Birmingham, 2013), working with staff in various universities to evaluate current and innovative practice in feedback on mathematical courses.

This project involved staff from science, technology, engineering and mathematics (STEM) disciplines in a variety of universities, including Russell group, research intensive and post-92 universities who were first invited to complete an online survey in which they described their typical practice, any unusual feedback activities, and their views on the advantages and disadvantages of these. Slightly more than 30 staff, mainly but not exclusively from mathematics departments, completed the survey, and as such the results provide a snapshot of some current practice, rather than an exhaustive study.

A smaller group in seven departments worked with the project to evaluate a current or novel feedback method. Evaluation typically took the form of an online or paper-based questionnaire for students, and discussion with the staff member (phone interviews or via email) about their experience of the particular type of feedback. The student questionnaire focused upon open-ended qualitative responses, which were designed to discover:

- a) What things the students considered as ‘feedback’.
- b) To what extent they used these, or found them helpful.
- c) What students actually do with the feedback they receive.

Approximately 850 students were involved across the sector in the feedback which was evaluated. However, since sample size in some departments was relatively small, and the students were self-selecting, the evaluation focuses on the qualitative responses to draw a picture of the student views, rather than quantitative data.

What makes effective feedback?

Before considering how to provide effective feedback, it is (perhaps surprisingly) necessary to discuss exactly what this means. Sadler (1989) identified three parts to effective feedback, namely that students must gain a knowledge of what is required, they must compare their own performance with the required standard, and they must do something which will (at least in part) close the gap between the two.

Whilst this may seem obvious to tutors, it bears repeating that the improvement to student learning comes almost entirely in the third of these three parts: the actions of the students to close the gap; and yet in most cases this is the part over which tutors have the least control. For new students, this most likely contrasts with their experience at school, where teachers are more likely to direct future students' further study in the subsequent lessons. Much feedback focuses particularly on the second part: identifying where they have made mistakes or fallen short of what is required. Whether this is effective will depend on what the student does next, and the question of how students might be helped to understand what further work they need to do is addressed later in this chapter.

Expanding on Sadler's three part list, and combining Gibbs and Simpson's (2004) examples of the different functions of feedback, and Nicol and Macfarlane-Dick's (2006) "*seven principles of good feedback practice*", the following list identifies 10 possible aims of feedback; no doubt there are others too.

Feedback may:

1. Clarify what good performance is.
2. Correct errors.
3. Develop students' understanding through explanations.
4. Prompt further study by the student, to provide opportunities to close the gap between the student's current work and the required standard, or to provide further learning beyond the work on which feedback is given.
5. Promote the development of generic skills by focusing on the use of skills more than on the content.
6. Deliver high quality information to students about their learning.
7. Promote awareness of the learning processes involved (so-called meta-cognition).
8. Facilitate the development of self-assessment and reflection.
9. Promote a dialogue between tutors and peers about the learning.
10. Encourage students to continue studying, particularly by encouraging positive motivational beliefs and self-esteem.

Nicol and Macfarlane-Dick (2006) provide some interesting discussion and strategies for developing good feedback practice under each of their seven principles, incorporated in the list above, which is well worth reading. In addition, let us consider some further specific points.

Firstly, the notion of viewing – and encouraging our students to view – feedback as a dialogue, rather than a transmissive model where tutors tell students how they performed, is valuable; indeed, one might go a stage further and suggest that students and tutors should see the work on which feedback is given as a part of a dialogue as well. Here again, the contrast with students' prior experience at school is important; with (generally) smaller class sizes, more interaction and more frequent classes, informal and frequent teacher-student dialogue in schools may be more the norm than it is in universities. Except in institutions where regular, very small group teaching is undertaken, in universities the formal feedback on student work is more likely to be a major part of the student-teacher dialogue. It is also worth noting that the nature of the dialogue might – indeed should – change as the student progresses through their university career, matures and becomes more independent in their learning. A model of feedback where tutors simply identify mistakes has little scope for development and is less likely to encourage students to take responsibility and become partners in the process.

This is, however, harder when the formative and summative aspects of assessment are combined, so that students have a motive for hiding any deficiencies they believe they have, and staff feedback may focus primarily on explaining the grade (of which, more later).

Secondly, the principle that teaching staff should encourage positive motivation bears repeating. Perhaps because of the nature of the subject, a page of perfect working might be rewarded with a single tick, whereas an error can result in an 'explosion' of red ink as the marker explains the error and corrects it. On one level this is reasonable; where there are errors a teacher will seek to identify them, and explain or direct the student to explore these further, whereas when the work is correct there is little to say. However, our students are people, and especially, the newer students are people who have just moved into an unfamiliar environment, academically and socially, probably with much less direct contact with teachers than they are used to, and within larger classes. For many, comparisons with their new peers contribute to a sense of insecurity. They are probably used to more positive feedback in schools than they receive at university. Perhaps the purpose of some feedback should simply be to encourage students to continue working. As one colleague commented:

“How often do we tell them that, actually, they're really rather good?”

In fact, even this may not be ideal. Both Gibbs and Simpson (2004) and Black and Wiliam (1998) talk about the importance of addressing the students' work, rather than their characteristics or their achievement compared with other students. Tutors should aim, they suggest, for feedback which tells students where they have done well, how they have gone wrong and what they should do about it, rather than telling them that they are in the bottom or top 20%, or that they have done well or badly. The latter can demotivate some students and create complacency in others, whereas feedback which is based upon a discussion of the content provides all students with action they should take to improve future work.

Gibbs and Simpson (2004) list ten conditions under which assessment supports learning, and it is a sign of the importance of feedback that seven of these relate to feedback; from these one may see three broad themes to 'good quality' feedback: timing, content/quality, and student engagement.

Timing

Firstly, issues of *timing*: feedback needs to happen often enough, and quickly enough. In particular, the practice in some university modules of a single large coursework assignment for which feedback is provided three weeks later, say, or worse, only after the course has finished, is more or less useless for the students' learning. Very few students will engage actively with the feedback given, except perhaps to understand the reasons for the mark they receive, and there is therefore little opportunity for students to 'close the gap' between their performance and the required standard.

Similarly, Chickering and Gamson (1987) name prompt feedback as one of their seven principles for good practice in higher education, and comment, *"assessment without timely feedback contributes little to learning"*.

For many students, feedback in university is much slower and therefore far less timely than they are used to in schools, and there are of course reasons why this is often the case. A common theme through the National HE STEM Programme feedback project research was the difficulty in providing feedback quickly, whilst at the same time maintaining quality:

"I feel that I could do so much more (and so much better) if only I had more time/resources."

Gibbs and Simpson (2004) suggest that timing may be more crucial than the quality:

"imperfect feedback from a fellow student provided almost immediately may have much more impact than more perfect feedback from a tutor four weeks later."

Gibbs & Simpson (2004: 19)

That said, it is of course worth considering the content and quality of the feedback, which is discussed in the next section.

Content and quality

The recent National HE STEM Programme feedback project revealed a clear notion of what constituted 'typical' feedback. Many colleagues would recognise the following description of a set of feedback given on a single assignment:

"Feedback on formal written assessed coursework is given in multiple ways for any one assignment. Generally all or most of the following are used for any one piece of work: (i) short comments on scripts (ii) model answers (iii) review of common errors in class (iv) written summary of common errors (v) follow up one-to-one discussion in practical classes following the return of work."

The rationale for such a varied approach is summed up neatly by this colleague:

"Some students respond better to some types of feedback than others. Giving feedback on the same piece of work in different ways is more likely to maximise the impact."

Within this 'bread-and-butter' approach, there is obviously some variation, which is discussed in more detail below. Beyond it, staff talked about a wide range of less typical activities. Some offer feedback on examination scripts, either generically or in rare cases individually to students. Many use 'spot tests' in class with instant marking (whether

online or not). Many mentioned the discussions in tutorial classes which offer more informal formative feedback on student work. In the following sections ‘typical’ feedback is discussed first, before considering a few less traditional approaches.

Individual written comments

Most staff write comments about student work, in one form or another. Most typically, this consists of handwritten comments on the actual script. Staff can view this as their main opportunity to “*provide detailed written comments on the student’s progress*” and, where this is true, students value this.

“Coursework feedback always helps me to see where I am going wrong in any topic and where I need to improve.”

However, staff have a number of concerns, which are often reflected in what students say. Firstly, staff identify a problem balancing the competing demands of providing high quality feedback, with the need to provide it in a timely manner:

“The amount of marking required means that it is not always possible to include enough feedback to make it effective.”

As a result, staff say that their comments are often very brief, or limited solely to identifying where the students have made an error. This is inevitably considered less helpful by students:

“Short statements like “wrong” and bits crossed out with no reason are particularly frustrating.”

Secondly, where marking is done by teams of markers, some staff worry about the variable quality of the feedback:

“I encouraged other tutors (I was the lecturer) to do [as much as I did] (I know for a fact that some did rather less than I do). In my 2nd year module, work was marked by graduate students – I failed to encourage them enough to give anything beyond ticks and crosses...”

and again this is reflected by what some students say:

“Some feedback from the assignment sheets were useful and very detailed however some tutors did not put in very much effort.”

Finally many staff report concerns about the level of student engagement with the feedback which is provided:

“I am sure that there are many students who basically don’t look beyond the mark.”

“Some students do not collect their work so never see the feedback.”

The issue of how students can be engaged more effectively in the third stage of the feedback process – action to close the gap – is discussed later.

Pre-prepared comments

As shown above, comments on student work are valued when they are sufficiently detailed, but time constraints often mean that they are brief. Some staff use their knowledge of typical student performance to create a numbered list of standard comments, so that they can refer students to particular comments which are relevant to them. Others have embedded a similar system into software where the tutor identifies the relevant comments for each student on screen, and the program generates a feedback sheet containing just those comments for each student, either on paper or electronically. Staff report that they consider the time spent preparing such a system worthwhile in the long run.

In-class comments

Most staff report that they try to supplement the brief comments on individual student work with generic feedback given in class. Although most students reported that these comments were helpful, none reported them as the most helpful. Here, small things might make a difference:

“Giving oral feedback before scripts are returned encourages students to listen more carefully rather than spending their time flicking through returned work.”

It is clear from the National HE STEM Programme feedback project that a great deal of the comments which staff typically provide – whether individual, pre-prepared or generic feedback in writing or in class – on student work consists of identifying student errors, and it is perhaps in the nature of mathematical arguments that this will remain a significant feature of our feedback. However, as outlined above, identifying errors is not the only purpose of feedback. Gibbs and Simpson (2004) note that feedback needs to be appropriate to the purpose of the assignment; in other words, if the aim is, say, to encourage students to continue studying, then some positive motivation is appropriate. Similarly, if students are expected to take some action to remedy their misunderstandings, the feedback might state what action would be appropriate.

Model solutions

Some staff members resist providing model solutions:

“No model solutions is part of a general campaign to convert students from the process-driven approach that seems to have been inculcated in school.”

and although other colleagues raise similar concerns, the majority who talked to us do provide them. Provided they are written in a suitable manner, model solutions may address the first possible feedback objective given above (clarifying what good performance is) although it is worth noting that without guidance, students may not understand why the solutions are better than their own attempt, particularly where staff are marking for aspects other than the basic mathematical method. For example, if staff place value on the style of the mathematical writing, or the development of a logical argument, students may only see that they have included all the same steps to a solution, without understanding the critical differences between their work and the tutor's. Staff saw solutions as a means of providing “uniformity”

of feedback when tutoring/marking is shared, satisfying student demand, and crucially as an effective means of encouraging students to reflect and to prompt further work:

“Providing model solutions encourages students to review their own work.”

To what extent is this true? Students in our survey reported heavy reliance on model solutions; at some institutions in our project, they were considered the most used, and best feedback available. One student’s description of how they used these is typical:

“[I] looked at how the model solutions showed something being done, I tried to understand it, then I did another similar question on my own without looking at solutions and then once I finished I compared my method and answers with the model solutions.”

This active engagement with model solutions contrasts with the much more passive reactions to individual comments written on scripts that were reported by many students, perhaps raising doubts about the value of the time spent on creating individual feedback.

Videos

A variation on written model solutions, which has been explored by some staff, involved producing videos of them deriving the solutions. Most commonly this is done with special pens with video cameras attached (‘pencasting’) or tablet computers (‘screencasting’) but other methods are available.

“The biggest advantage of the (new) current practice of providing screencast solutions is that it allows students to see the process of solution rather than simply seeing the solution in a completed form.”

If produced after marking student work, it is possible to include pointers about common errors or misunderstandings.

One project partner used these with a class of around 220 first year students on a calculus course, and amongst the 34 students who responded to our survey request, the videos were overwhelmingly popular. Almost all thought this was the best feedback they had, citing a variety of factors including the ability to go at one’s own pace, seeing the development of the written solution rather than just the finished product, and hearing commentary about typical problems. One student said:

“they gave a more detailed explanation of solutions, rather than just the solution itself.”

and another said it was:

“like having a personal tutor.”

There were a couple of dissenters who would have preferred a more conventional written set of solutions, but they were in a small minority.

It is noticeable that although these students have clearly used the videos, their response to them was sometimes very passive:

“[I] just watched through.”

although some were much more active:

“Before watching the video I read over my coursework again and then whilst watching the video compared it to how I had done the question and where it was similar or where I had gone wrong. I then watched the video again and wrote down notes of how to do the questions and additional comments and tips. After reading my coursework and watching the videos I discussed with my friends to talk about where I had gone wrong and where they had and to talk about the correct way that the question should have been completed.”

The contrast between these two student responses highlights a common theme; namely the difference in levels of student engagement. Staff have tried various approaches to encouraging the less engaged students to behave more like the most engaged, and we now consider some differing approaches to in-class activities; as we shall see, these aim to promote effective use of model solutions, assessment of written mathematics (including self-assessment), and discussion between students.

Student engagement with feedback

It is argued above that the main spur to student learning comes in the actions which the student takes *after* receiving feedback. Gibbs and Simpson (2004) argue that feedback needs to be received, taken notice of, and acted upon to have any impact.

The typical feedback described above will, for some students, succeed in prompting further work, and promoting appropriate reflection and self-awareness, (for example, by a comparison of their own marked work and model solutions). However, any experienced tutor will know that not all students will respond in this way. We have argued elsewhere (Robinson, Thomlison & Challis, 2010b) that lecturers’ experiences of university study are generally atypical, in that we usually had high level study skills and motivation which enabled our success. Tutors may wish for the same of their students, and whilst high expectations are entirely appropriate, we need to recognise that not all students are motivated to act this way.

More importantly, not all students *know* how to act this way. This is perhaps one of the key difficulties for new students arriving fresh from school. In schools many teachers plan subsequent lessons to include the action which is needed to ‘close the gap’ between what was required and what the student actually did. In universities, most tutors expect – or at least hope – that students will take this action in their own time and by their own initiative. Whilst this is undoubtedly desirable by the end of a student’s university life, it is clear that many students need guidance to develop the attitudes and skills over their time at university.

Some tutors in the National HE STEM Programme feedback project were attempting specifically to address what students do after receiving feedback.

Proformas/feedback sheets

Some staff use a proforma feedback sheet, rather than (or in addition to) comments on the student’s script. This can be a way of addressing more generic questions about how well the student has met the overall objectives. For example, work based around numerical solution of differential equations, say, might have an overall objective about exploring the difference

between exact and approximate solutions, which may not be directly addressed by marking which asks “did they answer question 2 correctly?”

One project partner developed this idea to try to increase student engagement with the feedback and to address two other aspects of feedback, namely to tell the student what was good about the work, and what they should do to improve the next time. This was intended to make the feedback more relevant and increase student engagement with it. This was viewed positively by most students, and the following comment is typical of the general response:

“The summary grids were very useful. I used [them] to look at what my coursework was missing, and which aspects of my assignments were poor. This allowed me to improve the weaker areas of my assignment and do any missing aspects of my assignments.”

In-class exercises

One lecturer introduced a classroom exercise asking students to identify all the errors in several pieces of mathematics based on a recent assignment, with added errors including incorrect notation and proof structure:

“The actual classroom activity produced excellent student engagement and students seemed to approach the activity as a challenge. The time spent by the students in analysing the mathematics was much longer than anticipated. Students seemed keen not only to identify the errors but also to explain why they were errors. In some cases the exercise contained partial solutions. Interestingly students identified the incomplete nature of these solutions as errors.”

Self-assessment

An alternative approach was designed to get students to assess their own work more effectively. After a test, scripts were returned to students with limited marking (ticks, crosses, circles around problem areas) but without a mark or any detailed comments. Instead they were given the model solutions and told to estimate their score, and try to repeat any questions on which they had done poorly.

Some students adopted a superficial approach to this:

“I counted a total of 13 ticks that were on my paper...”

but, perhaps spurred on by the natural desire to know their final score, most were much more active and positive. Many identified different aspects which they needed to work on, from the specific:

“I know now in future that I will pay more attention to the questions asked in the exam as well as read the instructions on the front of the exam paper.”

to the more general:

“[I] felt that not releasing the marks with the exam really helped. It made me look at my answers and properly assess that I need to do all the exercises and to not turn to someone else at the first sign of trouble. I need to become more independent with my learning.”

Team-based learning

The final example in this section has been termed team-based learning, and uses fellow students to provide instant feedback. At the start of a class, students were given a short individual test (in this case, multiple choice). After recording their answers, students are put into small groups and redo the test as a group. The tutor believed that this needs to be carefully introduced, and repeated regularly, to allow students to develop the confidence and experience to realise that deciding the correct answer requires discussion and consideration of each others' methods, but there were clear signs of this skill developing:

"After doing the test again with our groups I was surprised to see on a couple of occasions my group agreed on an answer but I disagreed and it was me who was right, which has given me a bit more confidence to speak up even when the majority think differently as usually I would doubt myself."

Generally the students were positive about this approach:

"Firstly we started by doing the test which tested our knowledge on the given material from the previous week... Once we had done the test and we got into our groups I found it helpful to be able to discuss my answers and got a higher score."

Indeed, the lecturer reported that everyone got a higher score in the group test, perhaps suggesting that all have learned something from the process.

Informal discussion between students

Whilst this type of exercise is intended to stimulate student discussion and involve all students in it, clearly many students already discuss their work informally with other students. In the student surveys as part of the feedback project, 'informal discussion between students' was included as one of the feedback methods which we asked students to rate. Almost all students said they had used this, and for a substantial minority of students it was their most used, or most helpful, feedback. Students like this because they feel less pressure, they develop solutions themselves, and (perhaps distressingly for professional teachers), they prefer the explanations:

"When it is your friends who you are discussing it with there is no element of pressure."

"I liked this method the best because it helped me to figure it out. Rather than being told what to do, you are developing an idea of how to solve questions."

"Other students can explain in a way you are more likely to understand."

Most staff would probably encourage student discussion, albeit with some reservations about collusion in summative coursework. However, whilst there is clear educational value in this, how do students view it? As one who rated it highly said:

"It wasn't really feedback, it was just discussion..."

Given the focus on NSS scores, this may well be a concern for many tutors.

Student awareness of the feedback process

At the start of the National HE STEM Programme feedback project, we quickly discovered that various institutions have responded to poor feedback scores in the NSS with an awareness campaign. As one staff member said:

“We don’t mark work anymore; we give feedback. We don’t talk to students; we give feedback. We don’t give out model solutions; we give them feedback sheets. I don’t have office hours; I have feedback hours. Basically it’s a big rebranding exercise.”

Whilst these institutions (or, crucially, the staff within them) may well have initiated other changes to their feedback, it is notable that such ‘rebranding’ alone does nothing to improve the students’ education, although it might work to improve NSS scores. In all our student surveys, students were asked to identify what feedback was available to them, before later presenting them with the list of feedback that the tutor said was available. In one module, students collectively identified 12 different types of feedback that were available, but individually most students mentioned only one or two of them, and at most they mentioned only half the feedback types.

This would suggest that using the word ‘feedback’ at every opportunity might be a successful policy, but one might hope that as teachers, we look to extract more educational benefit from a – perhaps cynical – university or departmental policy.

In other words, there is a step beyond “*tell them it’s feedback*” which would be to make explicit the importance of *doing something* with the feedback, and help students to develop the skills to know *what* to do:

“Allowing time for one-to-one discussion in a practical session immediately after the return of the work does make students engage with the feedback. This approach provides the opportunity to tell students that they are getting feedback on their work. This helps with understanding of the feedback process which students need to understand when they come to the point of filling surveys such as NSS...It does require additional work. But this seems worth the effort.”

Some lecturers may consider the need to engage with feedback to be obvious, although the concerns expressed by staff suggests that we know it is not obvious to all our students. Others may object that these are key study skills which mathematics lecturers ought not be required to teach. There are two responses to this: firstly, if their tutors don’t teach students the skills they lack, who will? Would staff and students alike not benefit from helping them to develop those skills? Secondly, many of the things staff hope students might do with feedback are key *mathematical* skills that professional mathematicians need: assessing one’s own work for errors, and redoing the work to correct mistakes; understanding written mathematics (in this case, model solutions or peer work); discussing how to approach a solution with peers and applying knowledge and logical argument to develop the correct method.

Time spent helping students to develop these skills, particularly early in a course, as they first make the transition to university study, could improve their engagement with the feedback, not just in the module where the work is done, but throughout their future work.

Conclusions

It is clear from the preceding sections that ‘feedback’ does not simply mean, for example, comments written on student assignments, although this may be the *first* thing that most staff and students alike think of. Rather, staff consider any of the means by which students might evaluate their progress and inform their future work. Certainly this includes comments and marks written by tutors, but would also encompass a wide range of other resources and activities, created or undertaken by staff or students or others. Thus self- and peer-assessment, for example, or activities designed to increase student engagement with the feedback, or informal comments made in class are all forms of feedback.

Both staff, and the key literature on the subject, identify three main aspects which need to be considered: the need for rapid, frequent feedback; the quality and nature of the feedback; and the action taken by students after receiving the feedback.

There is often a conflict between the need for quick turnaround, to provide students with feedback when it is still relevant, and the difficulty of providing high quality feedback. Some of the ways that partners in the National HE STEM Programme feedback project are trying to tackle this conflict have been presented here, although these ideas are by no means exhaustive (as a starting point, see the references given in this chapter for further ideas).

It can be seen from the discussions above that there is a tendency for much feedback to consist of two main parts: identifying the errors a student has made, and providing some exemplar of a correct answer as model solutions or examples given in class. It is worth repeating that whilst students with high level study skills and motivation are typically prompted by these both to reflect on their own work, and engage in further study, this reaction is by no means universal. However, staff need to recognise that many students, especially when they first come to university, do not possess the skills that they need to profit fully from university study, and if we are not prepared to help them develop these skills, who will? In-class activities to encourage them to act on the feedback provided, or specific advice given for further work, and positive encouragement will reap benefits throughout their university career.

Ideas for improving feedback

1. Be explicit with students about feedback; make clear that feedback is not just what the tutor writes on their work; encourage them to view it as part of a dialogue; crucially, make clear that they always need to act on the feedback.
2. Give them guidance on *how* they should respond. Particularly in the early stages of their university life, include required activities (for example follow-up assignments, or in-class activities) that encourage active engagement with feedback, and gradually develop the skills they need as independent learners.
3. Consider any means possible to provide some feedback quickly, and frequently. For example, short regular homework might be more effective for learning than fewer, larger, assignments. Where marking work fully is inevitably going to take more than a week, try providing alternative partial feedback, such as in-class discussion, peer or self assessment, or giving a couple of brief formative comments in advance of the detailed marking.

4. Model solutions – either on paper, via video or in class – can be effective feedback which can be provided as soon as students have handed in work. Their effectiveness is increased if – at least sometimes – they are linked to required activity to encourage students to actively engage with them.
5. Remember to include positive feedback as well as correcting errors, both as encouragement, and because students should know when they are doing something well and should keep doing it.
6. Focus on the work, rather than the student, and be specific about telling students how it is good, and how future work could be better. Avoid comparisons with other students.
7. Consider the use of formative-only feedback.

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Chapter 13: Progression within mathematics degree programmes

Tony Croft & Michael Grove

Abstract

Several independent research projects report that the enjoyment of mathematics by many undergraduate mathematicians decreases as they progress through their degree programme and this decrease is accompanied by increasing disillusionment and disengagement with their course and alienation from mathematics itself. These are students who *choose* to study mathematics at university and who are relatively well-qualified. Moreover, it is often the case that students who report such feelings are not failing students – indeed many are doing rather well. Of course, many other students find their undergraduate experience of mathematics to be extremely rewarding but the prevalence of studies reporting disaffection suggests that this is an issue worthy of exploration within a book on transitional issues affecting undergraduate mathematicians.

This chapter will review the evidence for this phenomenon and unpick the reasons students give for their changes in attitude to mathematics. After establishing the context for the chapter we present a brief review of the literature in this field. The evidence suggests that this state of affairs can be attributed, at least in part, to the mismatch between students' hopes, expectations and aspirations and the reality of learning mathematics at university level. Sometimes, traditional pedagogies and practices can exacerbate this situation. We will go on to provide several examples of ways in which some lecturers and departments have attempted to modify practices in order to improve the student experience of university mathematics. We summarise the findings of selective activities and projects that provide pointers in the hope that they might inspire or provoke a discussion amongst individual lecturers and more widely within departments about ways in which disillusionment, disengagement and alienation might be ameliorated so that the experience of undergraduate mathematics is truly rewarding for all who choose to study it.

Introduction and context

The ‘sophomore slump’ is a well-documented phenomenon whereby students in the second year of their university studies can feel the ‘second-year blues’. It has been the focus of a number of studies, although these have not specifically dealt with second year mathematics undergraduates. A recent, substantial contribution to the literature is that of Thompson et al. (2013) which is described as a first attempt in UK higher education to investigate the second year experience and issues associated with it. Of course students in their first year of study can be unhappy too. During the first year some fail to adjust to university life, both socially and academically. They perhaps struggle with the depth and breadth of the mathematical material that is expected of them, or fail to adapt to new approaches to learning. They might experience challenges living away from home for the first time or establishing new friendship groups. Such long-standing challenges at the school/university transition are not only well understood, but have received considerable attention: academically, through diagnostic testing (see for example MathsTeam, undated), first-year curriculum change (Savage & Hawkes, 2000:3), provision of additional support (see Chapter 4 in this book); and socially, through university experience days, buddying schemes, students union activities, halls of residence activities and so on. Moving into year 2, any initial ‘novelty’ and enthusiasm felt for university life has perhaps long-since waned; there is a realisation that the work is getting harder and requires an increasing proportion of a student’s time coupled with the expectation that students are more independent and mature learners. The stakes are also starting to get higher; in the majority of universities performance in the second year contributes towards final degree classification. Thompson et al. (2013) discuss the wide range of factors which contribute to disengagement and underperformance and argue that *“the second year is a pivotal stage in the student life cycle and that academic and support staff need to be sensitive to the complex range of issues students may face in their second year”*.

Whilst students from any discipline may experience this sophomore slump, for many mathematics undergraduates the problems can be severe and can persist through the remainder of their degree programme to the extent that they describe negative feelings about their mathematics degrees even after graduation (Goulding, Hatch & Rodd, 2003). We argue that there are reasons why mathematics students might be particularly affected:

- a) Some question the purpose of studying abstract concepts of which they have little understanding, and see little in the way of application of much of the mathematics they are learning; many come to realise, quite rightly, that they will never use many of these specific mathematical techniques again.

One might argue that this is true of many academic disciplines studied at university; however the evidence from a study we cite later suggests that an important reason for students choosing to study mathematics is because they believe the mathematics they will learn will be useful and that they will use it to solve real world problems and yet many find that this is not necessarily the case.

- b) As we cite below, students report feelings of alienation, they describe narratives of existing on the margins and not feeling part of a learning community.

We shall point to research which suggests that the ways in which mathematics teaching is traditionally ‘delivered’ to students can exacerbate such feelings for some, though not all.

- c) Less academic support may be available in year 2 during a period in which the mathematics may, depending upon the type of institution in which it is being studied, become increasingly formal and abstract. Whilst a reduction in support may be a feature of the second year for students of any discipline, mathematicians in particular may no longer have access to mathematics support centres (established principally for first year students) which they may have relied upon (sometimes heavily) during their first year. Students report that there is less in the way of traditional tutorial provision. Frequently, closed-book final examinations are the predominant mode of assessment.

We would like to emphasise that such negative feelings, whilst commonly reported, are not ubiquitous, and for many students, university mathematics continues to provide challenge, inspiration and a continuing love of the discipline. For these students the curriculum and the traditions and practices of university mathematics departments are not alien and the students thrive. The focus of this chapter though is not upon the students who are enjoying their courses and thriving mathematically – they will continue to do well. Rather, in the spirit of encouraging a more inclusive mathematical experience, we focus here on those who are not. Evidence that there are a sufficient number of such students to merit consideration of this issue at all can be found in studies such as that by Brown, Macrae, Rodd and Wiliam (2005) described in more detail below, which found that in a survey of undergraduate mathematicians (N=150) in two English universities, over 20% failed two or more modules (out of six) in their first semester examinations in year 2.

Literature review

In an effort to understand better the nature of the challenges at second year level we have reviewed a range of relevant research studies. From these we have extracted five themes which are common across many and which we discuss further. Of course, some of these themes will have resonance in disciplines other than mathematics, but where there are mathematics specific issues we draw attention to these:

1. The importance of a sense of belonging and inclusion in a peer or departmental community and the learning and teaching relationship between staff and students; alienated students refer to lecturers’ lack of interest in them.
2. The quality of teaching – often cited by students as a reason why they disengage; poor teaching means that advanced topics are made even harder to assimilate.
3. The nature of university mathematics and the pedagogic practices of departments and of individual staff members which can often fail to align with the preferred learning styles of students.
4. Many students are encouraged or choose to study mathematics because they were ‘good at it’ when at school. This may change when they get to university, when for some, an important motivating factor (i.e. success) is removed. Others, even very successful ones, do not have a strong interest in mathematics, but see the acquisition of a mathematics degree as an enabler for their future (non-mathematical) careers.

5. The apparent lack of relevance and application of much of the mathematics taught in a context wherein students have been encouraged to study mathematics because of its usefulness in life and in the workplace.

We now consider each of these themes in turn.

The sense of belonging and staff/student relationships

Our starting point is the study Student Experiences of Undergraduate Mathematics (SEUM) (Brown et al., 2005) which examined progress and changing attitudes to mathematics amongst a cohort of well-qualified single honours mathematicians in two research-intensive English universities. Feeling part of a mathematical community emerged as a crucial factor in the student experience, and in the SEUM project, this community focused on a particular physical space within one of the participating universities. Within this space students could not only work together but *“could ‘catch’ lecturers going to and from [their] offices, as well as make formal and informal appointments to discuss various concerns”*. So feeling part of a mathematical community involves both opportunities for interactions with other students and, crucially, with staff.

Croft, Solomon and Bright (2008b), motivated by the SEUM project, believed that there may be scope for improvement in mathematics support beyond year one of a degree programme. In their study, through focus groups they explored which forms of support are accessed by students, which forms they find inaccessible, their views on support provided formally and on the informal networks that they develop with their peers. Focus groups were held in two research-intensive universities A and B. At University A student volunteers were selected from amongst those who had achieved at least an upper second-class score (i.e. greater than 60% overall) in their first year. Two focus groups, each of five students took place in University A. At University B, all second year mathematics students were invited to volunteer and two groups of two and three students respectively were interviewed. All the students in the focus groups in University A were positive about their university experience. On the other hand, when discussing mathematics at university, students in the second made the point that they didn't feel part of the mathematics community:

“Here you're just kind of a number, or one of a crowd. You feel quite anonymous and have all this learning to do”.

Croft et al. (2008b:25)

When asked about their sources of support it became apparent that the support they were obtaining from their peers was highly valued. Students in both universities were asked to whom they would turn first for support:

“yeah I reckon my friends will be the first point of call”.

Croft et al. (2008b:26)

“[working with others] that's the most fruitful situation in which I work – and sit with a group of friends”.

Croft et al. (2008b:26)

Similarly, Brown et al. (2005) reported that students who had more positive attitudes to studying mathematics were those who shared their ideas and problems with other students.

In Mathematical Literacy – Developing identities of inclusion, Solomon (2009) has a chapter devoted to the experiences of undergraduate mathematicians beyond their first year. She reports upon how the students she interviewed positioned themselves in relation to mathematics, their tutors and peers on the course, for example Emma who always positions herself as ‘second-best’ but goes on to describe her tutor’s assessment of her:

“my tutor seems to have high expectations of me...my tutor has more faith in me than I do”.
Solomon (2009:125)

Solomon reports that in her studies, as in other work she refers to, the importance for women of relationships with tutors comes to the fore. Emma goes on:

“It helps because whenever I go to see him when I’m stuck he doesn’t think “well I’m not going to help here – she can figure it out for herself”, he’s always like, “I’m going to help her then she can get there””.
Solomon (2009:126)

Unfortunately, not all students reported the same supportive tutors – in the same study Jess reports:

“my tutor laughs at me”.
Solomon (2009:126)

Rather than being seen as humorous, her fellow students show expressions of sympathy. Even inadvertent or throw-away remarks can inflict damage in ways of which tutors are probably unaware.

Goulding et al. (2003) report a study in which they surveyed 173 students on ten postgraduate teacher training courses (PGCE). These students had attended 65 different universities for their first degree that was either mathematics or one strongly related to mathematics. They were questioned about their undergraduate experiences. Some described the challenge and rigour of university mathematics as interesting and rewarding. Thirty-nine students commented explicitly about their development of understanding during their undergraduate degree. The majority of these students described this as a struggle but one which eventually gave rise to feelings of satisfaction; however, almost as many had been unsuccessful, claiming to have achieved *“little or no understanding”*:

“it was a struggle to maintain enjoyment in the subject when I never felt completely satisfied I understood the concepts in front of me”.
Goulding et al. (2003:370)

Students in this study, as in the others, all recognise the need to work independently, but some felt abandoned, with others relying on informal peer support mechanisms:

“Most lecturers and my tutor were completely unapproachable – you couldn’t ask questions during lectures and felt very alone”.

Goulding et al. (2003:371)

The importance of supportive staff and peer communities should not be underestimated and is a recurring theme in many studies and one to which we shall return.

The quality of teaching

Inglis, Croft & Matthews (2012) surveyed the views of 428 mathematics graduates, 85% of whom had graduated from ‘research intensive’ universities. Their aim was to ask mathematics graduates who had been in the workplace for around two to three years to reflect on their undergraduate experience to ascertain the extent to which the undergraduate curriculum prepared students for the workplace. The issue of teaching quality attracted a number of suggestions although the overall message was variable: many students reported an excellent teaching experience, but others clearly felt there was much room for improvement. The general message was that participants would have liked to receive more in the way of teaching than lecturing, more opportunities for problem solving sessions, and more opportunities to discuss mathematics with their lecturers.

In the survey by Goulding et al. (2003) 23% chose to offer comments about the quality of teaching. Most were very critical of their teachers and their teaching methods, particularly those with an over-didactic style:

“To sit taking meaningless notes all day was just too much...I feel that lectures could have been replaced with handed out notes and some real teaching allowed to take place”.

Goulding et al. (2003:372)

As a consequence of this lack of “*real teaching*” feelings of frustration were common, with many reporting their undergraduate study as a negative or lonely experience:

“I hated it most of the time...it became difficult to enjoy some of the maths modules if you were struggling and couldn’t get any help”.

Goulding et al. (2003:373)

University mathematics, pedagogic practices and students’ learning styles

In their study of twelve first-year mathematics undergraduate students, also in a (different from SEUM) research-intensive English university, Daskalogianni and Simpson (2002) refer to “*cooling-off*”. Cooling-off students are those who come to university from school with a positive attitude to mathematics. They find themselves in a learning environment which is very different from the one in which they have previously functioned successfully:

“the experienced mismatch between their beliefs about the nature of mathematics and its rigorous university character soon makes them lose interest in mathematics and develop a negative attitude towards it.”

Daskalogianni & Simpson (2002)

They sum up:

“It is then up to the students themselves [our emphasis] to either recover from that [cooling off] phase or give up and lose not only their interest in mathematics but even their interest in the course and develop a cooling-out behaviour which is very intense in its signs, very difficult to change and might even result in drop-out from the course.”

Daskalogianni & Simpson (2002)

Writing about pedagogic practices Solomon (2009) notes that:

“the fast pace of delivery in lectures meant that [often] the students were simply writing notes without having the time to think about what they were writing. This appears to contribute to a sense of disempowerment.”

Solomon (2009:126)

This is particularly poignant given that lecturers say they want students to be independent, and at the same time (some) would appear to disempower the very students they want to be independent (see Chapter 6 in this book). So what might a good lecture look like? Students commonly cite a desire for a ‘good set of notes’ but Solomon’s student quotes offer some further insight:

“[a good lecture should leave the student] actually wanting to go home and do the homework rather than “I’ve got to do that homework that I know I’m not going to be able to do””.

Solomon (2009:126)

It is not about spoon-feeding or dumbing down. Students expect to have to work for their mathematics degree; they know they are expected to be independent, they do not expect to understand everything in a lecture; however they must be given sufficient support and encouragement to enable them to think that putting in effort on their own is going to be worthwhile. What constitutes ‘sufficient’ support is open to debate, and colleagues will have a range of opinions on this matter, but we would suggest a level of complacency amongst those who argue that there is little room for improvement in learning resources or teaching practices.

Some students find assessment practices too to be a source of alienation. It is interesting here to note the findings from the MUMAP (Mapping University Mathematics Assessment Practices) Project (Iannone & Simpson, 2013) which examined undergraduate mathematics assessment practices scrutinising 1843 modules from 43 degree courses. The project concluded that assessment of university mathematics in the UK is still dominated by closed-book final examination with over one quarter of modules assessed solely by closed-book examination, and about 70% of modules using closed-book examinations for more than 75% of the final module mark.

Students becoming demotivated when no longer successful or interested

Several studies found students who described how much harder they found it to understand new material in year 2. In Croft et al. (2008b) one student mentioned they had

much less supervision, tutorials were different in terms of group size and so they felt there to be less support:

“This is the first year I’ve gone into an exam and had questions I can’t actually answer, because I’ve always been able to do all of the paper. There’s a lot of material this time around, but unless you understand it, it’s not really interesting.”

Croft et al. (2008b:25)

The SEUM project report notes:

“success in the subject at A-level was the major reason for choosing mathematics as a single honours course. When some no longer perceived themselves to be particularly successful there was little to motivate them to continue studying except a need to maintain self-esteem and gain credentials.”

Brown et al. (2005:3)

Doubtless anyone reading this will expect material in the second year of a mathematics degree to become more demanding but the point is to recognise the challenges that this causes for students and to ensure that mechanisms are in place to develop the skills of independent learning that are required, to encourage students to find and develop their own supporting mechanisms and to develop into robust learners.

The apparent lack of relevance

Inglis et al. (2012) citing Thomlinson, Challis and Robinson (2009) compared the outcomes of a survey of the expectations of 223 incoming undergraduates with the results of a survey of 428 graduates. Thomlinson et al. (2009) had found that 93% of incoming undergraduates had expected to develop skill in applying mathematics to real world problems and 95% expected this skill to be important in their future lives. However Inglis et al. (2012) report that only 62% of graduates felt that this had been achieved during their undergraduate studies. One respondent wrote:

“My only issue relates to real world experiences as I feel that this is where mathematics students have been let down. There is little if any mention of how to link mathematics to the real world of work.”

Inglis et al. (2012:19)

This resonates with earlier findings in the SEUM study: with specific reference to the relevance of the content being studied, these students had been led to believe that it would be:

“useful for life in general and for possible future careers”.

Brown et al. (2005)

As the course progressed:

“many of these same students reported their disappointment that the mathematics they were now learning had become so pure as to have no possible application: “I thought I was lucky to realise the importance of maths and to be able to study it at university but now I can’t imagine ever using it in any job I’ll ever do...”

Brown et al. (2005:4)

However it is important to balance this view with alternative perspectives reported by some. For example:

“it is satisfying to learn mathematics for its own sake – don’t try and make it more applied.”
Inglis et al. (2012:22)

and:

“for me, the most worthwhile aspects of the degree were those modules I took because they interested me. Please include the pure modules that may not seem applicable to most careers.”
Inglis et al. (2012:22)

Of course, the nature of mathematics degrees varies greatly within the sector with some offering courses which are more formal and abstract, some offering quite different proportions of pure and applied modules, and others (many of the newer universities fall into this category) offering courses which are much more orientated to applications and to the workplace (see for example Robinson, Challis & Thomlinson, 2010). Furthermore, many of the exciting real-world applications of mathematics can only be accessed through postgraduate study and perhaps this is not always made clear. However, the point here is that perhaps the particular type of course is not always well communicated to prospective students and a question for the community is how the apparent mismatch can be addressed.

Having briefly reviewed a selection of the literature relevant to the second-year university mathematics experience, the following section considers some ways in which these issues might start to be tackled.

Enhancing the second year experience

In this section we highlight several initiatives that have been instigated to enhance the second year experience. We do not suggest that these provide panaceas for the solution of the problems highlighted, but they provide pointers to possible ways forward. They include creating spaces within which students are encouraged to work together and support each other to develop learning communities, establishment of formal peer support schemes, involving students in curriculum development, encouraging more opportunities for staff/student interactions, and increasing exposure to real-world applications and contexts.

Creating spaces for students within mathematics departments

A finding of the SEUM study related to the extent to which students felt part of a mathematical community. The project noted:

“in one institution, students had their own mathematics study area comprising three sub-areas: an open plan section with tables and chairs; an area with easy chairs and coffee tables; and a cafeteria. Off these areas were staff offices and students could ‘catch’ lecturers going to and from these offices, as well as make formal and informal appointments to discuss various concerns.”

Brown et al. (2005:6)

In the second institution there was limited social space for mathematics (and further, at that institution many students lived at home), and:

“this resulted in a sense of isolation with many of the students reporting difficulty in making friends.”

Brown et al. (2005)

In a survey by Croft, Grove and Bright (2008a) students were asked about the benefits of providing them with social learning space in the department:

“When you go to work [in a dedicated social learning space] the people in there are going to be doing the same stuff as you anyway so you get help with each other.”

Croft et al. (2008a:13)

In this student's university, space was then found within the School of Mathematics to provide a dedicated area for second and third year undergraduates to work. This reflects an increasing trend by universities, certainly within the UK, and one which seems to be welcomed by students, to provide more extensive social study space for their use.

Peer support

Often friendship groups formed in year 1 are maintained throughout the degree and are mutually supportive. However not all students make friends with students on the same course, and this can be a cause of academic isolation. The formalisation of peer support forms a way in which a department can offer support and at the same time enhance the student experience. This aspect featured in the results of the graduate survey by Inglis et al. (2012). Fifteen out of sixteen unsolicited references to peer support referred to the benefits:

“perhaps we should be allocated to study groups to assist those who don't naturally find friends on the course and end up working alone.”

Inglis et al. (2012:2)

A peer-assisted learning (PAL) scheme was implemented at Loughborough University in 2011/12 to address the difficulties experienced by second year students. Students who had successfully completed two specific second year modules volunteered and were provided with training to act as 'Student Leaders' in weekly PAL sessions. Around half of the second year cohort chose to engage with the scheme and those who attended the PAL sessions had higher achievement in their final module examinations even after controlling statistically for prior attainment and lecture attendance. Of course, not all students wanted to attend PAL sessions, some because they preferred to work alone and could succeed independently. Those that did attend found the PAL sessions to be informal and welcoming and involved discussion of mathematical topics that had previously been suggested by the PAL participants.

The preceding discussion highlights the non-homogeneity of student groups in terms of their preferred learning styles, and whether they prefer to learn mathematics with others or on their own. Attempts to address problems of disengagement and alienation must be sensitive to these differences and try to present a portfolio of opportunities sufficient to meet the needs of diverse cohorts.

Involving students in expert communities

In recent years there have been calls to the higher education community to involve students more in the planning and design of courses (see for example Kay, Marshall & Norton, 2007). The SYMBOL (Second Year Maths BeyOnd Lectures) project based at Loughborough University in 2011 did just that. Four students who had successfully completed the second year of the single honours course in mathematics were employed as interns for a six-week period during the summer. Their focus was upon two historically-problematic second year mathematics modules. They had several roles:

- To act as intermediaries between their peers and staff gathering information about teaching strategies that the students perceived to work well, suggesting improvements, and alternative ways of doing things.
- To work with staff, scrutinising the modules' resources and suggesting ways in which these might be improved.
- To develop further resources of their own, under the guidance of the module leaders, which would be (and were) made available to future cohorts.

Findings have been reported in Croft, Duah and Loch (2013) and include the ways in which the interns were inculcated into the mathematical community and the nature of their relationship with staff changed for the better. Through their mutual engagement staff and interns engaged in a mathematical discourse in ways that normal lectures and tutorials do not make possible: as well as building better relationships with staff and becoming part of a mathematical community the interns involved reported deeper mathematical understanding and confidence in their own abilities. Staff acknowledged that students can bring a new dimension to teaching and learning discourses:

"[Students probably have] more knowledge of where exactly the students are struggling. Pointing exactly at the right places where the lecturer might think the students have understood when they haven't actually because the feedback loop is not as closed as it should be."

Croft et al. (2013:1051)

In this sense, as well as providing opportunities for some second year students, we see the initiative as encouraging continuing professional development of staff and a means for at least some improvement in alignment between student and staff expectations.

In a more generic context while employability audits of undergraduate programmes have existed for some time, these have typically been undertaken by academic staff rather than students. The University of Exeter, in a student-led project undertaken in collaboration between students, graduate employers and staff (Cooper, 2012), carried out an audit of the extent to which support for the development of employability skills is embedded within the mathematics, computer science and engineering degree programmes. The process was overseen by a paid focus group of students working over a period of six weeks to audit current provision and make recommendations for how employability skills might be enhanced. All mathematics and engineering students (some 300 in total) from across all

year groups were invited to participate through a questionnaire to obtain a wider student perspective; 49 students responded.

Through a blog established in support of the project, updated by both staff and students, their views on the process were captured, and this was further supplemented by the creation of a 'Creative Learning Journey' which formed a multimedia staff and student perspective on the process. While established to help students develop their employability skills, through both undertaking a project and enabling revisions to existing provision to be suggested (for example one outcome has been the design of a mathematics group project module), students have been able to make a direct contribution to their learning experience which they have valued:

"Its [sic] been a very productive 6 weeks and we feel the contributions we have made will be valuable in improving the degree programme."

Prince & Neale (2011)

Through the National HE STEM Programme, the Student-Led Employability Audit was extended to five further universities, and at the University of Manchester this was implemented to audit undergraduate degrees within the School of Mathematics. A similar outcome to that at Exeter was that students produced a report, recommending both major and minor changes to the provision they experience, some of which were implemented in time for the 2012/13 academic year. The key outcome relevant here is the extent to which students felt involved in the process of developing and enhancing their own learning, recognised their role within the process, and had opportunities to engage in dialogue with staff about this.

While these two activities form specific examples, they represent a more general point of being approaches that enable students to contribute to the development of their own learning and feel that they are making a contribution to enhancing the practice of their department. It might be argued that the immediate benefits of such activities are limited to the relatively small number of students involved and do not transfer to the wider student body. But in fact we believe that some benefits do transfer, although perhaps in more subtle ways. Firstly, there is little doubt that this professional development experience for the small number of staff directly involved has given them new insights into the difficulties and perceptions of the students; in turn this is likely to impact gradually on their own teaching practices. There are also the benefits perceived by other (non involved) students in that the department is seen to be listening to their concerns, taking these seriously through meaningful action, and that modifications might ultimately be made as a consequence of their work.

Developing (informal) staff student communities

A number of mathematical sciences departments have established 'Maths Arcades' – drop in sessions where students can play a range of strategy games and puzzles with each other and with academic members of staff (see for example Bradshaw & Rowlett, 2012). Maths Arcades aim to simultaneously support those learners who may be struggling and test and challenge more confident learners. One of their key features, however, is to enable greater staff-student interaction in an informal, but mathematical, context.

While Maths Arcades are a new initiative, certainly within the UK, initial evaluations have shown that one of the key aspects students value is the increased opportunities they provide to interact with members of academic staff:

"I like to go to the Maths Arcade because all of my tutors attend it".

Bradshaw (2012:12)

"[It is] a really good way to meet people and get to know the lecturers in a more informal environment"

Bradshaw (2012:12)

And staff report complementary views:

"I thoroughly enjoyed playing against the students where they had as much chance of winning as I did. It allowed us to interact in a more relaxed but equally intellectual way."

Levesley (2012:23)

Maths Arcades have prompted students who have not engaged with drop-in mathematics support to seek support from tutors, and increased engagement between student cohorts has been reported by staff:

"...the more that students see mathematical conversations happening between staff and students, the more they realise that they too can ask questions... The games were used and maths conversations were started ("can I ask you a question?") by students who had not attended the traditional drop in sessions. There was also interaction between staff and second year students and between the year groups."

Barton (2012:27)

As a consequence of running Maths Arcades, universities have recognised the concept of games and puzzles to enhance student problem solving skills, and as such, have implemented these into the curriculum:

"In addition, the puzzles and games have been incorporated into a second year mathematics module that aims to develop problem-solving skills. Students study the various games and puzzles, in particular by playing the games against other students, and the discussion of the games themselves leads to understanding of the strategies required to play effectively."

Chadwick (2012:17)

Context and problem based learning

We have already commented that many, but by no means all, mathematics graduates would welcome increased exposure to more real-world applications of mathematics. There is no doubt that formulating and solving problems is a core part of all undergraduate mathematical sciences provision, but what is less well embedded, particularly when compared to disciplines like, for example chemistry (Belt & Overton, 2007), is the context within which mathematics is applied to solve problems (context based learning). Many of the recruitment talks and events students experience while making their university choices highlight examples of how mathematics can help solve key global challenges or,

for example, where mathematics can be applied in appealing fields such as sport or music technology, but quite often this is the last time such applications will be mentioned during their university studies; for some, this can lead to a mis-match between their expectations and the realities of studying mathematics.

Some mathematics departments have sought to implement ways for students to experience mathematics within a range of applied contexts. Placements, whereby students spend a period of time working within business or industry as part of their degree programme, have been offered for many years but as noted by the Science Council (Mellors-Bourne, 2011) while there is not generally an issue in the supply of placement opportunities an *“increasingly significant proportion of sandwich course students are reluctant to undertake long placements (partly because many HEIs charge at half fee level)”*. Despite this, placements offer a range of benefits including not only the chance for the student to develop essential employability skills, but they can also increase student motivation by demonstrating how they can apply their skills and knowledge successfully in business and industry. In the past, industrial placements would typically have been a year in duration but there now exist a range of models which allow students to access placement opportunities in a more flexible and tailored manner ((see Mann, 2012) for a review of models in mathematics).

Where it has not proved possible or practicable to implement placement models, mathematics departments have found alternative mechanisms for students to gain more context-based experience. A number of departments have, for many years, offered postgraduate level industry study groups whereby representatives from industry work with mathematicians in a structured manner on current workplace problems. Variations upon this model have been adopted by a number of mathematics departments where undergraduate students work collaboratively on employer-defined problems and case studies as part of their programmes of study. Students not only engage with current and meaningful real-world problems, but employers also typically have a role in contributing to delivery and supporting assessment, which means students develop relationships with employers in a manner that allows them to discuss the relevance and applications of mathematics. Recognising that not all mathematics departments have access to a wide range of employers, there has been significant work in developing a ‘bank’ of freely accessible industrial problems which academics can use with their students (see for example, Benjamin, Homer, Lawry & Rossiter, 2012). Such forms of problem based learning are more interactive, representing a move towards experiential rather than didactic learning, and as a consequence students have increased opportunity to engage with members of staff.

Concluding remarks

In this Chapter we have looked particularly at published evidence which suggests that the academic experience of many, but by no means all, second year mathematics undergraduates is less rewarding than it could be. Doubtless some of the issues we highlight affect students in other years and in other disciplines too. However our focus here has been the second year experience. We have suggested several ways in which this experience could be enhanced. We have provided details of these not as a prescription but rather with the intention that they encourage reflection by mathematics staff on practices within their own departments:

- Are there issues like the ones described here within my own department? How would I know?
- Is there scope for improvement of practice within my own department?
- Are the full range of students that we recruit best served by our teaching and learning practices, by our curriculum and by the supporting infrastructure we provide?
- When we promote our courses to prospective students are we clear about the nature of these courses and particularly what makes them distinctive in our institution?
- What steps, if any, are individuals and departments in other universities taking to address these issues?

If individuals, groups of staff and whole departments reflect seriously on these questions, and practices evolve accordingly, the experience of all their students will be enhanced. As many of the chapters in this book affirm, there is a growing corpus of literature and many colleagues and interest groups to support staff in this journey.

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Chapter 14: Employability for the workplace

Jeff Waldock & Stephen Hibberd

Abstract

High tuition fees lead to equally high student expectations, both of the qualification and the university experience. In challenging economic times these expectations are likely to also focus on graduate job prospects. Together with pressure from government, business and industry, and students (BIS, 2009a, 2009b, 2011; CBI, 2011; AGR, 2010), this has led to many higher education institutions adopting a renewed focus on the student experience and in engaging students as partners in learning.

This chapter will present the issues involved, summarising results from practitioner experience derived both from the published literature and from many of the projects carried out as part of the National HE STEM Programme. Successful curricular strategies for more effectively developing student employability are described.

Introduction

The importance of employability in mathematics degree programmes was recognised by the prominent inclusion of employability skills in the Quality Assurance Agency (QAA) for Higher Education Mathematics, Statistics and Operational Research (MSOR) Subject Benchmark Statement (QAA, 2007). In recent years there has been an increased recognition of the importance of professionalising teaching and learning practice in higher education mathematics (Cox, 2012), as evidenced for example by the introduction of the UK Professional Standards Framework (HEA, 2011) and the teaching fellowships and teaching development grants promoted by the Higher Education Academy. The pre-eminence of new measures by which applicants can select a university course – the National Student Survey, the Destination of Leavers from Higher Education (DLHE) survey, the Key Information Sets (KIS) and various university league tables – all add to the pressure to improve the overall student experience, and graduate employability (Hibberd & Grove, 2009).

In considering employability for the workplace, it is also necessary to recognise that future graduates will have much longer working lives, and 'portfolio' careers in which this workplace may have many forms. It has been estimated that as many as 25% of current jobs didn't exist 10 years ago – and the trend is likely to continue. To succeed in such a marketplace students need to be adaptable, engaging in professional development regularly to gain new skills.

Alongside this is the shift in the proportion of UK school leavers going to university from around 5% to 50%. The curriculum of many mathematics programmes is designed to prepare undergraduates for further study and research and has not changed to reflect the aspirations of this much larger cohort, many of whom undertake a mathematics degree in order to boost their career prospects. This is at odds with the fundamental approach adopted by many higher education institutions.

Some institutions tend to have a traditional approach to higher education, with the principal goals (or programme 'learning outcomes') being focused on the acquisition of subject knowledge, skills and their application. It is highly likely that in such an institution little support will be provided for staff who wish to build in employability through the taught curriculum – most academic staff will not recognise this as part of their role, and if students do receive any help in developing such skills it will be from extra-curricular sources, such as from the university careers unit. The institutional culture this leads to will militate against recognition or reward for academic staff who wish to incorporate elements of employability development into their teaching.

Other institutions may embrace the concept of developing student employability since this will support recruitment and help ensure the needs of the business and industrial community are met. Senior staff, often at pro-vice-chancellor level, have been appointed with specific responsibility for teaching and learning, the student experience and employability. Clearly, the culture in institutions such as these is much more supportive of developing student employability – both within the curriculum, and through co- and extra-curricular activities.

With one important measure by which universities are currently judged being graduate employment rates, all institutions are publishing top-level policy documents which include rhetoric that claims their courses develop employability. Changing the culture occurs slowly however, and many staff at institutions that have not previously focused on employability may still feel that a good degree from their university, together with an occasional careers presentation will be sufficient. Nevertheless there has been increasing investment to support employability initiatives at universities across the spectrum, holding out the promise for positive change.

A principal challenge for all institutions is to create an environment in which learning providers put employability enhancement at the heart of what they do, and recognise that some skills, practical job application skills for example, can be developed quickly and lead to short term gains in employment rates. Other important graduate attributes however, such as self-confidence, team-working and communication, take much longer but are equally in need of investment (Yorke, 2010). Institutions must therefore be willing to plan for long-term gains as well as 'quick wins'. It is also important to enhance the quality of

pedagogical approaches, and recognise the impact that co-curricular and extra-curricular activities have in developing skills and attributes that enhance employability.

For practitioners in higher education institutions supporting the development of student employability, the national and institutional context is subject to a high degree of variability and change, but while the external context is certainly important it is the local institutional context which directly frames the way that teaching and learning takes place and influences the pedagogical approaches to employability that may be developed.

Possibilities and constraints for work are felt at the local level, and understandings and definitions of employability are subject to institutional policies and cultures. It is a self-evident fact (but one that deserves emphasis) that if student employability is to be successfully developed within a particular institution, then a vitally important factor is the support provided by that institution. This includes direct support, in the form of staff resource, strategic support through corporate policy and course approval processes, and implicit support by virtue of the prevailing institutional culture. Conversely, it follows that if an institution does not value employability as a 'graduate output' then it is likely that an educational developer interested in promoting student employability, will have a tough task on their hands.

The institutional environment also impacts directly on models of delivery which range from a 'bolt-on' approach, where employability modules are provided as an optional extra, to an entirely embedded approach where employability is assessed and supported within the disciplinary curriculum.

Employability transitions in a mathematics programme

This book is concerned with transitions, which occur:

- **Into higher education**, including induction and the recognition of how university mathematics differs from A-level – being expected to tackle more open-ended tasks, use different study styles and work independently. University mathematics courses may, in addition to the acquisition of a whole host of advanced technical skills, address the expression of real world problems in mathematical form, making assumptions and approximations as necessary in order to make a problem tractable. Following the application of mathematical techniques to obtain a solution, students may also be expected to provide results in forms suitable for different audiences, and to communicate them in appropriate ways effectively (taking account of the approximations and assumptions made when formulating the mathematical representation of the original problem). An employable mathematics graduate is confident in applying their skills to new areas in this way.

Because the development of key employability skills such as self-confidence, communication and team-working takes a long time, it is important that students are made aware from the beginning of the course that certain course-related activities have been designed to specifically help them develop these skills. To stand the best chance of success, therefore, this should form part of transition into higher education.

- **Between levels** – students will be expected to develop increasing levels of sophistication. In terms of their employability, this will include increasing recognition of how curricular activities are building their skills towards graduate expectations, so that when necessary they are better able to evidence and articulate this development. Strategies for enabling this to take place effectively may include the provision of an employability ‘pathway’, perhaps in the form of a schematic diagram.
- **Out of higher education** and into employment, a process lately referred to as ‘outduction’. This can incorporate career management skills – practical job application skills – which are unlikely to be curricular, but may also include developing awareness of career pathways open to mathematicians. It may also include developing a digital online presence.

Strategies for implementation of these ideas are described below.

Strategies for change

This section aims to identify ways in which undergraduate mathematics programmes can change to develop student employability more effectively.

Employer engagement

For many UK higher education institutions, employer engagement has increasingly become a core part of the academic culture, the breadth of which has been demonstrated through a number of Higher Education Funding Council for England (HEFCE) funded projects (Tallantyre & Kettle, 2011). In recent years there has been a growth of whole curriculum-based approaches to developing graduate employability, and a rapid growth in enterprise education and entrepreneurship amongst the student body. Foundation degrees, providing qualifications with content closely defined by employers and building in workplace experience, have grown from 4,300 enrolments in 2001 to 103,000 by 2010 (Wilson, 2012:21), but foundation degrees in mathematics have met with little success.

As part of the government’s response to the Leitch Review (2006), HEFCE were asked by the Government *“to develop a new funding model that is co-financed with employers, achieves sustained growth in employer-based student places and introduces the principle of employer demand-led funding”* (DIUS, 2007:50). A Workforce Development Programme invested nearly £150 million between 2008-2011 in universities to develop their operations with employers around a new co-funding model (HEFCE, 2011:17).

There is a danger, however, in aiming to tailor a mathematics curriculum to target ‘skills that employers want’. Since mathematics graduates find employment in a wide range of areas, no one employer’s needs should be prioritised – a broad view of the needs of the sector as a whole has to be taken before designing curricula that address them.

A fundamental role of university academic staff is to design and deliver the academic curriculum. Furthermore, a mathematics degree programme should take account of the requirements of the QAA benchmark statement (QAA, 2007) and will be approved through institutional quality assurance procedures. This is a non-negotiable situation; the university is accountable for the standards of its awards and the quality of its provision. Programme

approval by the Institute of Mathematics and its Applications (IMA), providing it is followed by suitable training and experience, entitles a graduate who is a Fellow or Member of the IMA to apply for chartered mathematician status. This approval requires that two thirds of the topics on the course are mathematical, constraining the freedom of academic staff to design the degree programme. Although these requirements may appear to limit the freedom of employers to further influence the curriculum, experience indicates that there is sufficient common ground between employer needs, QAA benchmarks and IMA requirements for there to be little reason why employer needs, at the discretion of the university, cannot be integrated within the mathematics curriculum, if necessary through the creative use of learning, teaching and assessment strategies.

When designing a programme of study, academic staff must first articulate the type of graduate they are aiming for – which leads formally to a set of programme learning outcomes. In order to determine these learning outcomes, the views of the relevant stakeholders must be sought. These include employers – some of whom may be alumni of the department – who can advise on what skills they have found useful in their career so far.

Early in 2011 the Maths, Stats & OR Network (MSOR), as part of the National HE STEM Programme, held a ‘Higher Education Mathematics Curriculum Summit’ at which one of the principal outcomes was a set of 14 recommendations for priority activities in higher education mathematics curriculum development (Rowlett, 2011a), many of which relate to student employability. Since the summit brought together a highly representative sample of the higher education mathematics community, including senior staff from approximately 50% of universities in England and Wales offering mathematics degrees, these collectively-agreed recommendations have national sector-wide relevance. Following the summit, the National HE STEM Programme issued a call for activity proposals based on the recommendations, leading to a number of funded projects (Rowlett, 2011b). Throughout the rest of this chapter, reference will now be made to relevant recommendations and associated projects and activities.

Recommendation 14: “*Research to collect the feedback of graduates in employment on the mathematics higher education curriculum*”

A first approach to incorporating employer input into the curriculum at the design stage is to find out what skills and qualities they would like to see in a graduate from the programme. This can inform the design of the programme learning outcomes; a subsequent step would be to design the curriculum, and the associated learning, teaching and assessment strategy, which will deliver these outcomes. Some programmes (for example Chadwick et al., 2011) involve employers in the assessment of student project work.

Inglis, Croft and Matthews (2012) surveyed 428 mathematics graduates two years after graduation on the skills they gained in their degree and how useful they had been in their career so far. Results were that the vast majority recognised that their degree provided them with useful cognitive skills, but that the highly valued non-cognitive skills which incoming undergraduates expected to develop, such as making presentations, oral and written communication and team working, were missing. When asked what skill graduates wished they had had the opportunity to develop more during their mathematical studies, the most commonly selected was “*applying mathematics to the real world*”. Over 90% of incoming

undergraduates expected to develop this skill, whereas only around 60% of graduates believed that they had.

Davies and Marriott (2012) collected the views of employers and developed a list of employer expectations of the statistical knowledge, awareness and skills of STEM graduates. A number of other National HE STEM projects gathered information from employers and graduates in employment about what skills should be built into a mathematics degree, including those led by Bradshaw [Project 1], Cooper [Projects 5,6], Hind [Project 9], Levesley [Project 11], Messmer and Pugh [Project 12] and Golden [Waldock, 2011b:32-33].

From the student perspective, a vital thread of employer engagement is through the provision of possible career paths. While some subjects are clearly vocational, with a self-evident career path, this is not the case for mathematics. Statistically, a high proportion of mathematics graduates enter the general area of finance – although this itself covers a very wide area, including accountancy, banking, investment, credit risk, actuarial work and insurance – but there are mathematics graduates entering almost every profession. Apart from the significant minority who intend to teach, most mathematics undergraduates do not have a clear idea of their career path.

Further details of projects relating to employer engagement, funded through the National HE STEM Programme, are available in a collection of reports edited by Waldock and Rowlett (2012).

Curricular strategies

When designing a mathematics programme, one key factor is the set of attributes that a graduate of the programme should possess. Some of these will evidently be mathematical but – as made clear in the QAA MSOR benchmark statement – others will be generic. Furthermore, it is often the generic skills that have the greatest influence on employability. The most important of these skills are:

- Self-confidence – especially in tackling new challenges.
- Self-awareness – knowing what you are good at, and being able to articulate and evidence this.
- Problem-solving skills – knowing how to think creatively and apply existing knowledge and skills in new situations.
- The ability to communicate effectively and appropriately to different audiences.
- The ability to work successfully with others – having inter-personal skills.

How can the mathematics curriculum encourage the development of skills such as these without losing core content? Appropriate learning, teaching and assessment strategies can be chosen so that students gain generic and subject specific skills at the same time. Mathematical modelling, project work and examples in operational research provide vehicles for students to apply skills in new situations, think creatively and work in teams. Providing assessment tasks that include writing and presenting – in various forms – help

students gain confidence incrementally. Success at the practical application of mathematics leads to greater levels of confidence and a greater willingness to tackle new problems.

Being self-aware, recognising and being able to articulate and evidence skills, can be developed through reflective practice. Processes to support this – such as personal development planning – can also lead to greater levels of self-confidence since students will be encouraged to reflect on their progress and look objectively at their developing abilities. Most people value skills already acquired less than those they have yet to acquire, and students are no different. It can represent a huge boost to self-confidence for a student to review their progress over a year's study to see just how far they have come. A departmental approach that embraces students as partners in learning, explaining the rationale for the curricular structure and how tasks have been designed to help them develop a variety of skills throughout their course, is more likely to lead to greater levels of engagement, satisfaction and, ultimately, achievement.

Relevant to this is the view of current students. This includes the expectations of students before starting a degree and, since the ability to recognise the skills that a degree provides is important, the views of students currently studying for a degree. Hind [Project 9], Inglis et al. (2012) and Challis, Robinson and Thomlinson (2009) surveyed student views and Cooper [Project 5] has reported a project which developed a student-led employability audit toolkit. Student and graduate awareness of the skills they develop at university was explicitly addressed by Bradshaw [Projects 1, 2] and by Chatwin [Project 4].

Co-curricular strategies

Institutions that include employability as a 'core value' or 'mission' usually do so because graduate employability is a key metric that feeds into institutional rankings. They are therefore impatient for improvement. However, since employability depends in large measure on attributes as well as skills – self-confidence, communication and team-working as stated above, compared to the ability to present well at interviews – these improvements can take time. To make a significant change to attributes such as these requires developmental activities co-ordinated throughout a programme of study. While institutions may be willing to invest in longer-term developments such as these, they also need to demonstrate quicker 'wins', and so are keen to support specific high profile initiatives, which also demonstrate market distinctiveness for recruitment and publicity purposes.

Such initiatives can include leadership courses, job application and interview training courses, business start-up support to stimulate enterprise and entrepreneurship and on-campus employment opportunities. To raise the status of these initiatives, students should have to explicitly apply to them, to show commitment and recognition of the benefits to be gained. While such initiatives are welcome, it is important to remember that these will only ever benefit a relatively small number of students; to make a significant difference to institutional graduate employment (or employability) rates, longer term, programme-wide, changes need to be embedded into the taught curriculum.

Career Management Skills (CMS)

Employability involves far more than possession of the generic skills listed by graduate employers as attractive – graduates must be able to proactively navigate the world of work

and manage their own career throughout their working life (Bridgstock, 2009). If the higher education mathematics curriculum is to provide such graduates it must incorporate activities that will develop these skills.

Employers can provide help in several ways – for example by providing role models, offering visits and giving presentations. The provision of work based, or work related, learning opportunities – as discussed earlier – will also clearly help. One successful approach is through career mentoring, in which a student mentee is paired with a mentor in employment, ideally a graduate of the student's course. The mentor provides the student with guidance on all aspects of career planning, helping them understand a particular industry, and advice on how to successfully gain employment within it. A scheme of this type in mathematics at Sheffield Hallam University has been supported through the National HE STEM Programme (Baxter & Waldock, 2012). A student taking part in the scheme said *"I have learned that I am more confident than I thought [and] I felt my communicating skills, face to face, have improved greatly as a result of the scheme"*.

As part of the increased awareness of the importance of graduate employability, many universities are incorporating CMS into the higher education curriculum. These include the more practical aspects of job application, including recognising how to articulate and evidence your own skills in order to produce a good CV and to perform well in interviews.

"The CMS modules provide fantastic opportunities for students to develop practical skills that employers consistently look for."

Fiona Bousfield, Teach First (University of Manchester Careers Service, 2011)

A review of different approaches to the incorporation into the curriculum of CMS has been published by Evans (2008).

Discussion at the Higher Education Mathematics Curriculum Summit focused on the need to build an understanding of the development and culture of the subject:

Summit Recommendation 3: *"Develop a collection of teaching resources on the development of mathematics – stories from history and more recent development of the discipline. These should aim to counter a view of mathematics as a static, completed body of knowledge and instead encourage awareness of the process of doing mathematics. They should develop students' awareness of the culture of mathematics."*

In response to this, Mann (University of Greenwich) and Good (University of Birmingham) were funded to create materials on 'Being a Professional Mathematician' (Mann & Good, 2012). Audio interviews were recorded with eight contemporary mathematicians and with five historians who described the work of important mathematicians in the past, emphasising the skills required in the workplace and the importance of networking in professional practice. Teaching materials were produced which could form the basis for tutorial activities¹.

The National HE STEM Programme project led by Messmer and Pugh at the University of Leeds [Project 12] included a series of employer-led events promoted as *"November*

¹ The materials are available at <http://www.beingamathematician.org/>

is *Careers Month*". Attendance however was poor, with students appearing not to be interested in attending such co-curricular activities. Feedback indicated that the timing of these events was crucial and – interestingly – that all activities should form part of the taught curriculum. The second year module 'Maths at Work' was introduced in 2012/13, incorporating career development activities with employer input. A series of video interviews with students returning from work placement were made to increase awareness of possible career paths for mathematics graduates and to motivate school students.

Two projects from the University of Greenwich have been particularly successful in engaging students with employers. Bradshaw [Project 1] led a project working with a charity focusing on improving the prospects of black minority ethnic (BME) graduates. This involved a combination of events for students and training sessions for staff, who were learning how to deliver similar employability events in future. The aim was to make mathematics graduates more employable through master classes, workshops and CV clinics delivered by senior staff from major companies. Student engagement was enhanced both by tailoring the events to their needs and by the participation of staff, emphasising the importance placed on skill development. The project built a stronger relationship between staff and students, and helped staff understand the attitudes and fears of students preparing to graduate and face an uncertain future in adverse economic conditions.

Linked to this is a further project, also led by Bradshaw [Project 3] who used social networking tools to invite selected Greenwich graduates to an event at which they described their careers. The graduates were chosen to represent a range of careers to which students could aspire, and to demonstrate that it isn't only the top academic students who can get desirable jobs. One very positive result was that some of the least confident students, who previously would not have engaged, attended the event; it had a breakthrough effect, so that attendance at subsequent employability workshops was much higher than would have been previously expected.

With similar intentions, Good [Project 8] led a project explicitly aimed at sustainably embedding careers awareness as integral parts of the higher education mathematics curriculum at Birmingham. Nehemiah [Project 13] has led a STEM-wide project establishing an industry-led programme of lectures for undergraduates across Wales during the first half of 2012. Several case studies gathered through the project "*Developing Graduate Skills in HE Mathematics Programmes*" (Waldock, 2011a,b) address the theme of careers awareness and guidance.

- Walker (University of Manchester) described the practice of building CMS into the programme at Manchester through the mathematics-dedicated 'Calculating Careers' event (Waldock, 2011b:12-13).
- Waldock described the curricular strategies employed at Sheffield Hallam University to prepare second year undergraduates for applying for industrial placements (Waldock, 2011b:28-29).
- Golden and Roberts described how the University of the West of England builds CMS into the curriculum through the Graduate Development Programme (Waldock, 2011b:32-33).

- Graham and Millar explained how the mathematics degree at Plymouth embeds careers awareness (Waldock, 2011b:44-45).

Work experience

Work experience can make a significant difference to a student's chances of gaining graduate level employment. There is strong evidence to indicate that authentic work experience contextualises learning, has a strong influence on graduate employment and should be integrated into course curricula wherever possible (Pegg, Waldock, Hendy-Isaac & Lawton, 2012). Research overwhelmingly highlighted that employers, students, graduates and higher education institution representatives value work-based learning (such as placements and internships) as particularly effective approaches to promote graduate employability (Lowden, Hall, Elliot & Lewin, 2011). The Wilson review (Wilson, 2012:37) went as far as to state that the *"lack of work experience appears as a key barrier to young people, including graduates, in securing employment"*. It also points out that there has been a decline in this practice in recent years from 9.5% of the total full-time cohort in 2002/2003 to 7.2% in 2009/2010 and that in the UK most sandwich placements are provided by a small number of universities. Furthermore, as is discussed by Mann (2012), even when a placement programme is available it is often the case that only a small number of students take advantage of it.

Barriers to the take-up of sandwich provision include the time pressures of application, uncertainty in securing a placement, strong peer group pressure to opt out, and difficulties in finding a placement close to the university or parents' home. Despite this, some universities can still succeed in getting a significant proportion of sandwich students placed, suggesting that sustaining a sandwich course structure may be a consequence of university culture, strategy and investment. Aston University, for example, aims to have all students undertaking a placement year by 2020 (Higson, 2012). With the increase from 2012 in tuition fees, a number of universities have included a fee waiver for the placement year, providing an additional incentive for students to apply for a sandwich degree. Research evidence (Mendez and Rona, 2010; Driffield, Foster & Higson, 2011) suggests that students taking a placement year gain a better degree. Although students are in competition for work placements, and therefore those succeeding may be more likely to be higher achievers, Driffield et al. (2011) have carried out a statistical analysis to demonstrate that even allowing for this, placements enhance academic performance.

Clearly, universities have to invest to provide placement opportunities for students and developing and maintaining relationships with employers is not cost free. Good experiences however lead to repeat placements at a lower cost, and can deliver collaboration in other areas of university services. It is in the interests of students, employers, universities and government that barriers to increasing the number of students benefiting from sandwich placements are removed.

If graduate employability is of importance to course designers, it seems clear that the curriculum should embrace some form of work based, or work related, learning. There are a number of models to follow – and not all require a full sandwich year. There is clearly much to be gained from helping students recognise, and articulate, the skills they gain from all forms of work, including voluntary and casual work. Many universities now operate a skills

award which recognises this, but it is possible for the course curriculum to also incorporate explicit recognition of the skills acquired in the workplace, wherever that may be.

Summit Recommendation 6: “Pilot of undergraduate students gaining experience of working in industry through short term placements (e.g. 2 hours per week)”

Partly in response to this recommendation, Mann (2012) has reviewed current models of industrial placements for mathematics undergraduates, finding three basic models of delivery:

1. Sandwich placements, usually lasting 12 months, between the second and final year of study. The proportion of the year group taking a placement of this kind is typically quite small, however.
2. Placements in schools and colleges, either via the Undergraduate Ambassador Scheme or by other means. Hind [Project 9] reports results of a project that involved undergraduate students observing, preparing and delivering material to GCSE and A-level classes. Although not a full year placement, the students involved gained valuable school and classroom experience. A related project led by Ayres at the University of Reading (Ayres, Glaister & Smith, 2012) involves undergraduates undertaking small, paid, learning and teaching projects to create curricular materials for use on their course.
3. Short term work placements, such as vacation internships or via a credit-bearing final year module. These are usually unpaid. Some examples of such internships have been reported (Taylor, Bond, Anderson & Kennedy, 2012; Cooper, Gillard, Graham, White & Wilson, 2011), related to the **Summit Recommendation 13: “Fund undergraduate students to undertake focused summer intern projects within universities”**.

In general, employers are likely to prefer the year-long placement as students can be trained to fulfil a more specific role and it gives time to get a sufficient return on this investment. The student is also more likely to gain technical, professional and workplace skills.

The learning that takes place as a result of an industrial placement can extend beyond the end of the placement itself. There are a number of programmes which incorporate a final year module in which students reflect on the learning from placement. The project led by Mann (2012) included the development of a 30 credit final year ‘Mathematics Industry Placement’ module, for example.

The effectiveness of industrial placements cannot be taken for granted. As discussed by Chatwin [Project 4], it is clearly necessary to introduce institution-wide mechanisms for evaluating all aspects of placement activity including the sourcing and monitoring of placements and the support provided to students while on placement

Work-related learning

Realistic work experience, and the recognition of skills gained from it, has a major impact on success at achieving graduate employment. In a review of HE STEM Employability Skills, on behalf of the National HE STEM Programme, Toland (2011) points out that with the availability of industrial placements becoming more limited, alternative mechanisms for students to gain indirect experience of industry are of importance. These include industrial simulation, project-based learning, industrial group projects and case-

studies, site visits and field trips, the influence of part-time and mature students and student involvement with professional institutions.

Many programmes are developing elements of work-related learning to complement or act as a substitute for work-based learning. This has led to **Curriculum Summit**

Recommendation 4: “Develop a bank of industry-based problems, suitable for undergraduate students, developed in consultation with industry partners and vetted.”

Two National HE STEM Programme projects were supported in response to this recommendation. The first, Industrial Problem Solving for Higher Education Mathematics (IPSHE), involved the creation of an on-line resource, IPSHE, which makes available to the community a repository of real, documented, industrial problems categorised as introductory, intermediate or advanced, corresponding approximately to the three years of undergraduate study. There are currently (May 2014) 58 problems in the repository². The second project was Industrial Problems in Statistics for the HE Curriculum, Davies, Marriott and Martignetti (2012). This provides a freely available web-based repository of problems, with relevant data, in one particular industrial area.

Chadwick et al. (2012) raise a question about the degree of realism that work related learning should exhibit. They summarise the outcomes of a workshop involving input from universities and professional bodies, emphasising that for real learning to take place, work-related learning must be authentic. It recognises that institutional constraints will limit the degree to which authenticity can be achieved. However some departments have set up industrial study groups which allow students to develop mathematical modelling skills by working on live industrial problems (for example O’Brien, Mason, Charpin & O’Sullivan, 2011; Singh, 2011). Project work offers great scope for developing realistic work-related mathematical modelling (for example Robinson & Jaworski, 2010).

Curricular strategies to develop realistic workplace skills, particularly inter-disciplinary team working skills and enterprise, are rare, but one approach is an innovation at Sheffield Hallam University called the Venture Matrix (Waldock, 2011b:20-21). This involves teams of students forming business ‘ventures’, each offering a service or a project. Project ideas can also come from external organisations, in which case these are ‘live’ rather than simulated projects. The ventures offering services pitch to work on one of the projects and if successful, work together to deliver the service or product. The work contributes to validated modules, so is credit bearing and is assessed. Over 2000 students took part in the 2012/13 session.

A number of projects developed specific modules to provide work-related learning through student engagement with ‘real world’ activities:

- Chadwick developed a second year module delivered to mathematics undergraduates at Salford in which they work in groups to solve employer-defined problems. The industrial partners are also involved in the assessment of the student work, and a key lesson was that employer involvement is paramount for success. Students valued

² The repository is available at <https://wikis.bris.ac.uk/display/ipshe/Home>

the employer feedback, feeling that this made the module content real and relevant (Chadwick, Sandiford & Percy, 2011).

- The project led by Messmer and Pugh at Leeds [Project 12] worked with key employers to develop real-world learning activities. These are incorporated into a second year module (Maths at Work) in which students work in groups, developing team working skills, on a variety of industrially-related problems.
- At Leicester (Waldock 2011b:42-43), the mathematics programme includes an optional second year module 'Business Applications of Mathematics' which requires students to work on four case studies based on real life business problems. Students also reflect on the transferable skills they are developing whilst working through the case studies and are encouraged to consider how they can evidence these in the recruitment and selection process for graduate employment. The programme also now includes an employment-based project module, developed as part of a National HE STEM Programme project [Project 11].
- In a similar way, Good [Project 8] is developing a 10-credit module (Mathematics in Industry) for second year undergraduates at Birmingham.

Chatwin [Project 4] also uses local employers to provide 'real world' experiential learning tasks, and produces materials for employers to explain the benefits for them in getting involved in the HE mathematics curriculum. Easson [Project 7] brought together employers and academic staff to make specific links between the higher education curriculum and 'real world' applications, following on from materials developed at Queen Mary, University of London for the More Maths Grads project, such as the booklet *Where the maths you learn is used*.

Conclusions

There is recognition on all sides that universities and employers have much to gain from closer collaboration and in an era of increased reliance for university funding on student tuition fees, the need to make curricula relevant, and enhance graduate employability is becoming paramount. This Chapter has presented practice-based evidence drawn from published literature and from projects carried out as part of the National HE STEM Programme of strategies that have been successful, highlighting the importance of:

- Including work experience wherever possible.
- Including work-related learning activities through authentic mathematical modelling exercises and project work.
- Utilising employers and alumni – presentations, role models and case studies.
- Enhancing employability by modifying the learning and teaching strategy rather than losing core content.
- A whole-programme approach, building employability through coordinated activities developmentally, and being explicit about this to students.

- Reflecting on learning, recognising the skills gained from curricular activities, developing the ability to articulate this confidently, improving self-awareness and self-confidence.
- Recognising the market for your graduates, and designing courses accordingly.

While recognising the challenges faced by practitioners wishing to develop their courses or programmes to more effectively develop student employability, these approaches may provide some valuable ideas and motivators for change.

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Chapter 15 : The neurodiverse mathematics student

Clare Trott

Abstract

There is an increasingly diverse population of students entering mathematics courses in higher education. While this enriches the environment it can bring challenges that higher education institutions need to address. This chapter will introduce the idea of neurodiversity and provide further details about four types: dyslexia, dyspraxia, Asperger's syndrome and attention deficit hyperactivity disorder. The main focus of the chapter will be to consider the impact for neurodiverse students of four factors in undergraduate mathematics: notes, lectures, assessment and departmental provision. The chapter will explore the barriers that higher education institutions often pose for the neurodiverse student and suggest ways in which mathematics can be made more accessible and departments achieve greater inclusivity. The discussions are illuminated through vignettes of individual neurodiverse students and their journey through mathematics.

The legal context

In the U.K. the *Disability Discrimination Act* (DDA) (1995, 2001, 2005) made provision for Employment, Goods and Services, and Post 16 Education. According to the DDA, it is illegal to discriminate on the grounds of disability. It is mandatory for public sector bodies to identify how they will eliminate discriminatory barriers as well as how they will promote positive attitudes towards disability. The Act specified that:

"A person has a disability ... if he has a physical or mental impairment, which has a substantial and long-term adverse effect on his ability to carry out normal day-to-day activities."

DDA (1995:1.1)

"It is unlawful for the body responsible for an educational institution to discriminate against a disabled student in the student services it provides, or offers to provide."

DDA (1995:F3.28R(2))

“In relation to student services provided for, or offered to, students by it, disabled students are not placed at a substantial disadvantage in comparison with students who are not disabled.”

DDA (1995: F5.28T(1b))

The *Equality Act* (2010) built upon the foundations laid down by *The Special Educational Needs and Disability Act* (SENDA, 2001). It covered nine ‘protected characteristics’ including disability. Under parts 2 and 4 (SENDA) of the DDA (carried forward in the *Equality Act*), it is legally required for institutions to make “*reasonable adjustments*” for disabled students to enable access to their goods and services. In addition, institutions are required to ensure disabled students are not unjustly disadvantaged and are not less favourably treated by way of reasons related to their disability. Institutions are further required to put in place “*anticipatory measures*”. These might include, for example: the provision of assistive technology, adapting accommodation to meet individual needs, the provision of notes in an appropriate and accessible format, or arranging alternative examination and assessment arrangements for students where required.

The legislation is designed to ensure the removal of unnecessary barriers and promote best practice for the inclusion and equality of opportunity for all. Areas to be considered fundamental to this are those such as curriculum delivery, appreciation of different learning styles and provision of a variety of teaching methods. Indeed, putting in place good “*anticipatory measures*” will pre-empt potential barriers for disabled students so that “*reasonable adjustments*” can then be implemented more readily (see also Chapter 16 which considers technology for inclusive access within the mathematical sciences).

Terminology

Throughout this chapter, the term ‘neurodiversity’ (Oliver, 1990) will be used which is “*an umbrella term, for it encompasses a range of specific learning differences, including dyslexia, dyspraxia, dyscalculia, ADD, AD(H)D and Asperger’s*” (Grant, 2009:35). Neurodiversity is preferred to “*Specific Learning Difficulty*” (SpLD), since it perceives learning differences as a positive statement about differentiation, rejecting the notion that differences are dysfunctional (Grant, 2009). Such a stance clearly has implications for departments and curriculum delivery since all neurodiverse students must have same level of opportunity and choice as their neurotypical peers. The term ‘neurotypical’ refers to the non-neurodiverse.

In the UK, Disabled Students’ Allowances (DSAs) provide extra financial help to students with a disability, mental health difficulty, medical condition or specific learning difference, such as dyslexia or dyspraxia. They are designed to help meet the extra costs students can face as a result of their disability or specific learning difference. This includes specialist equipment needed for studying and a non-medical helper, such as a study support tutor (Loughborough University, 2013a). For neurodiverse students in general, specialist study support through DSA is predominantly focused on text-based material, time management and organisational skills. However, when supporting neurodiverse mathematics students it is preferable for the tutor to have some subject background since there are frequently specific issues associated with reading mathematics, writing mathematics and memory for mathematics. Mathematics study support follows the same ethos, principles and methods

as the text-based specialist study skills support, but focuses more on working with non-text material such as equations and proofs.

There are frequent overlaps of the neurodiversities but, in the context of this chapter, they will be considered individually. Each individual will have a unique profile of strengths and weaknesses. The Institute of Physics (2013) emphasises the importance of acknowledging and embracing the strengths and recognising and rewarding neurodiverse contributions and abilities.

Neurodiversity

This chapter will consider four neurodiversities: dyslexia, dyspraxia, attention deficit hyperactivity disorder (AD(H)D) and Asperger's syndrome. Each will now be considered, giving indications of some of the issues involved as well as the strengths which often accompany students with these neurodiversities.

Characteristics of dyslexia

"Dyslexia...mainly affects the development of literacy and language related skills. It is characterised by difficulties with phonological processing, rapid naming, working memory, processing speed"

(British Dyslexia Association, 2010).

In particular, working memory acts as an information-processing unit essential to learning particularly in conventional educational settings (McLoughlin, Leather & Stringer, 2002). The report of the working party on dyslexia in higher education (Singleton, 1999) notes some of the issues resulting from dyslexia include reading, retaining the meaning of text, difficulty gathering learned facts effectively in examinations as well as disjointed written work or omission of words. However, the report also highlights *"distinctive talents"*.

Cooper (2006) notes that dyslexic people are likely to think more visually while neuro-typicals would be more likely to think verbally. A dyslexic student might therefore approach a problem from a more visual perspective, make atypical connections and develop new ideas (Cooper, 2009).

Trott (2013) suggests that dyslexic students may be particularly drawn to STEM subjects, where the perception is that literacy skills are less demanding. She further notes the implications of the dyslexic strengths and weaknesses for mathematics, a logical and analytical system with hierarchical structures.

Characteristics of dyspraxia (Developmental Co-ordination Disorder, DCD)

"Dyspraxia affects the planning of what you do and how you do it. It is associated with problems of perception, language and thought."

(Dyspraxia Foundation, 2008).

Drew (2009) reports that as a consequence, the dyspraxic student is likely to spend longer on work and sometimes not hand it in on time. They may struggle to keep up in lectures and struggle to arrive in time. Issues with memory will result in slower retrieval of information especially when placed in a more stressful situation such as a test or examination. Further

issues may arise in holding a plan in the working memory while executing it (Drew, 2009). This will have implications for longer problem solutions in mathematics.

Dyspraxia is associated with inattentiveness and disorientation, together with ‘messy’ work and issues with illegible handwriting, typing and mouse control. This can sometimes be similar to dysgraphia, which is characterised by difficulties in writing (Nicolson & Fawcett, 2011). With reference to mathematics, Drew (2009) notes a tendency to reverse or mistype numbers, signs or decimal points as well as issues with drawing and spatial awareness, such as drawing tables and graphs. On numerous occasions in undergraduate mathematics, sketch graphs are given or expected to be produced by the student.

In contrast to the above, Drew (2009) describes the strengths that dyspraxic students bring to higher education. These strengths include persistence coupled with a strong determination to succeed. They are hard-working and highly motivated. As original thinkers, working ‘outside the box’ will often lead them to more creative solutions and imaginative answers to problems (Drew, 2009).

Characteristics of Asperger’s syndrome

Asperger’s syndrome is a form of autism. It is mainly characterised in three ways: communication, flexibility of thought and the social environment.

(Martin, 2009).

With regard to communication, Martin (2009:159) contends there is a “*need for order and clarity*” so that institutions communicate meaning with exactness and precision. Students with Asperger’s can find it more difficult to follow social conventions, so that they may appear to invade someone’s personal space or interrupt at a seemingly inappropriate moment. They may not be able to read and interpret facial expressions or emotions. The National Autistic Society (2013) emphasise the need to keep sentences short and be clear and concise. This is essential when giving instructions at both an academic and everyday level.

Martin (2009:160) emphasises the importance of “*a level of flexibility around the way services are delivered*”. Asperger’s students may be more inflexible in their thinking than their neurotypical peers and adhere to a routine. Adaptability is likely to be an issue, seeing or doing in the same way. This is likely to include Asperger’s students having difficulty seeing things from another point of view and empathising with their peers.

A social environment can be difficult. The Asperger’s student may also experience high sensitivity to even moderate levels of sensory stimuli and processing information from the social world can be challenging. It follows that friendships may be difficult to form as other people can seem confusing. Further, some neurotypical students may find it uncomfortable to work with Asperger’s students. This can lead to further isolation.

Asperger’s students can be very focused and engage whole-heartedly with their mathematics. They are likely to be logical in their thinking and can appreciate the details and complexities of mathematical arguments.

Characteristics of Attention Deficit Hyperactivity Disorder (AD(H)D)

AD(H)D has three key indicators: inattentiveness, hyperactivity and impulsivity. ADD is similar but without the hyperactivity indicator.

Inattentiveness or attention inconsistency (Hallowell & Ratey, 2006, cited in Colley, 2009) results in students becoming easily bored and distracted and so shifting from one incomplete task to another. Further, there may be planning or organisational barriers and a tendency to lose or forget equipment (BRAIN.HE, 2006).

“It is not that ADDers do not attend – they just attend to everything.”

Derrington (2005)

BRAIN.HE (2006) describes hyperactivity as being restless and fidgety, frequent talking and doing several things at once. The behaviours associated with impulsivity are interrupting others and difficulty awaiting a turn in a group, often making seemingly inappropriate comments.

Colley (2009) extols the strengths of students with AD(H)D. These include: creativity and originality, being good problem-solvers since they are often able to see the ‘bigger picture’. They are likely to have an ability to focus intensely for a time, have high levels of energy, be intuitive and adopt a ‘risk-taking’ approach that can lead to important discoveries (BRAIN.HE, 2006).

How to foster equity

This section focuses upon the ways in which barriers faced by neurodiverse students in mathematics can be addressed. Four areas will be considered: notes, lectures, assessment and departmental provision.

Notes

It is essential that neurodiverse students have access to lecture notes. It is good practice for these to be available in advance of the lecture. In particular, dyslexic or dyspraxic students will frequently struggle to read notes which are seen for the first time in a lecture. The expected reading speed of students in lectures is usually determined as that which is representative of their neurotypical peers. Furthermore, the neurodiverse student will need longer to process the information presented in the notes. It is not merely an issue of reading comprehension. Obtaining mathematical meaning from the text requires the different variables to be recognised, become related to each other and to, as yet, unknown data or further variables. Thus, more time is required to both read and process notes.

Many neurodiverse students will find writing notes difficult, both in terms of the handwriting and also the speed of writing. This is particularly relevant to both dyslexic and dyspraxic students. If required to take notes, they are likely to have many gaps and become increasingly illegible. Providing a neurodiverse student with full notes in advance of a lecture will enable them to read these in advance and keep pace with the class. It will also enable the student to listen more attentively, process the information, and avoid the struggle to write their own notes. In the same way, an AD(H)D student may also have gaps in their notes as a result of inconsistencies in attention during lectures. Providing notes will

ensure the neurodiverse student has a full and useful set to work or revise from. If different solutions to the same problem are possible, then these should also be provided as many neurodiverse students see a problem in different ways.

As a first year dyslexic mathematics student, Adam found it extremely difficult to take notes in his classes. He frequently missed out large sections. In a tutorial Adam found it impossible to keep pace with the problems and solutions discussed. When alternatives were offered he found this useful but when the solutions appeared electronically sometimes several days or weeks after the tutorial, only one version of a solution was given.

Neurodiverse students often experience visual stress when reading. Wilkins (1995) defined visual stress as the inability to see comfortably without distortion and discomfort. It affects approximately 12% of the general population but around 65% of people with dyslexia (Evans 2002). The result is that the text may appear to move around or appear to fall off the end of the page, making it difficult to access the material. However, there are a number of ways in which material can be presented in order to minimise visual stress. Such ways include changing the colour of the background or the font, altering the font size or font to a sans-serif font that is easier to read, increasing the line spacing and not justifying text which causes irregularity in the gaps between words. The exact specification of these factors will vary from student to student and be an individual preference. For example, one student may have a blue colour preference, experiencing less visual stress when using a blue overlay or blue background. A second student may find that green gives the same effect for them.

It is therefore important that electronic versions of all notes and handouts are made available to students and not as pdf documents. Pdfs create a barrier to making documents more accessible, particularly in terms of the ability to manipulate those features that help to reduce visual stress.

The organisation of the content of notes can also provide greater accessibility. Neurodiverse students need a well-defined structure with a logical flow through the material. It is important that diagrams, tables and charts are placed in the correct position in relation to the words in the text that link to the diagram and should be clearly labelled. Furthermore, if the notes are organised into manageable parts, they become more accessible to students with reading and processing issues. The use of a bullet point format is useful. Appropriate headings create greater clarity too. Such features enable the dyslexic or dyspraxic student to access the material more easily and also aid an AD(H)D student to focus their attention and aid concentration. The headings could be, for example, Definition, Theorem, Proof, Explanation, Limitations, Key Points and Worked Example.

The notes should always give clear definitions of all notation and terms used. Providing a separate list is particularly useful as a reference point for the neurodiverse student who may experience issues with working memory and need to more frequently check the meaning of each variable used.

Michael Faraday, who was believed to have had dyslexic tendencies, in 1851 wrote:

“Mathematical formulae, more than anything else, require quickness and surety in receiving and retaining the true value of the symbols, and when one has to look back at every moment to the beginning of a paper, to see what H or A or B mean, there is no making way. Still, though I cannot hold the whole train of reasoning in my mind at once, I am able fully to appreciate the value of the results...”

West (1991:107)

The provision of full notes ahead of lectures for the neurodiverse student is therefore an essential requirement. Considerable opposition is voiced in some communities, who postulate that such a provision would result in students not attending lectures and thereby missing further information. One solution suggested by Linehan (2012) is to provide neurodiverse students each with a memory stick containing a few weeks worth of notes. Regardless of the means, for the neurodiverse student the availability of notes that can be transformed into an accessible format (for technical and practical considerations regarding accessible formats see Chapter 16) means they can attend lectures and engage with the material in a more meaningful way.

Lectures

Another essential aspect to consider is the degree to which lectures are accessible to neurodiverse students. In the same way that notes allow the student to prepare, it is important for the neurodiverse student to know what the pre-requisite mathematics is so that they can recap these aspects. Retrieval of information from previous modules or from earlier in the current course may be an issue for some dyslexic and dyspraxic students who are likely to have issues with memory. By providing the opportunity to revise the required topics, the student will be more fully prepared. It is further recommended that in-built and frequent recaps will enable the student to fully assimilate material.

If material relates to real and practical scenarios, the neurodiverse student is likely to relate to it in a more tangible way. Dyslexics have strong visual memory and are readily able to visualise situations, often seeing solutions in their mind. They tend to be holistic rather than sequential in their approach. Cooper (2006) found that 80% of dyslexics preferred to solve problems in a visual way, as did 55% of non-dyslexics:

“...what should we do with students for whom the easy things are hard and the hard things are easy – those who naturally jump to the end and skip over the beginning, those who jump to the world of intuitive images without having mastered the basic elementary steps?”

West (1991:241)

Tom, a second year dyslexic student studying mathematics and physics, was able to visualise second order partial differential equations such as the heat equation in terms of real physics-based problems and the practical experiences and phenomena that he has observed in the lab. This helped him to ‘see’ the problem and more fully realise the mathematics. Tom used this visualisation at each step in the mathematical solution, ensuring it made sense to him.

It is recommended that non-linear formatting be employed whenever possible. This could include a tree diagram for partial differentiation (Trott, 2006a). Although accepting that mathematics essentially requires the development of a linear argument, a longer problem solution places a substantial load on the working memory since it is necessary to retain and manipulate several intermediate outcomes in order to achieve a final solution (Trott, 2003). This will discriminate against students with weaker working memories. As McLoughlin (2001:124) states: *“An inefficient working memory system will clearly undermine skill acquisition and learning...”*

Information presented in bullet point format enables the dyslexic student to *“keep in mind the steps of a procedure and follow it”* (Meehan, 2010:47). Further, Cooper (2009) maintains:

“...assuming that everyone can take in and remember sequences of information (particularly auditory information) is disabling and entirely unnecessary. The point is to understand and use, not to remember.”

Cooper (2009:70)

Colley (2009), writing in the context of AD(H)D, recommends providing videos of lectures. This allows the student to assimilate the material in smaller sections and focus on each part. It provides an aid to concentration.

Clear and unambiguous lectures are supportive for neurodiverse students, particularly for Asperger's students, who prefer *“clarity, predictability, logical explanations and tangible expectations”* (Martin, Beardon, Hodge, Goodley & Madriaga, 2008:10). This is because they *“contribute to security and underpin conditions for success”* (Martin et. al., 2008:10). Indeed such factors may be considered intrinsic to the subject of mathematics itself, although in practice, the teaching may not always concur.

A statement or slide at the beginning outlining the order of what is to follow is helpful as is a summary of the main points at the end. Further, the use of colour can greatly enhance the lecture, enabling the neurodiverse student to bring together those aspects that are similar. For example, coding the same variable in the same colour whenever it is met or using a colour for a frequently repeating equation or step can be helpful. The use of colour needs to be consistent throughout the module. Again, this will ease the load on the working memory for neurodiverse students.

In addition, the provision of a module or course glossary containing all new technical words together with their definitions, the notation used and its definition (particularly Greek characters) is good practice: for example the frequency, amplitude and period of a wave with its associated variable. Reducing the burden and the anxiety of having to recall this information will free-up working memory since anxieties deplete working memory resources (Beilock, 2008). This will then contribute to the student achieving their potential.

Alongside the lectures, the module is likely to have a recommended course textbook. Given the issues some neurodiverse students experience with reading, it is important that the chosen text not only fulfils the mathematical requirements but also presents a clear layout. It is often the case that mathematical textbooks intersperse text with non-text, both algebraic notation and calculations. Diagrams are frequently positioned out of sequence, perhaps on another page and have their own individual subtext in italic, smaller or different

font. It is this irregularity that makes it difficult for the dyslexic student to access the material (Henderson, 2012). Trott, (2012) states:

“Examples are often worked out in the middle of information or text on how to complete the calculation, so that if you have trouble reading the connecting text you may ignore both the text and the example.”

Trott (2012:25)

...a very able dyslexic mathematics student with a strong visual memory and a high level of numerical competence. However, she was a significantly slower reader compared to her peers and her reading was also very inaccurate. During reading she frequently lost her place, repeating or skipping lines. A specific issue for Kate was the use of a course textbook.

For example, half way down a page the text referred to a figure that was on the following page, or even a couple of pages further on. Turning over to study the required figure, Kate was unlikely to return to the point where the text had left off. The chapter was therefore rendered meaningless.

Finally in relation to lectures, Cooper (2009) states:

“Listening to long explanations is extremely difficult for most dyslexics who usually experience lectures as a poor form of education.”

Cooper (2009:68)

Assessment

For the majority of mathematics modules within higher education it is still the case that assessment is predominantly, and sometimes exclusively, by formal examination (Iannone & Simpson, 2012). A piece of coursework worth only 10% of the module is, at best, notional. A well-constructed piece of investigative coursework may allow the neurodiverse student to develop their own ideas outside of the constraints of the examination and demonstrate their underlying mathematical ability. However, its credit-bearing tariff means that, within the full picture, it is a minimal contribution.

All too often, coursework for mathematics modules is a class test akin to a mini examination or a computer assisted assessment. The latter, and sometimes the former, are marked solely on the accuracy of the answer submitted. These forms of assessment are likely to disadvantage neurodiverse students. In transferring from one medium to another, such as from the calculator to the test paper or screen, a dyslexic or dyspraxic student may struggle to copy correctly or may reverse digits (Trott, 2003). For AD(H)D students, Colley (2009:173) commented *“because focusing on a selected task will be hard, they will tend to make so-called ‘careless’ mistakes”*.

The main assessment tool remains the examination (Iannone & Simpson, 2012). It is pertinent to consider how far currently set papers reflect the intended learning outcomes of the module and how far they enable neurodiverse students to display their true

mathematical ability. Learning by rote and recall from memory are precisely those areas in which neurodiverse students are known to struggle.

Trott (2012) says:

“Definitions and theorems feature in many examination papers, often requiring students to state a particular definition or theorem, before going on to reason a proof. When statements of definitions or theorems (without proof) are assessed, it is a test of rote recall.”

Trott (2012:27)

And continues:

“While it is accepted that certain definitions and theorems are of prime importance, their rote learning appears to place at a disadvantage those students who find such learning difficult, but who fully comprehend the mathematics and who can develop the proof and utilise it appropriately.”

Trott (2012:27)

Indeed, without the recall of the theorem, a proof cannot be reasoned, thus incurring a double penalty. Indeed as Rob, a second year dyslexic mathematics student noted in 2006:

“There are about 60 theorems in this module, I cannot learn them!”

Trott (2013)

Rob is far from unique. John was another second year dyslexic student studying mathematics and accounting.

John's mathematical ideas, his engagement with his subjects and insight in problem solving were particularly exceptional and he had the ability to comprehend mathematical concepts easily. However, his retrieval from memory was particularly weak. He was repeatedly frustrated by examination questions that required him to state a definition or recall a formula that was not given in the standard student booklet provided. Consequently his marks did not reflect his ability.

John was however able to access the meaning of questions that many dyslexic students find difficult due to the wordiness of the context in which the mathematics is embedded. Such questions are more frequent in statistics and probability modules. Often the context contains irrelevant or spurious information that simply increases the load on the dyslexic reader. For an AD(H)D student, such irrelevancies can act as internal stimuli that are a distraction resulting in loss of focus (Colley, 2009).

Clear instructions regarding coursework are essential. Setting and hand-in dates should be clearly stated and adhered to. Martin (2009:159), referring to Asperger's, comments *“specificity regarding assignments may need to include detailed instruction about the process, as well as the desired outcome. Lack of clarity can cause anxiety”*.

John was holistic in his approach to problem solving. The issue was not how to solve a problem but rather how to document this in a way that conveyed meaning to the marker.

However, in contrast, Colley (2009) noted that some AD(H)D students may prefer examinations to coursework, which they may struggle to prioritise and organise in order to hit deadlines.

As mentioned above, it is far from good practice to award marks on the basis of answers only, but it is equally good practice to consider less well documented solutions. This may include mathematical arguments that may be less sequentially ordered or that skip over an expected step. Other issues arise in the marking of work presented by neurodiverse students, particularly with reference to copying and transcription errors as well as issues of the neat alignment of mathematical arguments. It can be argued that the essential meaning should remain clear, as is the case with a spelling error in an essay. Another way forward is to consider the question itself and try to introduce intermediate results of the form “*show that...*”. This result can then be used in subsequent parts of the question. This will act as a checkpoint for the student who has made a copying or transcription error.

Karl was a dyspraxic mathematics student in his first year of undergraduate study. His difficulties were particularly associated with handwriting and alignment. He used a computer for his assessments. However, his thinking processes and working out were paper-based and frequently illegible. Through his learning support tutor he was introduced to the idea of using squared paper with large squares for his digits or characters. This enabled Karl to line up his work in a more orderly manner. Matrices were particularly prone to alignment errors.

Good feedback is also essential. The neurodiverse student may well become frustrated by a single grade or mark without explanation. Good feedback should enable students to understand why a mark has been given and how to improve on this in future work. For a large class, it could be made clear to all students that such feedback can be given if requested. Furthermore, for some students, reading the information provided can be difficult and it is suggested that an audio recording supplement the written feedback.

Other forms of assessment should be considered. Group work can be advantageous to many students. However, students with Asperger’s syndrome who may struggle with peer relationships and communication are likely to find it difficult to take an active part in these projects since the dynamics of the group situation may be unclear to them. The challenge is to integrate the student into the group. Madriaga and Goodley (2010:122) state: “*Responding to the apparent isolating affects of AS is a key concern for practitioners and researchers*”. It is also the case that dyslexic and dyspraxic students may process information in different ways and neurotypical students are often unclear in their communications within the peer group situation. This has implications for peer group work to be fully inclusive, ensuring all students are an integral part of the work of the group. Thus the

neurodiverse student can feel side-lined. It is recommended that departments emphasise the need for inclusive group working and, where possible, try to smooth the group work process. Cliffe and Bradshaw (2012) outline a possible framework for minimising barriers to group work (see also Chapter 13) in mathematics. Factors include: providing explicit educational rationales, providing scaffolding, using carefully designed group allocation, including group working skills as learning objectives, ensuring explicit rules of engagement, transparent and fair marking and explicit reflection.

There are several possible alternative ways to assess mathematics. These include open book examinations, a portfolio of work, a presentation, a poster or a project report. Allowing students to choose the means by which they are assessed, Symonds (2009:252) contends that *“introducing new assessment tools is not readily welcomed because they are unfamiliar, but this is not a reason to withhold the offer of choice”*. However, offering a choice of assessment method provides inclusive rather than alternative assessment (Waterfield & West, 2009). An exemplar case study (Easterbrook, Parker, & Waterfield, 2005), although drawn from engineering, provides a good model. Student choice is based on *“their own perceived strengths and weaknesses”* and this is *“a key component of making assessment inclusive”* (Waterfield & West, 2009:2). Symonds (2009:252) further states: *“Assessment should be an integral part of the creative process of pedagogy”*. With this in mind, it is essential to select more inclusive forms of assessment that satisfy the intended learning outcomes and allow all students to demonstrate their mathematical abilities. However it is noted that there are clear implications for staff input and time.

Departmental provision

Earlier in this chapter it was noted that the selection of course text books is important in order to facilitate clarity and aid students who find reading more challenging. Furthermore, prioritised reading lists help guide the student to organise their reading and time management since it enables them to *“target the most important sources”* (Du Pre, Gilroy & Miles, 2008:41). The prioritised reading list should also indicate the location and availability of the book or journal online or in the library. It is important to work with the institution's library so that a well illustrated guide to the departmental section of the library is available that will further help neurodiverse students who may confuse reference numbers or who experience difficulty with finding their way round or following directions.

Departmental, if not institutional, provision should also include a recommended calculators policy that takes account of visual stress. Loughborough University implemented such a policy following research by Trott (2006b). This policy specifies those calculators that not only satisfy the requirement *“two-line scientific”* but also states the models that minimise visual stress.

Many neurodiverse students find it useful to have a framework to enable them to better structure project reports or extended pieces of essay-style coursework. These are often available through university libraries (for example Loughborough University, 2013b). Grant (2009) contends that mathematics and physics can appeal to students who struggle with essay writing. Further, Cooper (2009), states that for those who think holistically there is no beginning, middle and end. Exemplar reports (Drew, 2009) are particularly helpful; they combine generic report or essay writing with more subject specific skills.

Another issue that sometimes arises is concurrent deadlines for multiple pieces of coursework and problem sheets. Time management and the organisation of these can pose issues for the neurodiverse student. With reference to AD(H)D, Colley (2009) noted problems in sustaining attention and completing routine tasks.

Anton was a mathematics student with AD(H)D. He experienced particular issues with time management, organisation and concentration. Coping with lots of concurrent overlapping pieces was a concern as was hitting the necessary deadlines. Anton needed some reasonable adjustments in terms of spacing his pieces of coursework and support for managing deadlines.

Departmental mentoring is sometimes used to support students with similar issues to Anton, enabling the student to better prioritise work. With reference to Asperger's syndrome, Hughes, Milne, McCall, and Pepper (2010:12) state: "*Late submission of coursework can be frustrating for the academic and disastrous for the AS students if it is not recognised early enough*". They go on to suggest a mentor can be of benefit in helping the student to manage deadlines. The mentor provides a means for greater engagement with their studies.

It has already been seen how valuable it is to provide the student with a list or booklet containing all necessary formulae, theorems, technical terms, notation, standardised measures and a glossary of terms used throughout the course, as in the case of John discussed earlier. To avoid further misconceptions and confusion, it is advantageous to consider a standardisation of notation across a department. John encountered a variety of notation throughout his course. Often different notation was used to represent the same concept in separate modules. When recall of notation from memory is profoundly difficult within the constraints of one module, the enormity of this across a whole course becomes apparent.

Transitions into higher education and from higher education into the workplace can be difficult for neurotypical students. Therefore it follows that neurodiverse students will benefit from the implementation of policies and initiatives to help smooth the pathways. Martin (2009) emphasised the inappropriateness of some high stimulus fresher activities for Asperger's syndrome students. The provision of a quiet and low stimulus space is essential during induction week and beyond. This would also be beneficial to some AD(H)D students who experience distraction in multi-stimuli environments. Resources that aim to ease transitions include a series of videos to enable Asperger's students to experience a more rewarding transition into higher education (Hughes, Darwen, & Coleman, 2013). Such an initiative from The Department of Physics at the University of Manchester seeks to lay inclusive foundations.

Departments of Mathematics can often provide a supportive learning environment for the neurodiverse student. A well-informed designated member of staff acting as a departmental disability co-ordinator will provide a central point of contact for the student and enable any ensuing issues to be managed in a timely and effective manner, liaising between the support services, academic colleagues and students. The department as a whole should be a place where neurodiverse students are welcomed and supported at all times:

“...learning support is an intrinsic part of teaching and if teaching/assessment methods do not anticipate the diversity of learning styles and stances and are therefore not supportive of learning then any changes may need to start with those methods...”

Herrington (2001:172)

Checklist

From the points raised in this chapter, a checklist has been formed (Table 15.1). These can be addressed at both the individual and the departmental level. When using the checklist, try to focus on each point and consider how far this aspect of good practice is currently achieved and what measures could be put in place to make the environment more inclusive for the neurodiverse student.

Notes: Do you...	Yes	No
Provide full notes?		
Offer alternative routes/solutions where possible?		
Provide electronically available notes/handouts ?		
Provide accessible format, for example not pdf?		
Give well-structured and logical notes?		
Have clearly labelled diagrams in correct position?		
Use bullet points where possible?		
Use clear headings throughout?		
Provide definitions of all notation and terms used?		
Lectures: Do you...		
Provide a list of pre-requisite mathematics?		
Build in frequent recaps?		
Use real and practical scenarios where possible?		
Employ non-linear mind-maps or similar?		
Have video recordings of lectures?		
State the aims and summarise key points?		
Use consistent colour where possible?		
Provide a glossary of all notation and terms used?		
Recommend a text book with a clear layout?		
Assessment: Do you...		
Give due weighting to coursework other than tests/examinations?		
Not use ‘answer only’ coursework?		
Write exam papers so that rote recall is not required?		
Consider the ‘wordiness’ of questions?		
Give clear instructions for coursework, including hand-in?		

Provide feedback as an audio recording when requested?		
Offer a range of assessment to demonstrate ability?		
Try to remove barriers to group work for neurodiverse students?		
Departmental Provision: Do you...		
Prioritise reading lists?		
Have a departmental guide to your section of the library?		
Have departmental exemplars to guide essay-style work?		
Have a departmental calculator policy with advice for dyslexic learners?		
Have a departmental policy to ensure notation consistency?		
Provide departmental initiatives to aid the transition to higher education?		
Have a designated departmental disability co-ordinator?		

Table 15.1: A checklist for supporting the neurodiverse learner.

Concluding remarks

There is little doubt that an increasing number of neurodiverse students are now enrolling on undergraduate degree courses. This chapter has endeavoured to describe four common neurodiversities and some consequential transitional issues for neurodiverse students as they enter university. Consideration of points raised in this chapter and the accompanying checklist will enable both new and experienced mathematics lecturers to adopt good practice in support of such students both at an individual and departmental level.

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Chapter 16:

Creating an accessible learning environment: Anticipating and resolving practical barriers

Emma Cliffe

Abstract

Increasing access for disabled students to higher education is a priority for higher education institutions and their associated departments within the UK; this is captured in both educational practice and equal opportunities legislation. Mathematics poses specific accessibility challenges beyond those considered in generic good practice. Traditionally, mathematics teaching depends on handwritten delivery, printed mathematical material, use of pen and paper and a broad range of mathematical software. This leads to practical barriers for many disabled students.

Mathematics students' experience at university differs substantially from school. Students will have previously been taught in small classes, perhaps with in-class support near their current level of mathematical fluency. These classes provide regular opportunities to clarify understanding with a teacher and typically refer to a small number of resources. Finally, students in schools interact with only a small number of peers and staff and may use no mathematical software. In this environment, the use of ad hoc methods of communicating mathematics, for those students who cannot use paper and pen effectively, are easier to accommodate; providing a limited range of textbooks in alternative formats may also be straightforward. Hence, students may reach the university with limited experience of: accessing a large corpus of mathematical knowledge; independently producing mathematical documents using assistive technology; and, learning in large groups without direct interaction with staff. Students may therefore face practical barriers to accessing mathematics at this transition. This chapter considers how such barriers may be anticipated and resolved so as to create an accessible learning environment.

Background

Revisiting the legal framework

As noted in Chapter 15 there is a legal requirement to anticipate and resolve barriers to accessing the learning environment in a proactive manner. In addition to anticipating barriers staff should be aware of additional adjustments that may be reasonable. The modes in which mathematics is communicated, including handwritten delivery, mathematical documents, books, slides and websites, use of pen and paper when working on regular exercises and use of a range of specialist mathematical software, can present specific access challenges to mathematics educators. For instance, the Quality Assurance Agency (QAA) for Higher Education states in the Mathematics, Statistics and Operational Research (MSOR) Subject Benchmark Statement (QAA, 2007) that seeing extended arguments developed in ‘real-time’, with appropriate emphasis, benefits students and hence that the *“traditional board-based lecture continues to have substantial merit”*. This pedagogic good practice enables making the desired engagement with concepts and logical arguments.

In contrast, accessibility guidance such as the Equality and Human Rights Commission (2011) observe that a common anticipatory adjustment is that of providing resources in advance, in electronic formats. In non-mathematical subjects this may be a relatively simple matter of providing electronic resources the lecturer has already created, such as slides or handouts, unaltered. The use of ‘real time’ exposition may mean that no notes exist which can be converted for disabled students in advance and this creates something of a contradiction for mathematics educators. Even if a lecturer has produced notes, mathematical and other scientific equations rendered in common electronic formats are not always accessible or students may not have experience in accessing or converting mathematical resources using assistive technology.

The *MSOR Subject Benchmark Statement* (QAA, 2007) also notes the importance of *“regular opportunities to undertake appropriate exercises”* and, in some MSOR programmes, of direct experience of mathematical computing. Students who cannot effectively use pen and paper may use assistive technology or human support to work with mathematical text as well as with mathematical software. Students may not have acquired these skills prior to university, or the skills which they have, such as the use of mathematical Braille, may be difficult to use effectively in the new learning environment.

As the above examples illustrate, resolving barriers in a university mathematical learning environment, and hence effectively meeting the legal requirement, involves a level of planning which may not be required in a less mathematical subject.

Disabled students in higher education mathematics

A disabled student may draw on support from a needs assessor, assistive technology trainer, disability adviser, study, communication and examination support workers and document conversion staff. A computing services department, e-learning team and librarians are involved in ensuring access to computer facilities, virtual learning environments and print and electronic information sources. Most of these professionals will not have studied mathematics and, not unreasonably, may assume that generic approaches to access remain valid. Understanding of the nature of mathematics, how it is communicated, taught and

assessed, rests with the subject department and hence input from mathematics staff can be crucial. However, mathematics staff are likely to have only limited knowledge in the domains of the professionals listed above. Not unreasonably, they may assume that the student has been provided with assistive technology, support and advice appropriate to the nature of their studies and the ways in which mathematics content is communicated.

Applicants and their advisers may make assumptions about the university mathematics learning environment, the skills required and strategies which might be appropriate. The university learning environment is superficially similar to that experienced in school mathematics but the skills and strategies a student may have used at school do not necessarily scale or carry over the transition.

Faulty assumptions made by staff and students can form barriers to communication and successful collaboration between different parties; this may mean that difficulties are not anticipated or successfully resolved. For instance, in addition to recommending equipment, non-medical helper support (study support workers) and allowances, for a UK student the Disabled Students Allowance (DSA) Needs Assessment also makes specific recommendations for adjustments by the institution. The needs assessment typically takes place prior to entry or early in the first term. For students who may experience mathematics-specific barriers it is key that the needs assessor and, ideally, the applicant/student, have a clear understanding of how mathematics is taught and assessed at university. This would need to include the specialist skills used and which software, hardware, human support and departmental adjustments may be effective in a university mathematics learning environment. However, DSA needs assessors are unlikely to have studied mathematics and may not be aware of the subject specific barriers or resolutions which are appropriate. As a result an applicant/student who has not been prepared for this transition by their school or university may be provided with ineffective technology and support for accessing the learning environment.

Accessing mathematical studies: Practical issues and concerns

In this section we discuss the real-world problems presented to disabled students in mathematics and how practical issues and concerns arise as they enter university.

Students with print-disabilities

A student is said to have a print disability if they are unable to read standard print effectively. In this chapter we focus on the practical issues and concerns that a student with a print disability might have. Chapter 15 provides a wider view of difficulties experienced by neurodiverse students, who may experience some level of print disability.

As outlined by Rowlett (2011:179–185) and recently highlighted by the Royal National Institute of Blind People (RNIB) (Cryer, 2013) a significant barrier for students with a print disability is the delivery of mathematical learning resources in inaccessible formats. Lecturers are often unaware their resources are inaccessible: *“I thought a PDF would be accessible since it is an electronic format”*. Some students require access to materials in Braille (Whapples, 2007; Maddox, 2007; Cliffe & Rowlett, 2012) and may use a Braille display and screenreader either for Braille mathematics or direct access to LaTeX code (Cliffe,

2009; Cliffe & Withington, 2013). They may require diagrams in a tactile form or text description. Students with partial sight, neurodiversity or other print disabilities may require large or 'clear' print (RNIB, 2011) and text which can be read aloud. Further detail and references related to required formats are given by Cliffe and White (2012). Clear/large print may involve quite individualised font, colour and layout changes and students may also need to restructure, or move, text to improve legibility and to add their own annotations. To meet such needs editable formats and formats which can be adapted by the reader are required. This is also noted in Chapter 15.

Staff are unlikely to provide alternative text descriptions for diagrams in most formats. Equations in common formats can be inaccessible (Cliffe, 2012; Hughes & Leavitt, 2013). This includes PDF documents, images of equations, Word, PowerPoint, OpenDocument, print and electronically captured handwriting which are all inaccessible via screenreaders used by students with visual impairment and difficult to convert into Braille. The formats are also not directly accessible to text-to-speech software used by some students with neurodiversity, restricted mobility, mental health, fatigue and pain conditions. Equations may be enlarged/magnified and may allow font, layout and colour scheme changes, editing and annotation in some formats but in others this is not possible or the result is illegible. PowerPoint, PDF documents, images, print and handwriting, are both inaccessible and cannot be correctly converted by any automated method to an accessible format. In such situations, English text may be recognised by Optical Character Recognition (OCR) packages or handwriting recognition systems and an accessible version may then result. However, standard OCR and handwriting recognition systems cannot cope with equations.

In general, even with access to a suitable underlying editable format, students and staff may have difficulty creating accessible versions unless the original was carefully authored (Cliffe & White, 2012; Hughes & Leavitt, 2013). Such conversions rely on access to a variety of software and hardware, some of which is unlikely to be provided under the Disabled Students Allowance or available at an institution. Finally, some mathematical resources are accessible but students may have previously only learnt to use assistive technologies and conversion tools to access English text rather than equations and do not know how to use such formats. Students who use assistive technology, such as screenreaders, may find that their technology does not work with mathematical software required for their course or they may find it difficult to locate guides that describe how to learn to use the software.

As a consequence of difficulties in converting documents and accessing software students may need to work with human readers and alternative formats support staff. Staff who do not have a mathematical and technical background may find it difficult to provide appropriate support.

Most students with a print disability will have some level of difficulty reading from a board, visualiser, tablet output or other display of handwritten or print material in a lecture room (Cliffe & Rowlett, 2012; Cliffe & White, 2012). Students can lose track quite rapidly if they are not able to read the board clearly, or at all, and provision of full notes is required to enable the student to follow the lecture. One student said *"Without the lecture notes there is no point in being there [in lectures]"*. It is also important that notes are kept up-to-date as *"the confusing part is when lecturers change the order of things round completely"* and *"the*

lecturer often strays from the official course notes". For neurodiverse students (see Chapter 15) the provision of notes alters the main activity in which they are engaged: from copying precisely, maintaining place in the text, or retaining definitions in working memory, to the desired engagement with concepts and logical arguments.

Finally, for students accessing mathematical text via a linear format such as speech, Braille or very large print, large two dimensional structures, long equations and arguments place a high burden on working memory. This will impact on the amount of time it can take to extract information.

Students with hearing impairment

Students with a hearing impairment may refer to themselves as deaf or Deaf. Students who are Deaf are members of the Deaf community and are likely to use sign language, rather than English as their first or preferred language. Students may lipread, use a lipspeaker or a British Sign Language (BSL) interpreter depending upon their first/preferred language. They may also use audiological equipment to transmit the speaker's voice to them with greater clarity. It is not possible to lipread or use communication support while also taking notes and students may have both specialist note-taking and communication support funded through the DSA. Depending on the communication modes in use, a student may lose some or all verbal information given while the lecturer is facing the board and BSL interpretation involving several equations can become difficult to follow due to the spatial nature of the language (Mole & Peacock, 2005). Communication support workers may have a limited mathematical background and may have difficulty accessing and relaying the vocabulary, symbols, structures or meaning in mathematical language.

Students who are D/deaf may have greater difficulties following the real-time exposition of arguments particularly as at university classes become larger and there are fewer opportunities to clarify understanding in real time. Students may experience practical difficulties or fatigue from working to follow exposition, but also may have difficulties with the use of English language (Mole & Peacock, 2005). Generic practices known to assist include the use of good communication strategies; ensuring the student's attention does not have to be split between two places (for example reading from the board and lipreading); provision of full notes prior to class; provision of visual organisers; using a collaborative, case study, problem-solving approach where possible, and pre-teaching (or, at least, enabling preparation) of specialised vocabulary (Mole & Peacock 2005; Easterbrooks, Stephenson & Mertens, 2006).

Finally, audio or video resources may be difficult to access and transcripts or subtitles will need to be provided. The spoken form of mathematics can be ambiguous and so these may need to be produced by staff who are mathematically fluent.

Students who cannot effectively use pen and paper

Changes in computer use mean that, in less mathematical subjects, students may make relatively little use of pen and paper when producing work. In mathematics, the relative difficulty of producing electronic mathematical text means that much working will be completed on paper. Students may experience difficulties using pen and paper for a variety of reasons (Cliffe, 2013). Students with restricted mobility, fatigue or chronic pain conditions may have difficulties using pen and paper or may rapidly tire due to the

quantity of handwritten material that a mathematics degree can entail. Some neurodiverse students have difficulty producing legible handwriting in a reasonable time-frame (see Chapter 15). Students with mental health conditions have also reported difficulties working in handwriting, for example due to the need to produce ‘perfect’ output. Finally, students with visual impairment may or may not use pen and paper effectively. Some will read and write in Braille and may use a mathematics Braille code. Others may produce some form of electronic text (which may be non-standard) or use a scribe.

Students who use a computer may use assistive and/or mathematical software to produce text when doing or writing mathematics and when using mathematical software. They may use keyboard-only entry, pointer-only entry (perhaps with adapted keyboard or pointer control), speech input or predictive interfaces. While they may have learnt to use such systems to produce English text in school they may not have been taught to produce symbols and formulae, or to use mathematical software, even supposing this is possible. Some students may complete all A-level work by dictation to a scribe, via a personal shorthand (possibly electronic) which was transcribed or in Braille. Such students may need time and input to acquire an effective method of producing, independently, text which can be then be read by staff.

Producing mathematical text using the above methods will generally be slower than the average handwriting speed. For students producing mathematical text via a linear format such as dictation, Braille or in very large print, producing long equations/arguments, large two-dimensional structures and diagrams places a high burden on working memory. This burden on working memory can impact on the mathematical thought process which will further increase the amount of time it can take to work through a problem or write up work.

Students are likely to have note-taking support. This note-taker needs to produce notes in a suitable electronic format to ensure that the student can access the notes if they cannot read standard print and to enable electronic annotation and copy and paste of the notes using a computer.

Summary of practical barriers

1. Disabled students use a variety of reading formats and assistive technology. Course materials may not be available in suitable formats. Students, and any communication or note-taking support, need access to full, up to date, lecture notes in advance in a suitable format. Some students require additional preparatory materials and adapted learning environments.
2. Students may not yet have learnt to use technology to access mathematical resources which are accessible and may not have learnt an effective method to produce mathematical text which can be read by staff and support workers.
3. Support workers such as readers, document conversion, communication support, scribes and note-takers may not have a mathematical background.
4. Dictating to a scribe, production of electronic mathematical text and reading mathematics in linear formats (speech, Braille or very large print) is relatively slow and places a significant burden on working memory. This latter can interfere with the mathematical thought process resulting in still slower working speeds. Students may

complete fewer regular exercises and take longer to produce written work. Note-takers may have difficulty taking full electronic notes in 'real time'.

5. Not all mathematical software works well with specialist input and output technologies such as voice recognition or screenreaders. A specialist calculator may also be required.

Resolving practical barriers and anticipatory approaches in mathematics

In this section we discuss methods by which the above practical difficulties may be resolved and the planning and infrastructure required to facilitate this.

The primary practical barrier: Failure to anticipate and plan

Due to the above practical barriers, the transition between learning environments is so steep that for many disabled students specific preparation for the university learning environment is required. Such preparation is known to be highly successful for some students. For example, two students with visual impairment were guided in preparing for the university learning environment throughout A-level study and went on to achieve high results, enjoy their studies and progress to postgraduate level (Irving & Williams, 2007; Spybey, 2012; Williams & Irving, 2012; Cliffe & Withington, 2013). The annual International Conference on Computers Helping People With Special Needs (ICCHP) summer university provides students with visual impairment and their teachers with the opportunity to start making this preparation for those who have funding (ICCHP, 2010). Where a student does not have access to such specialist preparation a pre-entry working partnership between the student, their school or college and the university is required to ensure that the student and department are 'up and running' with suitable methods for communicating mathematical text and diagrams (Cliffe & Withington, 2013). Such partnerships might be undertaken in conjunction with university support or widening participation staff.

If a student and department are not 'up and running' prior to arrival, the student may be unable to access the learning environment during the crucial early months of their degree. It is known that many students in mathematics can have difficulties making the transition to higher level study (Thomlinson, Challis & Robinson, 2009) and barriers to accessing the learning environment will further impede successful transition.

Where a student has prepared for the university environment in school they may make early contact with a department or institution. Reasonable queries include whether notes are available and in which formats. Prospective students and their advisers may feel that there has been a failure to anticipate and plan if there appears to be uncertainty about making this basic adjustment or if responses are generic and do not appear to take into account the mathematical nature of the course. Such responses may lower aspirations, impact on the widening participation goals of the institution and on the public image of the department (Cliffe & Withington, 2013). Hence, departments and their institutions should provide realistic and specific information when meeting with prospective students, including at open days.

The primary practical barriers are therefore a failure to anticipate and plan how resolutions will be achieved even if they are not yet required and a failure to work with individual

students to put plans into action. Such work minimises costly and possibly ineffectual adjustments which are time-consuming for lecturers to put in place after a student arrives. Failures to anticipate and plan can be based on faulty assumptions, as noted in the background to this chapter, which do not take into account the specialist modes of communication used in mathematics and the differences between school and university learning environments.

Resolving practical barriers: Approaches and required infrastructure

In this section the available technology and human support choices relevant to mathematics are summarised.

Accessible mathematical documents

Mathematics departments may not be required to hold and deliver accessible formats directly. It is enough for lecturers or departmental support staff to produce formats which can be converted. Responsibility for converting a document or working with a student to use such a format may well be the remit of other staff, who may require input on working processes and sources of information.

Which formats can be converted change over time. At the time of writing PDF and PowerPoint documents containing mathematics cannot be converted to suitable formats. The three basic formats which can be converted are (see below for an explanation of ‘well-structured’):

1. Well-structured word processed formats in which equations can be exported as MathML (W3C, 1994); these include well-structured OpenOffice ODT documents and Word documents in which MathType (Design Science, 2009) has been used as the equation editor.

These are editable formats which can be used directly or adapted to their requirements by some students. They can be converted to MathML enabled web pages by LibreOffice (ODT) or MathType (DOCX), respectively. MathType can also be used as a route to conversion to MathML enabled DAISY (Digital Accessible Information System) or EPub3 e-book format. Such MathML enabled formats contain enough structure for equations to be correctly read aloud. The exact method by which to achieve this is likely to change over time but was until recently usually achieved mainly via assistive technology products which have MathPlayer (Soiffer, 2005) support. Word documents which have not been directly authored with MathType may be converted or adapted as required by MathType but errors may be introduced. Documents may be converted to LaTeX, for students with visual impairment who read LaTeX source, and Braille. For technical information please refer to Hughes and Leavitt (2013) and Cliffe (2012).

2. Well-structured LaTeX using limited packages: in the general case LaTeX cannot be converted – authoring and use guidelines should be followed (Cliffe, 2012). Note the structure provided by LaTeX is required; in general, TeX cannot be converted to other formats.

This is an editable format which can be used or adapted directly by some students via the ‘what you see is what you get’, LaTeX editor BaKoMa (BaKoMa Soft, 2012) or by those students with visual impairment who read LaTeX directly (Irving & Williams, 2007; Williams & Irving, 2012; Cliffe, 2009). The format can be converted via TeX4ht (Gurari, 2009) amongst others (Stamerjohanns et al., 2009) to MathML or to MathJax (Cervone, 2012) enabled web pages, and so can be read aloud. Large print hard-copy may be converted by rewriting LaTeX with scripts to use the breqn package (Høgholm, 2007). Documents may be converted to Braille; see (Cliffe, 2012) for further technical information relating to the above conversions.

3. For web-based resources, well-structured MathJax (Cervone, 2012) enabled HTML pages with equations encoded as LaTeX or MathML: the format may be of particular importance in computer aided assessment and virtual learning environments which support MathJax and page authoring (for example Moodle). The format can be produced given either of formats 1 or 2 (Cliffe, 2012).

MathJax will render in all browsers and the format can be used directly by many students. However, this format is relatively young: editing environments and print output are currently limited. MathJax contains enough structure for equations to be correctly read aloud. Currently, MathJax can be read aloud via ChromeVox (Google, 2013) or via a screenreader or text-to-speech product with MathPlayer support and versions of Internet Explorer prior to IE10 (at time of writing). MathJax using the HTML-CSS renderer, in any browser, supports reflow of equations with font size changes for students reading on screen large print, but this cannot be used to make sound hard copy.

As also noted in Chapter 15, all documents should be ‘well-structured’ but here this is a technical requirement of the conversions. By well-structured we mean the document is formed following generic good practice guidelines and encodes headings (for example using styles in Word or section and theorem commands in LaTeX), uses formatted lists, etc. General guidelines for Word (JISC TechDis, 2013) and other formats (UKAAF, 2009) are available. LaTeX is designed to be written in a well-structured manner but the author is not required by the processor to do so. Guidelines on producing accessible LaTeX and Word containing equations were produced as part of a recent National HE STEM Programme project (Cliffe & White, 2012; Cliffe, 2012).

Departments or institutions may need to deal with externally produced resources in PDF, for example if a lecturer sets required reading only available in such a format. The only current approaches are to use InftyReader (InftyReader Group, 2002), Optical Character Recognition software for scientific documents and/or MaxTract in the case of PDF (Sorge, 2012). Both pieces of software are capable of recognising a variety of symbols and structures in PDF but the error rates can be quite high and are very document dependent. Errors in single symbols can radically alter meaning in mathematical text so all output must be checked and corrected by mathematically fluent support staff. Similar methods may need to be available for students and their support workers to use on recommended and independent reading.

Creating mathematical output using assistive technology

Due to the specialist nature of the text being produced it may be argued that mathematics departments have a role in ensuring that all students, mathematics staff and some support workers acquire an effective and efficient method by which they can create electronic mathematical documents. This may be via provision of taught classes or learning resources. Departments may need to be ready to cover or advise on alternative and accessible methods of text production and this may require collaboration with assistive technologists.

Which formats can be produced using alternative interfaces and assistive technologies change over time. At the time of writing the following options might be considered:

1. Students who can produce text effectively using a keyboard may work in either a word processed or LaTeX format.
 - MathType (Design Science, 2009) should be available as an alternative equation editor in Word. This enables conversions if necessary and students can input equations via keyboard only (LaTeX and shortcuts) or mouse only. The interface can be enlarged for use in large print and adapted mouse interfaces such as gazetrackers.
 - Students will benefit from using a WYSIWYG LaTeX system such as BaKoMa (BaKoMa Soft, 2012) or ChattyInfty (Komada, Yamaguchi, Kawane & Suzuki, 2006), for students who require speech and Brailled LaTeX output. Some students with visual impairment who read LaTeX source directly may use Emacs with the LaTeX-access scripts (Irving & Williams, 2007; Williams & Irving, 2012).
2. Students who require voice input and who are Dragon Naturally Speaking users will need an additional approach for equations. Currently this will be MathTalk (Metroplex Voice Computing, 2003) with Scientific Notepad. TalkMaths (Wigmore, Hunter, Pfluegel & Denholm-Price, 2009; Wigmore, Pfluegel, Hunter, Denholm-Price & Colbert, 2010) aims to provide a more natural interface but is still under development.
3. Students who can use only an adapted pointing device may use MathType with Word (see 1) or may use Dasher with the LaTeX alphabet (The Dasher Project, 2006). MouSense (Colleyeder Ltd. & NIPG Group, 2013) and other adapted input devices (The Dasher Project, 2011), for instance, can be successfully used with both.
4. Some students can produce handwriting but find a supportive environment to be useful when working through problems before formal write up. This might include use of a SmartPen (Livescribe, 2007) so that recorded speech can be linked to personal shorthand. This is also possible using a tablet and some notes applications. Systems that support handwriting recognition enable handwriting to be easily edited, manipulated and searched (equation support remains error prone but is under development).

Working with support workers

Due to the specialist modes of communication needed it may be argued that mathematics departments have a role in the design of appropriate mechanisms for support workers to use and in the provision of resources, advice and additional training for support workers. This would likely be in collaboration with disability service departments.

Development and documentation of support practices which are successful in mathematics change over time. At the time of writing documentation of successful practice is quite limited. The three areas in which there is some documentation are:

1. Working with an amanuensis:
 - The handbook for spoken mathematics (Chang, 1983)
 - Mathspeak (Nemeth, 1995)
2. Working with a study skills tutor:
 - See Chapter 15
3. Working with British Sign Language interpretation:
 - Sign language glossaries: EngineeringSigns (University of Wolverhampton, 2005a), ScienceSigns (University of Wolverhampton, 2005b) and the Scottish Sensory Centre BSL Glossary for STEM (Scottish Sensory Centre, 2005)

Required infrastructure

To deliver the above some infrastructure is needed:

- Software to perform conversions and make use of the output should be available.
- Guidance for authoring and converting mathematical documents including lists of symbols, notation and structures likely to be used throughout the course of study should be available.
- Guidance on how to work with support workers in mathematical learning contexts should be available.
- Applicants, their advisers and support workers should have access to information on the available formats for reading and creating documents including software requirements and information on forms of human support which are effective in the learning context.
- A mechanism for keeping up to date with developments is needed.

As yet, guidance and infrastructure at a national level is limited and fragmented although some resources do exist (AccessMSORWG 2009; Waterfield & Draffan 2012; Open University 2013).

Anticipating practical barriers: Collaborative approaches

Each department will need to:

- Explore the mathematics-specific practical barriers in the specific learning context.
- Establish how to resolve these barriers if required.
- Embed changes to infrastructure, policy or processes required to deliver future resolutions in a timely manner.
- Enable individual students and their support to plan how they will interact with the department and infrastructure.

The requirement for positive attitude and awareness of what is involved across the entire organisation is noted by Naysmith (Cliffe & Withington, 2013).

Some institutions may use formal project work within an inter-disciplinary team to guide wholesale changes in infrastructure, policy and processes.

Portland Community College (Hughes & Leavitt, 2013) released two full-time mathematics faculty members from a portion of their teaching load for one semester. They worked with staff specialists in learning technology, staff in disability services, external sources of advice and received input from an ‘end user’ with visual impairment and a strong educational and technical background. Through this approach they captured the barriers and also specific practice which would enable access to mathematics for a variety of students. The importance of working with a collaborative team is noted in the video and accompanying resources as without the input from an end user the project *“would not have been as successful as it was, it would not have been what it was...we were so off track [... the one solution considered] was not the solution for everyone and so without Maurice [end user] our results would have been completely different, and honestly? They would have been wrong”* – Scott Leavitt, Mathematics Tutor (Portland Community College, 2013). This anticipatory collaboration uncovered specific recommendations for resolutions which can be reached; input into policy and infrastructure required for future resolutions; guidance for staff including technical specifics and clarity on the responsibilities and required understanding of staff in various roles. They conclude that *“Every student’s needs must be addressed individually; however, there are some common practices that can help to support a wide range of students”*. The Open University has run similar projects (Cooper, Lowe & Taylor, 2008; Naysmith, 2011).

One-off project work which results in substantive changes in policy and processes may not be feasible or necessarily required outside of distance learning contexts. At the University of Bath students and staff with mathematical, technical and disability support backgrounds have evolved approaches to improve provision over an extended period (Cliffe, 2009; Cliffe & Doyle, 2009; Cliffe & Rowlett, 2012; Cliffe & White, 2012; Cliffe, 2012; Cliffe & Withington, 2013). Departments may wish to refer to these resources for technical information on production of notes, lists of resources and papers which outlined approaches useful to a variety of students. The framework given by Cliffe and Withington (2013) will assist departments working with visually impaired applicants. A similar framework, though with different specialist considerations (as outlined) might be used when planning with other disabled students. The same document also makes institutional and national level recommendations. These, as well as some points extracted from the work at Portland and the Open University, can be used to describe an idealised approach.

A checklist for departments: an idealised approach

The list below is aimed at departments and their institutions although an individual lecturer with a technical interest may be able to undertake some points independently. Chapter 15 provides a checklist for enabling neurodiverse students with points for individual lecturers and this should also be consulted.

Identify mathematics staff to work with support professionals and with disabled applicants/students with time set aside to:

- Form working partnerships with other professionals.
- Attain a core understanding of the practical and technical barriers which may arise.
- Source local, national and international sources of information and advice.
- Develop anticipatory recommendations for the teaching and learning context.
- Work with individual students to meet their needs pre- and post-arrival.

Audit each programme to quantify the practical barriers, required skills and technologies at each level. Consider:

- Barriers formed by individual teaching and learning practices. Can staff access development and guidance in accessible mathematics teaching?
- Accessibility of internally produced resources. Have you catalogued holdings or production of formats which cannot be converted?
- Availability and cost of expertise and resources required to create formats which can be converted. Is there agreed separation of responsibilities in a known process for conversions?
- Availability and cost of provision to ensure students acquire effective skills which permit them to produce work. Who will deliver this?
- Accessibility of course specific software. Is accessibility part of the procurement policy?
- Accessibility of external resources including e-books. Have you rated your reading list for accessibility and publisher responsiveness? Is there local expertise to convert external resources if necessary?
- Required skills of support workers, for example in terms of mathematical background and knowledge of technology and alternative formats. How would quality be ensured in such services?

Share information from the above audit:

- Internally to disability advisers and support workers.
- Externally so it may be accessed by applicants and their advisers.

After the audit:

- Work to hold an agreed minimum of lecture notes in formats which can be converted into various alternative formats. Ensure processes to convert resources which can be converted flow smoothly.
- Work to hold training materials for students and support on acquiring effective skills which permit production of mathematical documents.
- Work with disability advisers and alternative formats services to source, recruit and train mathematically fluent support workers as required.
- Consider creating policy or guidelines as to when lecturers might highlight reading and exercise priorities so as to enable students to focus their attention if their reading

or working speed is slower due to working memory burdens or if all documents and diagrams cannot be converted.

- Form a pre-entry working partnership with applicants, their school or college, a disability adviser and the student's needs assessor. Use the outcomes of the audit to work with individual students to ensure they are 'up and running' before they reach university.
- Locate, as required, role models and specialist careers advice including disabled graduates and older students in the mathematical sciences.
- Continue to work collaboratively with other professionals to improve access to software and external resources, to locate sources of information and to maintain expertise in the above areas.

Conclusion

To ensure that practical and technical barriers at the transition between school and university are anticipated and resolved presupposes a level of infrastructure and planning which may not be required in less mathematical subjects. Understanding of the nature of mathematics, how it is communicated, taught and assessed, rests with the subject department and hence a collaborative approach with emphasis on planning and clear communication between students, mathematics staff and support professionals at school and university level underlines good practice in this area.

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Chapter 17: Gender and university mathematics teaching

Melissa Rodd

Abstract

Making the transition from school to university mathematics is a major identity investment for all students and, although there have been significant increases in the proportion of female students studying mathematics at university in recent years, in some institutions university mathematics still exudes an aura of male culture. This chapter is written for university mathematics teachers who want to encourage both female and male undergraduates. It argues that gender is a central aspect of a person's identity, yet gender does not automatically impact on participating in learning mathematics in a stereotyped way.

The notion of *pedagogic voice* is introduced as a way of thinking about establishing learning environments which attend to relational aspects of teaching. Relational aspects of teaching develop through personal connections with students and their mathematics and empowers learners from a range of identities. Quotations from mathematics undergraduates' mathematical biographies are given throughout the chapter. These quotations are presented to communicate students' reasons for coming to university to study mathematics and thus something of their identification with the subject. Each quotation is presented without reference to the gender of the undergraduate who wrote it. The purpose of veiling the gender of the speaker is to help the reader to reflect on what might be gendered.

Establishing a *pedagogic voice* includes the university teacher establishing a voice to speak to the students, as well as learning to listen. In such a relational learning environment, developing an authentic teaching voice contributes to a new lecturer's identity as university mathematician.

"For me, maths has always been something that comes with a feeling of comfort as well as being relatively easy to comprehend. I think I have this attitude for a number of reasons, but I am going to focus on two in particular; the first is that my mum, who I am very close to and have a lot of respect for, was always very competent with numbers and used them with ease in everyday life as well as at work. Another very important

factor was that most of the time the concepts have been easy to grasp and in the case where I have had to work hard to understand something, the final understanding comes with a great sense of satisfaction and achievement which is more than enough incentive to continue on to the next problem undaunted.

At A level I found that maths was the only subject that I could spend a substantial amount of time doing and not get bored or distracted. I had thoroughly enjoyed the change in maths from GCSE to A level. It was though I had just peaked [sic] into a whole new world of mathematics, one that was much more abstract and yet still applicable to real life and I wanted to find out more.”

3rd year undergraduate mathematics student A1¹

Introduction

This chapter’s focus is on gender issues in teaching mathematics in higher education. It has been written for university lecturers in mathematics, particularly those moving from research-only positions to posts that include teaching. The stated focus already raises the question ‘Are there gender issues in teaching mathematics at the university?’ Assuming some gender issues exist, are they to be celebrated or avoided? How can a new lecturer be helped to find a way to teach with a sensitivity to students’ identities generally – gender, is at heart, central to identity – within a wider social context that includes the student as consumer and a discourse of fairness?

The route offered here is to introduce an appreciation of *pedagogic voice* as a way for thinking about student-teacher communication. *Pedagogic voice* is a notion, more fully explained below, that develops from *student voice* and that positions relationships as central to the educational enterprise. Extracts from narratives from third year undergraduate mathematicians, one of which heads this introduction, exemplify *student voice* in this undergraduate mathematics education context.

The outline of the rest of the chapter is as follows: some contemporary gender issues relevant to undergraduate mathematics education are raised, then the notion of *pedagogic voice* is introduced for this context. A brief analysis of mathematics undergraduates’ narratives follows and finally the relational aspect of *pedagogic voice* is explored further which includes suggestions for the lecturer’s reflection and professional development. Several quotations from undergraduate mathematicians are given throughout the chapter. These quotations are presented in order to illustrate the complexity of what has brought young adults to university to study mathematics and also to nudge the reader into wondering something about their author (that might concern gender) and thus develop the attribute of listening to students’ voices.

The central purpose of this chapter is to explain and discuss: (1) that gender is a complicated and multi-faceted aspect of identity for both student and teacher, and (2) that

.....
1 ‘A1’ indicates this student was on the mathematics education course register from the year 2012/13 and is the first quotation from that year used in this article, Bj indicates the jth quotation from a student from the year 2011/12 etc.

successful teaching-learning scenarios are relational, so part of a teacher's job is to hear the students' voices – including those that are like his/hers and those that are different.

Gender and undergraduate mathematics

"I moved to a Sixth Form College so I could take further maths. Many in my form there were also taking further maths and my form tutor was also my maths teacher. We were all far removed from the typical mathematician stereotype and I made some good (and unlikely) friends. Now after my second year at university my thoughts about maths have changed. I know I am not a born mathematician; the exams for me have been extremely challenging, and I spend a lot of time pursuing other creative interests. But I remain with the feeling that maths is 'my thing' and I hope that [as a school teacher] I will be able to use the skills and knowledge I have gained over the years to help other people get to grips with the subject, as challenging, difficult and rewarding as it is."

3rd year undergraduate mathematics student E1

From where do we find a contemporary take on gender issues in undergraduate mathematical education? As gender is a part of a person's identity both biologically and socially, it is reasonable to attend to how gender dimensions of personality play out in the maths undergraduate field. Gender is a multi-faceted and contested concept and this is not the place for a comprehensive review of how gender impacts university mathematics education. Nevertheless, statistics tell something. Female participation in undergraduate courses has risen over the past few decades: from 41% in 1971 to 56% in 2007 (Earlham, 2013). From 2006 until 2012, the proportion of female UK-based mathematical sciences undergraduates has remained steady at just under 40% (HESA, 2007, 2012). However, the rise in participation has not yet flowed through the whole system, for example, the London Mathematical Society (2013) recently reported that 'at the top' 94% of professors of mathematics are male.

Is the roughly 40% proportion of female undergraduates sufficient for female students to not be inhibited by their gender when they arrive at university to study mathematics? However, a question of such generality cannot be addressed in terms of averages and trends, despite arising from attention to a statistic – namely the proportion of female mathematics undergraduates. This is because those female mathematics students arrive at university from such different backgrounds and preparations. Recent research from the Understanding Participation rates in Mathematics and Physics (UPMAP) project did not find strong gender factors at the undergraduate level (for example Rodd, Mujtaba & Reiss, 2010), but it did reveal students' conceptions of their preparation for mathematical study. For instance, one female undergraduate told of the support her girls' boarding school had given her, another young woman was aware of her competitive nature, particularly with her two older sisters, and was eager to 'prove herself' in an intellectually tough environment. However, other students, male and female, not well-prepared for the lack of strong scaffolding they had been given at school and had come to rely on, did not flourish in the challenging environment of independent mathematical enquiry and learning.

A thorny problem for practitioners in education is how to use statistics, personal experience and qualitative accounts to improve practice. Being in a 40% minority probably does not

feel as intimidating as being in a 4% minority. But who can say for a given individual? The young woman who has come from an all girls' school to a place where 60% are male might well have more apprehension about gender than the young woman who has been in a significant gender minority when doing her sixth form studies in an elite boys' school that accepts exceptional girls in Year 12 and then finds herself as one of two females in a 50-student strong option class at university. As another of our UPMAP undergraduates said *"it is more interesting being a girl doing physics than being a girl doing English"*. Being able to shine in the spotlight is a feature available to females and males and there are now many well-established female mathematicians.

And yet, the hinterland of mathematical personalities is not female, despite recognition of historical figures like Somerville, Noether, Lovelace and Kovalevsky. As part of the Students' Experiences of Undergraduate Mathematics (SEUM) project (for example Rodd, 2009), a lecturer was observed in a large first year class and, pausing from the explicitly mathematical exposition, he asked the students if they knew of any 'women mathematicians'. This particular lecturer was a staunch supporter of female participation in mathematics, yet by the reference 'women mathematicians' did he position females as alien to the mathematical enterprise? Or does this male lecturer who raises awareness of women's contribution to this mathematical enterprise, in a spirit of encouragement and of countering the image of the 'Men of Mathematics' poster that still graces some mathematics departments' corridors, contribute to his undergraduates' wider mathematical education? The role of the lecturer who is the personification of mathematics is a subtle one and there are no stock answers.

Another question to consider though, is 'What is a 'typical mathematician stereotype' that the student E4 refers to?' An extensive review of literature of attainment (DfES, 2007) reinforced the image of girls as conscientious and risk averse and boys as adventurous and more likely to be of extreme attainment. And yet the bulk of the female and male distributions (on attainment, disposition and participation) overlap considerably! Patterns in participation and achievement will continue to be researched, reported and will contribute to images of what it is to be a maths student (for example Epstein, Mendick, & Moreau, 2010). Despite self-positioning away from a 'stereotype', student E1 does consider maths as *"my thing"* and so integral to identity.

Stereotype threat is a notion that was initially developed to account for poor performance of groups not generally associated with excelling in a certain arena, for example, the stereotype 'girls don't do math' depresses girls' performance in mathematics tests (for example Spencer, Steele & Quinn, 1999). More recently, psychologists have investigated stereotype threat as a barrier to participation (Murphy, Steele & Gross, 2007). An example of a potential stereotype threat relevant to female undergraduate participation is Simon Baron-Cohen's attribution of a propensity to 'systematise' as 'male' (Baron-Cohen, 2003). Baron-Cohen defines *"systemizing"* as that which a *"brain ... [which is] hard wired for understanding and building systems"* does (Baron-Cohen, 2003:1); to be a systematiser means that you like to order things, design and run systems. Many of these dispositions are suitable ones for mathematical activity. Baron-Cohen has found that systematising is a positive trait of many individuals on the autistic spectrum but sadly he attributes this systematising ability to something he defines as the 'male brain' and thus these positive

mathematical dispositions are associated with male-ness, despite his saying “*I would weep...if a reader took home the message that...all women have lower systemizing skills.*” (Baron-Cohen, 2003:183). It is the discourse that contributes to stereotype threat and social identity formation and potentially inhibits female participation (Rodd, 2006) in a subliminal way. This example has been presented to show how gender stereotyping about what it takes to do mathematics can be unintended, so, lecturers starting to develop a *pedagogic voice* that affords multi-gender participation need to understand how their own gender identity impacts on their teaching and even ‘positive’ interjections (like that of the lecturer illustrated above) may inadvertently position female as ‘other’ than standard when it comes to being a mathematician.

Gender is a complicated and multi-faceted aspect of identity for both student and teacher.

Pedagogic voice in undergraduate mathematics

“It is one of those things, that even when I encounter difficulty in it, I enjoy it. Despite being better at science than maths, I knew my heart had settled on mathematics. Further Mathematics was not being taught at my college however I was fortunate: the Further Mathematics Network was giving the opportunity to college students to take on AS-Further Mathematics outside of college hours. My Further Mathematics teacher was an amazing mathematician who was very caring and loving. I thoroughly enjoyed Further Mathematics and that only increased the zeal in me for going further in the subject.

It’s fascinating how at the skeleton of many of the established systems today that keep things organised – such as technology, society, economics etc – there emerges the beautiful science, mathematics.”

3rd year undergraduate mathematics student C1

Initially a tool for instigating social change, *student voice* has become an important aspect of contemporary education. All lecturers in UK higher education institutions will be required to run evaluations and invite student feedback. In this sense, listening to *student voice* is now an established part of a lecturer’s practice. However, the lecturer ‘hears’ *student voice* through a pedagogical encounter. Thus Arnot and Reay (2007) move to consider *pedagogic voice* following, for example, Bernstein (1990). They believe that in any teaching environment there is an interplay of power relations and that this interplay creates conditions for particular “*voices*” to emerge. Indeed, ‘voice’, from their point of view should be understood as a technical term: “*voice uniquely represents the relation between the school and the social structure. It is a pedagogic voice, the voice of the pedagogy*” (Moore, Arnot, Beck & Daniels, 2006:4). In the context of undergraduate mathematics education, the ‘school’ is not only the particular university with its own modules and curriculum pathways, but also the wider, global school of academic mathematics which is shared internationally. This trans-national school of academic mathematics has its own social structure and undergraduates are the novices. In order to draw these initiates into the mathematical world, the university teacher develops relationships with the students and draws the students into relationship with the mathematics. Students’ mathematical talk and other mathematics-orientated interactions, with teachers and with peers, is a function of the *pedagogic voice* of their learning environment, and that which is sayable is controlled (often

subliminally) by those who are powerful in that environment. For example, there are classes during which only the lecturer speaks, there are seminars where only highly achieving students feel empowered to be able to ask questions and there are classes that specifically cater for those students who don't understand easily and explanations can 'start from the beginning'. Central to the educational enterprise advocated here in this chapter is that the *pedagogic voice* respects wider social identities of, for example, class, race or gender. And in particular, that power is distributed such that the female has voice.

In practice, *pedagogic voices* of undergraduate mathematics education are brought into existence via the ways people teach undergraduate mathematics. Lectures, computer interfaces, assignments, tutorials – which are examples of institutional and cultural approaches employed and adapted by individuals – are examples of the particular social and technical infrastructures on/by which relationships are made that enable mathematical ideas to be communicated and taken in by learners. An example of *pedagogic voice* for undergraduate mathematics education is given by Povey, Angier and Clarke (2006) through their 'storying' of one of their undergraduate students. In their account, the metaphor "*spacious*" is used in various contexts – for the mathematical tasks, for the relationships developed, for mathematics generally. This metaphor points to an example of the *pedagogic voice* invoked/generated by their course which enabled a female mathematics undergraduate to be heard (their 'hearing' of their student Joanne is the centre of their paper). This was because in their context, tasks could be worked on without a priori delineation, relationships could attend to issues beyond the classroom and mathematics could be envisaged as a broad and inviting discipline. Power was granted such that (a particular) female voice was able to be heard.

A contrasting example of undergraduate mathematics education is given by Nardi's study of mathematics undergraduates' tutorials at their Oxbridge Colleges where students are tutored in pairs. In Nardi's study, the pedagogic encounter is couched in terms of transitioning from school to university, where undergraduates are to be inducted into a new culture of formal mathematics by an expert. In contrast to the 'spacious' learning environment discussed above, in this context, an appropriate metaphor might be 'focused'. This is because undergraduates are taught in tutorials of two, the scope of the mathematics is limited to a challenging yet extensive syllabus and the methods are rigorous and formal (Nardi 1996). Power in this scenario is invested in the authority of the lecturer who is the personification of the abstract mathematics, and the abstract mathematics, its results, symbols and methods, is the product on which the undergraduates are focused. The particular tutor's personality, interest in students and teaching skills will vary considerably and hence there will be a wide range of examples of *pedagogic voices* from Oxbridge tutorial teaching scenarios, yet all will be pitched to the overarching and well-established elite university undergraduate curriculum.

These two examples are presented to indicate that the particular university context, the particular course or topic, the particular department and the particular tutor or lecturer produce distinctive *pedagogic voices*.

Successful teaching-learning scenarios are relational, so part of a teacher's job is to hear students' voices – including those that are similar to his/hers and those that are different.

Undergraduates' voices and analysis

"From a very young age it was always clear that Mathematics is my forte. I have always been confident and comfortable with numbers and problem solving. I see Mathematics as a continuum, linking Pure modules with Applied, such as: binomial expansions linked into Statistics and calculus into Mechanics. I am a logical thinker and find the way one theory can lead to another through analysis and communicating mathematical ideas fascinating."

3rd year undergraduate mathematics student A2

One of the contributions to the establishment of a *pedagogic voice* is in hearing students describe the reasons behind their decision to enrol on an undergraduate mathematics programme. In Povey, Angier and Clarke (2006) their *pedagogic voice* communicates/ gives space for Joanne's reasons for participation. From one of the courses we teach, an undergraduate mathematics education module for maths or physics undergraduates from neighbouring universities, we also have had an opportunity to hear why young people (all of whom quoted were under 25) come to be studying mathematics. Early on in the course, students write about 500 words of mathematical autobiography²; these narratives represent an aspect of student voice on participation. While course participants are skewed to those interested in teaching as a career or education more generally, nevertheless, the range of issues raised gives an opportunity to listen out for a gendered theme to do with undergraduate participation. The extracts from students throughout this chapter have been taken from these 'mathematical journeys'.

Reading the undergraduate mathematicians' narratives on why they chose mathematics, the themes for their positive participation can be loosely grouped into three main categories³. These are to do with the subject itself, issues about how they subjectively experience maths and issues to do with how mathematics is seen or functions in the world. Specifically these three main themes have subsections as listed below:

1. Nature of mathematics

- Logical
- Numbers not words
- Puzzle/challenge
- Right or wrong

2 This mathematics education elective has run for the past six years and we have a sample (n=108) of these narratives.

3 Results presented here are an aspect of our reflection as practitioners (that is being these undergraduates' teacher for their mathematics education module); the essays are not a set of data generated from a funded/formal research project where objective or generalisable results are sought, but as part of a course. This makes a difference to the way we read them and to how they were produced: they are part of the *pedagogic voice* this course was creating.

2. Affective issues, inner

- People/relationships (only mentioned were teachers and parents)
- Satisfying
- Calming/confidence-making
- Identity/special

3. Societal issues, outer

- Useful
- Attainment
- Fun lessons

The most dominant themes over both genders for ‘why I study mathematics’ were the affective issues of *identity* and *people*, as exemplified in all the extracts already quoted. A majority of students cited one of the subthemes of the nature of mathematics theme – that is, they said they had an attraction to the *logical*, *not-in-words*, *puzzle* or *definite* aspect of mathematics. Here is an extract which expresses delight in maths-as-puzzle as well as the fun of how useful it is together with the importance of a special person:

“I like how at first mathematics is not clear, it’s like a puzzle which needs unravelling. Once this puzzle is unravelled, a new and fascinating world is uncovered. This world is absolutely perfect. When I was younger, my mum inspired me in mathematics by telling me about the kinds of things mathematicians solve in the real world such as how mathematics is used to see a 3D image of a patient’s heart on a computer. My first year complex analysis has applications in engineering and building spaceships!”

3rd year undergraduate mathematics student B1

The issue of their mathematical capability was also mentioned by the majority, in some cases this was linked strongly to identity, as in the extract at the beginning of this section. It was also linked to *attainment* like being ‘good at maths’ or, as mentioned in the extract below, achieving an award; this extract also speaks about ‘specialness’ and the usefulness of mathematical talent in communication within teaching others:

“Ever since I was a child I loved the clarity and exact nature of Mathematics. For some reason it matched perfectly my personality and made me appreciate it even more. I began solving problems for a magazine and because of my intense interest in sending solutions, I was delighted to receive a special award, which I view as proof that Mathematics can bring satisfaction and happiness as a child.

Many of my friends used to ask me for help. I soon became a kind of peer tutor. I really enjoyed helping them because I love to see their satisfaction when they find the solution.”

3rd year undergraduate mathematics student B2

The quotations reveal something of each undergraduate’s identification with mathematics which motivated their commitment to studying the subject at degree level. The gender of the author is not revealed! This is to prompt the reader to consider whether these undergraduates’ conceptions of mathematics or their motivations to study the subject are gendered.

Undergraduates' mathematical identities are to be nurtured throughout their course; the gender aspects of these mathematical identities are complex, personal and often unpredictable.

Pedagogic voice in undergraduate mathematics education: Reflection and practice

Teaching is central to a university lecturer's job, yet teaching skills are not central to securing such an appointment. It is a candidate's scholarly research skills and depth of involvement with the subject that wins a lecturing job. Although a scholar's deep involvement with their subject is a personal thing and not easy to parade, it is this personal depth of involvement that is central to inspiring university teaching (Rodd, 2003).

The following extract from John Williams's 1965 novel 'Stoner', relates his protagonist William Stoner's awareness of his authentic voice as a teacher emerging. The character Stoner is a lecturer in English at a mid-Western US university in the 1930s and he has been a university teacher for ten years prior to his reflections in this extract:

"During the Christmas holiday, ... William Stoner... began to understand that it might be possible for him to be a good teacher.

He was ready to admit to himself that he had not been a good teacher. Always, from the time he had fumbled through his first classes of freshman English, he had been aware of the gulf that lay between what he felt for his subject and what he delivered in the classroom... Those things that he held most deeply were most profoundly betrayed when he spoke of them to his classes; what was most alive withered in his words; and what moved him most became cold in its utterance...

But during the weeks that Edith was in St Louis, when he lectured, he now and then found himself so lost in his subject that he became forgetful of his inadequacy, ... he became so caught by his enthusiasm that he... ignored the lecture notes that usually guided his talks. At first he was disturbed by his outbursts, as if he presumed too familiarly upon his subject, and he apologized to his students; but when they began coming up to him after class, and when in their papers they began to show hints of imagination and a tentative love, he was encouraged... The love of literature, of language, of the mystery of the mind and heart showing themselves in the minute, strange and unexpected combinations of letters and words, in the blackest and coldest print – the love that he had hidden as if it was illicit and dangerous, he began to display, tentatively at first, and then boldly and then proudly."

Williams (1965/2012:112-3)

This extract gives the reader a glimpse of how Stoner was developing his voice as a teacher that enabled student participation at the social, emotional and intellectual levels. The medium gives an example of *reflection* on developing an authentic voice; the message for mathematics lecturers in the UK today is that students still get inspired by genuine enthusiasm communicated in a lecturer's *practice*: mathematics' 'blackest and coldest print' can be exciting, satisfying and communicated with passion.

We do not know the gender of the students who came 'up after class', nor the gender of those in whose writings Stoner noticed 'hints of imagination'. As 'Stoner' is a story, so

it is our imaginations who might suppose that the ‘after class’ students were those who identified with their lecturer, so likely to be male, or heterosexually attracted to him, thus female. These stereotypic categories of student still exist, but encouraging student communication in these ways is to be avoided rather than reinforced. Universities are more inclusive of identities (class, race, sexuality, gender, etc.) now than was the case in 1930 mid-West America. Students come from a wide range of backgrounds (and the law requires no discrimination), so how can the contemporary lecturer create a *pedagogic voice* so that the full range of students feel able to come ‘up after class’ (that is develop relationships; today this might be achieved by social networking) and show ‘hints of imagination’?

Let us consider this through an example of ‘female voice’ within a university mathematics department’s *pedagogic voice*. As part of the *Students’ Experiences of Undergraduate Mathematics* (SEUM) project, individual students were interviewed up to four times over the course of their three-year degree programme and the research team attended their lectures. The SEUM project aimed to find out what was important to students and what was helpful to their progression and participation. One of the results from this research was on the nature of some young women undergraduates’ participation and the SEUM research team’s not having noticed these quiet, academically successful young women as readily as the interacting and questioning young men were noticed (Rodd & Bartholomew, 2006).

Without attributing a negative connotation to either word, we characterised these young women’s participation as ‘invisible and special’. ‘Invisible’ as we observed no interaction with lecturers and ‘special’ as, when interviewed, they communicated that they believed they had mathematical talent that had been evident from an early age. Because this ‘invisible and special’ form of participation arose from noticing female students, it appeared as gendered-female. Yet just as a female student can participate in a stereotyped gendered-male way (for example Mendick 2005), an ‘invisible and special’ form of participation is possible for a male student. This draws attention to forms of participation being (metaphorically) gendered, yet students of any gender identification could potentially participate in this way. Despite this opening up of gender positioning, it is worth noting that female students’ concerns will sometimes differ from those of their male peers, for example, in pregnancy, hormonal and other changes can have a big affect on prioritisation, or when a sexist stereotype threat has damaged an individual’s self-concept as a mathematician.

Returning to the query posed above: ‘How can a lecturer create a *pedagogic voice* so that the full range of students feel empowered to develop relationships around mathematics learning?’, the emphasis has shifted from the gender identity of the student to: what forms of participation are available to students of various identities (some of which an individual student might want to keep from public display) within the *pedagogic voice* being developed in the learning environment? In particular, an ‘invisible and special’ voice⁴ is available and empowering when the engaged learner is comfortably quiet and securely confident.

4 Metaphors of hearing and seeing are mixed here but these metaphors both represent perception generally.

In conclusion

To develop a *pedagogic voice*, the university lecturer also has to develop a voice – a teacher's voice – to communicate with learners in order to promote the students' relationships with undergraduate mathematics. This voice, as the character Stoner was discovering, is personal, hence an aspect of the lecturer's identity. While we do not hear Stoner dwell on the gendered nature of his identity as a lecturer, personal gender identity and how this is expressed in communicating the subject is a point for reflection for university teachers today. The personal nature of the teacher's voice, including this gender dimension, is recognised in professional development programmes. For example, those run by the Higher Education Academy which integrate the sharing of experience into professional development⁵. Indeed, as part of professional development, a university teacher is required to be inclusive with respect to student diversity.

By encouraging university teachers to reflect on gender-identity and gendered approaches to participation in undergraduate mathematics, hopefully, the lecturer is positioned to imagine, plan for and enact practices that are personally authentic, that facilitate good student learning and participation and contribute positively to her or his identity as a university mathematician.

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A note about the quotations

The short quotations given within this chapter have been chosen to represent the range of themes and styles of talking. All the quotations given in the body of the chapter were from female undergraduate mathematicians. Below are some quotations from male students (again third or final year undergraduates). Over the eleven extracts, there are representatives from the range of social classes and from five ethnic groupings. Gender differences in their reports of why they continue with maths at university are hardly discernable (although, now the gender of each author has been revealed, one is likely to project gender-stereotyped characteristics on to that author's writing⁶).

6 We gave some mathematics education colleagues a set of quotations from these narratives – without saying whether a female or male student was author – as expected, colleagues were not able to attribute the gender of the author reliably.

"It's not hard to pinpoint the beginning of my mathematical journey; my Mum studied maths at Cambridge and has spent her entire life teaching maths to secondary school students. It's hard for me to remember my early development as a mathematician; most of my very early memories either revolve around the classic feelings of a simple satisfaction at correctly answering questions and the lack of sympathy when other students didn't necessarily take to new concepts with as much ease. I do, however, have fond memories of my Mum teaching me concepts slightly outside of the standard curriculum and being fascinated at the way that no matter how far I went there was always more to be found out."

C2

"I had always been an inattentive child at school and it was the corridor for me during maths lessons in Y10 and 11. But I continued studying mathematics at my local college of further education. I spent most of those two years in the library just trying to keep up with the workload. It was within those four walls of the library that I discovered something new about the world. Some things are true. The proofs laid out in the maths books were, to my mind, true. Here is a statement, and this is why it is incontestably so. This was a powerful revelation to me."

A3

"I liked studying mathematics from a young age. The first time I realised it was when I had to learn all the times tables from one to twelve. I remember writing all of them down on one piece of paper and noticing patterns in each set. I found it amazing that these sets of numbers adhered to certain rules and I wanted to find out about it. I think the reason I liked doing maths was because it follows logic and makes sense. It's not someone's opinion; it doesn't involve having to add some form of creativity, it's logic in its most fundamental form."

C3

"The presence of mathematics in my life has developed significantly as I've grown older. My earliest memory of mathematics is not a very positive image. In the Y6 SATS mathematics was my lowest scoring subject. At secondary school I had a private tutor that allowed me to ask questions I would not have asked in a classroom; most of my questions contained the word 'why?' With the help of my tutor I obtained 98% in GCSE Maths so there was a great transition within the five years from the end of primary school to the end of secondary school. My tutor suggested I think about a career in teaching so in Y13 I decided to take on this new challenge and tutor GCSE students. Teaching mathematics opened my eyes to how difficult it can be for first time students to learn the course."

E2

"As a child I loved origami – the idea that you could create an infinity of shapes, forms and structures from a single material triggered my imagination. We moved to the beach and my mathematics enveloped me; whether it was the spiralling evolution of a conch or the proportions of the breaking wave, mathematics seemed to be everywhere."

F1

Chapter 18: Developing mathematics teaching: what can we learn from the literature?

Stephanie Treffert-Thomas & Barbara Jaworski

Abstract

In this chapter we address the extensive literature which can inform the teaching of mathematics drawing on our own experience of using and finding value in the literature to enhance our own knowledge and practice in teaching mathematics at university level. Three areas of literature are recognised and addressed: *professional literature*, in which we gain insights into the ways in which other teachers/lecturers have thought about their teaching and the approaches/strategies and frameworks they have used; *research literature* which offers *what is known*, findings from research that can enable more informed approaches to teaching; and *pedagogical literature* that deals overtly with developing and enhancing teaching, through the lecturer engaging with new ideas (for example those offered in the professional literature), attending to research findings or using specific tactics or teaching approaches recommended by the authors. Many examples are provided, both of particular sources in the literature and of approaches to learning and teaching in mathematics, with an extensive reference list. This approach is intended to be informative to mathematics teachers at university level who look for knowledge and ideas to inform their teaching and support its development.

Introduction

There is a wealth of literature focusing upon learning and teaching in mathematics. Much of it focuses at school level, although there is a growing literature base focused upon higher education. Some of the latter has resulted from research which has been conducted into a range of issues in learning and teaching in areas of mathematics at higher levels such as in calculus, linear algebra, or abstract algebra. There are more studies focusing on issues in students' learning than on issues in teaching. Research studies are conducted to address

clearly defined research questions and according to recognised methodological principles. We refer to this as *'the research literature'*. In other cases we see the reflections of people who are teaching at this level, often mathematicians, who write to communicate their thinking about teaching and their students' learning, and to share aspects of their practice. We refer to this as *'the professional literature'*. Both kinds of literature are informative to people teaching at this level who are interested in developing their practice.

There is a third kind of literature which we refer to as *'the pedagogical literature'* in mathematics. This refers to writing in which the focus is on *approaches* to mathematics teaching and the learning of students, with some analyses of the ways in which approaches can contribute to effective teaching and learning. The writers synthesise from the research and professional literature to offer a commentary on teaching and learning in mathematics and on ways of developing teaching. This book in its entirety is an example of such literature.

In this chapter, we focus on these kinds of literature and the ways in which they can be used to help practitioners – those who work with students and teach mathematics at higher levels – to become more informed about learning and teaching and to develop their practice. We are ourselves teachers of mathematics at this level and are interested in developing our own practice. We have also conducted research into mathematics teaching in a university. As researchers we have needed to read what others have written in both the professional and research literature and to base our research on what is already known. Here, we draw on this experience. We refer to research in which we have studied the teaching of linear algebra, and therefore provide examples of the literature we have read relating to linear algebra learning and teaching, and ways in which it can inform practice in the teaching of linear algebra. We also look beyond linear algebra to literature relating to other areas of mathematics and to the pedagogical literature which discusses the learning and teaching of mathematics and its development.

The issues we are addressing

Why should we want to consult the literature anyway? Why might it be important?

In 1982, the authors of the Cockcroft report into the teaching of mathematics in schools claimed *"Mathematics is a difficult subject both to teach and to learn"* (Cockcroft, 1982:67). They reasoned that mathematics is a *hierarchical subject* and that *"ability to proceed to new work is very often dependent on a sufficient understanding of one or more pieces of work which have gone before"*. However, learning mathematics is important because *"mathematics provides a means of communication which is powerful, concise and unambiguous"* (Cockcroft, 1982:1). We see these quotations to be just as relevant to the learning and teaching of mathematics in higher education as they are to school mathematics. In 2000, Savage and Hawkes addressed *"The Mathematics Problem"* relating to mathematics in higher education. Their report states:

"Evidence is presented of a serious decline in students' mastery of basic mathematical skills and level of preparation for mathematics-based degree courses. This decline is well established and affects students at all levels. As a result, acute problems now confront those teaching mathematics and mathematics-based modules across the full range of universities."

Savage & Hawkes (2000:ii)

Perkin, Bamforth and Robinson (2010:1) acknowledge that “*Since the 1990’s there has been well documented concern and evidence regarding the declining mathematical ability of students entering numerate courses at university, see: LMS, IMA and RSS (1995), Sutherland and Pozzi (1995) and, Savage, Kitchen, and Sutherland et. al., (2000)*”. More recent evidence shows that students entering university with A grades at A-level achieve these grades through a highly instrumentalised learning and teaching approach in which the questions addressed in examinations can be tackled through memory and rote learning with little demand for problem solving and higher level thinking skills (Minards, 2012). Thus, those teaching mathematics in higher education have to work with students who are ill-prepared for university-level study.

So, how do we approach teaching mathematics for such students? In what ways do teaching approaches take into account the prior experiences and levels of preparedness of the students to whom they are addressed? University lecturers acknowledge, and research shows, that traditional lecturing based on mathematical abstraction and formalism causes severe difficulty and alienates some students who experience it (Alsina, 2001; Artique, Batanero & Kent 2007; Solomon, 2007; Nardi, 2008). Yet, the traditional lecture is the most common approach to teaching mathematics in higher education as new lecturers typically teach the way they were taught (Felder, 1993; and see also Chapter 5).

In many universities now, new lecturers are inducted to university teaching via some pedagogical course taught by a university teaching centre. Often the pedagogy of such a course is rather general since it applies to lecturers in all subjects. We know mathematics lecturers who disdain general pedagogy because they cannot see how it applies to teaching mathematics. The mathematics discipline requires students to learn the nature of mathematics with its characteristics of abstraction and formalism and the centrality of proof, however, it is not clear what are the best ways of supporting this learning. Often lecturers cannot see beyond the formal traditional ways – the ways they were taught themselves – of presenting their subject in a lecture theatre. Those who are keen to find alternative ways of teaching their students sometimes look for advice from the literature. We were fortunate enough to work with a mathematician who was in this position.

Mathematician *M*, in the early years of his teaching career was asked to teach the first semester of a two-semester module in linear algebra to a first year cohort of (200+) mathematics students. In his first delivery of the module, he followed very closely the way the module had been taught by his predecessor, but was very dissatisfied with the way it went and the outcomes for his students. He had read an article in the professional literature (Uhlig, 2003) which inspired him to redesign his teaching in order to achieve better outcomes for his students. At this point we asked if we could work with him to study his teaching approach, and he accepted.

A research study of the teaching of linear algebra

The research was qualitative, interpretative and developmental. We were outsider researchers and *M* was an insider researcher exploring his own practice and learning from his inquiry. We observed (and audio-recorded) all of *M*’s teaching sessions (lectures and tutorials) and recorded extensive conversations with him before, after and between teaching

sessions. We collected and analysed data formally and rigorously, and developed theory (see Jaworski, Treffert-Thomas & Bartsch, 2009; Thomas, 2012). *M* reflected on his teaching in conversation with us and considered modifications to his teaching as a consequence of his reflection. During this process, we recommended some articles from the literature that could be useful to him in reflecting upon and (re)designing his teaching (for example Dorier, 1998, on the role of formalism in the teaching of linear algebra). On several occasions he read and later commented on these articles in our research meetings.

As a normal part of a research study we explored the relevant literature in both research and professional domains. We discuss here, briefly, key items from the literature and explain the ways in which they informed our study of the teaching and learning of linear algebra.

There were some recurring themes within the body of the literature that we looked at. Some articles focused on the content matter that students ought to learn and hence lecturers needed to teach (Carlson, 1993). Some authors discussed students' difficulties with linear algebra (Dorier, 2000) and what could be done to address these (Hillel, 2000). Some authors presented ideas for developing resources, in particular incorporating technological tools and devices in teaching (Sierpiska, 2005; Berry, Lapp & Nyman, 2008), while others discussed different approaches to teaching (Dorier, Robert, Robinet & Rogalski, 2000a; Uhlig, 2002, 2003) including more student-centred approaches (Uhl, 1999).

We discuss each of these aspects in more detail. Since these were situated in the subject area of linear algebra, this discussion is by nature subject specific to linear algebra. However, we hope that the reader can see that this discussion has relevance in other areas of mathematics and might adapt our approach to research in their own field.

Content

When teaching any area of mathematics lecturers will consider what their students will bring into the classroom by way of previous knowledge. Any changes in the post-16 school curriculum will therefore impact directly on a first year course at university. Since linear algebra is not usually taught at pre-university level (at least not in the UK where this study took place) the subject matter should be new to all students. Having said this there are some aspects such as matrix operations, calculating determinants and eigenvalues and eigenvectors that some students in the UK may have studied at school or sixth form college when taking an additional A-level course in mathematics. At our university this additional knowledge was not assumed and hence linear algebra was designed as a course to be taught 'from scratch' by lecturer *M*. It is fair to say that most mathematics undergraduate courses will offer new content for students as well as new ways of looking at and working with mathematics.

There are many examples in the literature where authors discuss the subject matter to be taught in a first year linear algebra course. Our study of the literature included an early publication by the LACSG, the Linear Algebra Curriculum Study Group, set in a US context. The group drew together the concerns raised over students' ability to cope with an undergraduate course in linear algebra as a result of changes in the school curriculum (Carlson, Johnson, Lay & Porter, 1993). The group suggested a core syllabus for a linear algebra course that consisted of matrix algebra, the solution of linear equation systems, determinants, and eigenvalues and eigenvectors, as well as the fundamental concepts of

subspace, basis, span, linear dependence, and so on, in the context of the vector space $\mathbb{R}^n$. They also stated explicitly that students should be introduced to problem-solving situations.

This list represented what mathematician M himself considered important. Furthermore this list of topics is shared by many authors in the literature and appears as the content page in many textbooks on linear algebra (for example Poole, 2006). Thus we concluded that linear algebra was one of those topics that had a universally accepted character in terms of content. This is not necessarily the case in other areas of mathematics.

The main disagreement in the literature, if we can call it that, surrounded the context in which linear algebra concepts were to be first introduced. Some authors preferred the most general and abstract vector spaces while others preferred to reduce the context to the vector space $\mathbb{R}^3$ only. The reasons lay with the target audience: engineering students, for example, might not require the more abstract theory so that teaching it may not only be unnecessary but also hinder students' progression and understanding (Stewart & Thomas, 2009).

One important source of information when first teaching an undergraduate course at university are literature reviews. These are articles that synthesise research in an area of mathematics teaching and learning, and not only present research outcomes and conclusions but invariably contain information regarding content (for example Britton & Henderson, 2009, for linear algebra).

Students' difficulties

When considering students' learning of linear algebra almost all authors agreed that students find linear algebra a difficult subject to study at university (Carlson, 1993; Hillel & Sierpinska, 1994; Dorier, Robert, Robinet & Rogalski, 2000b; Berry et al., 2008). Authors also agreed on the sources and the kind of difficulties students experience with linear algebra. This may not be the case with all mathematical topics but the difficulties mentioned were identical to the ones that mathematician M identified in his own students.

Students typically are not accustomed to working with a theoretical framework that is as highly conceptual in nature as linear algebra. The body of formal and interrelated knowledge requires abstract ways of thinking and behaving that M 's students had not met before they arrived at university. It is the first time that students must learn about axioms and algebraic structures such as vector spaces. It does not fit well with students' previous experience of mathematics at school where knowledge of computations and procedures often sufficed to solve the problems presented. It is also a time when students are required to learn a whole new language to express the ideas in linear algebra, "*comparable with learning a foreign language*" (Artique, Chartier and Dorier, 2000:256).

Dorier (2000) presented a comprehensive account of students' cognitive difficulties with linear algebra and included a historical overview of the development of the subject. Linear algebra is a tool and a body of knowledge that arose from considering problems of linearity. Dorier argued that students encountered the same difficulties that mathematicians had in creating linear algebra as a unifying theory. He argued that what took mathematicians years to develop, students were expected to accomplish in the period of a few months, but the intellectual leaps required were the same.

Sometimes lecturers notice in their students the kinds of difficulties described above. Sometimes the brevity of lectures and the brevity of contact make it difficult to get to know students. Literature can make the lecturer or teacher of mathematics aware of the kinds of difficulties that can arise so that he or she may be more prepared and more able to respond in a manner that helps students to move forward.

Mathematician M noticed these difficulties with students that he taught in a small group (usually a maximum of six to eight students), a particular feature of the first year structure at our university. Here he noticed some severe difficulties that students seemed to have with linear algebra. He concluded that his teaching approach, a deductive style, stating definitions and theorems first, followed by a proof, hindered students' progress and decided to change his style.

He discussed this with us, the researchers, in our weekly meetings. As time went by he became well versed in the literature into students' difficulties with linear algebra and used this knowledge to good effect in lectures, for example spending more time discussing carefully chosen examples, both with individuals and collectively.

How to overcome students' difficulties

There were many examples in the literature that suggested ways of addressing students' difficulties with linear algebra.

Dorier and Sierpinska (2001) commented on lecturers' use of multiple representations such as geometric explanations of subspace alongside algebraic understandings. They argued, as did Hillel (2000), that students were not able to follow the switching between representations, often losing the thread of the explanation altogether. They suggested that lecturers should make their students aware when switching between representations.

Geometric reasoning and visualisation in linear algebra were at the heart of many articles in the literature. This is hardly surprising as linear algebra and geometry can be seen as two sides of the same coin. However, research with students found that geometric insights can both help and hinder students' developing understanding (Gueudet-Chartier, 2004). If studying linear algebra involved a context up to $\mathbb{R}^3$ only, then students benefited greatly from geometric insights (Stewart & Thomas, 2009). When abstraction beyond $\mathbb{R}^3$ became necessary, geometry and visualisations hindered weaker students who took the geometric language of linear algebra too literally (Hillel, 2000). At the same time it helped stronger students who became more secure in their knowledge through making the connections between $\mathbb{R}^3$ and abstract vector spaces (Sierpinska, 2005).

Our discussion in the paragraph above provides an example where the literature points to ways of teaching that depend on the body of students and the course to be taught. It requires the lecturer to make an informed decision on the most appropriate action in light of the students that he or she will be teaching.

The examples in the literature vary greatly and suggest many possibilities beyond mathematical content and delivery. Some authors emphasise everyday language and encourage journal writing for students as a means of engaging with the concepts in linear algebra (Hamdan, 2005). Another avenue suggested was to develop teaching that involved the use of technology and technological devices. Hillel (2001) reported on the use of a

computer algebra system for exploring the concepts of linear algebra. Using software often involved changing the learning context from whole-class teaching to lab-style tutorials, with students working on computers. This is closely related to teaching styles and teaching approaches which we discuss in more detail below.

Teaching style

One more resource in dealing with students' difficulties lies with approaches to teaching and teaching style. The traditional way to present mathematics in lectures is what we call 'top-down' – definitions and theorems are presented first, followed by examples and exercises. There is little opportunity for students to build own understandings since all theory is presented to them and for them. Some authors advocate this style as their preferred method. For example, Wu (1999) compares lecturing to a performance on stage aimed at capturing students' imagination. Pritchard (2010) also favours teaching via lectures. He states that lectures, when delivered well, present effective communication of mathematical ideas and furthermore, that students like them.

Students at university represent a diverse body and are probably more diverse now than has ever been the case. It is unlikely that one approach will fit all. We find examples in the literature that describe alternatives to the traditional lecturing style. These include 'bottom-up' approaches that stress intuition and exploration before the formal theory of linear algebra is introduced. Mathematician *M* fell into this category. He redesigned his teaching in order to engage students more, and more actively in lectures. As we have said, he was inspired by the article written by Frank Uhlig (2003), a mathematician who advocated an essentially 'bottom-up' approach to teaching linear algebra based on posing questions such as 'What happens if...?' 'Why does it happen?' 'How do different cases occur?' Uhlig used these questions to make students think about the mathematics and recognise generalities for themselves leading to a more gradual understanding of the generalisation and abstraction processes in mathematics. Uhlig's approach relied heavily on matrix algebra which was not a central feature in mathematician *M*'s approach. For his own teaching *M* developed a structure that used *examples* as starting points from which to introduce concepts and theorems in linear algebra. This involved students working on the examples first by themselves or in small groups within a lecture in order to 'develop a feel for what is going on' as *M* often used to say (see Thomas, 2012).

Our aim in presenting these examples of teaching is to encourage the reader to reflect on their own practice and to open up possibilities for acting in the classroom, as indeed we do ourselves. The suggestions are not intended to be prescriptive. They are certainly not exhaustive. But these themes and suggestions recurred as themes in the literature in relation to other undergraduate mathematics courses (more of this later). We hope that the reader may take inspiration from some of them.

In following up on some of the ideas in the literature, *M* came to understand students' difficulties better. He spent more time repeating explanations or using different words as Uhlig had done. This was probably the first and most immediate result from his reading. Understanding of this kind can motivate a teacher to find different explanations or approaches to teaching a topic.

The literature in other areas of mathematics and its learning/teaching

As well as literature relating to linear algebra, we have found literature regarding the learning and teaching of students in calculus, abstract algebra, group theory, differential equations, vector calculus, analytic geometry, functions in two variables, and proofs. Much of this literature is professional literature, although there is some that reports research. Some literature is relatively old, showing that mathematicians have been addressing teaching and learning issues for many years. For example, Freedman (1983:641) described a way of teaching *abstract algebra* that included making students “*play a more active part in their own education*” echoing statements made by the lecturer in our linear algebra study.

Several examples show that issues raised in relation to linear algebra are issues more generally. Staying within the context of abstract algebra, Cullinane (2005:339) expressed a desire “*to connect what [students] are learning to their prior experiences ... in high school*”, recognising a lack of connection between school mathematics and mathematics in higher education. Other issues concern students’ engagement with abstract algebra and the value of communication on a number of levels. Miller and Madore (2004) gave examples of in-class group projects while Warrington (2009) provided students with a practical out-of-class assignment as an introduction to abstract algebra. Leganza (1995) described the use of three specific writing assignments in an abstract algebra course. Several authors considered the use of technology in their teaching. Perry (2004) described his teaching using software entirely throughout the course while Leron and Dubinsky (1995) used interactive programming activities with students. Blyth and Rainbolt (2010:217) described three examples that were designed to help students “*discover classical theorems*” (in abstract algebra) “*with the help of leading questions and the software*”. Comparing these sources with those we have found relating to linear algebra, we see that the themes are the same: connecting with previous knowledge, engaging with interactive or group work, use of technology, written assignments, use of questions and questioning.

In relation to set theory, Padraig (Padraig & McLoughlin, 2010) discussed an inquiry-based approach to his teaching combined with a method called the Moore Method involving questions and questioning (which we discuss further below). He showed why and how he used this approach to teach students the beginnings of set theory. He argued that his pedagogic approach was equivalent to the approach that he took in his own research in mathematics. This echoes sentiments expressed by Uhlig (2002, 2003) in relation to linear algebra.

Some literature is more mathematically focused with discussions centering around the mathematics. Quinn and Rai (2010), for example, discussed the mathematics behind the form of the solution to differential equations and provided some reasoning for the ‘method of variation of parameter’. Several articles in *The American Mathematical Monthly* presented discussions of the *mathematics* in an approach rather than the *pedagogy* of the approach. For example, Ebanks (2012) about the Mean Value Theorem, Thomson (2007) about the Riemann and Lebesgue integral (and how another integral altogether would make more sense for teaching and for students), O’Bryant, Reznick and Serbinowska (2006) on the limit of a sum, Cobb and Moore (1997) about the mathematics of statistics. Some articles

include the historical development of a piece of mathematics such as a theorem or a conjecture which provides a rich starting point for thinking about teaching.

Most of the examples referenced above form part of what we have called the professional literature relating to mathematics; they have been published in just a few journals such as *Primus*. There are other publications and periodicals that have *mathematics* as their main focus but will from time to time include articles that relate to teaching and learning issues in mathematics. Examples include publications by the Mathematical Association of America, the European Mathematical Society and The Institute of Mathematics and Its Applications.

We turn now to literature in the domain of mathematics education which, largely, draws upon research to discuss issues in and approaches to learning and teaching mathematics and develops associated theory. A seminal work, published in 1991, draws together key elements of the work of researchers in *Advanced Mathematical Thinking* (AMT) (Tall, 1991). This book is special in a number of ways. It emerged from a working group at the annual PME (International Group for the Psychology of Mathematics Education) Conference focusing on AMT, it had a strong emphasis on psychological theory linked to mathematical learning, and it reported research in areas of AMT. The cognitive theory on which ‘mathematical thinking’ is based, derives from the work of Jean Piaget (for example Piaget, 1950; Piaget & Inhelder, 1973) and underpins psychological constructs such as *concept image*, *concept definition* and *reflective abstraction* which provide insights into ways in which learners construct mathematical concepts. It takes seriously the difficulties that students face in learning mathematics, with discussion of *cognitive obstacles* which impede a conceptual appreciation of mathematics: these are mathematical concepts which challenge the learner’s intuition, for example the notion of *limit* which (some) students see as an approximate process in which the limit is never attained. It discusses research into students’ mathematical thinking in areas such as functions, limits and analysis, recognising the centrality of generality, abstraction and proof. For example, taking linear algebra as an example Tall (1991:11) writes:

“The terms ‘generalization’ and ‘abstraction’ are used in mathematics both to denote processes in which concepts are seen in a broader context and also the products of those processes. For instance we generalize the solution of linear equations in two and three dimensions to n dimensions and we abstract from this context the notion of a vector space. In doing so two very different mental objects are produced: the generalization $\mathbb{R}^n$ and the abstraction, a vector space V over a field F .”

Tall (1990:11)

Because of the fundamental nature of the mathematical concepts and the psychological theory, most of this book is still relevant today for those who teach mathematics at this level.

The importance of mathematics learning and teaching in higher education was further recognised at the end of the 1990s with an international study conference sponsored by the International Commission on Mathematical Instruction (ICMI). Following the conference a ‘study volume’ was published entitled *The Teaching and Learning of Mathematics at University Level* (Holton, 2001). Papers in the volume were based on presentations from the conference and included both professional and research contributions. In the opening chapter, Claudi Alsina addresses, “*Why the professor must be a stimulating teacher*”. The author critiques a

range of “myths” about teaching, such as the top down approach, deductive organisation, ‘researchers make the best lecturers’, and goes on to discuss “*a new paradigm of teaching mathematics at university level*” (Alsina, 2001:7). He writes that university teaching of mathematics should be taken as seriously as is research in mathematics, that teachers can benefit from efficient training and that innovative teaching approaches, including use of technological tools, will have positive effects on learning. The book is wide ranging, written by mathematics educators and mathematicians, and addressing research, teaching, assessment, technology, teacher education, and a range of mathematical topics such as linear algebra and calculus. As well as teaching future mathematicians, it addresses also the teaching of mathematics to future engineers and scientists.

We end this section with reference to two important areas of pedagogy in the learning and teaching of mathematics in higher education. They are important because they have been influential among some mathematicians in offering different ways of thinking about encouraging students’ conceptual engagement with mathematics.

The work of Robert Burn focusing on approaches to teaching related to promoting students’ learning of the concepts in focus

Burn was a mathematician and mathematics educator in Cambridge and then in Exeter. He proposed “*A Pathway*” into an area of mathematics in focus, for example, “*A pathway into number theory*” (Burn, 1982). A ‘pathway’ is a bottom up (rather than top down) approach; it consists of a sequence of problems with which students would engage and through which a student would climb towards the standard theorems in the mathematical topic. Burn (1982:1) writes, “*Time and again, it was the exploration of special cases which illuminated the generalities for me. ... [the pathway] follows more closely than usual the natural process of discovery and puts logic in its proper place*”. Burn quotes Hadamard as having said, “*The purpose of rigour is to legitimate the conquests of the intuition, and it never had any other purpose*” (Burn, 1982:1).

In Burn’s book on number theory, the first chapter consists of 67 problems followed by ‘Notes and answers’ and a ‘Historical note’ and this sets the pattern for future chapters. The problems of Chapter 1 take the learner through the division algorithm, the Euclidean algorithm, unique prime factorisation, the infinity of primes and Mersenne primes, proving theorems as part of the building process. Lest the reader thinks that there is something special about number theory, for which this approach might work differently than for other areas of mathematics, Burn’s two further volumes (1985, 1992) offer “*A Pathway to Geometry*” and “*Steps into Analysis*”.

The Moore Method

This section is written with the help of a personal communication from Chris Sangwin, who has used the Moore Method in his teaching and written about its use (for example Sangwin, 2011). It is adapted from an unpublished document (Sangwin, 2013) entitled “*Choosing Problems for a Moore Method Course*”.

1 A copy of which can be obtained from the author via email: C.J.Sangwin@lboro.ac.uk.

The Moore Method is a type of enquiry based learning (EBL) developed by the influential Texan topologist Robert Lee Moore (1882–1974) for university mathematics courses. The central tenet of the Method is learning by problem solving, where students answer questions that accumulate into theorems and lemmas. The essence of the Method is to engage students via the following process:

1. Mathematical problems are posed by the lecturer to the whole class.
2. Students solve the problems independently of each other.
3. Students present their solutions to the class, on the board.
4. Students discuss solutions to decide whether they are correct and complete.

Solutions are not imposed by the lecturer, who acts as facilitator by offering comment and guiding discussion. Moore is quoted as saying (Parker, 2004:vii) *“That student is taught the best who is told the least”*. As Mahavier (1999) said of Moore: *“Moore helped his students a lot but did it in such a way that they did not feel that the help detracted from the satisfaction they received from having solved a problem. He was a master at saying the right thing to the right student at the right time.”*

Colleagues varied aspects of the Method, with some encouraging students to work as a group, both answering questions and formulating research topics of their own. Such teachers encouraged alternative solutions to be presented and discussed, helping students refine their sense of aesthetics, and providing other strategies for subsequent problems. In all forms, a key aspect is that it is the students’ responsibility to solve the problems for themselves. And, in all versions, *the group* criticises these solutions and ultimately, together with the teacher, decides if a solution is complete and correct.

The pedagogical literature

In what we have written above, we have provided examples of literature which addresses pedagogy in mathematics learning and teaching in higher education. What we are calling *pedagogical literature* goes beyond the exposition of approaches to teaching and learning, or research findings in teaching and learning, to discuss the nature of teaching and learning and to analyse ways in which teaching can be designed and improved to enhance mathematical learning opportunities for students. This literature draws on both professional and research literature and there are elements of it in the sources discussed above (Burn, 1982; Tall, 1991; Holton, 2001; Sangwin, 2011).

Fundamentally, pedagogic literature analyses approaches to teaching mathematics in higher education, discusses how such approaches do or can contribute to students’ learning of advanced mathematics and provides guidance to those teaching at this level. The authors of such books or articles are often mathematics educators who teach advanced mathematics and who reflect critically on the nature of their teaching and how it develops. Often they work closely with mathematicians and are able to take mathematicians’ views into account in their analyses. The following are some examples of what is available, not a comprehensive list.

In *Fundamentals of Teaching Mathematics at University Level*, Baumslag (2000) presents his perspective on what it means to teach at university level and ways in which those teaching at this level can/should think about teaching. The book ranges over issues related to systems of education, approaches to teaching, organisation and management of teaching and assessment, and advice for the lecturer. Thus it deals with the practice and processes of teaching within a general framework of the demands of the system and principles of education.

On the other hand, Khan's (2001) *Studying Mathematics and its Applications* is a study guide for undergraduate students studying mathematics as part of a degree in mathematics, science or engineering. The author writes, "*We will focus here exclusively on the distinctive approaches needed to study mathematics and its applications. What you will not find here is more general advice on time management, positive thinking and so on*" (Khan, 2001:xiii). Since this book is written for students, it might be asked why we consider it as pedagogic literature; the reason is that as teachers we are fundamentally concerned about the learning of our students and any approaches by which students can learn more effectively are likely to be of interest to us. For example, in a section labelled "*Skills*" the author addresses, using examples, thinking visually, and coping with symbols; under "*Tasks*", he focuses on solving problems, applying mathematics and constructing proofs. We suggest that all of these constitute areas in which we should like our students to gain understanding and fluency and therefore it is valuable to be better informed on how to encourage students in these areas of their study.

In a book entitled *Mathematics Teaching Practice* Mason (2002) writes explicitly for lecturers at university or college level. He takes the perspective that "*mathematics has to be learned by actively engaging with it*" (Mason, 2002:v). For this purpose he recommends a set of "*tactics*", specific acts on the part of the lecturer designed to "*stimulate students to take the initiative, to act upon the mathematical ideas and make sense of them and not just attempt to master a succession of techniques*" (Mason, 2002:1-2). Mason encourages readers to reflect on their own teaching and identify what they do, how they do it and in particular what they value about the teaching process. The book is organised around aspects of teaching, such as lecturing, tutoring, constructing tasks, marking and commenting, offering and discussing tactics in all of these areas. However, the first chapter relates to the students: it focuses on "*Student Difficulties with Mathematics*" highlighting the kinds of difficulties that lecturers typically notice and offering tactics to address them. Tactics include *using common errors; specialising, generalising and counter examples; boundary examples; say what you see; reconstructing definitions*; and many more. In the chapter on *Lecturing*, tactics include *talking in pairs; pausing; presenting the essence of a theorem; invoking mental imagery; being human*; and many more. Tactics are explained and discussed and readers are drawn into thinking about how they might themselves use such tactics with their own students. The book as a whole is a pedagogic resource which might well form the basis of a training course for new lecturers in mathematics.

Ideas from Mathematics Education (Alcock & Simpson, 2009) is also written explicitly for those teaching undergraduate mathematics with "*an accessible introduction to some ideas from mathematics education research*" (Alcock & Simpson, 2009:1). The authors focus explicitly on Definitions, Mathematics Objects, and Reasoning Strategies, relating these to findings from research and related areas of theory such as Concept Image and Definition,

APOS (Action-Process-Object-Schema) theory, and Semantic and Syntactic Strategies. Explaining research findings and exemplifying theoretical perspectives, the authors address issues in learning and teaching with guidance as to how these inform those teaching at this level.

Nardi's (2008) *Amongst Mathematicians, Teaching and Learning mathematics at University Level* is a rather different sort of book from those mentioned above. It presents the results of several years of research in which the author has worked closely with and studied the practices of mathematicians as teachers of mathematics along with the learning of their students. The book is presented as a series of dialogues between a *mathematician* and a *researcher in mathematics education* (RME). The two together focus on data from mathematics learning and teaching sessions and dialogues between students and their tutor. For example, on page 172, the focus is on the duality of *function as a process* and *function as an object*. They focus on a piece of dialogue which addresses the importance of *domain* when constructing the *inverse* of a function, and the inverse function theorem. The mathematician presents the mathematical thinking behind the theorem while RME plays a 'devil's advocate' role representing the student. "*So here I am, solving a differential equation and, until now whenever I solve an equation, the answer is a number. And all of a sudden, you are asking me to solve an equation and the answer is a function, or a family of functions. I wouldn't be sure what you mean! Functions are formulas I use to evaluate y in terms of x . They are not solutions to equations*" (Nardi, 2008:172). The reader's attention is drawn to the student perspective within the formality of mathematical exposition and thinking; this enables a broadening of the teacher's perspective, beyond the mathematics, to the leaps of conceptualisation required of students.

We end this section with reference to some activity and research at our own university focusing on the teaching of mathematics and on developing a discourse on mathematics teaching. Based on the idea that we learn from each other in discussion of mathematics and the ways we teach it (the focus of many of the books mentioned above) we instituted a series of seminars called the 'How we Teach' seminars. In each seminar, one teacher of mathematics (in some cases a mathematician, in others a mathematics educator) gave a presentation of some aspect of his or her teaching, and the thinking associated with it. Those attending listened, raised questions and discussed issues arising from the presentation. Through this activity, there were two important outcomes: first, we all gained from hearing a colleague's experiences and discussing issues, enhancing our own thinking and possibilities for practice; second, we were brought closer as a community, sharing perspectives, trusting each other, and supporting less experienced colleagues. The seminar series was accepted within the University's New Lecturers Course, as a medium through which new mathematics lecturers could demonstrate their growth of understanding about teaching – preparing and offering a seminar and reflecting on its outcomes, as well as reflecting on other seminars attended. The first 20 of these seminars were video recorded and 10 of them analysed to elicit aspects of what was presented and what was discussed. Several papers resulted from this analysis (for example Jaworski & Matthews, 2011; Matthews & Jaworski, 2012).

In conclusion

This chapter has been about the literature that is available to lecturers and others teaching mathematics in higher education and has suggested ways in which readers might use this literature to inform their own teaching. We have distinguished particularly the professional, research and pedagogic literature. Each of these has something specific to offer. From the professional literature, we gain insights into the ways in which other teachers/lecturers have thought about their teaching and the approaches/strategies and frameworks they have used. The research literature offers us *what is known*, findings from research that can enable more informed approaches to teaching. The pedagogical literature deals overtly with developing and enhancing teaching, through the lecturer engaging with new ideas (for example those offered in the professional literature), attending to research findings or using specific tactics or teaching approaches recommended by the authors. We have also referred to some of our own experiences in using the literature in a research study with a mathematician teaching linear algebra, and in working with colleagues to generate a pedagogical discourse and increase awareness of teaching processes. In the end, we hope to have shown that there is a wealth of literature on which teachers/lecturers can draw to guide and inform their teaching. While this can be an individual initiative, it can be especially fruitful to join with colleagues to read and discuss what is offered.

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Chapter 19:

Embedding a scholarly element into your teaching

John O'Donoghue & Olivia Fitzmaurice

Abstract

Educational and economic priorities have forced a reappraisal of mathematics teaching at all levels of the education system in many developed countries in recent years. Mathematics teaching at all levels has been found wanting and heavily criticised for badly serving the educational needs of students and the economic requirements of developed economies.

This chapter deals with aspects of this reality by focusing upon university mathematics teaching and how it might be improved through direct practical action by practitioners viz. university mathematics teachers. Our contention is that a scholarly approach to university mathematics teaching will result in an improvement in outcomes for students. Further, a characterisation of a scholarly approach to university mathematics teaching is developed using different perspectives of the role. It is argued that the tools offered by these different perspectives can be harnessed by university mathematics teachers to embed a scholarly approach to their own teaching. A case study encompassing an important aspect of university mathematics teaching is presented for the reader's consideration.

Introduction

Background and context

Many developed countries, including the UK, USA, New Zealand, Australia and Ireland, acknowledge a 'mathematics problem' (London Mathematical Society, 1995; Committee on Undergraduate Science Education, 1999; Task Force on the Physical Sciences, 2002). A problem that appeared initially as a localised problem at the transition from school to higher education, that is the mathematical under-preparedness of students entering higher education, has emerged as a system-wide challenge of major proportions. It is the case that many students, who enter higher education in these developed countries are mathematically under-prepared for the demands of their mathematics courses and in their

future careers (Expert Group on Future Skills Needs, 2008; Vorderman, Porkess, Budd, Dunne & Hart, 2011).

This is a significant problem for governments and their policy makers in economic and education ministries in the developed world because mathematics is an integral part of their economic strategy and central to its success. In these countries, mathematics has been identified as underpinning the knowledge economy and science, technology, engineering and mathematics, the so-called STEM disciplines, are viewed as important pillars of economic policy. Increasingly, this perspective informs and drives mathematics education at all levels in developed countries and it gains an urgent and compelling dimension when it is linked to national competitiveness. However, the educational argument for the study of mathematics in its own right is also compelling and must not be overlooked in the face of economic priorities. Ernest (1991) offers a useful classification in terms of five ideologies including the Old Humanist and Utilitarian perspectives. In this context, the Old Humanist position holds that the study of mathematics is intrinsically valuable while the goal of the Utilitarian position is to provide for the needs of the economy. A comprehensive discussion of the goals of mathematics education is developed in Niss (1996). A concern must be to maintain a balance between mathematics education as a public good and serving the utilitarian demands of modern economies.

Thus one can see that there is a convergence of national economic and education policies leading to a strong focus upon the STEM disciplines, and mathematics in particular as an underpinning discipline. This immediately raises the question of the adequacy and quality of mathematics teaching on both sides of the school-university transition, even if no shortcomings had previously been identified. Research in mathematics education confirms that the quality of the student experience in mathematics is directly related to the quality of teaching (Fennema & Franke, 1992). Improving the quality of university mathematics teaching for STEM students and other students studying components of mathematics as part of their undergraduate programmes is among the current major challenges facing higher education.

The university mathematics landscape

Mathematics education at university level is complex and multifaceted. It does not comprise a single discipline in the traditional sense nor is it a monolithic enterprise. Broadly speaking it includes:

- Mathematical Sciences including Mathematics (Pure and Applied)
- Statistics
- Operations Research
- Others (for example Financial mathematics)

Figure 19.1 (overleaf) is an attempt to capture a useful depiction of the nature of the university mathematics landscape as a backdrop for this discussion.

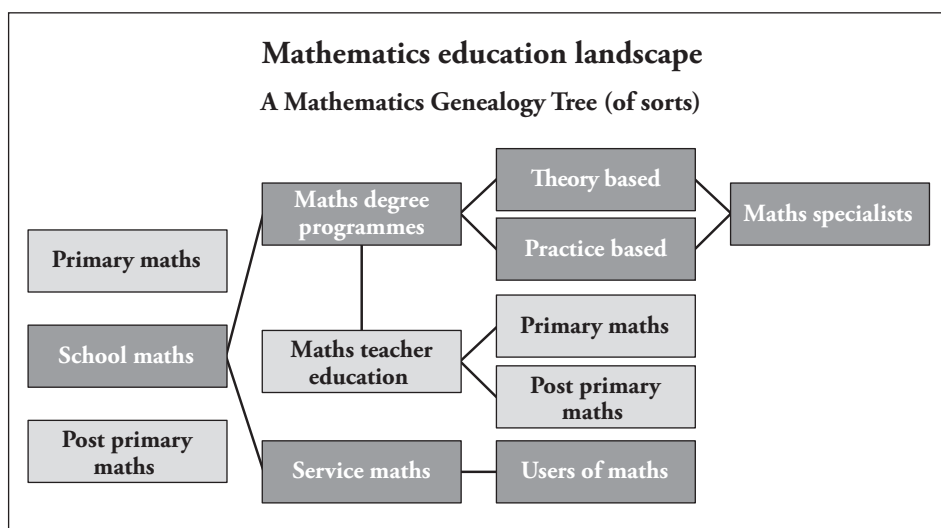


Figure 19.1: A mathematics genealogy.

University mathematics education is aimed at educating and training several kinds of professionals not all of whom are immediately visible. The list below is meant to be informative not exhaustive:

- Professional mathematicians who are generally visible and recognisable.
- Mathematics professionals who are all but invisible in the workforce because their job titles hide their mathematical background and expertise.
- Other professionals who use mathematics such as engineers, scientists and technologists.
- Mathematics teachers.

The challenge across universities generally is to improve mathematics teaching in all its various offerings including single and joint honours degrees, service mathematics and university-led mathematics teacher education.

In this chapter, we offer perspectives and insights on improving university mathematics teaching in a particular area, namely service mathematics teaching, and show how different perspectives and tools from mathematics education can assist individual attempts to improve mathematics teaching in this domain; this was and continues to be, a journey towards embedding a scholarly element to teaching.

Multiple perspectives on the role of the mathematician

Those engaged in teaching mathematics in universities arrive at their task from a number of different perspectives. Here two well known and easily recognisable perspectives, the mathematician and the mathematics educator are investigated for their impact on their owners' practice of mathematics teaching and student outcomes. We argue that explicit awareness of these perspectives is necessary, but not sufficient, to embed a scholarly element into one's teaching at university level. One can treat these perspectives as singly embodied

in a matching number of individuals employed to teach mathematics in the university, or as being overlaid to some extent in a single individual, which is more likely to be the case given the demands in modern universities. There is some value in treating each perspective separately to investigate its nature and what it might offer.

The mathematician

The role of the university mathematician has high visibility even if it is not widely understood by the outside world. The job description, *inter alia*, includes teaching, research and course development. This brings us directly to consideration of the university mathematician as a teacher and course developer. There is a considerable body of literature in mathematics education research that supports the view that effective teachers need to be well versed in their discipline and in other areas such as curriculum and pedagogical knowledge (for example Shulman, 1986; Fennema & Franke, 1992; Rowland, Huckstep & Thwaites, 2005). Teaching and course development are clearly educational matters that are less highly valued or adequately provided for in the mathematician's education and training or ongoing professional development, unlike research training which is prized and not left to the vagaries of 'on the job' experience. It has also been recognised that the research mathematician is not necessarily the best teacher of mathematics and may not even be a good teacher of mathematics. In modern universities the demands of undergraduate mathematics programmes are far removed from the current research interests and papers of researchers (Alsina, 2001). The London Mathematical Society (2010:Recommendation 3) in its *Teaching Position Statement* acknowledges that this gap is widening when it states "*Masters degrees in mathematics should not necessarily be obliged to reach the frontiers of knowledge*".

Mason (2003:7) is also clear that "*Teaching well requires expertise different from that required to be a creative mathematician*". In extreme circumstances the perspective of the mathematician rarely extends beyond the discipline of mathematics, a current research agenda driven by issues other than education, and personal interests and tastes. This perspective barely acknowledges students as individual learners of mathematics with individual learning styles, needs and motivation and directs little or no specific attention to them as learners.

The mathematics educator

The mathematics educator on the other hand, is committed to teaching and learning and developing students as learners of mathematics. The interaction between discipline knowledge and pedagogical knowledge is appreciated as is the role of the teacher of mathematics in effective instruction. This perspective and expertise is often not highly valued in the higher education mathematics community where research prowess is still the benchmark of success. Often there may be little support in university mathematics departments for mathematics staff who have a strong interest in teaching mathematics. These members of staff are left to improve their practice through their own efforts, very often without departmental support and sometimes in the face of active opposition. Some signs of change are evident in this state of affairs as persistent criticism of mathematics education in schools, colleges and higher education by industrialists and other stakeholders has led to initiatives to improve mathematics education in higher education, for example the establishment of Centres for Excellence in Teaching and Learning (UK), Mathematics

Learner Support (UK and Ireland), and opportunities for early career mathematicians to avail themselves of qualifications in teaching on a voluntary basis (UK and Ireland).

Higher education mathematics as a mathematics education task

University mathematics teaching is an important enterprise for educational and economic reasons and represents a significant challenge for all engaged in it. In the face of big challenges it is useful to think ‘outside the box’. But thinking outside the box means getting outside this particular box (traditional university mathematics teaching), and looking into some ‘other boxes’. This activity could also be described as ‘enlarging the universe of discourse’. Mathematics education as a research domain is concerned with issues and problems related to the teaching and learning of mathematics. The suggestion here is that there is an advantage in enlarging the universe of discourse to the field of mathematics education, which includes the discipline of mathematics as a sub-domain with several other disciplines including education, psychology, cognitive science, history, philosophy and others. The advantage lies in the fact that this enlarged universe offers new perspectives, intellectual frameworks, methodologies and tools to address our challenge, namely university mathematics teaching (O’Donoghue, 1997; Gill, 2006). In this context it is beneficial to consider a point made by Heid (2010) and apply it to the teaching of mathematics within higher education. The practice of university mathematics teaching should be infused with research from mathematics education and complete the cycle of practice informed by research and research informed by practice.

We discuss an area of university mathematics teaching in later sections that has been subjected to this kind of analysis. Those treating university mathematics teaching as a mathematics education task can be seen to have added a scholarly dimension to their own teaching. As another example, the reader might consider whether this chapter as written could have been possible without moving outside mathematics as a discipline and adopting a mathematics educator/researcher perspective.

Scholarship and higher education mathematics teaching

Scholarship

The measure of one’s worth and contribution as a professional academic within higher education is based upon scholarship defined by sustained productive engagement in scholarly activities associated with the discipline and the post. It is not uncommon to find a separation between scholarship and research in discussions about the role of academics within higher education. The idea of looking for different origins for scholarship and research is unhelpful as it creates an unnecessary dichotomy between them.

In this context, research is a scholarly activity and may indeed be viewed as the highest form of scholarship for an academic. There are other forms of scholarly activity including university teaching. If university teaching is to be accorded status in higher education then it must be seen as a significant form of scholarship and meet the normal criteria for scholarly activity. This is possible in a context, such as that outlined by Boyer (1990), that sees teaching as a form of scholarship.

Boyer (1990) proposed a model of academic scholarship based on an extended view of scholarship incorporating four forms: scholarship of discovery, scholarship of integration, scholarship of application and scholarship of teaching. All four forms are to be seen as separate but overlapping functions of the model and taken to be integral to it as opposed to being derived from the model. Discovery and application are familiar forms as they correspond to the usual interpretation accorded in higher education, that is original research and its applications. But scholarship also involves integration of new and existing knowledge by finding connections and bridging the theory-practice divide. The work of the scholar also involves communicating their knowledge to students in effective ways, thus extending scholarship to teaching. While Boyer's work was well received in the USA and has there shaped an important ongoing debate, the model is not unproblematic. For example, the model offers no assessment guidelines for determining quality for any form of scholarship, and no compelling characterisation of the scholarship of teaching. Despite these shortcomings, which meanwhile have received considerable attention within the USA debate, Boyer's extended definition of scholarship has much to commend it for higher education globally. Samuels and Patel (2010) applied this conception of scholarship to mathematics support in UK universities to good effect.

Research versus teaching

How should university academics spend their time? The standard accepted response is that they should pursue scholarly activity across three domains identified as teaching, research and administration. Rarely are these domains accorded equal status by individual academics or their institutions; instead a recognisable hierarchy exists in practice that puts research at the top followed by teaching and then administration. This hierarchy influences the relative emphasis academics place on activities in these domains as they go about their careers. This influence finds expression in the research versus teaching debate that exists throughout British and Irish universities and further afield in higher education institutions across the world. However, it has been evident for some time now that teaching is regaining ground in universities as a valued activity, partly because government has funded significant teaching and learning initiatives, for example Centres for Excellence in Teaching and Learning in the UK, National Science Foundation funded science, technology, engineering and mathematics (STEM) projects in the USA, National Centre for Excellence in Mathematics and Science Teaching and Learning in Ireland (NCE-MSTL); and partly because academics themselves, influenced by Boyer's work, have been re-appraising their functions and priorities.

Praxis versus research in university mathematics teaching

While these factors are constants in the debates across all faculties and disciplines, they seem to be accentuated in the STEM disciplines and in mathematics in particular. It is evident in these disciplines that research is viewed as the highest form of scholarly activity. On the other hand, university mathematics teaching in its current state must be viewed as a professional practice carried out mainly by professional mathematicians who do not engage systematically in scholarship and research into teaching mathematics. Mason (2003) has a great deal to say about the practice of mathematics teaching in higher education and

his excellent book should be compulsory reading for all aspiring mathematics lecturers and those seeking to improve their practice.

Generally speaking, high calibre scholarly activity associated with mathematics teaching is accorded low status in university mathematics departments compared to high calibre mathematics research. The relative standing of research is reinforced by practices and activities in higher education such as the Research Assessment Exercise (Higher Education Funding Council for England, 2008) (UK), and the periodic peer review of departments by expert panels (Ireland) where research is a benchmark for success. Institutional and faculty status and funding are greatly affected by these mechanisms within higher education. For these reasons and others, it is unlikely that parity will ever be achieved but one is optimistic that mathematics teaching will gain in status from contributions by leading mathematicians and mathematics educators as mathematics education matures as a field of research, and a better balance in favour of mathematics teaching is achieved in universities.

Discipline knowledge and teaching of mathematics

Scholarship in university mathematics teaching is not a soft option. Firstly, it demands scholarship of the discipline of mathematics as a prerequisite. Strong disciplinary knowledge of mathematics is required for successful teaching but there is an additional disciplinary demand for those pursuing excellence in teaching that involves integrating, synthesising and transforming mathematical knowledge in order to make it available to learners. This in turn demands a sustained intellectual engagement with the discipline of mathematics and with the teaching of mathematics. This debate finds expression in the mathematics education/education literature under teacher knowledge for teaching that in turn lays particular emphasis on subject matter knowledge and pedagogical content knowledge. The position for primary (Ball, Thames & Phelps, 2008) and secondary school mathematics teaching (Fennema & Franke, 1992) is well articulated by various researchers but does not extend to higher education mathematics teaching. However, this work has potential for adaptation and offers a useful framework to apply to university mathematics teaching.

Mathematics teaching and scholarship

Scholarship within a discipline presumes a deep comprehensive knowledge of the discipline by the individual scholar. When the scholarship of teaching university mathematics is the focus of attention, then the individual scholar, in this case the university mathematics teacher, must combine multiple perspectives and expertise as previously discussed. One can postulate that this is not always successful because mathematicians do not fully engage with the mathematics educator role for a variety of reasons that may be associated with motivations and dispositions, their perception of their role as university mathematicians, departmental pressures to undertake research, and policies on promotion. It is unlikely that low-level intellectual engagement with the mathematics educator role will lead to anything but poor teaching due to weak reinforcement of the multiple perspectives. One can further speculate that threshold values need to be achieved in the intellectual engagement with each role/perspective for positive reinforcement to occur. In the main, university mathematics teachers can be expected to surpass threshold values in intellectual

engagement with their discipline but fall short when it comes to intellectual engagement with the teaching of mathematics.

Is it possible to build a profile of appropriate scholarship to meet the required specification? We contend that it is indeed possible and desirable to do so and our profile of scholarship of teaching university mathematics entails:

- Command of the discipline of mathematics acquired through sustained study and scholarship in mathematics.
- The systematic, sustained and considered application of learning and teaching principles to the teaching of university mathematics.
- Expertise in course design, development and evaluation.
- Research in the teaching and learning of mathematics.
- An ability to harness the potential of digital technologies for mathematics teaching and learning.

On a practical level what does scholarly activity in this area entail? This is an important consideration because academic credibility hinges on proposing well defined scholarly activities that are acceptable within the wider academic community and especially to mathematicians. Such a list cannot be exhaustive but must be reasonably comprehensive to provide a sound basis for judgement and comparison, and is offered for another important reason: to give visibility to scholarly activities that give meaning to scholarship in university mathematics teaching.

Scholarly activities aligned with the teaching of university mathematics include the identification and localisation of good practices in university mathematics teaching and learning, leading to better educational outcomes for university mathematics students. Reviewing and updating existing courses, and developing new courses to meet new demands are important activities at university level. The systematic application of relevant educational principles and theories to mathematics teaching is important and will be visible in new approaches to teaching, for example teaching large groups, peer teaching, problem based learning, assessment for learning, and the application of digital technologies. Potentially all these activities may be captured and presented in print and media outputs such as programme descriptors and learning outcomes, and syllabuses, course materials including notes, assessments and projects, and may include the internet and web based sources and materials. Original and applied research in the teaching and learning of mathematics should be pursued and disseminated through normal academic outlets and vehicles including scholarly peer reviewed journals, books and book chapters, and conference presentations.

Mathematics textbook writing is a very significant scholarly activity associated with university mathematics teaching. The UK boasts many exemplary examples of this kind of scholarship including the pioneering work of Bajpai and associates at Loughborough University starting in the 1970s, books by Stroud, and more recent works by Croft and Davison. Newer textbooks integrate digital resources and come with DVDs and web based support.

This list is not exhaustive and one should recognise the concept of scholarly activity continues to evolve as can be seen by considering the case of technology. Digital technology has forced an ongoing re-appraisal of scholarly activity in all disciplines, for example, reflection on teaching mathematics using technology or the new roles for both teachers and learners that have been afforded through the use of technology. Nevertheless, a set of scholarly activities can be identified and used to develop and inform the practice of university mathematics teaching.

Developing a scholarly approach to university mathematics teaching

The university mathematics teacher

Very often the role of the university mathematics teacher is seen as uni-dimensional and concerned only with discipline knowledge and expertise. This is an incomplete picture of a role that is essentially multi-dimensional. Scholarship in teaching mathematics is an essential aspect of the role and the university mathematics teacher must take steps to actively combine in their professional persona and practice the roles of mathematician and mathematics educator. Fortunately, these roles are mutually supportive. Effective mathematics teaching at all levels is based on sound disciplinary knowledge but while the mathematician's command of mathematics is necessary for good teaching, it alone is not sufficient (Fennema & Franke, 1992). The sufficiency condition can only be met in combination with the mathematics educator role. This casts the university mathematics teacher as a scholar practitioner because of the heavy practitioner emphasis of the work that must be informed by serious sustained scholarship. Mason (2003) recognises that creativity in the complementary disciplines of mathematics and mathematics education is very different, but argues that teaching promotes awareness and sensitivity to the structure and history of mathematics taught that in turn has the possibility of informing the lecturer's research in both disciplines. Borovik (2011) adds weight to this argument by instancing specific cases in the development and use of computer packages where teaching and learning has benefited from research in mathematics and vice versa.

The scholarly activity of university mathematics teaching from an educational viewpoint is based upon a comprehensive knowledge of learners, and teaching and learning principles and their application to mathematics teaching at university level. The practice of university mathematics teaching must be informed by research in mathematics education and mathematics. Thus it is clear that scholarly activity in university mathematics teaching encompasses research in the teaching and learning of mathematics. The teaching of mathematics is about *more* rather than *less* mathematical understanding and insight, and must lead to engagement with scholarly activities associated with university mathematics teaching.

Journey from practice to research informed practice: a case study in higher education mathematics

University mathematics teaching through new lenses

As the 'mathematics problem' is one that revolves around the teaching and learning of mathematics in general, perhaps it would make more sense to re-name it the 'mathematics

education problem' (O'Donoghue, 1997). The advantage of this is it allows the researcher/practitioner to view mathematics as a curriculum issue, thus focusing on issues other than content (learning outcomes, assessment and teaching methodologies) in order that problems and challenges can be adequately analysed and addressed. Using mathematics education as a lens it is possible to bring new intellectual tools to bear on the problem, for example, Brousseau (didactical contract) and Bernstein (curriculum theory).

Viewing service mathematics through a mathematics education lens

Within every mathematics classroom an implicit contract exists between a teacher and their students (Brousseau, 1997). The expectations held between students and their teacher/lecturer determine this *didactical contract* within the classroom. Brousseau (1997), Bloch (2002) and Sbaragli (2004) contend that this contract impacts greatly on the teaching and learning that occurs and therefore has a crucial role in the investigation and development of effective conditions for the teaching and learning of mathematics.

Students' perceptions of mathematics can be quite restricted as a result of their school experiences. They are exposed to little in the way of problem solving or applying mathematics, and their mathematical knowledge is often deficient on their departure to third level education (Institute of Mathematics and its Applications, 1995; London Mathematical Society, 1995; O'Donoghue, 1999; Faulkner, Hannigan & Gill, 2010; Gill, O'Donoghue, Hannigan & Faulkner, 2010). Research indicates that the norm for many is a teacher-led classroom; being told what mathematics they need to know, how they need to tackle examinations, and being provided with several, similar examples to practise (Murphy, 2002). This results in a very narrowly focused didactical contract resulting in poor motivation, lack of understanding, and an inability to apply mathematics to solve problems.

The didactical contract can be observed through identification of the tasks lecturers and students participate in during the mathematics class (Bloch, 2002). If all parties are willing and students are mature enough to assume responsibility, the didactical contract can be broken. In this way new situations are created which counter the clauses in their previous contract and thus behaviours are then altered accordingly (Sbaragli, 2004).

We have used this framework to analyse what might be termed 'service mathematics' classes, that is the teaching of mathematics within other disciplines such as science, technology, engineering or business, in Irish universities. One of the aims of this study was to identify the didactical contract between students and lecturers in service mathematics classrooms. What emerged was a narrowly defined contract with very little interaction or learning in lectures and tutorials (Gill, 2006). Lecture style was principally 'chalk and talk' with few resources exploited other than the white/blackboard and overhead projector. There was very little interaction or questioning in lectures with almost no relevant real-life mathematical applications included. There was a distinct mismatch in expectations uncovered between students and lecturers. Where lecturers expected students to assume responsibility for their own learning, students expected lecturers to 'spell out' everything within lectures and relating to examinations, a practice to which they had become accustomed at secondary school. Once the contract details were identified, the information was used to help determine the kind of mathematical and didactical work necessary to

ensure a smoother transition from the second level classroom to the third level lecture and provide students with an enhanced mathematics education.

The concepts of classification and framing were fashioned as vehicles for methodically analysing pedagogic practices (Bernstein, 1996). Classification relates to the selection and organisation of knowledge to be taught and, more specifically, the strength of the boundaries between categories. Strong classification therefore implies strong insulation or rigid boundaries between categories, while weak classification indicates permeability or weak insulation. Pure mathematics, for example, is strongly classified (Fitzsimons, 2002). Robertson (2008) explains that this is because it comprises a specialised language and rule system that must be mastered hierarchically to efficiently work or succeed within the subject. This characteristic separates or insulates it from outside influences; an information silo, so to speak. The Irish Mathematical Society (2008) stresses the importance of relevance within service mathematics modules in order to motivate students. In order to achieve this, they recommend lecturers liaise with client departments and utilise applications relevant to specific disciplines and student backgrounds. The boundaries between service mathematics and other disciplines should therefore be more permeable than those of pure mathematics which is associated with strong insulation.

Framing is concerned with the analysis of pedagogic practice. Thus framing provides a means of describing the communication of knowledge between the teacher/lecturer and the student. This includes a scrutiny of the relationship between lecturers and students and the levels of control possessed by the lecturer and/or students over issues such as selection, sequencing, pacing of content and evaluation/assessment. Framing values may also vary, for example it is possible to have weak framing over selection of communication and strong framing over pacing. The spectrum of framing as it relates to assessment is discussed by Robertson (2008).

Bernstein (1996) defined two types of educational transmission. If classification and framing are weak, there is what he termed “*invisible pedagogic practice*”. In this case, students have more control over classroom practice. In contrast, a situation where classification and framing are strong is referred to as “*visible pedagogic practice*”, where the lecturer has more control over the selection, organisation, sequencing and pacing of knowledge and the rules governing learning are explicit. Bernstein stated that in cases like this, students will struggle to be creative.

These ideas of Bernstein were applied in an Irish study we conducted (Gill, 2006). This qualitative study in Irish universities suggests that as a discourse, service mathematics is less strongly classified than pure mathematics. While there was no external input from industry or employers into content taught or applications used, lecturers of service mathematics interviewed across all seven Irish universities stated that content is negotiated between mathematics and client departments. However we applied the concept of classification to content taught within the modules. It was observed in one particular case where the topics comprised functions, trigonometry, differentiation, exponential laws, linear equations, vectors and complex numbers that the topics are taught in a segmentally organised manner. When one topic is taught, there may be some basic application and then the lecturer will move on to the next topic where there is seemingly no link to the previous one. The ‘walls’ between mathematical topics seem to be impermeable. Despite the fact

the 'strong classification' sounds like a positive thing, in service mathematics it is not. Each mathematical topic should be linked to the next or previous one and also to applications. If students are shown this, they are more likely to understand how mathematics can be successfully utilised to solve problems in other areas.

Service mathematics lecturers were observed to have complete control on what is to be taught, the sequence, pacing and mode of transmission in lectures. Therefore, there is strong framing in service mathematics lectures on all counts resulting in visible pedagogic practices. Tutors and teaching assistants also have sole control over how tutorials are to be taught. Lecturers determine the sequencing of material in tutorials as they follow the same order of the content taught in lectures. The students have more say in the pacing of tutorials. There is strong framing over sequencing and choice of communication but weak framing over pacing. This would result in a pedagogic practice somewhere between visible and invisible.

Bernstein's (1996) and Brousseau's (1997) work allows one to approach the selection and delivery of content in a scholarly fashion. Bernstein's theory provides a method for analysing the 'what' and the 'how' of classroom practice while Brousseau's theory helps one uncover the covert practices which impact on our teaching and student learning.

Service mathematics in our university comprises five cognate strands: Computer mathematics, business mathematics, science mathematics, technology mathematics and engineering mathematics. All strands are lectured by mathematicians from the department of mathematics and statistics and tutorials are taught by postgraduate students from within the same department. Each module consists of 3 lectures and 1 tutorial on a weekly basis in first year, and reduced to 2 weekly lectures and 1 tutorial from the second year onwards. There is annual collaboration at head of department level where feedback is given and received and changes are made accordingly. A mathematics learning centre, (MLC) managed and directed by ourselves, is situated in the department and caters mostly for students from service mathematics programmes. While change within service mathematics modules, based on research conducted by the authors is desirable, the current teaching and liaison arrangements make it difficult to effect change. The authors however are committed to change in their own teaching and application of theory within the MLC.

Expectations and behaviours are discussed between students and staff so the didactical contract is made explicit on day one in the MLC. The Centre endorses a self-directed learning policy and is intended as a support to lectures, rather than a replacement. This is established with all students on their first visit either to the drop in centre or support tutorials.

The support tutorials organised and taught by the MLC are more student-led in terms of the content and pace of delivery and so are more weakly framed than their traditional tutorials. Students on various programmes outline areas of difficulty and this dictates what is taught in MLC tutorials. These tutorials generally have up to 15 students in each class, as opposed to up to 50 in regular tutorials, so lend themselves more to active participation and questioning which is more student-friendly.

Research undertaken in the MLC indicates that these mathematics education strategies to enhance the students' learning experience have been worthwhile and effective (Carroll & Gill, 2012).

Concluding remarks

This chapter invites the reader to accompany the authors on a mathematical journey that is based on our own experiences of teaching university mathematics, reflections on our teaching, and researches in mathematics education over several years. With hindsight it is easy to interpret this journey as embedding a scholarly element to our teaching. The hope is that the chapter will contribute to others' foresight when they make their own choices as mathematicians and mathematics educators.

In addition to the case study presented, the chapter offers a framework for analysis of the role of university mathematics teacher that is a composite of experiences, insights, reflections and research. This, we believe, is a contribution to a scholarly analysis of the role of university mathematics teacher that might be used to develop a case for greater recognition and status for university mathematics teaching and what it entails.

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Transitions in Undergraduate Mathematics Education

When studying mathematics during single or joint honours mathematics courses, undergraduate students experience a number of ‘transition points’ or ‘transition periods’. These are times when many students experience difficulty applying or developing their mathematical knowledge, or adapting to changed learning methods and processes during their higher education studies.

Written to meet the needs of university lecturers, teachers and tutors, this book forms a guide to understanding key issues, good practices and developments in learning and teaching in mathematics within higher education. Each Chapter is focused around an important transition point and written in a style that brings together published and evidence-based literature from across the higher education sector, analysing this in a scholarly manner to identify practical recommendations and ‘tips’ for both new and more experienced higher education practitioners alike.

The Editors

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